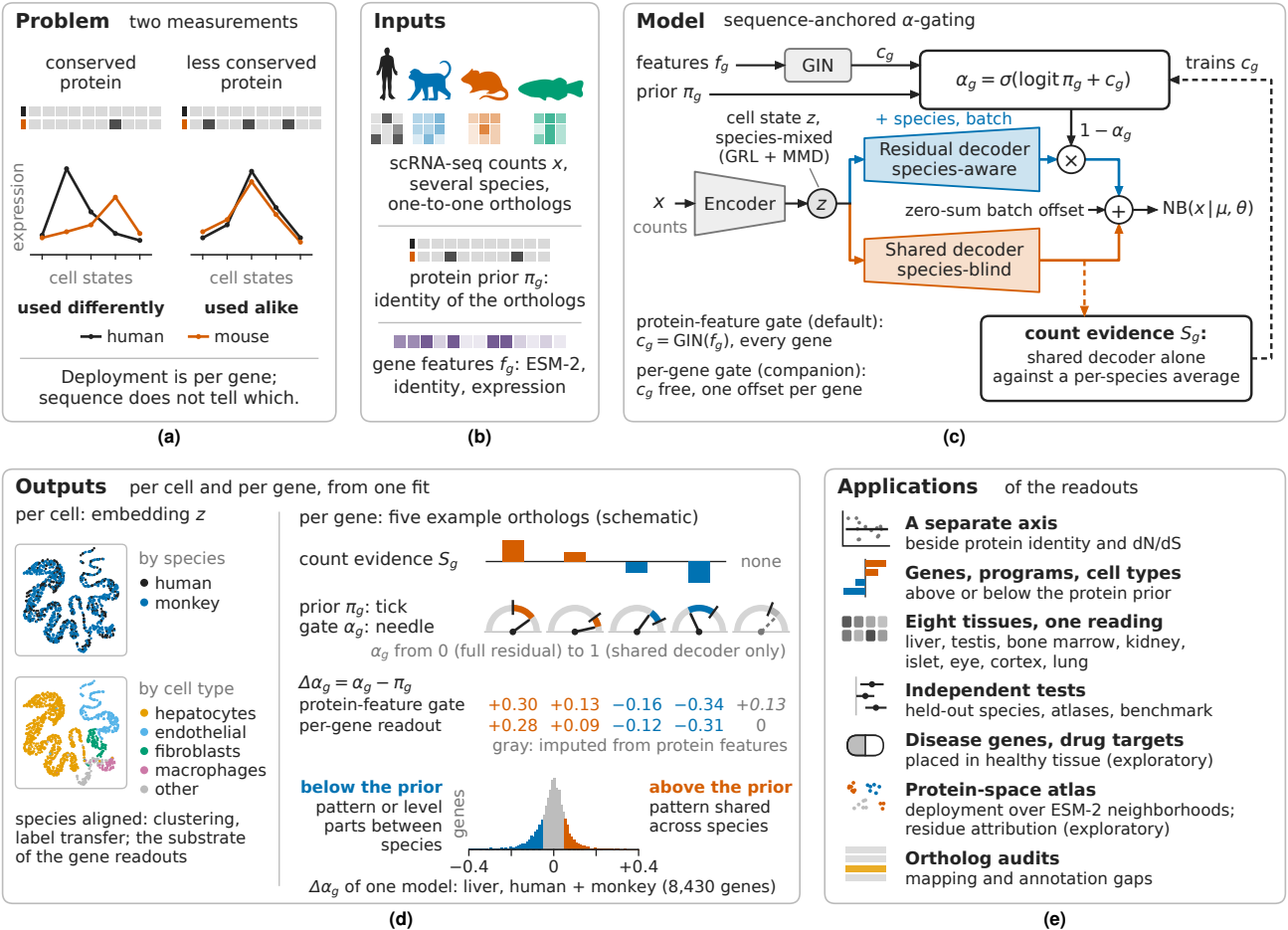
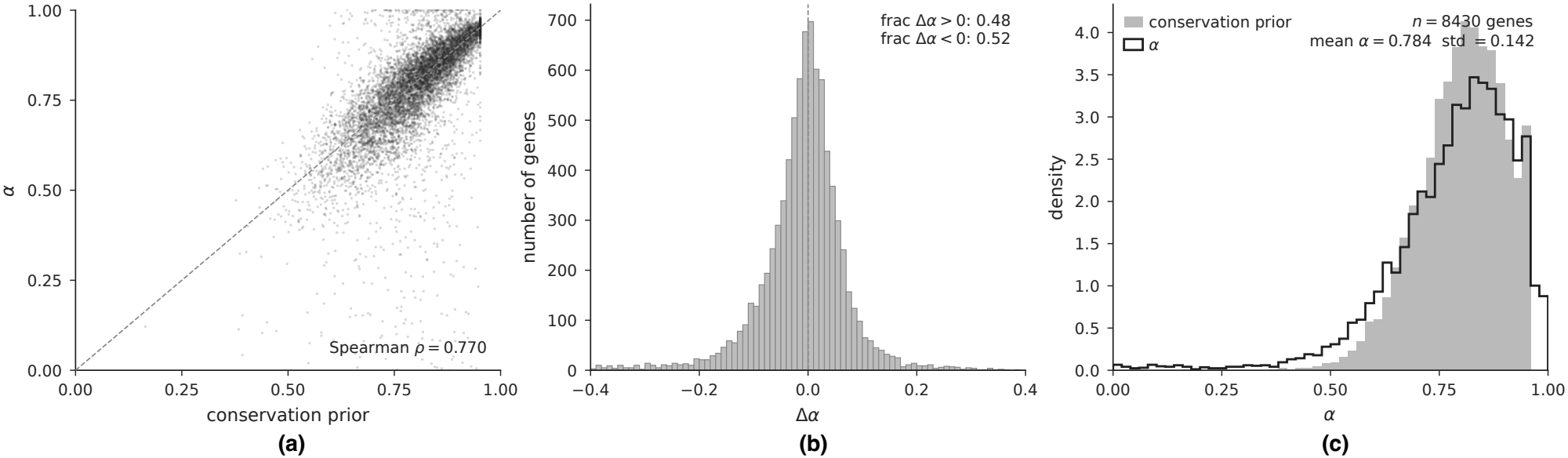
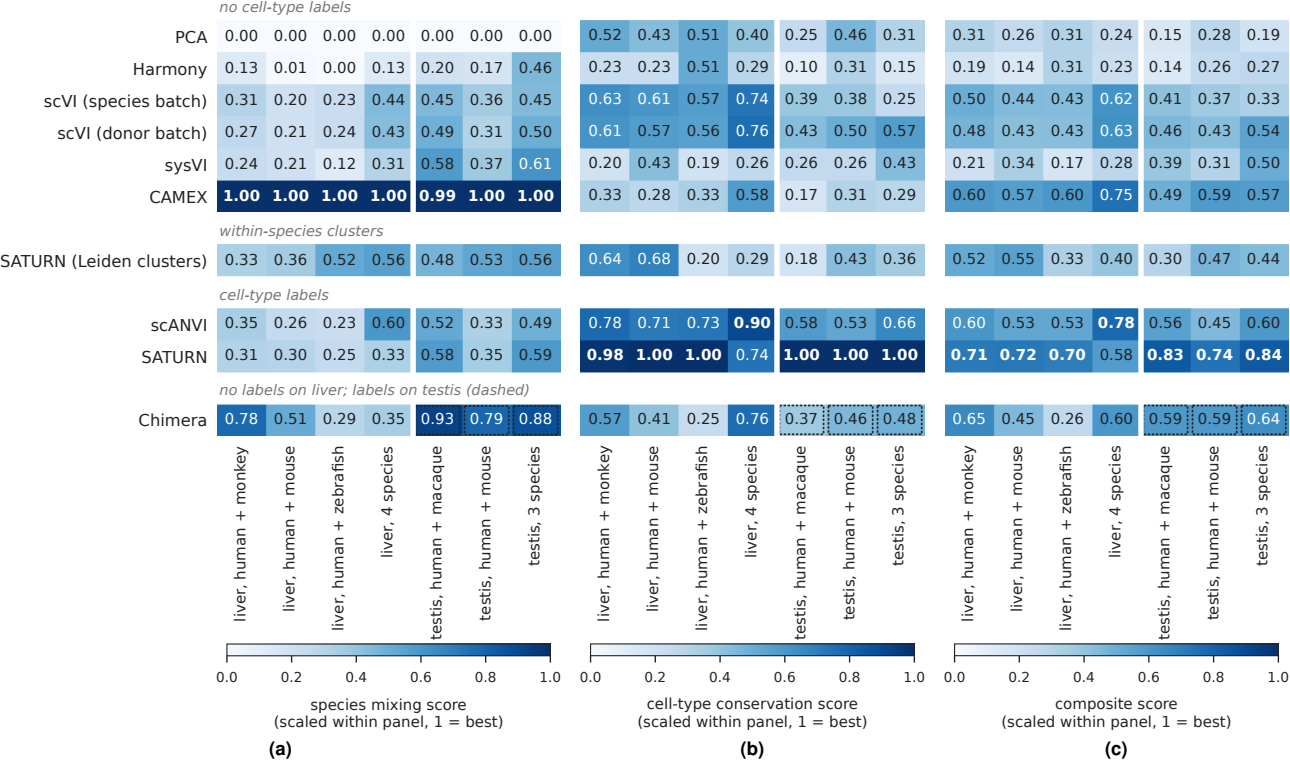
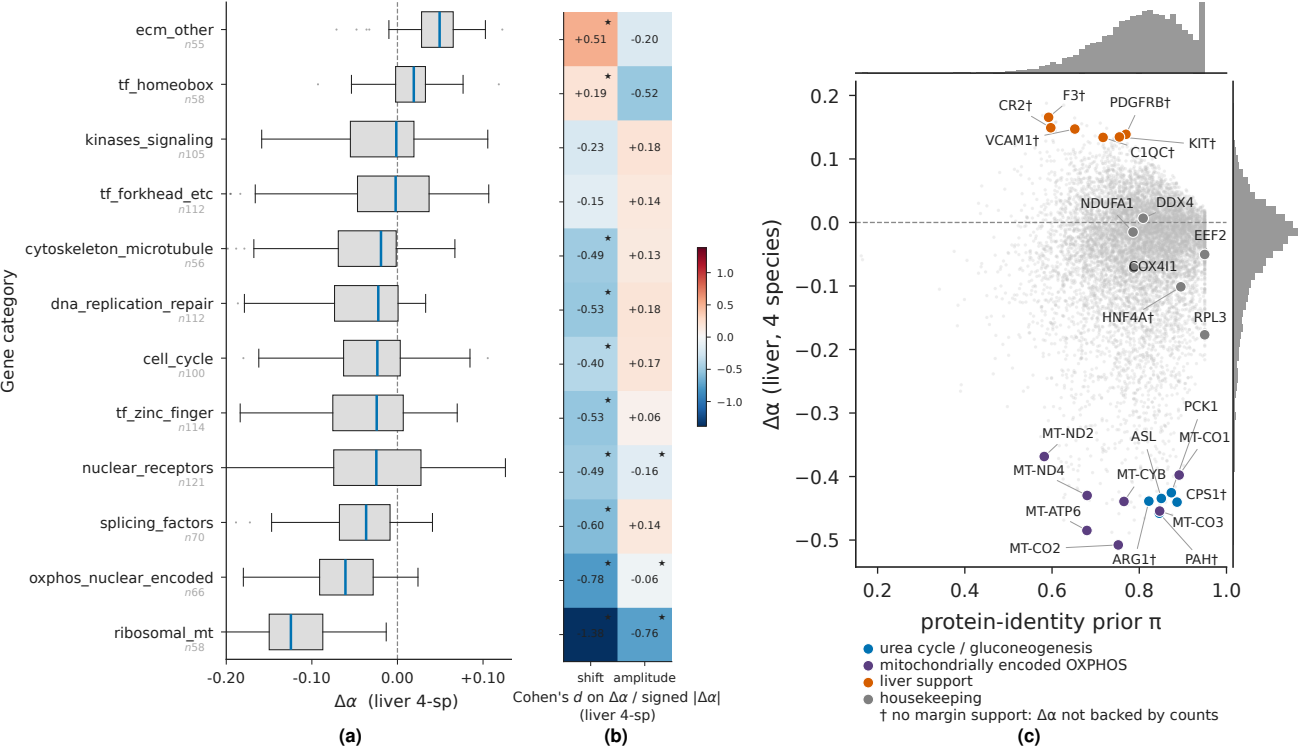
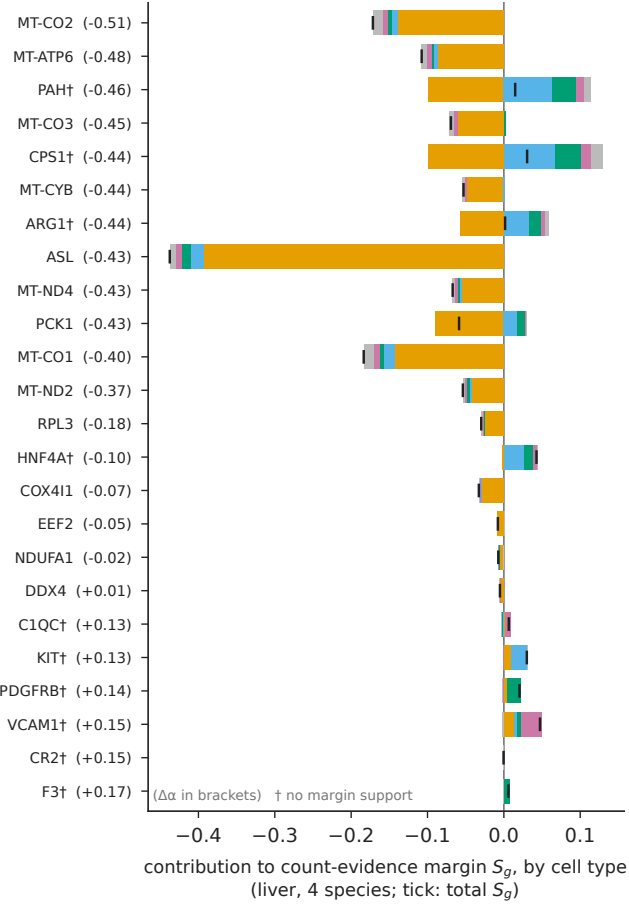

Problem two measurements
conserved protein
less conserved protein
expression
cell states
cell states
used differently
used alike
human
mouse
Deployment is per gene; sequence does not tell which.
(a)
Inputs
scRNA-seq counts x, several species, one-to-one orthologs
protein prior π_g: identity of the orthologs
gene features f_g: ESM-2, identity, expression
(b)
Model sequence-anchored α-gating
features f_g
prior π_g
GIN
c_g
$\alpha_g = \sigma(\text{logit}\,\pi_g + c_g)$
trains c_g
cell state z, species-mixed (GRL + MMD)
+ species, batch
$1 - \alpha_g$
Residual decoder species-aware
x counts
Encoder
z
zero-sum batch offset
NB(x | μ, θ)
Shared decoder species-blind
protein-feature gate (default): $c_g = \text{GIN}(f_g)$, every gene
per-gene gate (companion): c_g free, one offset per gene
count evidence S_g: shared decoder alone against a per-species average
(c)
Outputs per cell and per gene, from one fit
per cell: embedding z
by species
human
monkey
by cell type
hepatocytes
endothelial
fibroblasts
macrophages
other
species aligned: clustering, label transfer; the substrate of the gene readouts
per gene: five example orthologs (schematic)
count evidence S_g
none
prior π_g: tick
gate α_g: needle
α_g from 0 (full residual) to 1 (shared decoder only)
$\Delta\alpha_g = \alpha_g - \pi_g$
protein-feature gate +0.30 +0.13 −0.16 −0.34 +0.13
per-gene readout +0.28 +0.09 −0.12 −0.31 0
gray: imputed from protein features
below the prior
pattern or level parts between species
genes
above the prior
pattern shared across species
−0.4
0
+0.4
$\Delta\alpha_g$ of one model: liver, human + monkey (8,430 genes)
(d)
Applications of the readouts
A separate axis
beside protein identity and dN/dS
Genes, programs, cell types
above or below the protein prior
Eight tissues, one reading
liver, testis, bone marrow, kidney, islet, eye, cortex, lung
Independent tests
held-out species, atlases, benchmark
Disease genes, drug targets
placed in healthy tissue (exploratory)
Protein-space atlas
deployment over ESM-2 neighborhoods; residue attribution (exploratory)
Ortholog audits
mapping and annotation gaps
(e)

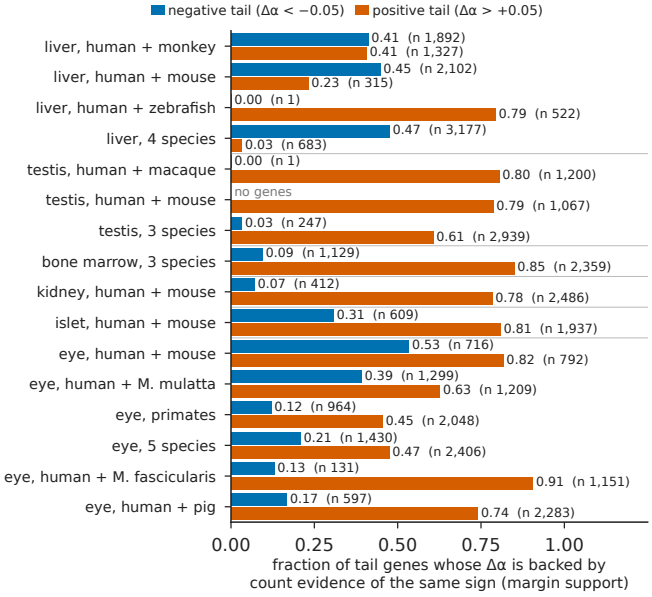

1.00
0.75
0.50
0.25
0.00
α
0.00
0.25
0.50
0.75
1.00
conservation prior
Spearman ρ = 0.770
(a)
700
600
500
400
300
200
100
0
number of genes
frac Δα > 0: 0.48
frac Δα < 0: 0.52
−0.4
−0.2
0.0
0.2
0.4
Δα
(b)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
density
conservation prior
α
n = 8430 genes
mean α = 0.784 std = 0.142
0.00
0.25
0.50
0.75
1.00
α
(c)


no cell-type labels
PCA 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.52 0.43 0.51 0.40 0.25 0.46 0.31 0.31 0.26 0.31 0.24 0.15 0.28 0.19
Harmony 0.13 0.01 0.00 0.13 0.20 0.17 0.46 0.23 0.23 0.51 0.29 0.10 0.31 0.15 0.19 0.14 0.31 0.23 0.14 0.26 0.27
scVI (species batch) 0.31 0.20 0.23 0.44 0.45 0.36 0.45 0.63 0.61 0.57 0.74 0.39 0.38 0.25 0.50 0.44 0.43 0.62 0.41 0.37 0.33
scVI (donor batch) 0.27 0.21 0.24 0.43 0.49 0.31 0.50 0.61 0.57 0.56 0.76 0.43 0.50 0.57 0.48 0.43 0.43 0.63 0.46 0.43 0.54
sysVI 0.24 0.21 0.12 0.31 0.58 0.37 0.61 0.20 0.43 0.19 0.26 0.26 0.26 0.43 0.21 0.34 0.17 0.28 0.39 0.31 0.50
CAMEX 1.00 1.00 1.00 1.00 0.99 1.00 1.00 0.33 0.28 0.33 0.58 0.17 0.31 0.29 0.60 0.57 0.60 0.75 0.49 0.59 0.57
within-species clusters
SATURN (Leiden clusters) 0.33 0.36 0.52 0.56 0.48 0.53 0.56 0.64 0.68 0.20 0.29 0.18 0.43 0.36 0.52 0.55 0.33 0.40 0.30 0.47 0.44
cell-type labels
scANVI 0.35 0.26 0.23 0.60 0.52 0.33 0.49 0.78 0.71 0.73 0.90 0.58 0.53 0.66 0.60 0.53 0.53 0.78 0.56 0.45 0.60
SATURN 0.31 0.30 0.25 0.33 0.58 0.35 0.59 0.98 1.00 1.00 0.74 1.00 1.00 1.00 0.71 0.72 0.70 0.58 0.83 0.74 0.84
no labels on liver; labels on testis (dashed)
Chimera 0.78 0.51 0.29 0.35 0.93 0.79 0.88 0.57 0.41 0.25 0.76 0.37 0.46 0.48 0.65 0.45 0.26 0.60 0.59 0.59 0.64
liver, human + monkey
liver, human + mouse
liver, human + zebrafish
liver, 4 species
testis, human + macaque
testis, human + mouse
testis, 3 species
0.0 0.2 0.4 0.6 0.8 1.0
species mixing score
(scaled within panel, 1 = best)
cell-type conservation score
(scaled within panel, 1 = best)
composite score
(scaled within panel, 1 = best)
(a)
(b)
(c)

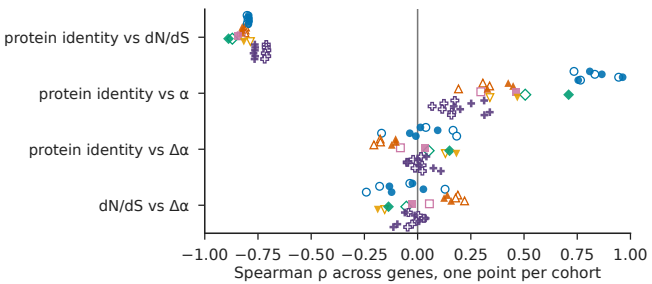

Gene category
ecm_other
n55
tf_homeobox
n58
kinases_signaling
n105
tf_forkhead_etc
n112
cytoskeleton_microtubule
n56
dna_replication_repair
n112
cell_cycle
n100
tf_zinc_finger
n114
nuclear_receptors
n121
splicing_factors
n70
oxphos_nuclear_encoded
n66
ribosomal_mt
n58
-0.20
-0.10
0.00
+0.10
Δα (liver 4-sp)
(a)
+0.51
-0.20
+0.19
-0.52
-0.23
+0.18
-0.15
+0.14
-0.49
+0.13
-0.53
+0.18
-0.40
+0.17
-0.53
+0.06
-0.49
-0.16
-0.60
+0.14
-0.78
-0.06
-1.38
-0.76
shift
amplitude
1.0
0.5
0.0
−0.5
−1.0
Cohen's d on Δα / signed |Δα|
(liver 4-sp)
(b)
Δα (liver, 4 species)
protein-identity prior π
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
0.2
0.4
0.6
0.8
1.0
CR2†
F3†
PDGFRB†
VCAM1†
KIT†
C1QC†
NDUFA1
DDX4
EEF2
COX4I1
HNF4A†
RPL3
PCK1
MT-ND2
ASL
MT-CO1
MT-ND4
MT-CYB
CPS1†
MT-ATP6
MT-CO3
MT-CO2
ARG1†
PAH†
urea cycle / gluconeogenesis
mitochondrially encoded OXPHOS
liver support
housekeeping
† no margin support: Δα not backed by counts
(c)

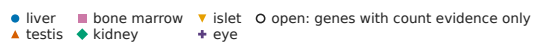

MT-CO2 (-0.51)
MT-ATP6 (-0.48)
PAH† (-0.46)
MT-CO3 (-0.45)
CPS1† (-0.44)
MT-CYB (-0.44)
ARG1† (-0.44)
ASL (-0.43)
MT-ND4 (-0.43)
PCK1 (-0.43)
MT-CO1 (-0.40)
MT-ND2 (-0.37)
RPL3 (-0.18)
HNF4A† (-0.10)
COX4I1 (-0.07)
EEF2 (-0.05)
NDUFA1 (-0.02)
DDX4 (+0.01)
C1QC† (+0.13)
KIT† (+0.13)
PDGFRB† (+0.14)
VCAM1† (+0.15)
CR2† (+0.15)
F3† (+0.17)
(Δα in brackets) † no margin support
−0.4 −0.3 −0.2 −0.1 0.0 0.1
contribution to count-evidence margin S_g, by cell type
(liver, 4 species; tick: total S_g)
hepatocytes (weight 0.74)
endothelial (weight 0.12)
stellate / fibroblasts (weight 0.07)
Kupffer cells and macrophages (weight 0.03)
other cell types

(d)

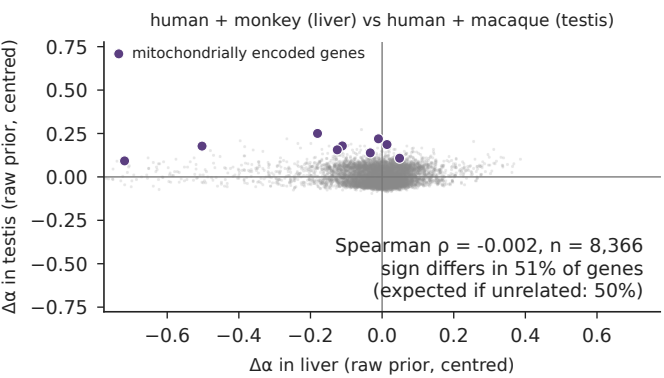

negative tail (Δα < −0.05)
positive tail (Δα > +0.05)
liver, human + monkey 0.41 (n 1,892) 0.41 (n 1,327)
liver, human + mouse 0.45 (n 2,102) 0.23 (n 315)
liver, human + zebrafish 0.00 (n 1) 0.79 (n 522)
liver, 4 species 0.47 (n 3,177) 0.03 (n 683)
testis, human + macaque 0.00 (n 1) 0.80 (n 1,200)
testis, human + mouse no genes 0.79 (n 1,067)
testis, 3 species 0.03 (n 247) 0.61 (n 2,939)
bone marrow, 3 species 0.09 (n 1,129) 0.85 (n 2,359)
kidney, human + mouse 0.07 (n 412) 0.78 (n 2,486)
islet, human + mouse 0.31 (n 609) 0.81 (n 1,937)
eye, human + mouse 0.53 (n 716) 0.82 (n 792)
eye, human + M. mulatta 0.39 (n 1,299) 0.63 (n 1,209)
eye, primates 0.12 (n 964) 0.45 (n 2,048)
eye, 5 species 0.21 (n 1,430) 0.47 (n 2,406)
eye, human + M. fascicularis 0.13 (n 131) 0.91 (n 1,151)
eye, human + pig 0.17 (n 597) 0.74 (n 2,283)
0.00 0.25 0.50 0.75 1.00
fraction of tail genes whose Δα is backed by
count evidence of the same sign (margin support)

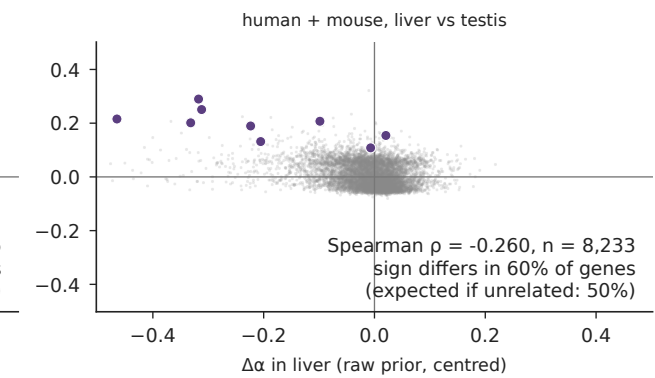
(e)


protein identity vs dN/dS
protein identity vs α
protein identity vs Δα
dN/dS vs Δα
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
Spearman ρ across genes, one point per cohort

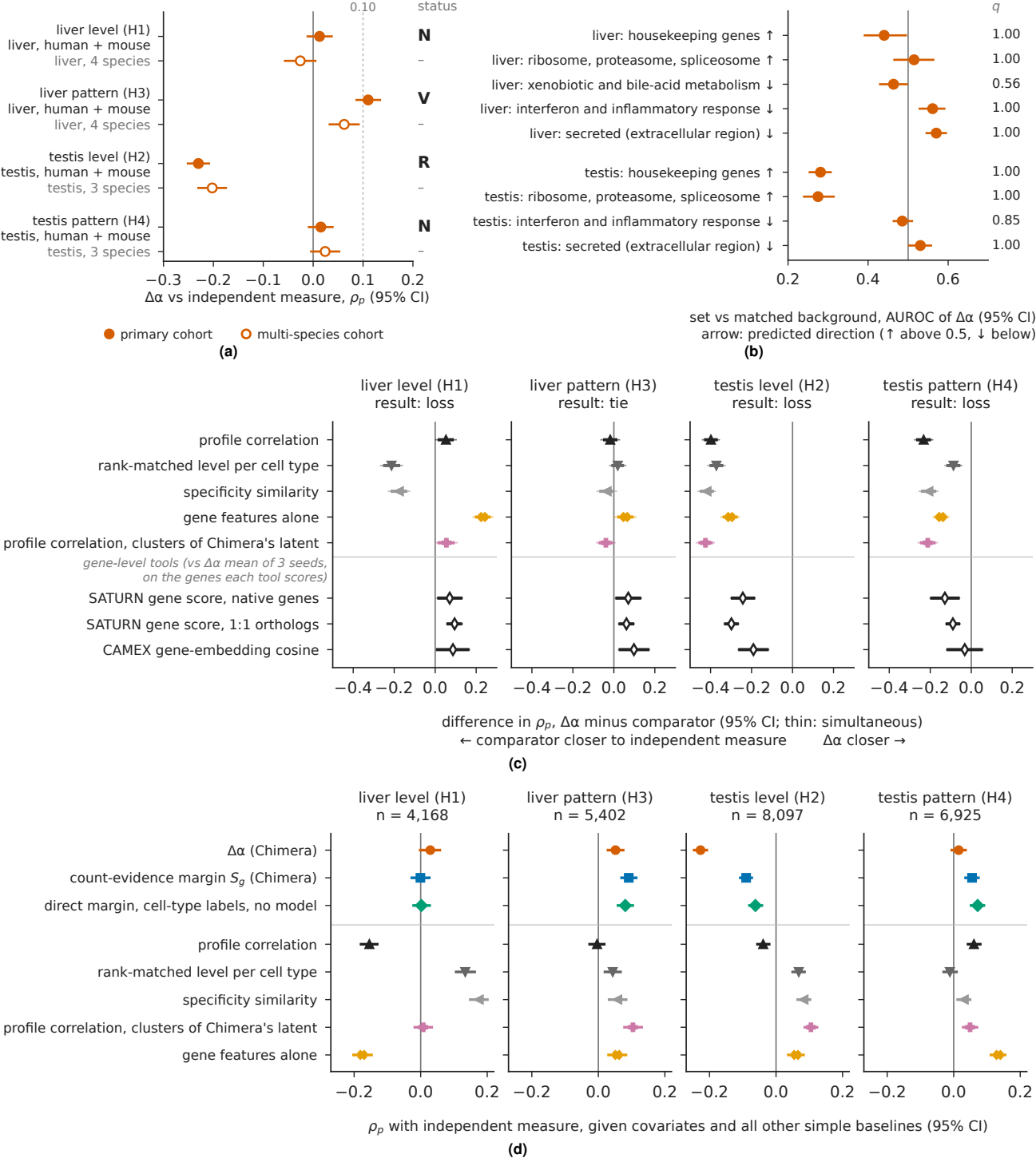

liver
bone marrow
islet
open: genes with count evidence only
testis
kidney
eye

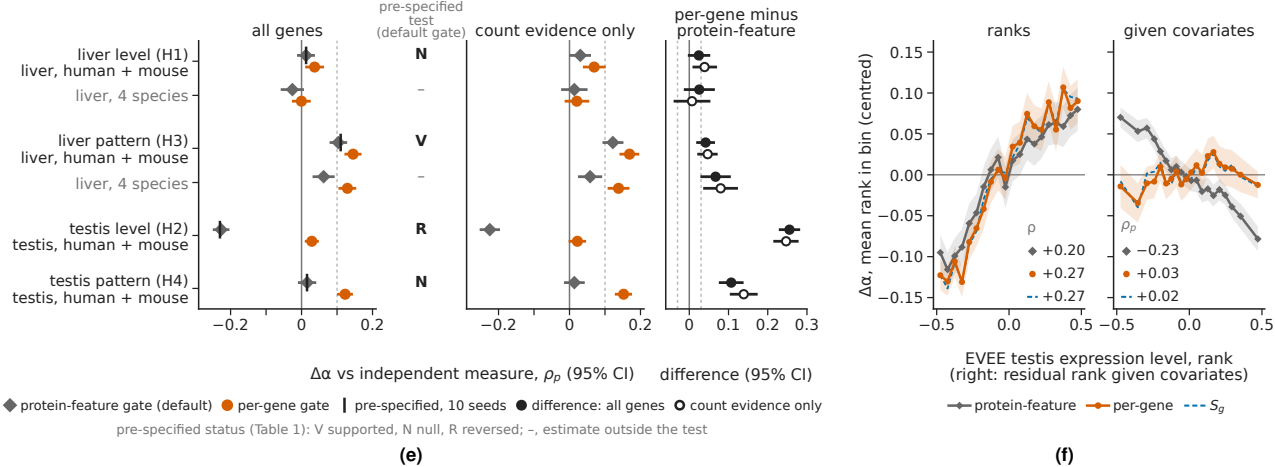
(g)

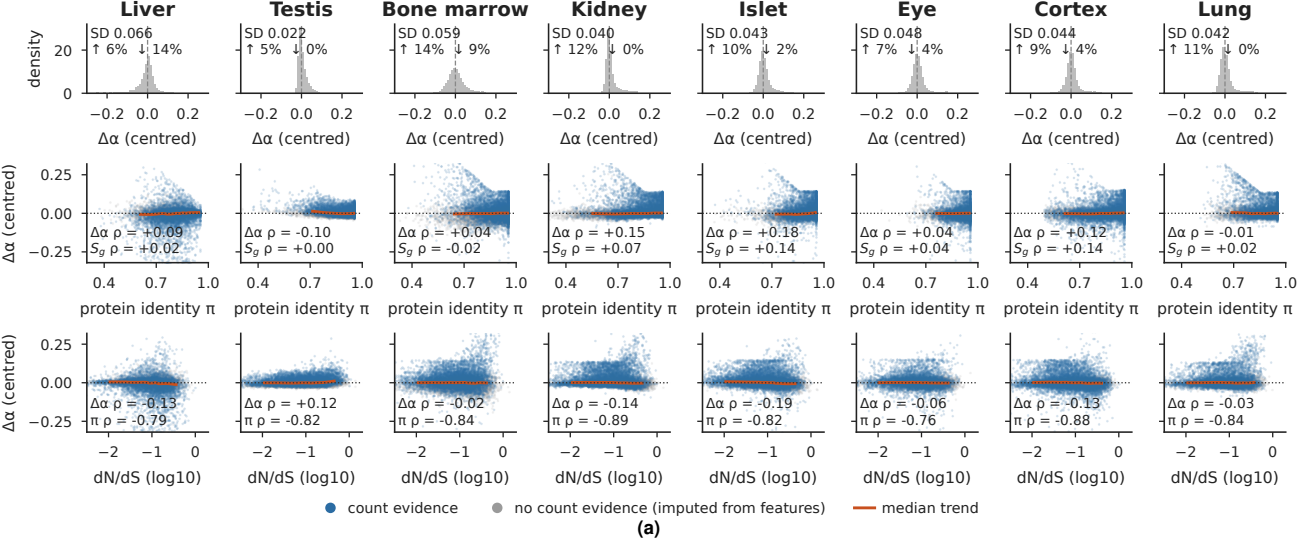

human + monkey (liver) vs human + macaque (testis)
mitochondrially encoded genes
Δα in testis (raw prior, centred)
0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75
Spearman ρ = -0.002, n = 8,366
sign differs in 51% of genes
(expected if unrelated: 50%)
−0.6 −0.4 −0.2 0.0 0.2 0.4 0.6
Δα in liver (raw prior, centred)


human + mouse, liver vs testis
0.4 0.2 0.0 −0.2 −0.4
Spearman ρ = -0.260, n = 8,233
sign differs in 60% of genes
(expected if unrelated: 50%)
−0.4 −0.2 0.0 0.2 0.4
Δα in liver (raw prior, centred)

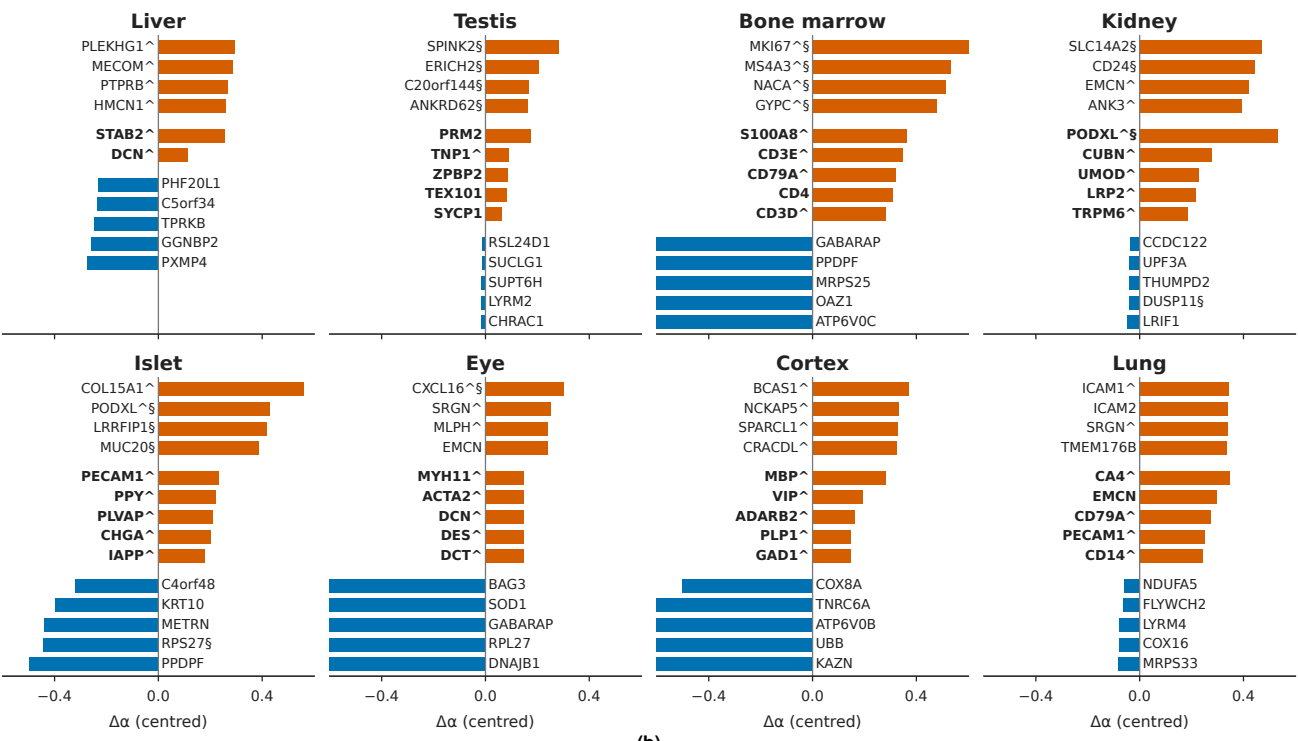
(f)


0.10
status
liver level (H1)
liver, human + mouse
liver, 4 species
N
–
liver pattern (H3)
liver, human + mouse
liver, 4 species
V
–
testis level (H2)
testis, human + mouse
testis, 3 species
R
–
testis pattern (H4)
testis, human + mouse
testis, 3 species
N
–
−0.3 −0.2 −0.1 0.0 0.1 0.2
Δα vs independent measure, ρ_p (95% CI)
primary cohort
multi-species cohort
(a)
q
liver: housekeeping genes ↑ 1.00
liver: ribosome, proteasome, spliceosome ↑ 1.00
liver: xenobiotic and bile-acid metabolism ↓ 0.56
liver: interferon and inflammatory response ↓ 1.00
liver: secreted (extracellular region) ↓ 1.00
testis: housekeeping genes ↑ 1.00
testis: ribosome, proteasome, spliceosome ↑ 1.00
testis: interferon and inflammatory response ↓ 0.85
testis: secreted (extracellular region) ↓ 1.00
0.2 0.4 0.6
set vs matched background, AUROC of Δα (95% CI)
arrow: predicted direction (↑ above 0.5, ↓ below)
(b)
liver level (H1)
result: loss
liver pattern (H3)
result: tie
testis level (H2)
result: loss
testis pattern (H4)
result: loss
profile correlation
rank-matched level per cell type
specificity similarity
gene features alone
profile correlation, clusters of Chimera's latent
gene-level tools (vs Δα mean of 3 seeds, on the genes each tool scores)
SATURN gene score, native genes
SATURN gene score, 1:1 orthologs
CAMEX gene-embedding cosine
−0.4 −0.2 0.0 0.2
difference in ρ_p, Δα minus comparator (95% CI; thin: simultaneous)
← comparator closer to independent measure
Δα closer →
(c)
liver level (H1)
n = 4,168
liver pattern (H3)
n = 5,402
testis level (H2)
n = 8,097
testis pattern (H4)
n = 6,925
Δα (Chimera)
count-evidence margin S_g (Chimera)
direct margin, cell-type labels, no model
profile correlation
rank-matched level per cell type
specificity similarity
profile correlation, clusters of Chimera's latent
gene features alone
−0.2 0.0 0.2
ρ_p with independent measure, given covariates and all other simple baselines (95% CI)
(d)

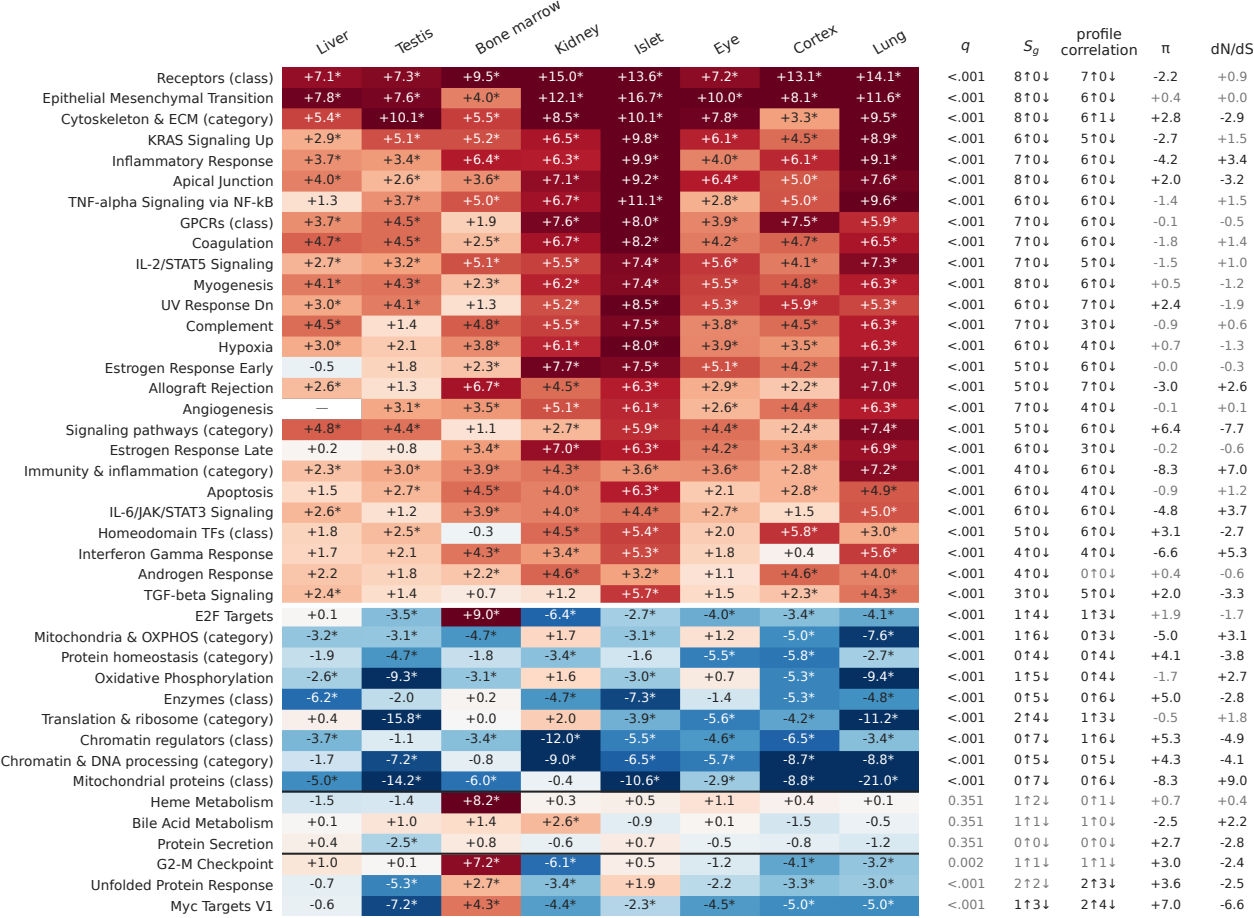

all genes
pre-specified test (default gate)
count evidence only
per-gene minus protein-feature
liver level (H1)
liver, human + mouse
liver, 4 species
liver pattern (H3)
liver, human + mouse
liver, 4 species
testis level (H2)
testis, human + mouse
testis pattern (H4)
testis, human + mouse
N
–
V
–
R
N
−0.2
0
0.2
0
0.1
0.2
0.3
Δα vs independent measure, ρ_p (95% CI)
difference (95% CI)
protein-feature gate (default)
per-gene gate
pre-specified, 10 seeds
difference: all genes
count evidence only
pre-specified status (Table 1): V supported, N null, R reversed; –, estimate outside the test
(e)
ranks
given covariates
Δα, mean rank in bin (centred)
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
ρ
+0.20
+0.27
+0.27
ρ_p
−0.23
+0.03
+0.02
−0.5
0.0
0.5
EVEE testis expression level, rank
(right: residual rank given covariates)
protein-feature
per-gene
S_g
(f)
protein-feature
per-gene
ρ = 0.62
ρ = 0.99
ρ = 0.48
ρ = 0.99
liver
testis
Δα, rank
0
0.5
1
S_g of the same fit, rank
(genes with count evidence)
(g)
testis, spermatocyte genes
spermatocyte peak, 1,103
with count evidence
strict peak, 166
with count evidence
hepatocyte catabolic genes
liver, 4 species
programme, 63
core genes, 15
liver, human + mouse
programme, 63
core genes, 15
−4
−3
−2
−1
0
matched excess, SD units (95% CI)
protein-feature
per-gene
S_g
(h)
Δα vs dN/dS
liver, human + mouse
liver, 4 species
testis, human + mouse
−0.3
0
0.3
Δα vs protein prior π
liver, human + mouse
liver, 4 species
testis, human + mouse
Spearman ρ across genes
all genes
count evidence
(i)


Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
density
SD 0.066 ↑ 6% ↓ 14%
SD 0.022 ↑ 5% ↓ 0%
SD 0.059 ↑ 14% ↓ 9%
SD 0.040 ↑ 12% ↓ 0%
SD 0.043 ↑ 10% ↓ 2%
SD 0.048 ↑ 7% ↓ 4%
SD 0.044 ↑ 9% ↓ 4%
SD 0.042 ↑ 11% ↓ 0%
Δα (centred)
protein identity π
dN/dS (log10)
Δα ρ = +0.09 Sg ρ = +0.02
Δα ρ = -0.10 Sg ρ = +0.00
Δα ρ = +0.04 Sg ρ = -0.02
Δα ρ = +0.15 Sg ρ = +0.07
Δα ρ = +0.18 Sg ρ = +0.14
Δα ρ = +0.04 Sg ρ = +0.04
Δα ρ = +0.12 Sg ρ = +0.14
Δα ρ = -0.01 Sg ρ = +0.02
Δα ρ = -0.13 π ρ = -0.79
Δα ρ = +0.12 π ρ = -0.82
Δα ρ = -0.02 π ρ = -0.84
Δα ρ = -0.14 π ρ = -0.89
Δα ρ = -0.19 π ρ = -0.82
Δα ρ = -0.06 π ρ = -0.76
Δα ρ = -0.13 π ρ = -0.88
Δα ρ = -0.03 π ρ = -0.84
count evidence
no count evidence (imputed from features)
median trend

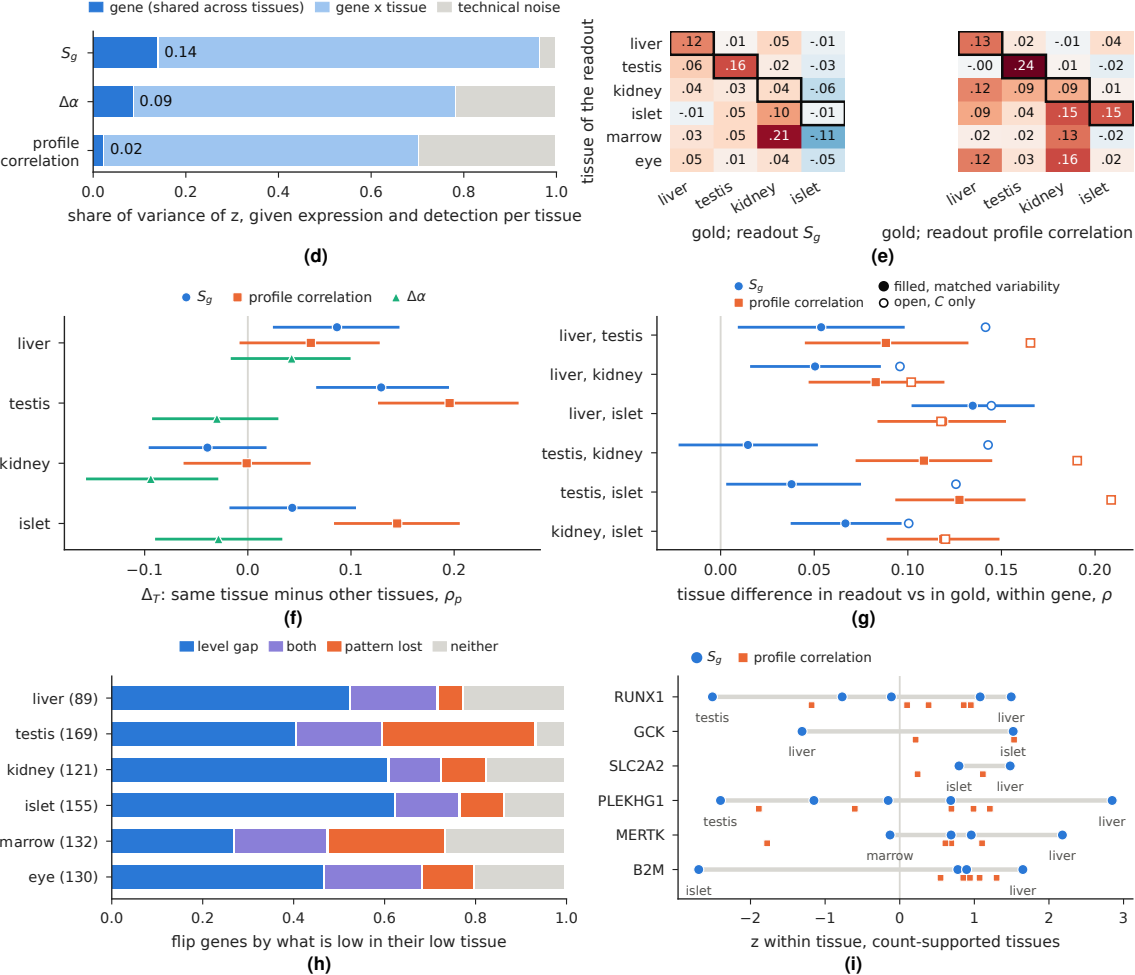
(a)

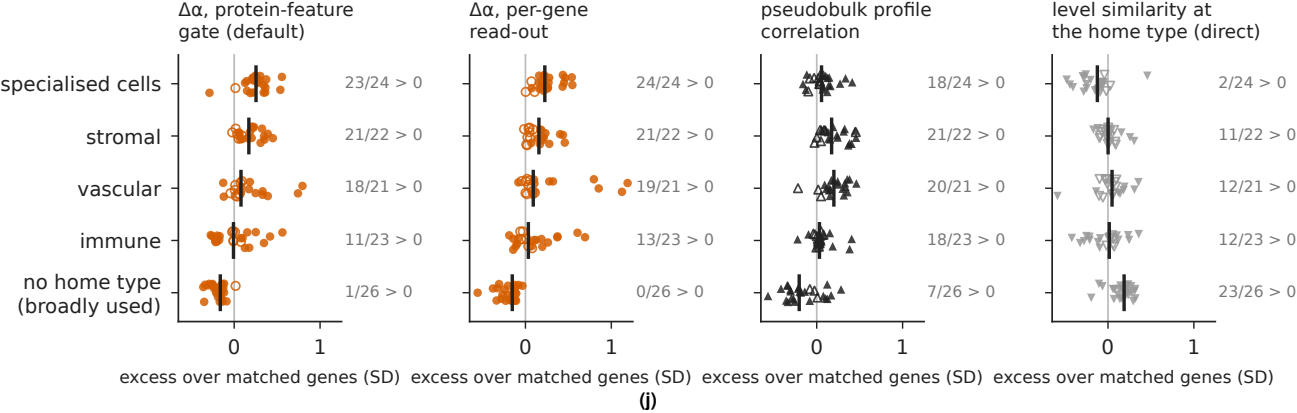

Liver
PLEKHG1^
MECOM^
PTPRB^
HMCN1^
STAB2^
DCN^
PHF20L1
C5orf34
TPRKB
GGNBP2
PXMP4
Testis
SPINK2§
ERICH2§
C20orf144§
ANKRD62§
PRM2
TNP1^
ZPBP2
TEX101
SYCP1
RSL24D1
SUCLG1
SUPT6H
LYRM2
CHRAC1
Bone marrow
MKI67^§
MS4A3^§
NACA^§
GYPC^§
S100A8^
CD3E^
CD79A^
CD4
CD3D^
GABARAP
PPDPF
MRPS25
OAZ1
ATP6V0C
Kidney
SLC14A2§
CD24§
EMCN^
ANK3^
PODXL^§
CUBN^
UMOD^
LRP2^
TRPM6^
CCDC122
UPF3A
THUMPD2
DUSP11§
LRIF1
Islet
COL15A1^
PODXL^§
LRRFIP1§
MUC20§
PECAM1^
PPY^
PLVAP^
CHGA^
IAPP^
C4orf48
KRT10
METRN
RPS27§
PPDPF
Eye
CXCL16^§
SRGN^
MLPH^
EMCN
MYH11^
ACTA2^
DCN^
DES^
DCT^
BAG3
SOD1
GABARAP
RPL27
DNAJB1
Cortex
BCAS1^
NCKAP5^
SPARCL1^
CRACDL^
MBP^
VIP^
ADARB2^
PLP1^
GAD1^
COX8A
TNRC6A
ATP6V0B
UBB
KAZN
Lung
ICAM1^
ICAM2
SRGN^
TMEM176B
CA4^
EMCN
CD79A^
PECAM1^
CD14^
NDUFA5
FLYWCH2
LYRM4
COX16
MRPS33
Δα (centred)

(b)

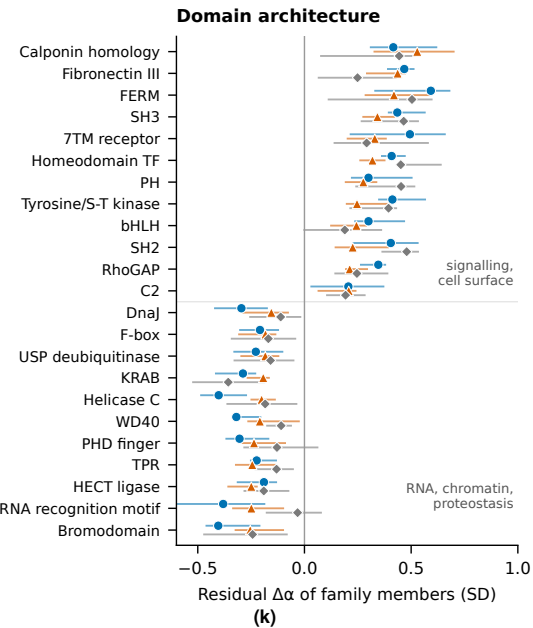

Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
q
S_g
profile correlation
π
dN/dS
Receptors (class) +7.1* +7.3* +9.5* +15.0* +13.6* +7.2* +13.1* +14.1* <.001 8↑0↓ 7↑0↓ -2.2 +0.9
Epithelial Mesenchymal Transition +7.8* +7.6* +4.0* +12.1* +16.7* +10.0* +8.1* +11.6* <.001 8↑0↓ 6↑0↓ +0.4 +0.0
Cytoskeleton & ECM (category) +5.4* +10.1* +5.5* +8.5* +10.1* +7.8* +3.3* +9.5* <.001 8↑0↓ 6↑1↓ +2.8 -2.9
KRAS Signaling Up +2.9* +5.1* +5.2* +6.5* +9.8* +6.1* +4.5* +8.9* <.001 6↑0↓ 5↑0↓ -2.7 +1.5
Inflammatory Response +3.7* +3.4* +6.4* +6.3* +9.9* +4.0* +6.1* +9.1* <.001 7↑0↓ 6↑0↓ -4.2 +3.4
Apical Junction +4.0* +2.6* +3.6* +7.1* +9.2* +6.4* +5.0* +7.6* <.001 8↑0↓ 6↑0↓ +2.0 -3.2
TNF-alpha Signaling via NF-kB +1.3 +3.7* +5.0* +6.7* +11.1* +2.8* +5.0* +9.6* <.001 6↑0↓ 6↑0↓ -1.4 +1.5
GPCRs (class) +3.7* +4.5* +1.9 +7.6* +8.0* +3.9* +7.5* +5.9* <.001 7↑0↓ 6↑0↓ -0.1 -0.5
Coagulation +4.7* +4.5* +2.5* +6.7* +8.2* +4.2* +4.7* +6.5* <.001 7↑0↓ 6↑0↓ -1.8 +1.4
IL-2/STAT5 Signaling +2.7* +3.2* +5.1* +5.5* +7.4* +5.6* +4.1* +7.3* <.001 7↑0↓ 5↑0↓ -1.5 +1.0
Myogenesis +4.1* +4.3* +2.3* +6.2* +7.4* +5.5* +4.8* +6.3* <.001 8↑0↓ 6↑0↓ +0.5 -1.2
UV Response Dn +3.0* +4.1* +1.3 +5.2* +8.5* +5.3* +5.9* +5.3* <.001 6↑0↓ 7↑0↓ +2.4 -1.9
Complement +4.5* +1.4 +4.8* +5.5* +7.5* +3.8* +4.5* +6.3* <.001 7↑0↓ 3↑0↓ -0.9 +0.6
Hypoxia +3.0* +2.1 +3.8* +6.1* +8.0* +3.9* +3.5* +6.3* <.001 6↑0↓ 4↑0↓ +0.7 -1.3
Estrogen Response Early -0.5 +1.8 +2.3* +7.7* +7.5* +5.1* +4.2* +7.1* <.001 5↑0↓ 6↑0↓ -0.0 -0.3
Allograft Rejection +2.6* +1.3 +6.7* +4.5* +6.3* +2.9* +2.2* +7.0* <.001 5↑0↓ 7↑0↓ -3.0 +2.6
Angiogenesis — +3.1* +3.5* +5.1* +6.1* +2.6* +4.4* +6.3* <.001 7↑0↓ 4↑0↓ -0.1 +0.1
Signaling pathways (category) +4.8* +4.4* +1.1 +2.7* +5.9* +4.4* +2.4* +7.4* <.001 5↑0↓ 6↑0↓ +6.4 -7.7
Estrogen Response Late +0.2 +0.8 +3.4* +7.0* +6.3* +4.2* +3.4* +6.9* <.001 6↑0↓ 3↑0↓ -0.2 -0.6
Immunity & inflammation (category) +2.3* +3.0* +3.9* +4.3* +3.6* +3.6* +2.8* +7.2* <.001 4↑0↓ 6↑0↓ -8.3 +7.0
Apoptosis +1.5 +2.7* +4.5* +4.0* +6.3* +2.1 +2.8* +4.9* <.001 6↑0↓ 4↑0↓ -0.9 +1.2
IL-6/JAK/STAT3 Signaling +2.6* +1.2 +3.9* +4.0* +4.4* +2.7* +1.5 +5.0* <.001 6↑0↓ 6↑0↓ -4.8 +3.7
Homeodomain TFs (class) +1.8 +2.5* -0.3 +4.5* +5.4* +2.0 +5.8* +3.0* <.001 5↑0↓ 6↑0↓ +3.1 -2.7
Interferon Gamma Response +1.7 +2.1 +4.3* +3.4* +5.3* +1.8 +0.4 +5.6* <.001 4↑0↓ 4↑0↓ -6.6 +5.3
Androgen Response +2.2 +1.8 +2.2* +4.6* +3.2* +1.1 +4.6* +4.0* <.001 4↑0↓ 0↑0↓ +0.4 -0.6
TGF-beta Signaling +2.4* +1.4 +0.7 +1.2 +5.7* +1.5 +2.3* +4.3* <.001 3↑0↓ 5↑0↓ +2.0 -3.3
E2F Targets +0.1 -3.5* +9.0* -6.4* -2.7* -4.0* -3.4* -4.1* <.001 1↑4↓ 1↑3↓ +1.9 -1.7
Mitochondria & OXPHOS (category) -3.2* -3.1* -4.7* +1.7 -3.1* +1.2 -5.0* -7.6* <.001 1↑6↓ 0↑3↓ -5.0 +3.1
Protein homeostasis (category) -1.9 -4.7* -1.8 -3.4* -1.6 -5.5* -5.8* -2.7* <.001 0↑4↓ 0↑4↓ +4.1 -3.8
Oxidative Phosphorylation -2.6* -9.3* -3.1* +1.6 -3.0* +0.7 -5.3* -9.4* <.001 1↑5↓ 0↑4↓ -1.7 +2.7
Enzymes (class) -6.2* -2.0 +0.2 -4.7* -7.3* -1.4 -5.3* -4.8* <.001 0↑5↓ 0↑6↓ +5.0 -2.8
Translation & ribosome (category) +0.4 -15.8* +0.0 +2.0 -3.9* -5.6* -4.2* -11.2* <.001 2↑4↓ 1↑3↓ -0.5 +1.8
Chromatin regulators (class) -3.7* -1.1 -3.4* -12.0* -5.5* -4.6* -6.5* -3.4* <.001 0↑7↓ 1↑6↓ +5.3 -4.9
Chromatin & DNA processing (category) -1.7 -7.2* -0.8 -9.0* -6.5* -5.7* -8.7* -8.8* <.001 0↑5↓ 0↑5↓ +4.3 -4.1
Mitochondrial proteins (class) -5.0* -14.2* -6.0* -0.4 -10.6* -2.9* -8.8* -21.0* <.001 0↑7↓ 0↑6↓ -8.3 +9.0
Heme Metabolism -1.5 -1.4 +8.2* +0.3 +0.5 +1.1 +0.4 +0.1 0.351 1↑2↓ 0↑1↓ +0.7 +0.4
Bile Acid Metabolism +0.1 +1.0 +1.4 +2.6* -0.9 +0.1 -1.5 -0.5 0.351 1↑1↓ 1↑0↓ -2.5 +2.2
Protein Secretion +0.4 -2.5* +0.8 -0.6 +0.7 -0.5 -0.8 -1.2 0.351 0↑0↓ 0↑0↓ +2.7 -2.8
G2-M Checkpoint +1.0 +0.1 +7.2* -6.1* +0.5 -1.2 -4.1* -3.2* 0.002 1↑1↓ 1↑1↓ +3.0 -2.4
Unfolded Protein Response -0.7 -5.3* +2.7* -3.4* +1.9 -2.2 -3.3* -3.0* <.001 2↑2↓ 2↑3↓ +3.6 -2.5
Myc Targets V1 -0.6 -7.2* +4.3* -4.4* -2.3* -4.5* -5.0* -5.0* <.001 1↑3↓ 2↑4↓ +7.0 -6.6

(c)

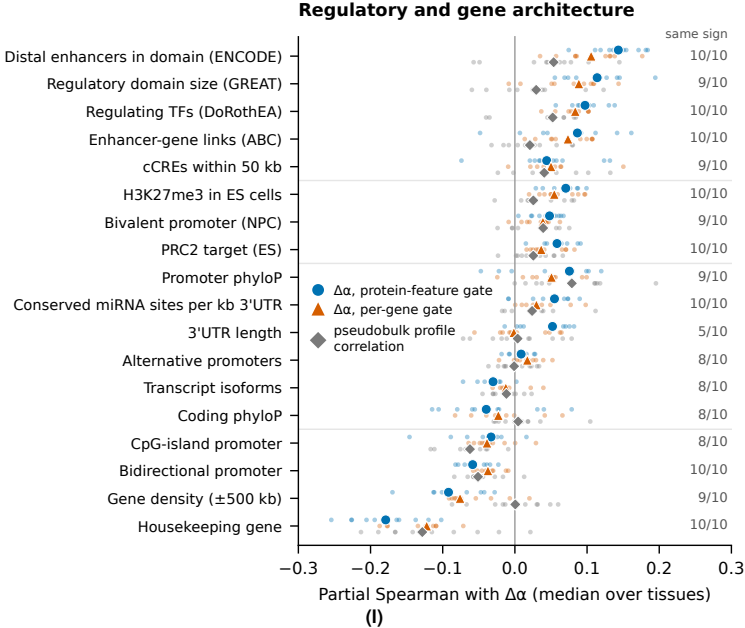

gene (shared across tissues)
gene x tissue
technical noise
S_g
$\Delta\alpha$
profile correlation
0.14
0.09
0.02
0.0
0.2
0.4
0.6
0.8
1.0
share of variance of z, given expression and detection per tissue
(d)
tissue of the readout
liver .12 .01 .05 -.01
testis .06 .16 .02 -.03
kidney .04 .03 .04 -.06
islet -.01 .05 .10 -.01
marrow .03 .05 .21 -.11
eye .05 .01 .04 -.05
liver testis kidney islet
gold; readout S_g
.13 .02 -.01 .04
-.00 .24 .01 -.02
.12 .09 .09 .01
.09 .04 .15 .15
.02 .02 .13 -.02
.12 .03 .16 .02
liver testis kidney islet
gold; readout profile correlation
(e)
S_g
profile correlation
$\Delta\alpha$
liver
testis
kidney
islet
−0.1
0.0
0.1
0.2
Δ_T: same tissue minus other tissues, ρ_p
(f)
S_g
profile correlation
filled, matched variability
open, C only
liver, testis
liver, kidney
liver, islet
testis, kidney
testis, islet
kidney, islet
0.00
0.05
0.10
0.15
0.20
tissue difference in readout vs in gold, within gene, ρ
(g)
level gap
both
pattern lost
neither
liver (89)
testis (169)
kidney (121)
islet (155)
marrow (132)
eye (130)
0.0
0.2
0.4
0.6
0.8
1.0
flip genes by what is low in their low tissue
(h)
S_g
profile correlation
RUNX1
testis
liver
GCK
liver
islet
SLC2A2
islet
liver
PLEKHG1
testis
liver
MERTK
marrow
liver
B2M
islet
liver
−2
−1
0
1
2
3
z within tissue, count-supported tissues
(i)


Δα, protein-feature gate (default)
Δα, per-gene read-out
pseudobulk profile correlation
level similarity at the home type (direct)
specialised cells
stromal
vascular
immune
no home type (broadly used)
23/24 > 0
21/22 > 0
18/21 > 0
11/23 > 0
1/26 > 0
24/24 > 0
21/22 > 0
19/21 > 0
13/23 > 0
0/26 > 0
18/24 > 0
21/22 > 0
20/21 > 0
18/23 > 0
7/26 > 0
2/24 > 0
11/22 > 0
12/21 > 0
12/23 > 0
23/26 > 0
0
1
excess over matched genes (SD)

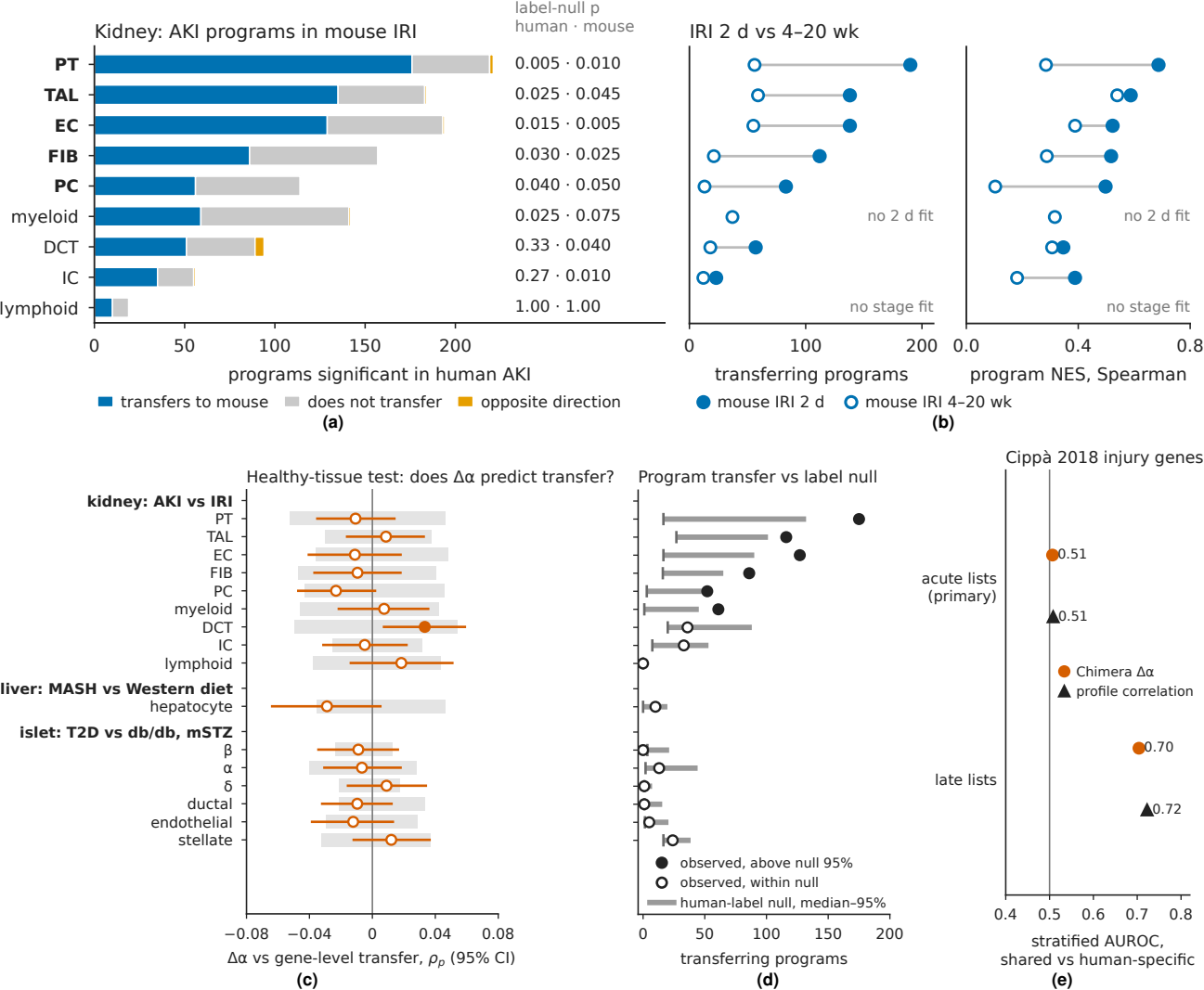
(j)

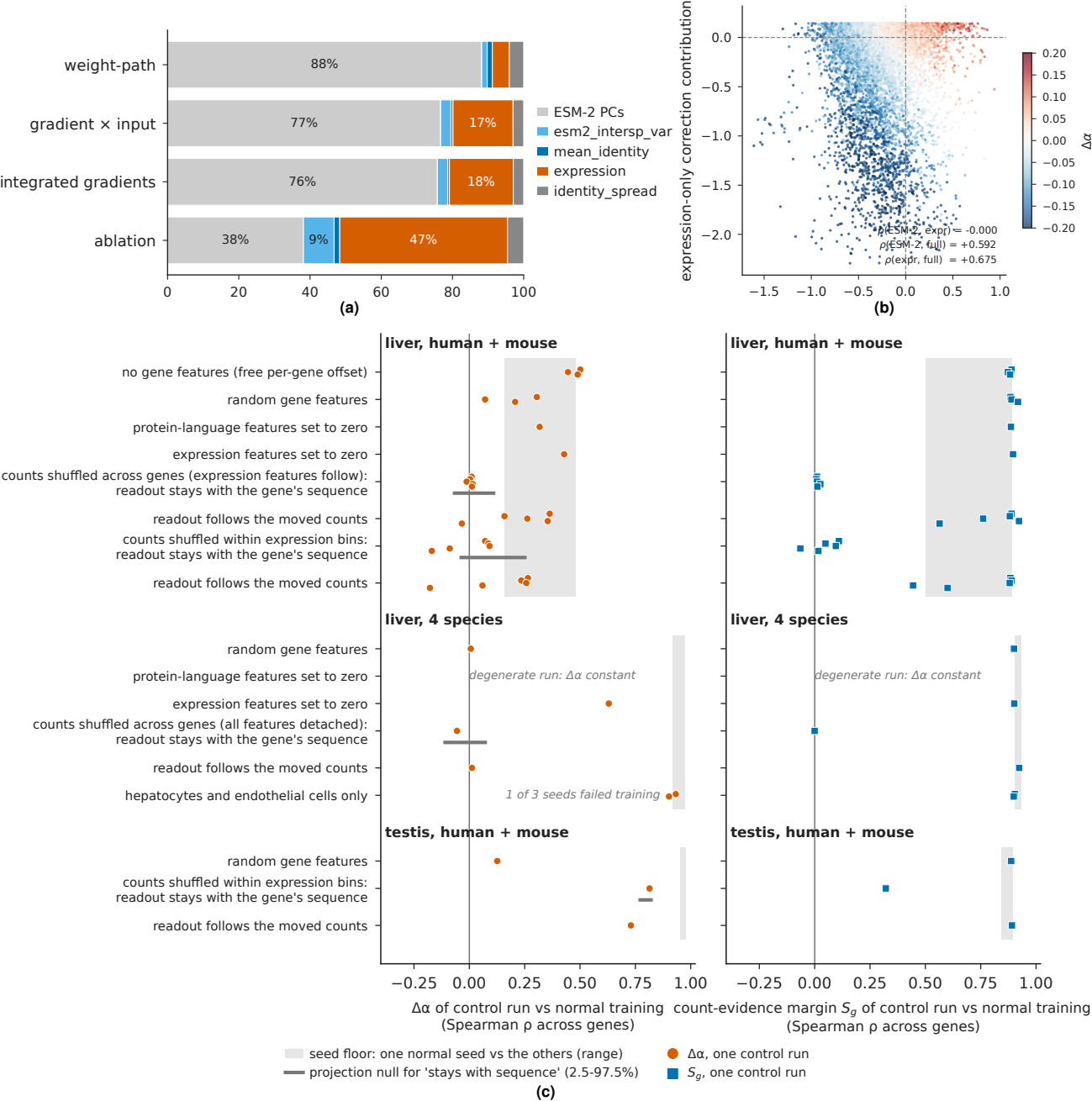

Domain architecture
Calponin homology
Fibronectin III
FERM
SH3
7TM receptor
Homeodomain TF
PH
Tyrosine/S-T kinase
bHLH
SH2
RhoGAP
C2
DnaJ
F-box
USP deubiquitinase
KRAB
Helicase C
WD40
PHD finger
TPR
HECT ligase
RNA recognition motif
Bromodomain
signalling, cell surface
RNA, chromatin, proteostasis
−0.5
0.0
0.5
1.0
Residual Δα of family members (SD)

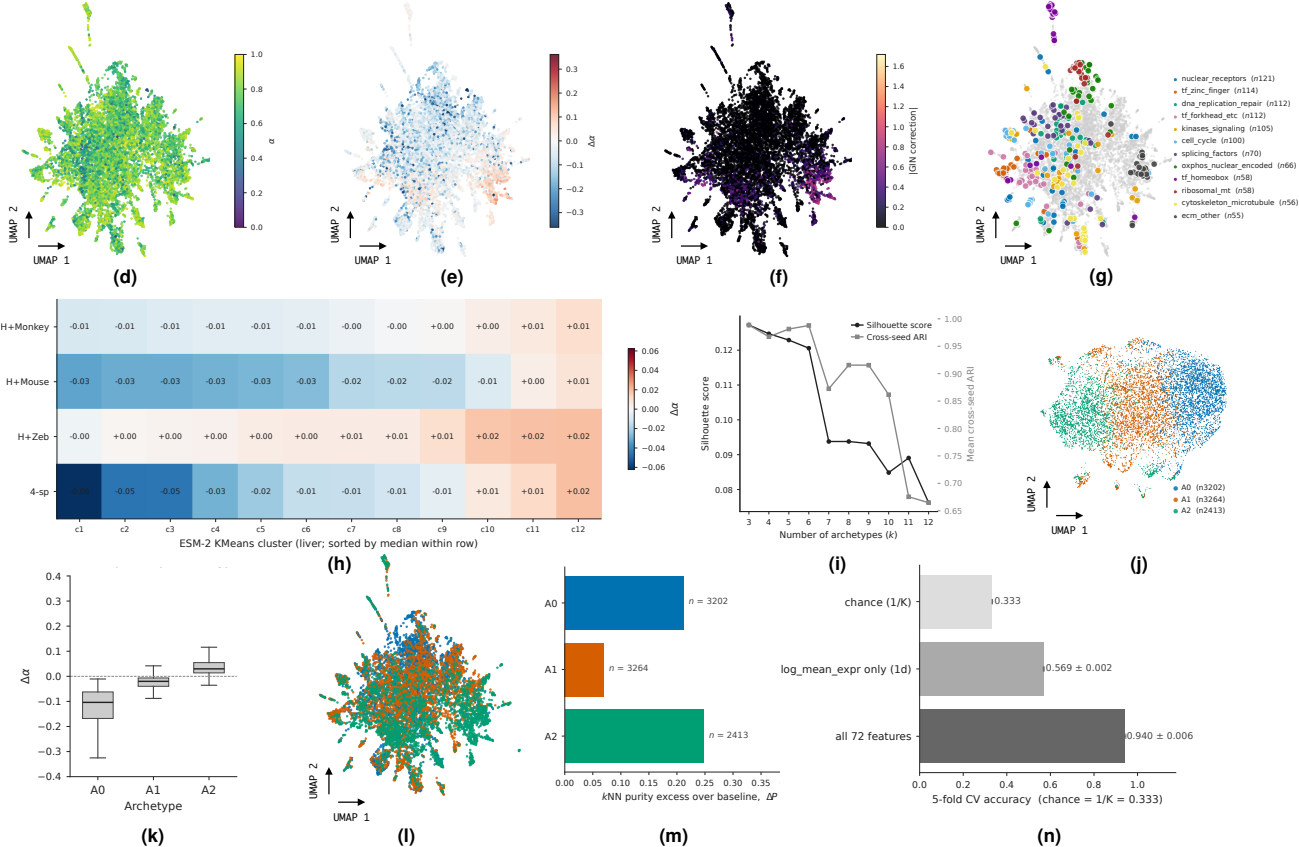
(k)

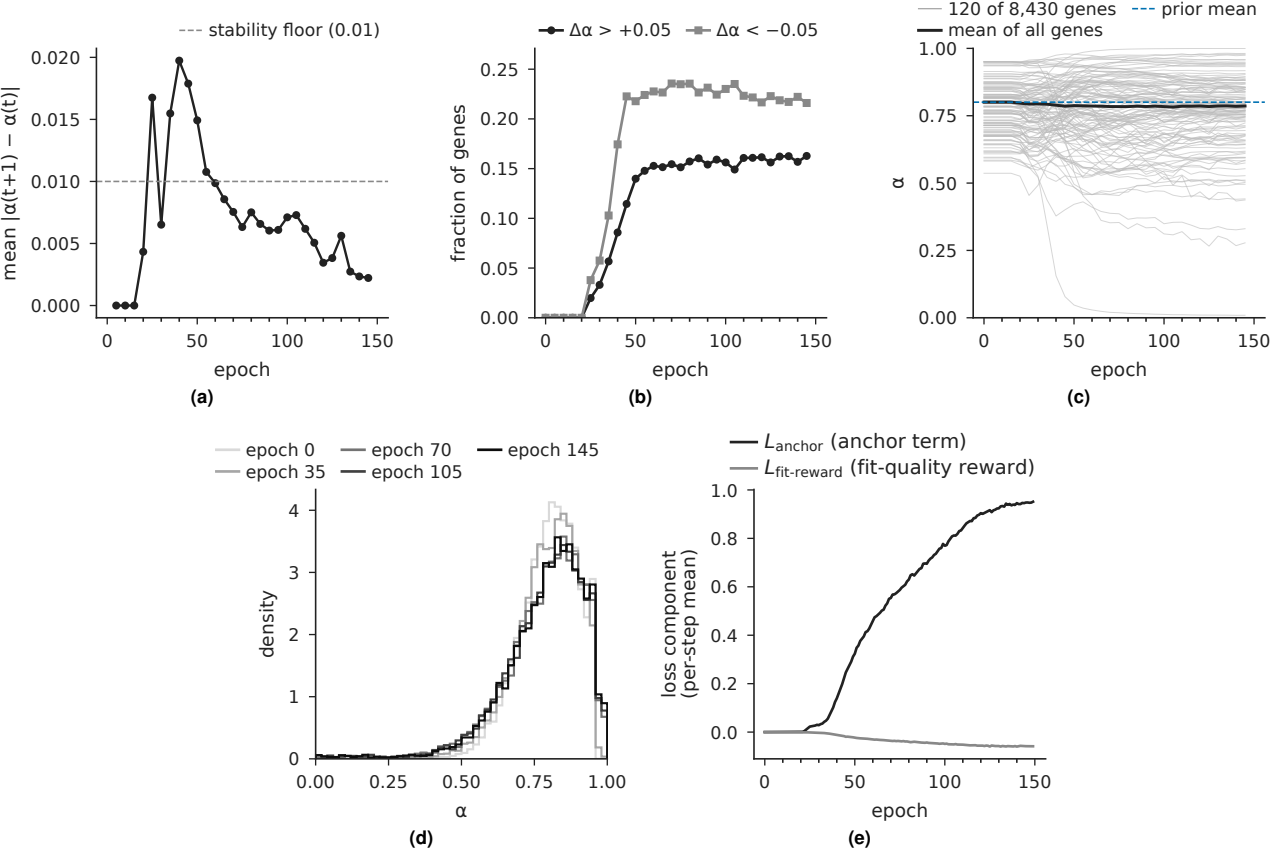

Regulatory and gene architecture
same sign
Distal enhancers in domain (ENCODE)
Regulatory domain size (GREAT)
Regulating TFs (DoRothEA)
Enhancer-gene links (ABC)
cCREs within 50 kb
H3K27me3 in ES cells
Bivalent promoter (NPC)
PRC2 target (ES)
Promoter phyloP
Conserved miRNA sites per kb 3'UTR
3'UTR length
Alternative promoters
Transcript isoforms
Coding phyloP
CpG-island promoter
Bidirectional promoter
Gene density (±500 kb)
Housekeeping gene
10/10
9/10
10/10
10/10
9/10
10/10
9/10
10/10
9/10
10/10
5/10
8/10
8/10
8/10
8/10
10/10
9/10
10/10
Δα, protein-feature gate
Δα, per-gene gate
pseudobulk profile correlation
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Partial Spearman with Δα (median over tissues)

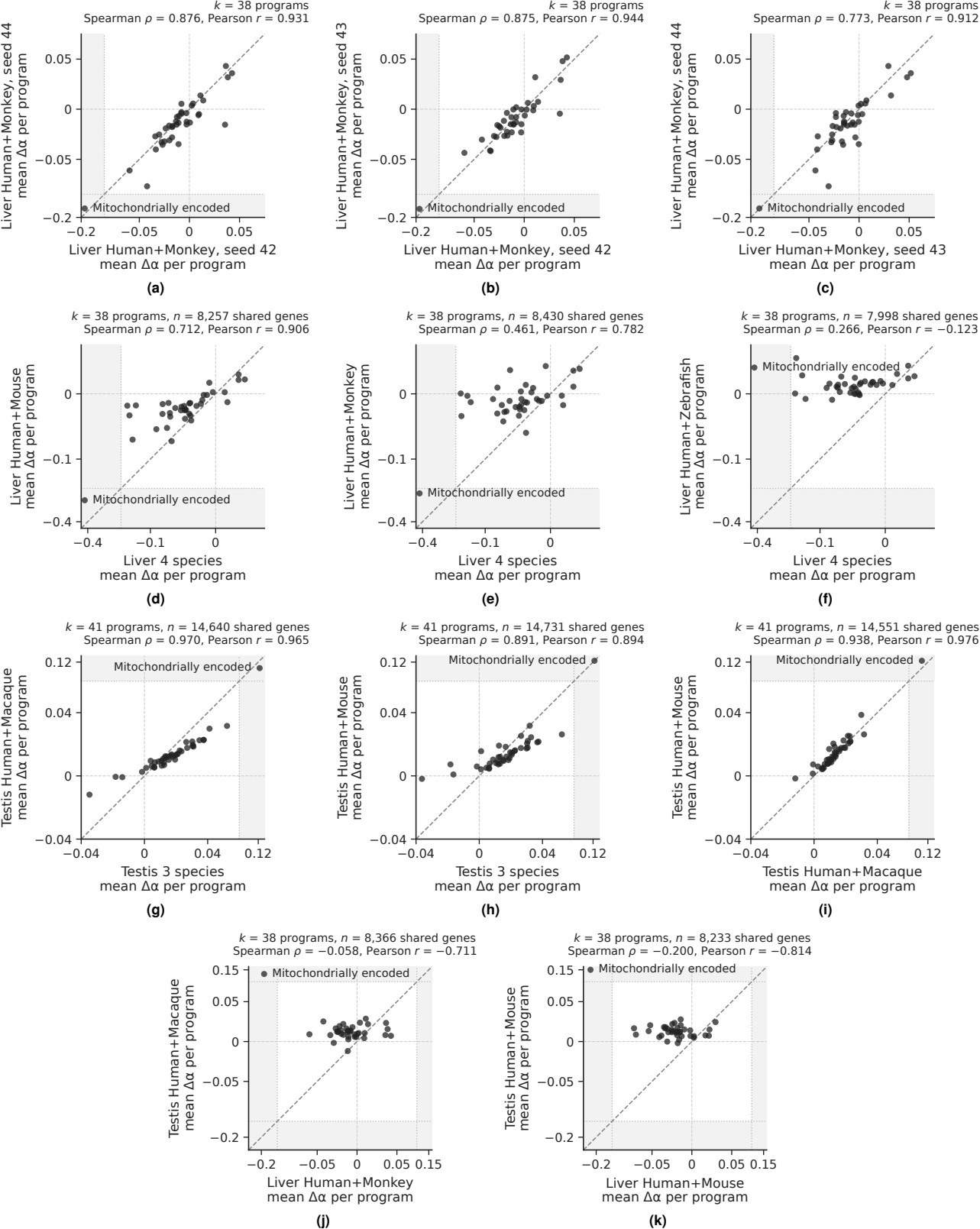
(l)

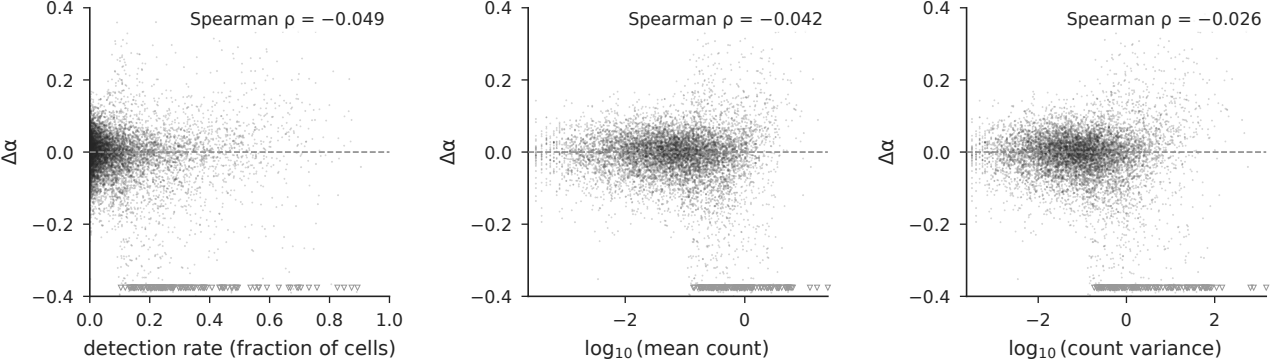

Kidney: AKI programs in mouse IRI
label-null p
human · mouse
PT
TAL
EC
FIB
PC
myeloid
DCT
IC
lymphoid
0.005 · 0.010
0.025 · 0.045
0.015 · 0.005
0.030 · 0.025
0.040 · 0.050
0.025 · 0.075
0.33 · 0.040
0.27 · 0.010
1.00 · 1.00
0
50
100
150
200
programs significant in human AKI
transfers to mouse
does not transfer
opposite direction
(a)
IRI 2 d vs 4–20 wk
no 2 d fit
no stage fit
no 2 d fit
no stage fit
0
100
200
transferring programs
0.0
0.4
0.8
program NES, Spearman
mouse IRI 2 d
mouse IRI 4–20 wk
(b)
Healthy-tissue test: does Δα predict transfer?
kidney: AKI vs IRI
PT
TAL
EC
FIB
PC
myeloid
DCT
IC
lymphoid
liver: MASH vs Western diet
hepatocyte
islet: T2D vs db/db, mSTZ
β
α
δ
ductal
endothelial
stellate
−0.08
−0.04
0
0.04
0.08
Δα vs gene-level transfer, ρ_p (95% CI)
(c)
Program transfer vs label null
observed, above null 95%
observed, within null
human-label null, median–95%
0
50
100
150
200
transferring programs
(d)
Cippà 2018 injury genes
acute lists
(primary)
late lists
0.51
0.51
Chimera Δα
profile correlation
0.70
0.72
0.4
0.5
0.6
0.7
0.8
stratified AUROC,
shared vs human-specific
(e)

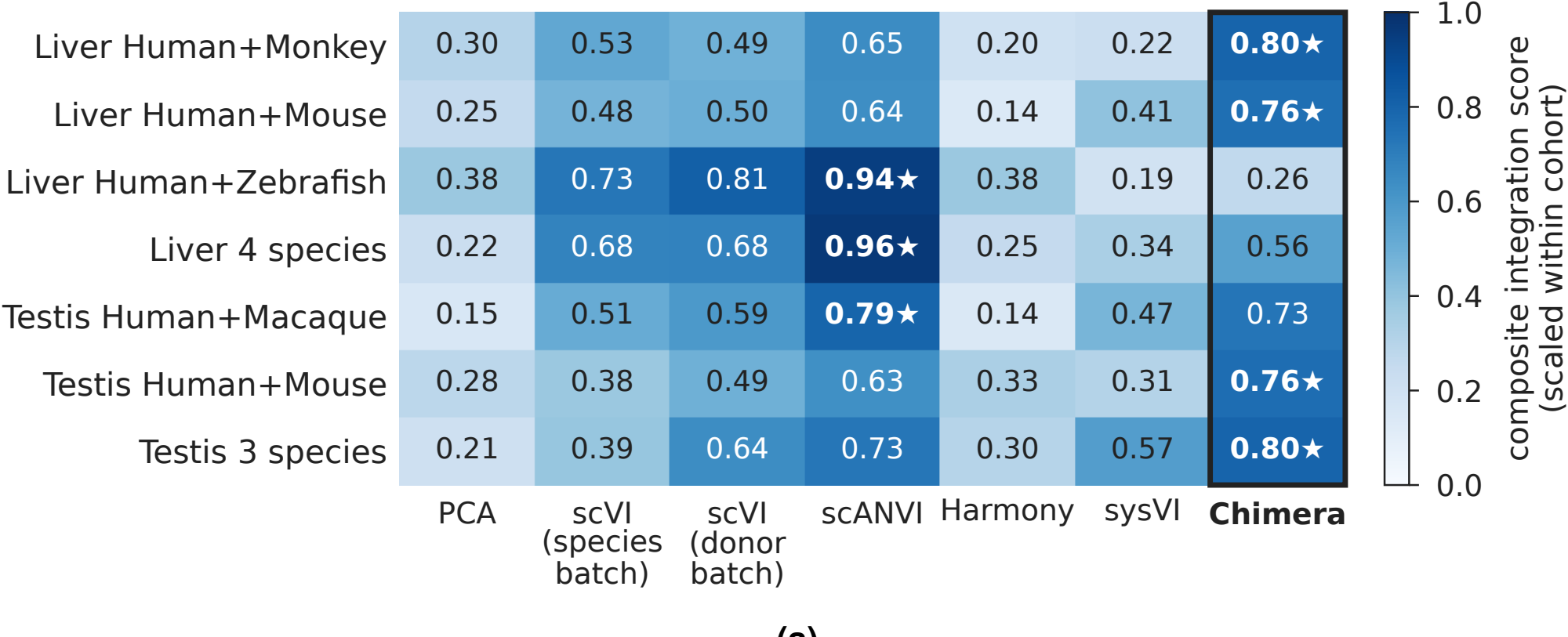

weight-path
gradient × input
integrated gradients
ablation
88%
77%
76%
38%
9%
17%
18%
47%
0
20
40
60
80
100
(a)
ESM-2 PCs
esm2_intersp_var
mean_identity
expression
identity_spread
expression-only correction contribution
0.0
−0.5
−1.0
−1.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
ρ(ESM-2, expr) = -0.000
ρ(ESM-2, full) = +0.592
ρ(expr, full) = +0.675
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Δα
(b)
liver, human + mouse
no gene features (free per-gene offset)
random gene features
protein-language features set to zero
expression features set to zero
counts shuffled across genes (expression features follow): readout stays with the gene's sequence
readout follows the moved counts
counts shuffled within expression bins: readout stays with the gene's sequence
readout follows the moved counts
liver, 4 species
random gene features
protein-language features set to zero
expression features set to zero
counts shuffled across genes (all features detached): readout stays with the gene's sequence
readout follows the moved counts
hepatocytes and endothelial cells only
degenerate run: Δα constant
1 of 3 seeds failed training
testis, human + mouse
random gene features
counts shuffled within expression bins: readout stays with the gene's sequence
readout follows the moved counts
−0.25
0.00
0.25
0.50
0.75
1.00
Δα of control run vs normal training (Spearman ρ across genes)
count-evidence margin S_g of control run vs normal training (Spearman ρ across genes)
seed floor: one normal seed vs the others (range)
projection null for 'stays with sequence' (2.5-97.5%)
Δα, one control run
S_g, one control run
(c)


UMAP 1
UMAP 2
α
Δα
|GIN correction|
nuclear_receptors (n121)
tf_zinc_finger (n114)
dna_replication_repair (n112)
tf_forkhead_etc (n112)
kinases_signaling (n105)
cell_cycle (n100)
splicing_factors (n70)
oxphos_nuclear_encoded (n66)
tf_homeobox (n58)
ribosomal_mt (n58)
cytoskeleton_microtubule (n56)
ecm_other (n55)
(d)
(e)
(f)
(g)
H+Monkey
H+Mouse
H+Zeb
4-sp
c1 c2 c3 c4 c5 c6 c7 c8 c9 c10 c11 c12
ESM-2 KMeans cluster (liver; sorted by median within row)
(h)
Silhouette score
Cross-seed ARI
Mean cross-seed ARI
Number of archetypes (k)
(i)
A0 (n3202)
A1 (n3264)
A2 (n2413)
(j)
Archetype
(k)
(l)
n = 3202
n = 3264
n = 2413
kNN purity excess over baseline, ΔP
(m)
chance (1/K)
log_mean_expr only (1d)
all 72 features
0.333
0.569 ± 0.002
0.940 ± 0.006
5-fold CV accuracy (chance = 1/K = 0.333)
(n)

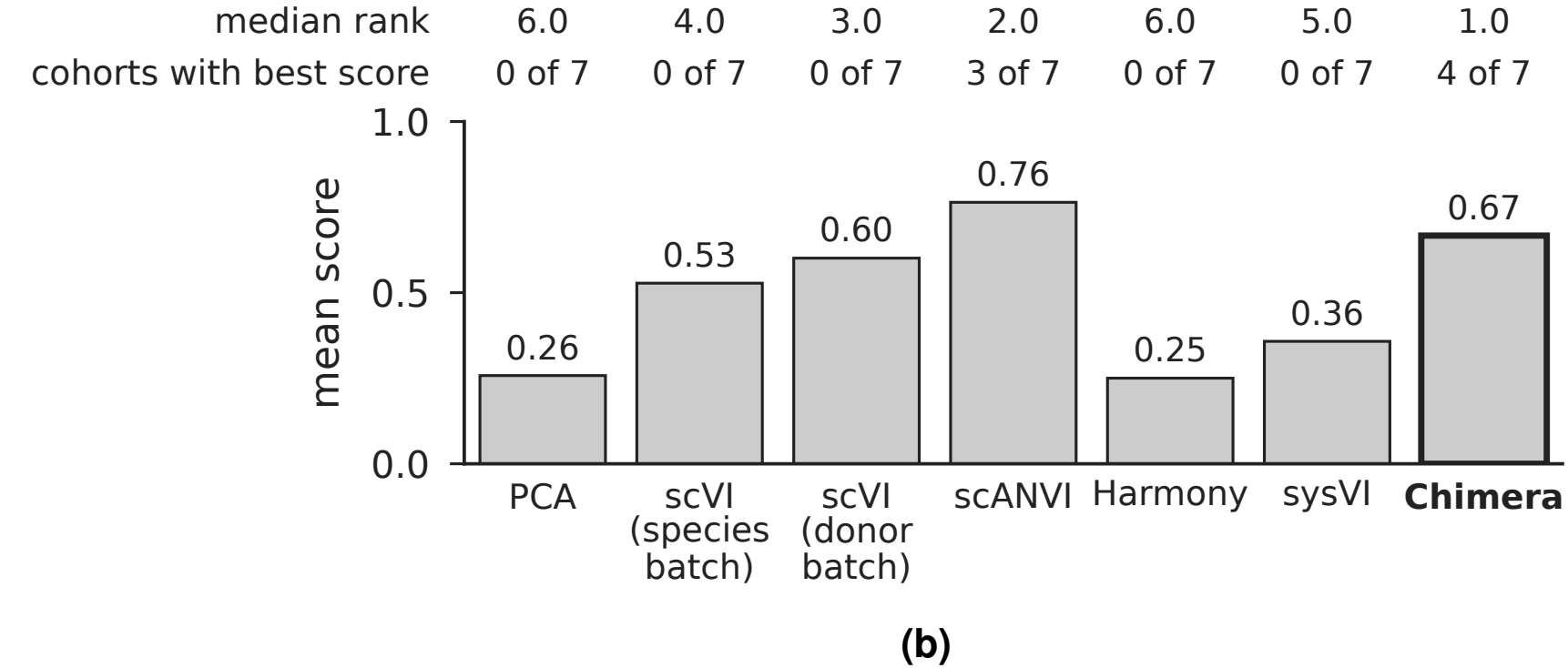

stability floor (0.01)
mean |α(t+1) − α(t)|
epoch
(a)
Δα > +0.05
Δα < −0.05
fraction of genes
epoch
(b)
120 of 8,430 genes
prior mean
mean of all genes
α
epoch
(c)
epoch 0
epoch 35
epoch 70
epoch 105
epoch 145
density
α
(d)
L_{anchor} (anchor term)
$L_{\text{fit-reward}}$ (fit-quality reward)
loss component (per-step mean)
epoch
(e)


k = 38 programs
Spearman ρ = 0.876, Pearson r = 0.931
Liver Human+Monkey, seed 44 mean Δα per program
Liver Human+Monkey, seed 42 mean Δα per program
Mitochondrially encoded
(a)
k = 38 programs
Spearman ρ = 0.875, Pearson r = 0.944
Liver Human+Monkey, seed 43 mean Δα per program
Liver Human+Monkey, seed 42 mean Δα per program
Mitochondrially encoded
(b)
k = 38 programs
Spearman ρ = 0.773, Pearson r = 0.912
Liver Human+Monkey, seed 44 mean Δα per program
Liver Human+Monkey, seed 43 mean Δα per program
Mitochondrially encoded
(c)
k = 38 programs, n = 8,257 shared genes
Spearman ρ = 0.712, Pearson r = 0.906
Liver Human+Mouse mean Δα per program
Liver 4 species mean Δα per program
Mitochondrially encoded
(d)
k = 38 programs, n = 8,430 shared genes
Spearman ρ = 0.461, Pearson r = 0.782
Liver Human+Monkey mean Δα per program
Liver 4 species mean Δα per program
Mitochondrially encoded
(e)
k = 38 programs, n = 7,998 shared genes
Spearman ρ = 0.266, Pearson r = −0.123
Liver Human+Zebrafish mean Δα per program
Liver 4 species mean Δα per program
Mitochondrially encoded
(f)
k = 41 programs, n = 14,640 shared genes
Spearman ρ = 0.970, Pearson r = 0.965
Testis Human+Macaque mean Δα per program
Testis 3 species mean Δα per program
Mitochondrially encoded
(g)
k = 41 programs, n = 14,731 shared genes
Spearman ρ = 0.891, Pearson r = 0.894
Testis Human+Mouse mean Δα per program
Testis 3 species mean Δα per program
Mitochondrially encoded
(h)
k = 41 programs, n = 14,551 shared genes
Spearman ρ = 0.938, Pearson r = 0.976
Testis Human+Mouse mean Δα per program
Testis Human+Macaque mean Δα per program
Mitochondrially encoded
(i)
k = 38 programs, n = 8,366 shared genes
Spearman ρ = −0.058, Pearson r = −0.711
Testis Human+Macaque mean Δα per program
Liver Human+Monkey mean Δα per program
Mitochondrially encoded
(j)
k = 38 programs, n = 8,233 shared genes
Spearman ρ = −0.200, Pearson r = −0.814
Testis Human+Mouse mean Δα per program
Liver Human+Mouse mean Δα per program
Mitochondrially encoded
(k)

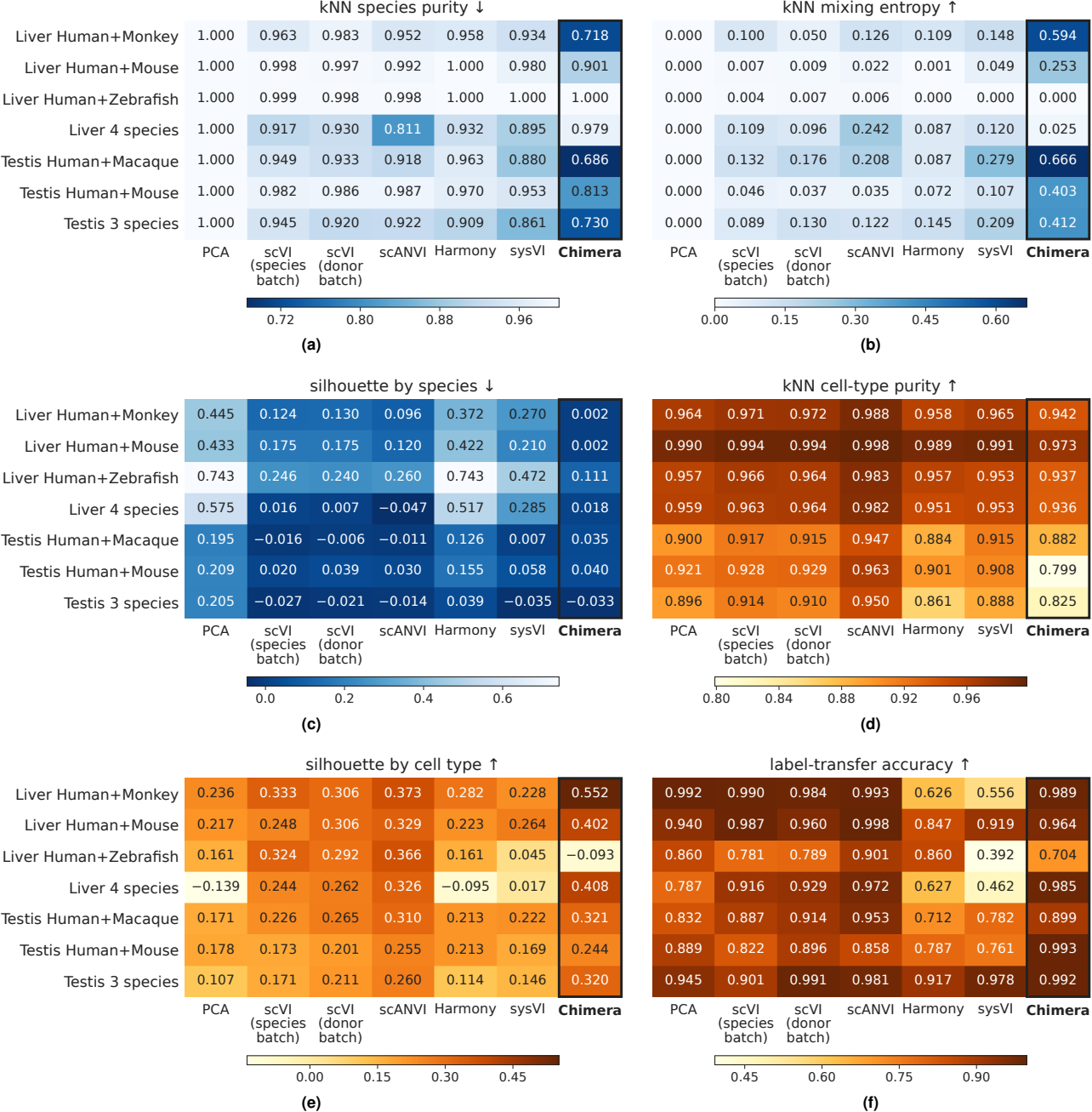

Spearman ρ = −0.049
Δα
0.4
0.2
0.0
−0.2
−0.4
0.0
0.2
0.4
0.6
0.8
1.0
detection rate (fraction of cells)
(a)
Spearman ρ = −0.042
Δα
−2
0
log10 (mean count)
(b)
Spearman ρ = −0.026
Δα
−2
0
2
log10 (count variance)
(c)

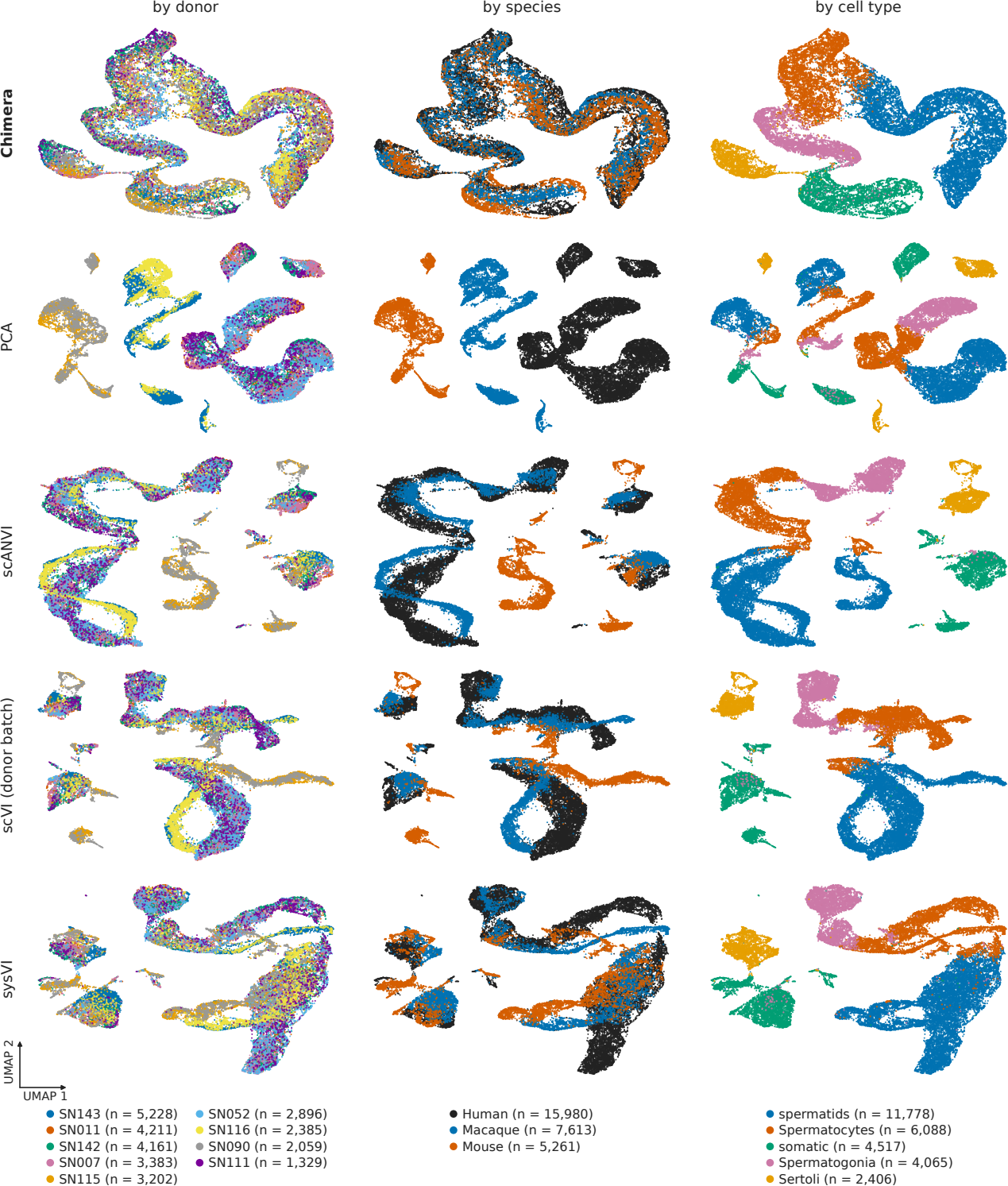

Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
0.30 0.53 0.49 0.65 0.20 0.22 0.80★
0.25 0.48 0.50 0.64 0.14 0.41 0.76★
0.38 0.73 0.81 0.94★ 0.38 0.19 0.26
0.22 0.68 0.68 0.96★ 0.25 0.34 0.56
0.15 0.51 0.59 0.79★ 0.14 0.47 0.73
0.28 0.38 0.49 0.63 0.33 0.31 0.76★
0.21 0.39 0.64 0.73 0.30 0.57 0.80★
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
composite integration score (scaled within cohort)
1.0
0.8
0.6
0.4
0.2
0.0

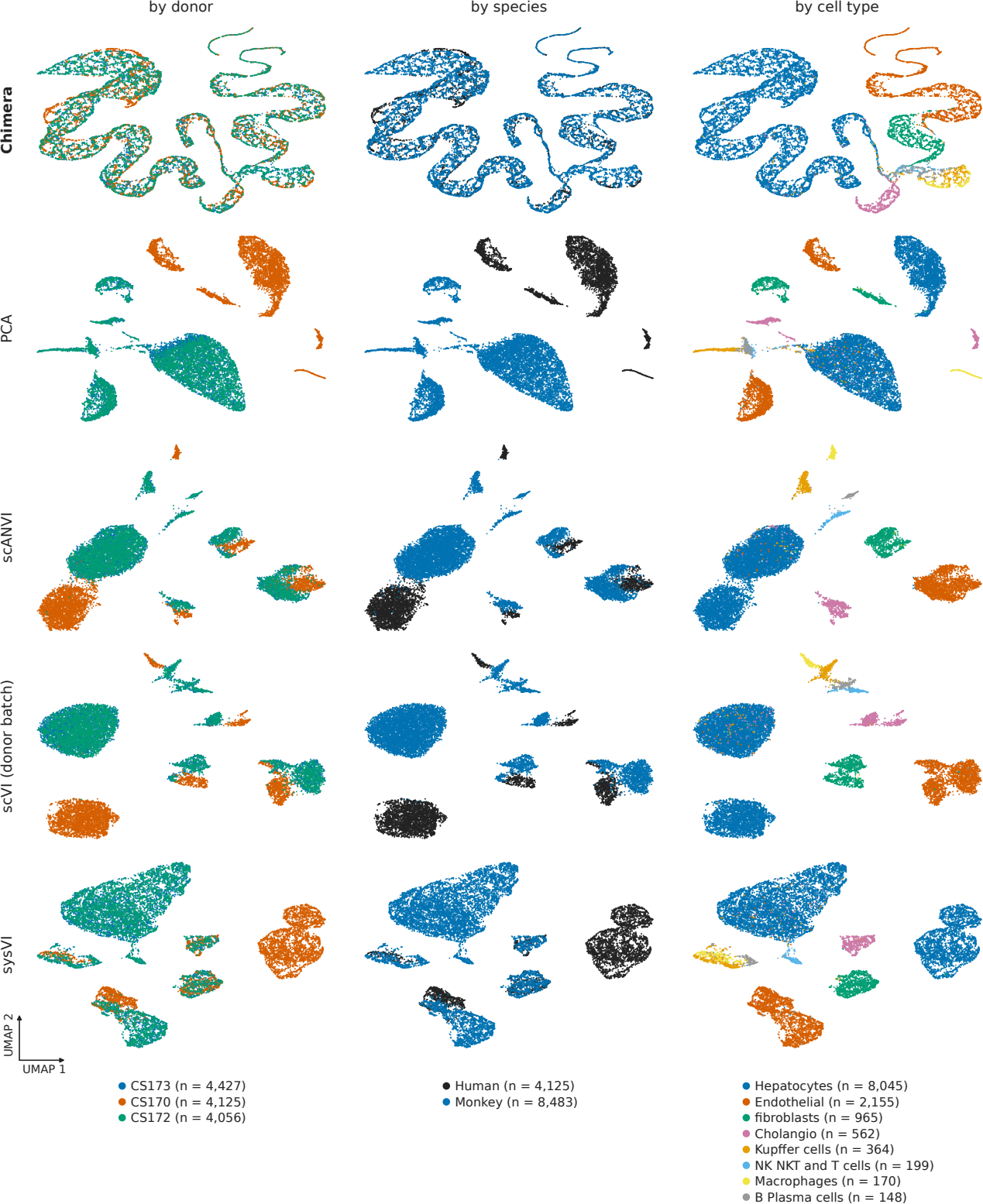
(a)

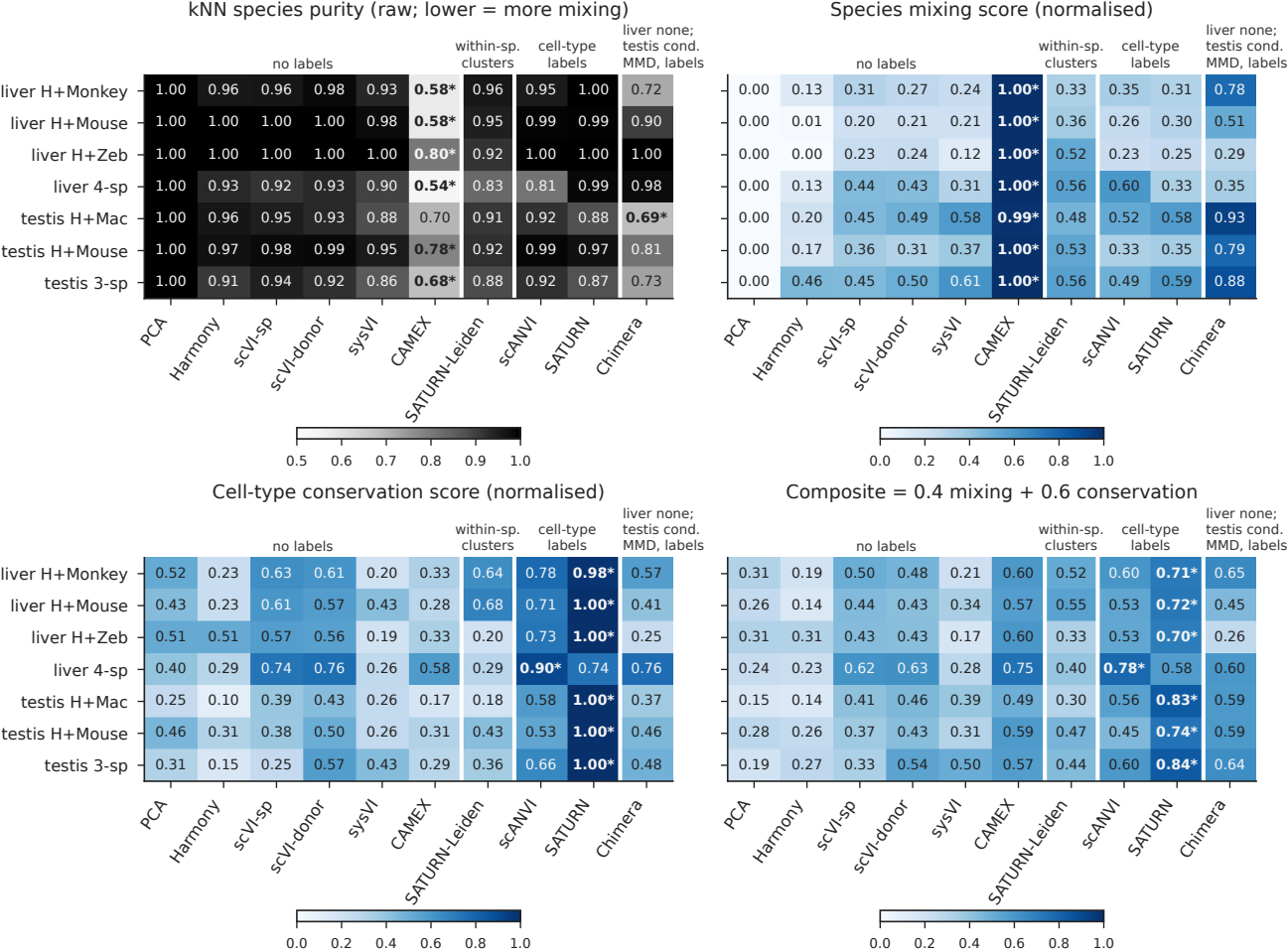

median rank 6.0 4.0 3.0 2.0 6.0 5.0 1.0
cohorts with best score 0 of 7 0 of 7 0 of 7 3 of 7 0 of 7 0 of 7 4 of 7
mean score
1.0
0.5
0.0
0.26
0.53
0.60
0.76
0.25
0.36
0.67
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera

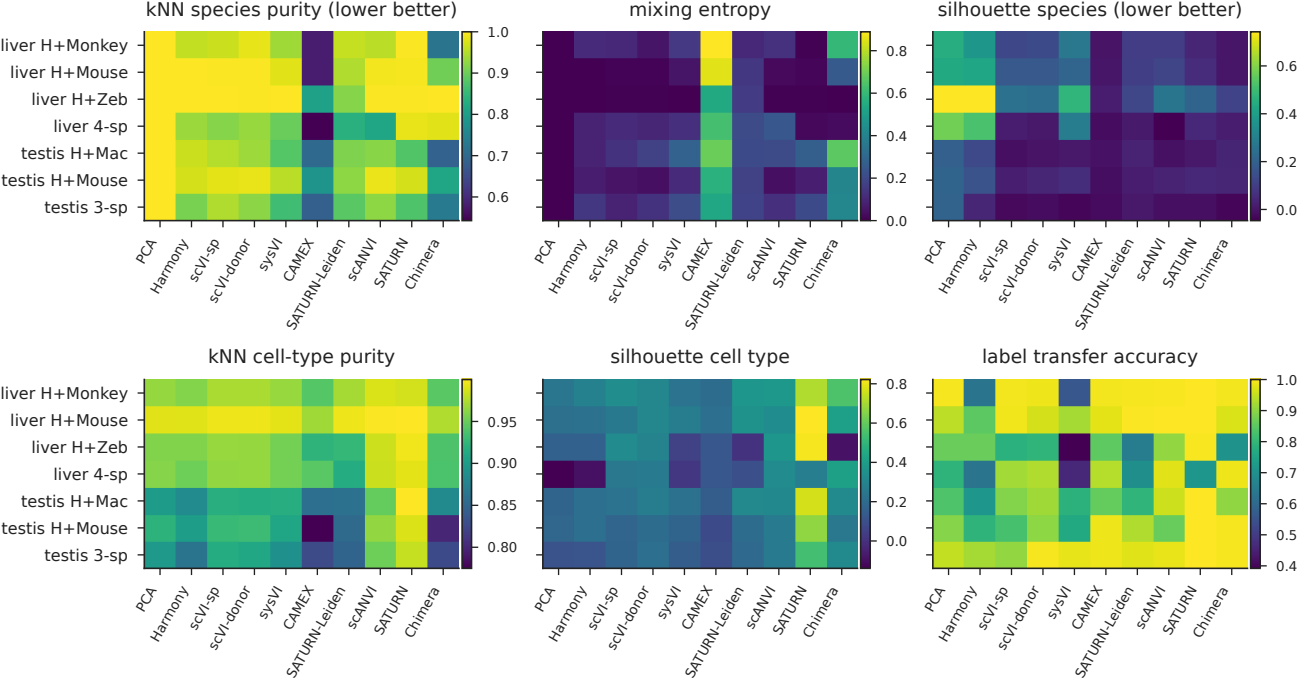
(b)

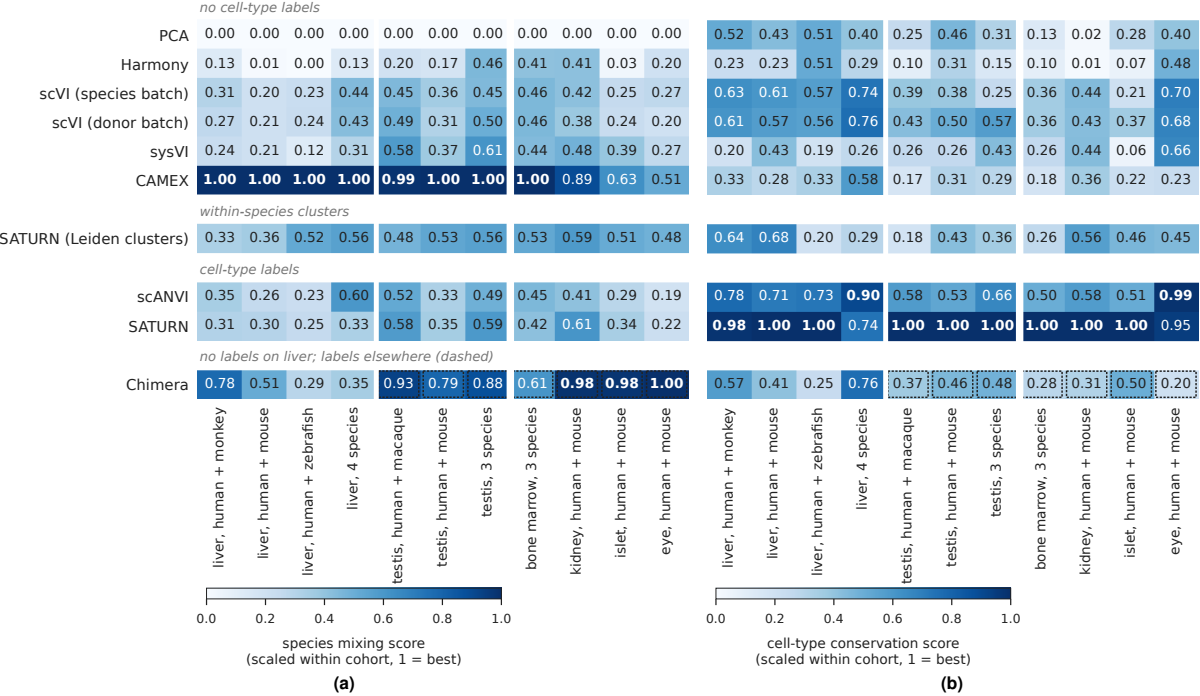

kNN species purity ↓
Liver Human+Monkey 1.000 0.963 0.983 0.952 0.958 0.934 0.718
Liver Human+Mouse 1.000 0.998 0.997 0.992 1.000 0.980 0.901
Liver Human+Zebrafish 1.000 0.999 0.998 0.998 1.000 1.000 1.000
Liver 4 species 1.000 0.917 0.930 0.811 0.932 0.895 0.979
Testis Human+Macaque 1.000 0.949 0.933 0.918 0.963 0.880 0.686
Testis Human+Mouse 1.000 0.982 0.986 0.987 0.970 0.953 0.813
Testis 3 species 1.000 0.945 0.920 0.922 0.909 0.861 0.730
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.72 0.80 0.88 0.96
(a)
kNN mixing entropy ↑
0.000 0.100 0.050 0.126 0.109 0.148 0.594
0.000 0.007 0.009 0.022 0.001 0.049 0.253
0.000 0.004 0.007 0.006 0.000 0.000 0.000
0.000 0.109 0.096 0.242 0.087 0.120 0.025
0.000 0.132 0.176 0.208 0.087 0.279 0.666
0.000 0.046 0.037 0.035 0.072 0.107 0.403
0.000 0.089 0.130 0.122 0.145 0.209 0.412
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.00 0.15 0.30 0.45 0.60
(b)
silhouette by species ↓
Liver Human+Monkey 0.445 0.124 0.130 0.096 0.372 0.270 0.002
Liver Human+Mouse 0.433 0.175 0.175 0.120 0.422 0.210 0.002
Liver Human+Zebrafish 0.743 0.246 0.240 0.260 0.743 0.472 0.111
Liver 4 species 0.575 0.016 0.007 −0.047 0.517 0.285 0.018
Testis Human+Macaque 0.195 −0.016 −0.006 −0.011 0.126 0.007 0.035
Testis Human+Mouse 0.209 0.020 0.039 0.030 0.155 0.058 0.040
Testis 3 species 0.205 −0.027 −0.021 −0.014 0.039 −0.035 −0.033
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.0 0.2 0.4 0.6
(c)
kNN cell-type purity ↑
0.964 0.971 0.972 0.988 0.958 0.965 0.942
0.990 0.994 0.994 0.998 0.989 0.991 0.973
0.957 0.966 0.964 0.983 0.957 0.953 0.937
0.959 0.963 0.964 0.982 0.951 0.953 0.936
0.900 0.917 0.915 0.947 0.884 0.915 0.882
0.921 0.928 0.929 0.963 0.901 0.908 0.799
0.896 0.914 0.910 0.950 0.861 0.888 0.825
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.80 0.84 0.88 0.92 0.96
(d)
silhouette by cell type ↑
Liver Human+Monkey 0.236 0.333 0.306 0.373 0.282 0.228 0.552
Liver Human+Mouse 0.217 0.248 0.306 0.329 0.223 0.264 0.402
Liver Human+Zebrafish 0.161 0.324 0.292 0.366 0.161 0.045 −0.093
Liver 4 species −0.139 0.244 0.262 0.326 −0.095 0.017 0.408
Testis Human+Macaque 0.171 0.226 0.265 0.310 0.213 0.222 0.321
Testis Human+Mouse 0.178 0.173 0.201 0.255 0.213 0.169 0.244
Testis 3 species 0.107 0.171 0.211 0.260 0.114 0.146 0.320
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.00 0.15 0.30 0.45
(e)
label-transfer accuracy ↑
0.992 0.990 0.984 0.993 0.626 0.556 0.989
0.940 0.987 0.960 0.998 0.847 0.919 0.964
0.860 0.781 0.789 0.901 0.860 0.392 0.704
0.787 0.916 0.929 0.972 0.627 0.462 0.985
0.832 0.887 0.914 0.953 0.712 0.782 0.899
0.889 0.822 0.896 0.858 0.787 0.761 0.993
0.945 0.901 0.991 0.981 0.917 0.978 0.992
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.45 0.60 0.75 0.90
(f)

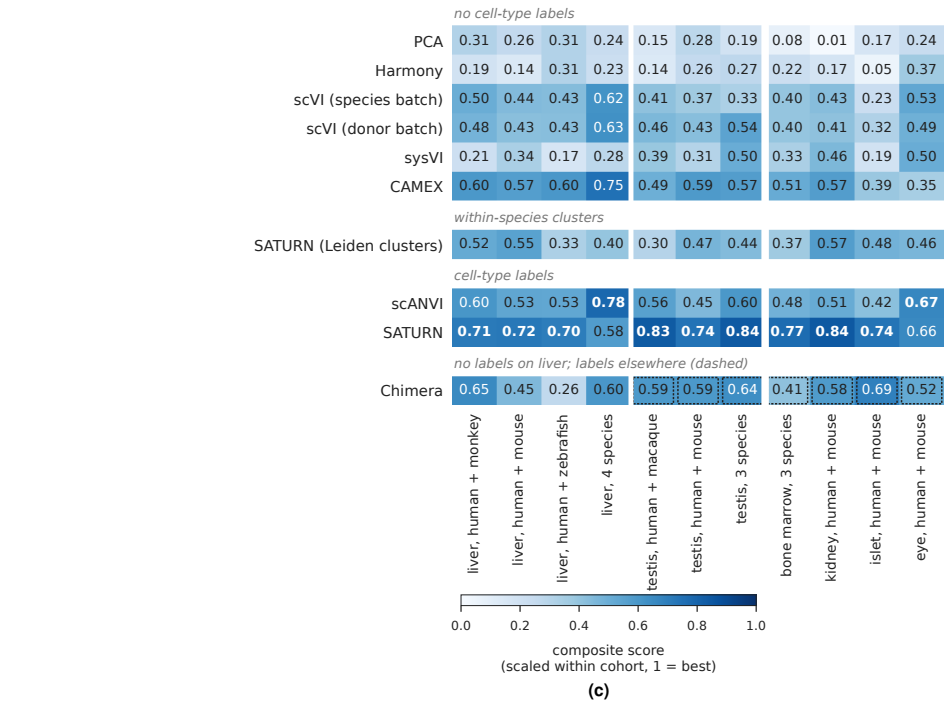

by donor
by species
by cell type
Chimera
PCA
scANVI
scVI (donor batch)
sysVI
UMAP 2
UMAP 1
SN143 (n = 5,228)
SN011 (n = 4,211)
SN142 (n = 4,161)
SN007 (n = 3,383)
SN115 (n = 3,202)
SN052 (n = 2,896)
SN116 (n = 2,385)
SN090 (n = 2,059)
SN111 (n = 1,329)
Human (n = 15,980)
Macaque (n = 7,613)
Mouse (n = 5,261)
spermatids (n = 11,778)
Spermatocytes (n = 6,088)
somatic (n = 4,517)
Spermatogonia (n = 4,065)
Sertoli (n = 2,406)

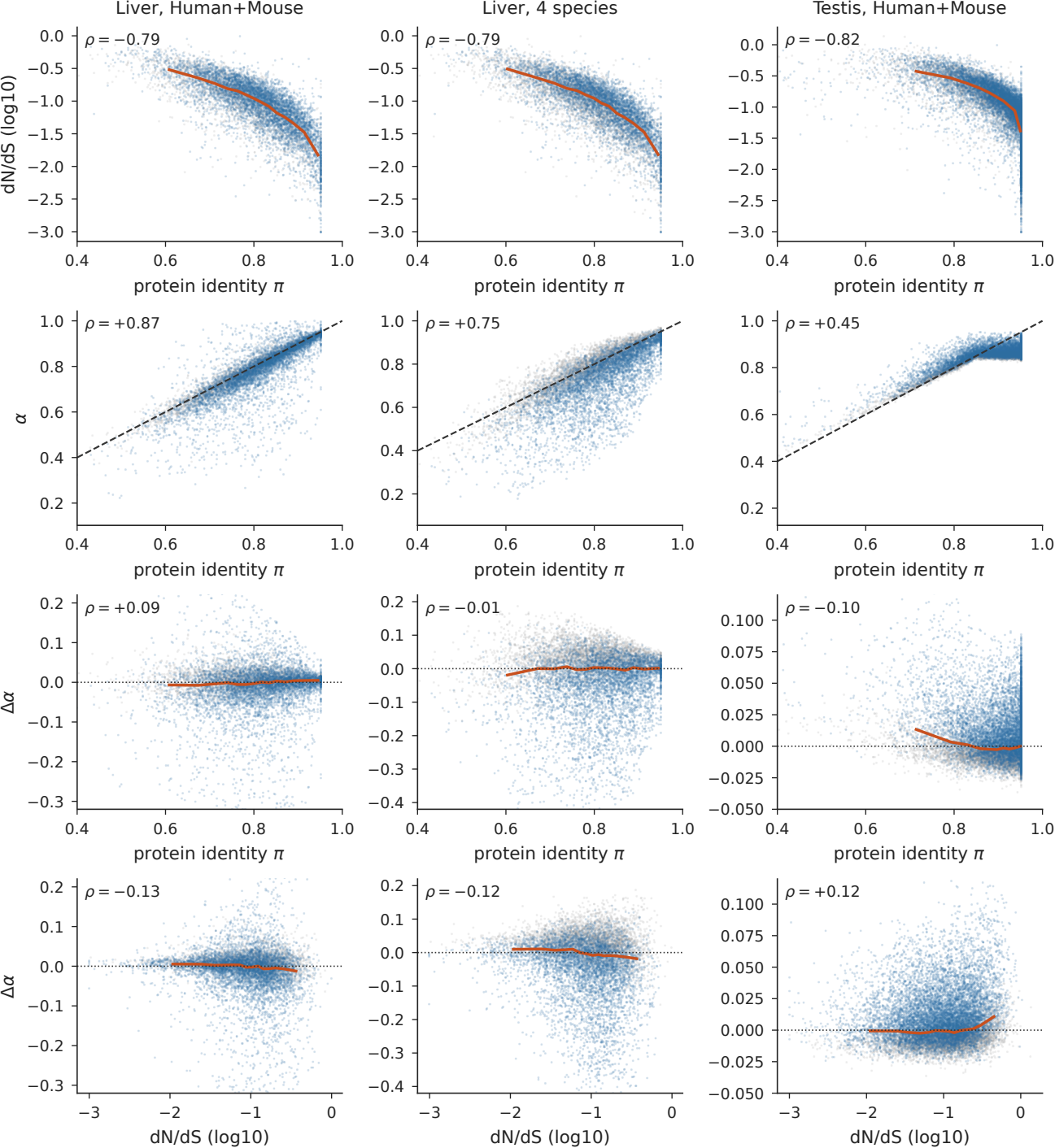

by donor
by species
by cell type
Chimera
PCA
scANVI
scVI (donor batch)
sysVI
UMAP 2
UMAP 1
CS173 (n = 4,427)
CS170 (n = 4,125)
CS172 (n = 4,056)
Human (n = 4,125)
Monkey (n = 8,483)
Hepatocytes (n = 8,045)
Endothelial (n = 2,155)
fibroblasts (n = 965)
Cholangio (n = 562)
Kupffer cells (n = 364)
NK NKT and T cells (n = 199)
Macrophages (n = 170)
B Plasma cells (n = 148)

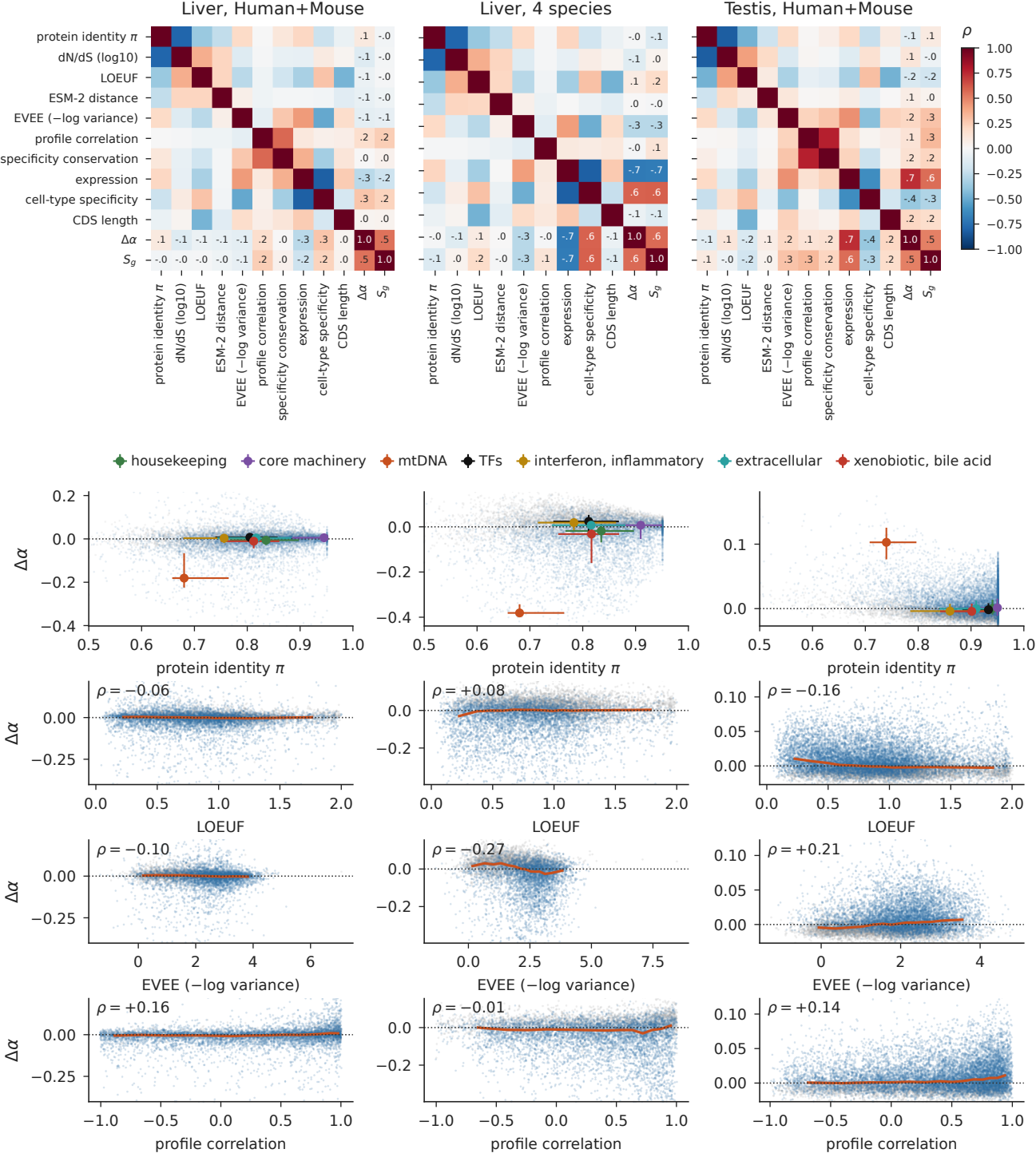

kNN species purity (raw; lower = more mixing)
Species mixing score (normalised)
Cell-type conservation score (normalised)
Composite = 0.4 mixing + 0.6 conservation
no labels
within-sp. clusters
cell-type labels
liver none; testis cond. MMD, labels
liver H+Monkey
liver H+Mouse
liver H+Zeb
liver 4-sp
testis H+Mac
testis H+Mouse
testis 3-sp
PCA
Harmony
scVI-sp
scVI-donor
sysVI
CAMEX
SATURN-Leiden
scANVI
SATURN
Chimera
0.5 0.6 0.7 0.8 0.9 1.0
0.0 0.2 0.4 0.6 0.8 1.0

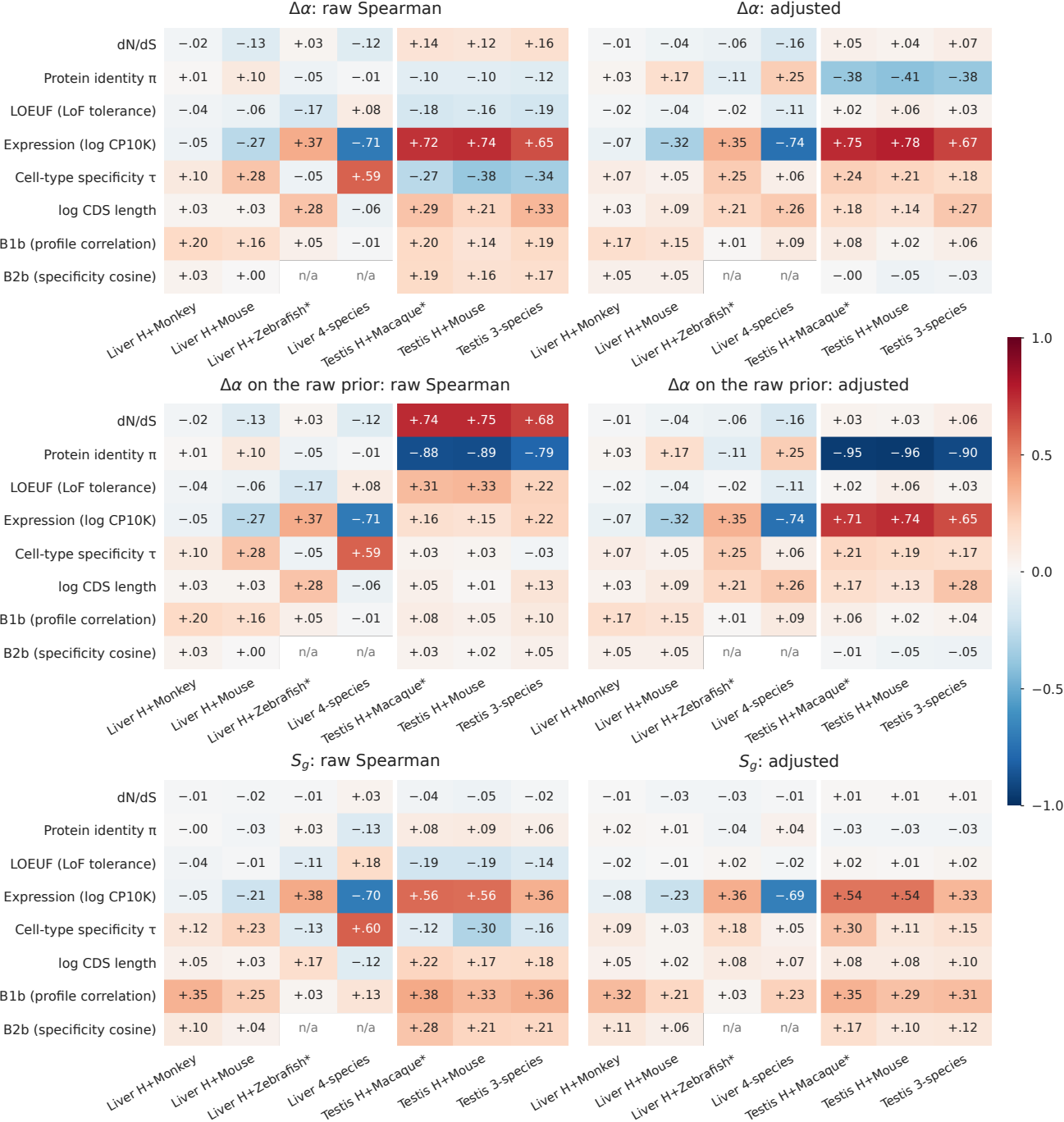

kNN species purity (lower better)
mixing entropy
silhouette species (lower better)
kNN cell-type purity
silhouette cell type
label transfer accuracy
liver H+Monkey
liver H+Mouse
liver H+Zeb
liver 4-sp
testis H+Mac
testis H+Mouse
testis 3-sp
PCA
Harmony
scVI-sp
scVI-donor
sysVI
CAMEX
SATURN-Leiden
scANVI
SATURN
Chimera


no cell-type labels
PCA 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.52 0.43 0.51 0.40 0.25 0.46 0.31 0.13 0.02 0.28 0.40
Harmony 0.13 0.01 0.00 0.13 0.20 0.17 0.46 0.41 0.41 0.03 0.20 0.23 0.23 0.51 0.29 0.10 0.31 0.15 0.10 0.01 0.07 0.48
scVI (species batch) 0.31 0.20 0.23 0.44 0.45 0.36 0.45 0.46 0.42 0.25 0.27 0.63 0.61 0.57 0.74 0.39 0.38 0.25 0.36 0.44 0.21 0.70
scVI (donor batch) 0.27 0.21 0.24 0.43 0.49 0.31 0.50 0.46 0.38 0.24 0.20 0.61 0.57 0.56 0.76 0.43 0.50 0.57 0.36 0.43 0.37 0.68
sysVI 0.24 0.21 0.12 0.31 0.58 0.37 0.61 0.44 0.48 0.39 0.27 0.20 0.43 0.19 0.26 0.26 0.26 0.43 0.26 0.44 0.06 0.66
CAMEX 1.00 1.00 1.00 1.00 0.99 1.00 1.00 1.00 0.89 0.63 0.51 0.33 0.28 0.33 0.58 0.17 0.31 0.29 0.18 0.36 0.22 0.23
within-species clusters
SATURN (Leiden clusters) 0.33 0.36 0.52 0.56 0.48 0.53 0.56 0.53 0.59 0.51 0.48 0.64 0.68 0.20 0.29 0.18 0.43 0.36 0.26 0.56 0.46 0.45
cell-type labels
scANVI 0.35 0.26 0.23 0.60 0.52 0.33 0.49 0.45 0.41 0.29 0.19 0.78 0.71 0.73 0.90 0.58 0.53 0.66 0.50 0.58 0.51 0.99
SATURN 0.31 0.30 0.25 0.33 0.58 0.35 0.59 0.42 0.61 0.34 0.22 0.98 1.00 1.00 0.74 1.00 1.00 1.00 1.00 1.00 1.00 0.95
no labels on liver; labels elsewhere (dashed)
Chimera 0.78 0.51 0.29 0.35 0.93 0.79 0.88 0.61 0.98 0.98 1.00 0.57 0.41 0.25 0.76 0.37 0.46 0.48 0.28 0.31 0.50 0.20
liver, human + monkey
liver, human + mouse
liver, human + zebrafish
liver, 4 species
testis, human + macaque
testis, human + mouse
testis, 3 species
bone marrow, 3 species
kidney, human + mouse
islet, human + mouse
eye, human + mouse
0.0 0.2 0.4 0.6 0.8 1.0
species mixing score
(scaled within cohort, 1 = best)
(a)
0.0 0.2 0.4 0.6 0.8 1.0
cell-type conservation score
(scaled within cohort, 1 = best)
(b)

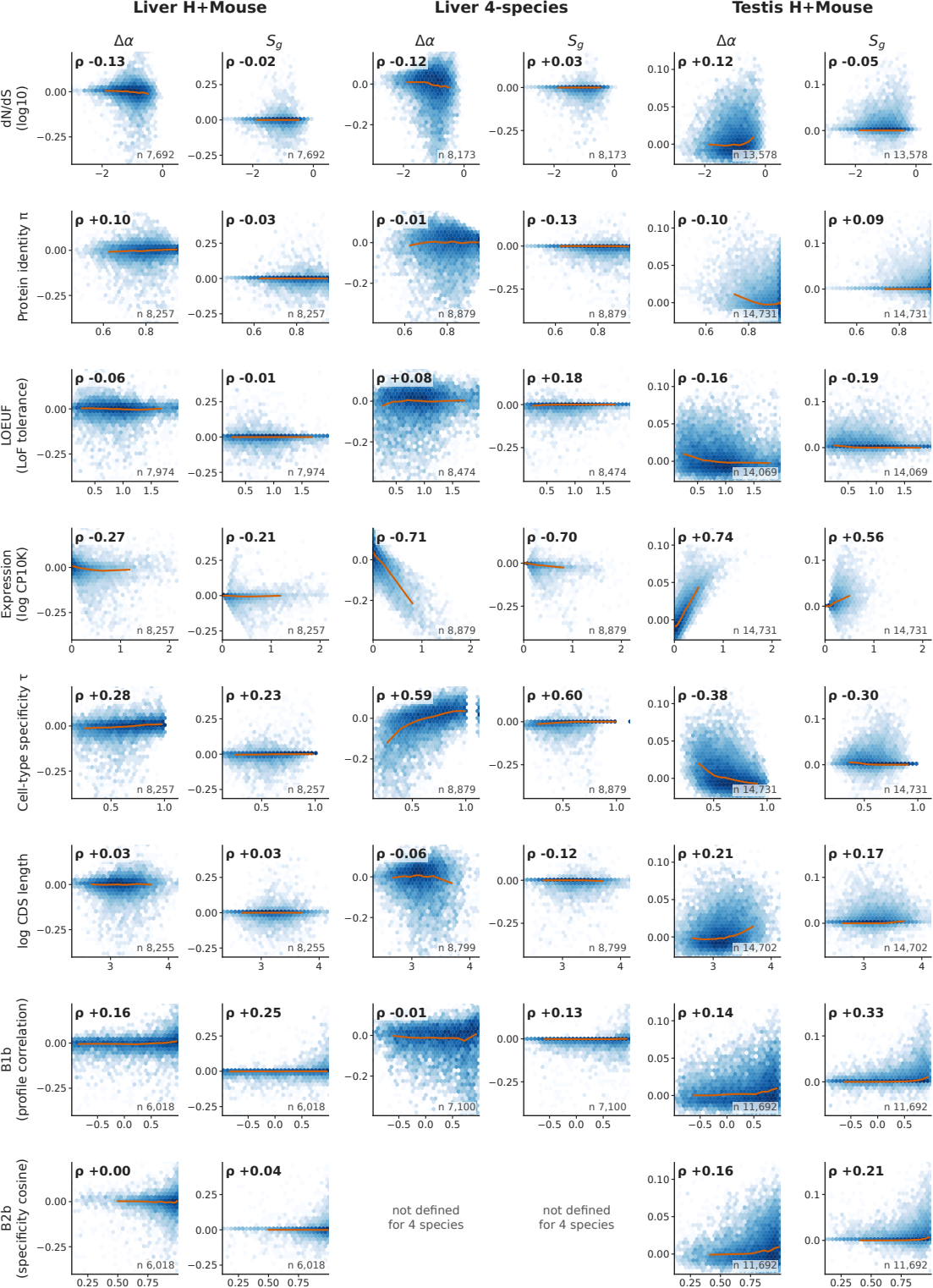

no cell-type labels
PCA 0.31 0.26 0.31 0.24 0.15 0.28 0.19 0.08 0.01 0.17 0.24
Harmony 0.19 0.14 0.31 0.23 0.14 0.26 0.27 0.22 0.17 0.05 0.37
scVI (species batch) 0.50 0.44 0.43 0.62 0.41 0.37 0.33 0.40 0.43 0.23 0.53
scVI (donor batch) 0.48 0.43 0.43 0.63 0.46 0.43 0.54 0.40 0.41 0.32 0.49
sysVI 0.21 0.34 0.17 0.28 0.39 0.31 0.50 0.33 0.46 0.19 0.50
CAMEX 0.60 0.57 0.60 0.75 0.49 0.59 0.57 0.51 0.57 0.39 0.35
within-species clusters
SATURN (Leiden clusters) 0.52 0.55 0.33 0.40 0.30 0.47 0.44 0.37 0.57 0.48 0.46
cell-type labels
scANVI 0.60 0.53 0.53 0.78 0.56 0.45 0.60 0.48 0.51 0.42 0.67
SATURN 0.71 0.72 0.70 0.58 0.83 0.74 0.84 0.77 0.84 0.74 0.66
no labels on liver; labels elsewhere (dashed)
Chimera 0.65 0.45 0.26 0.60 0.59 0.59 0.64 0.41 0.58 0.69 0.52
liver, human + monkey
liver, human + mouse
liver, human + zebrafish
liver, 4 species
testis, human + macaque
testis, human + mouse
testis, 3 species
bone marrow, 3 species
kidney, human + mouse
islet, human + mouse
eye, human + mouse
0.0 0.2 0.4 0.6 0.8 1.0
composite score
(scaled within cohort, 1 = best)
(c)

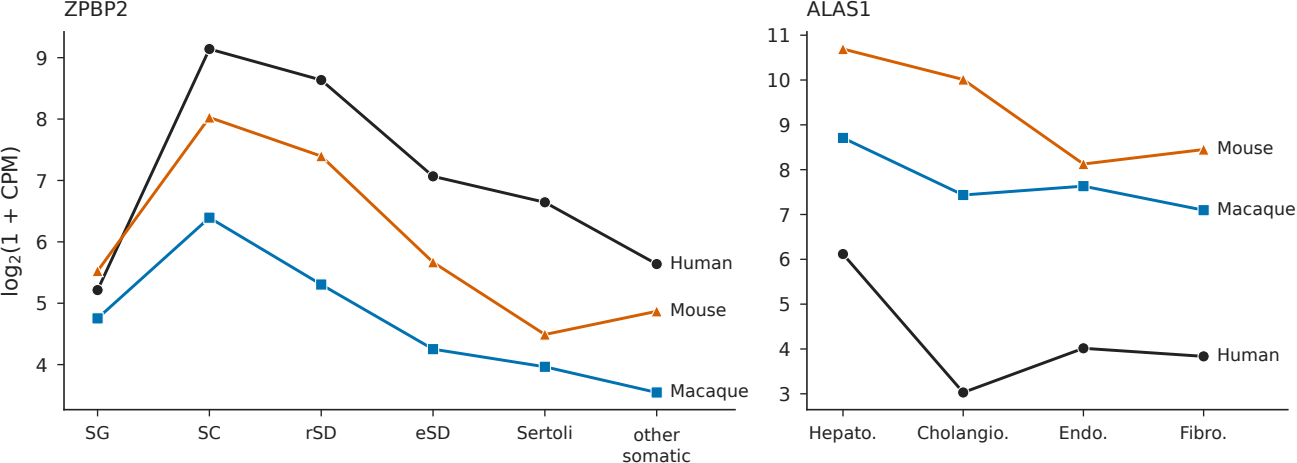

Liver, Human+Mouse
Liver, 4 species
Testis, Human+Mouse
dN/dS (log10)
α
Δα
ρ = −0.79
ρ = −0.79
ρ = −0.82
ρ = +0.87
ρ = +0.75
ρ = +0.45
ρ = +0.09
ρ = −0.01
ρ = −0.10
ρ = −0.13
ρ = −0.12
ρ = +0.12
protein identity π
dN/dS (log10)

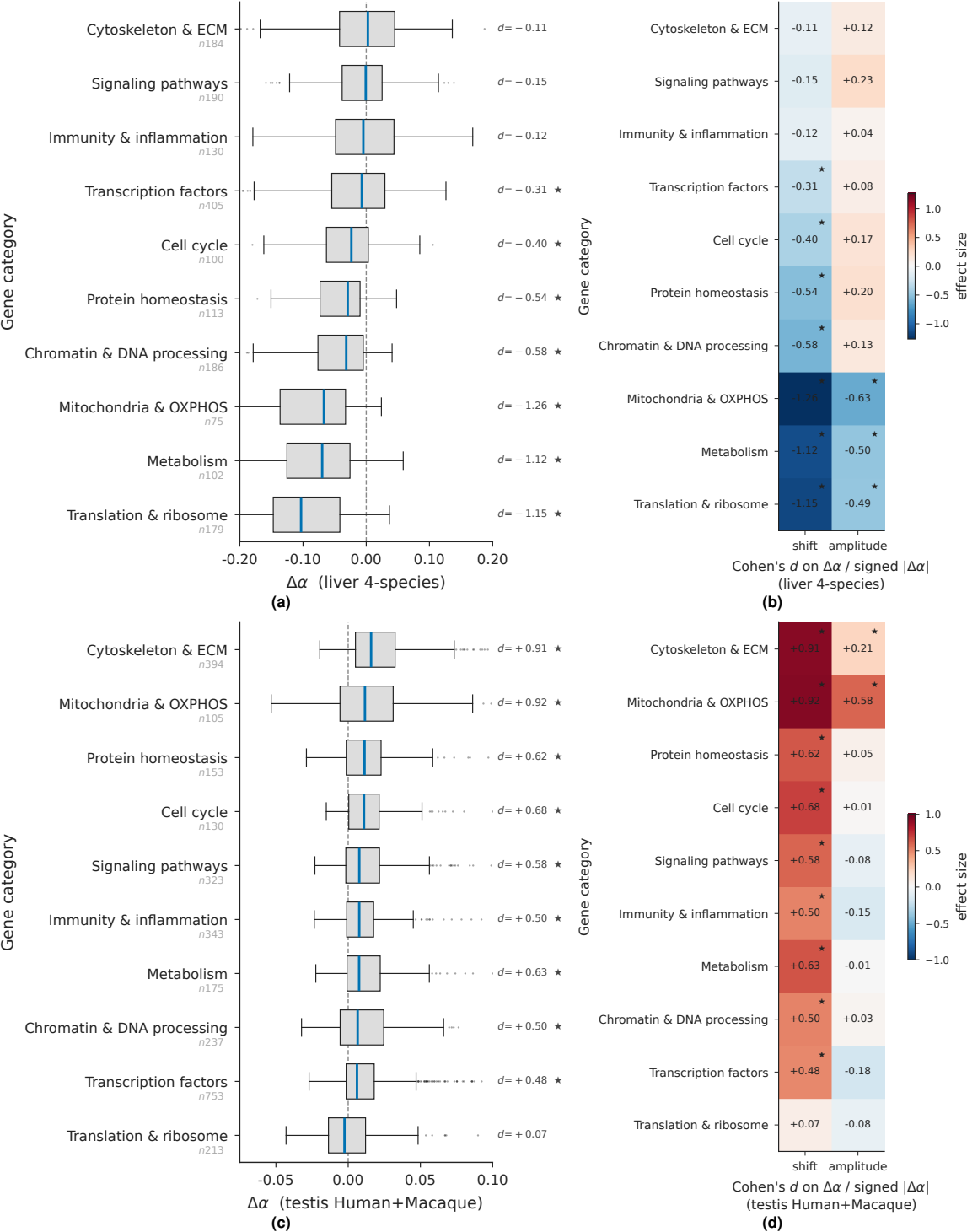

Liver, Human+Mouse
Liver, 4 species
Testis, Human+Mouse
protein identity π
dN/dS (log10)
LOEUF
ESM-2 distance
EVEE (−log variance)
profile correlation
specificity conservation
expression
cell-type specificity
CDS length
Δα
S_g
ρ
housekeeping
core machinery
mtDNA
TFs
interferon, inflammatory
extracellular
xenobiotic, bile acid
ρ = −0.06
ρ = +0.08
ρ = −0.16
ρ = −0.10
ρ = −0.27
ρ = +0.21
ρ = +0.16
ρ = −0.01
ρ = +0.14

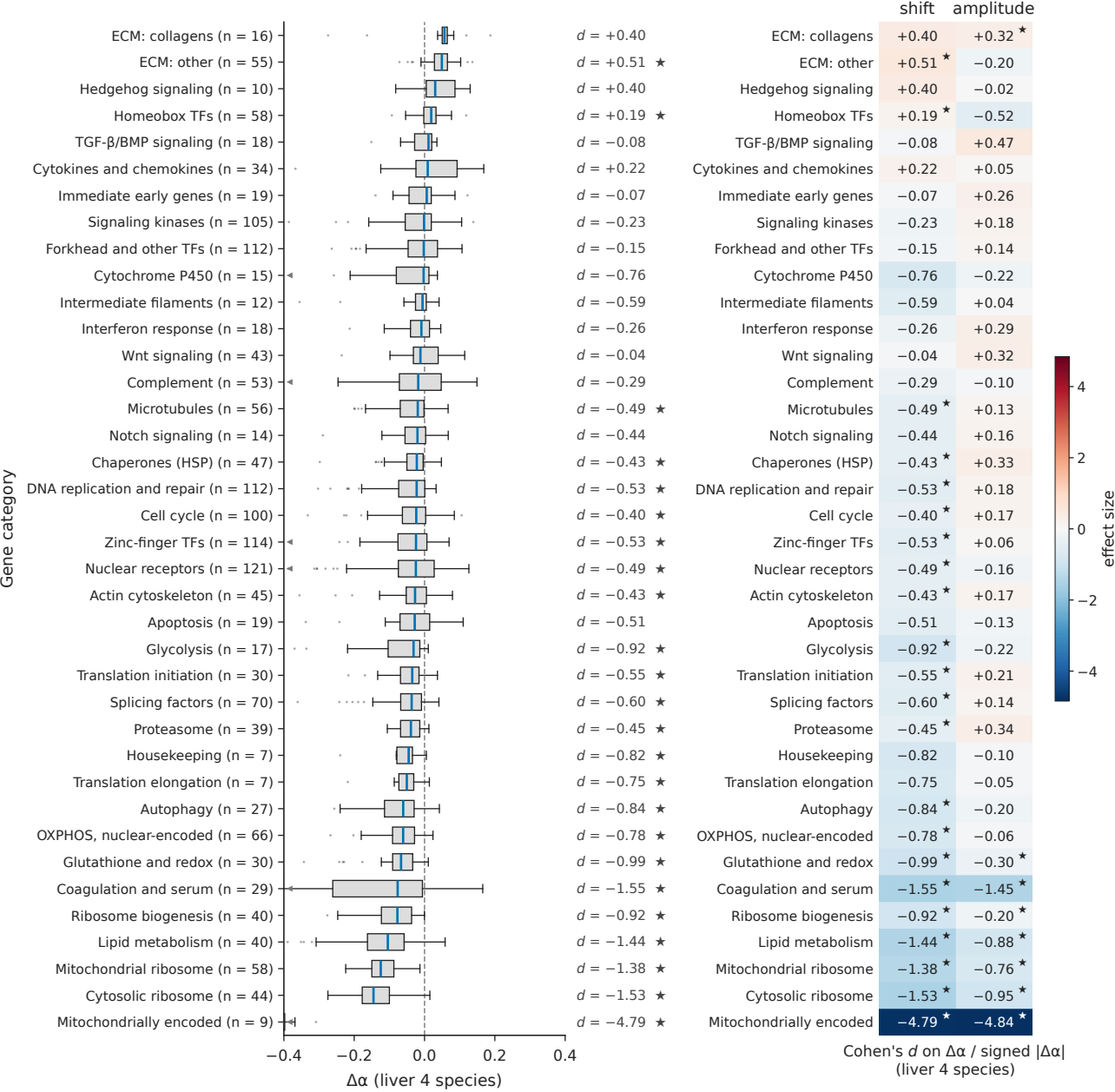


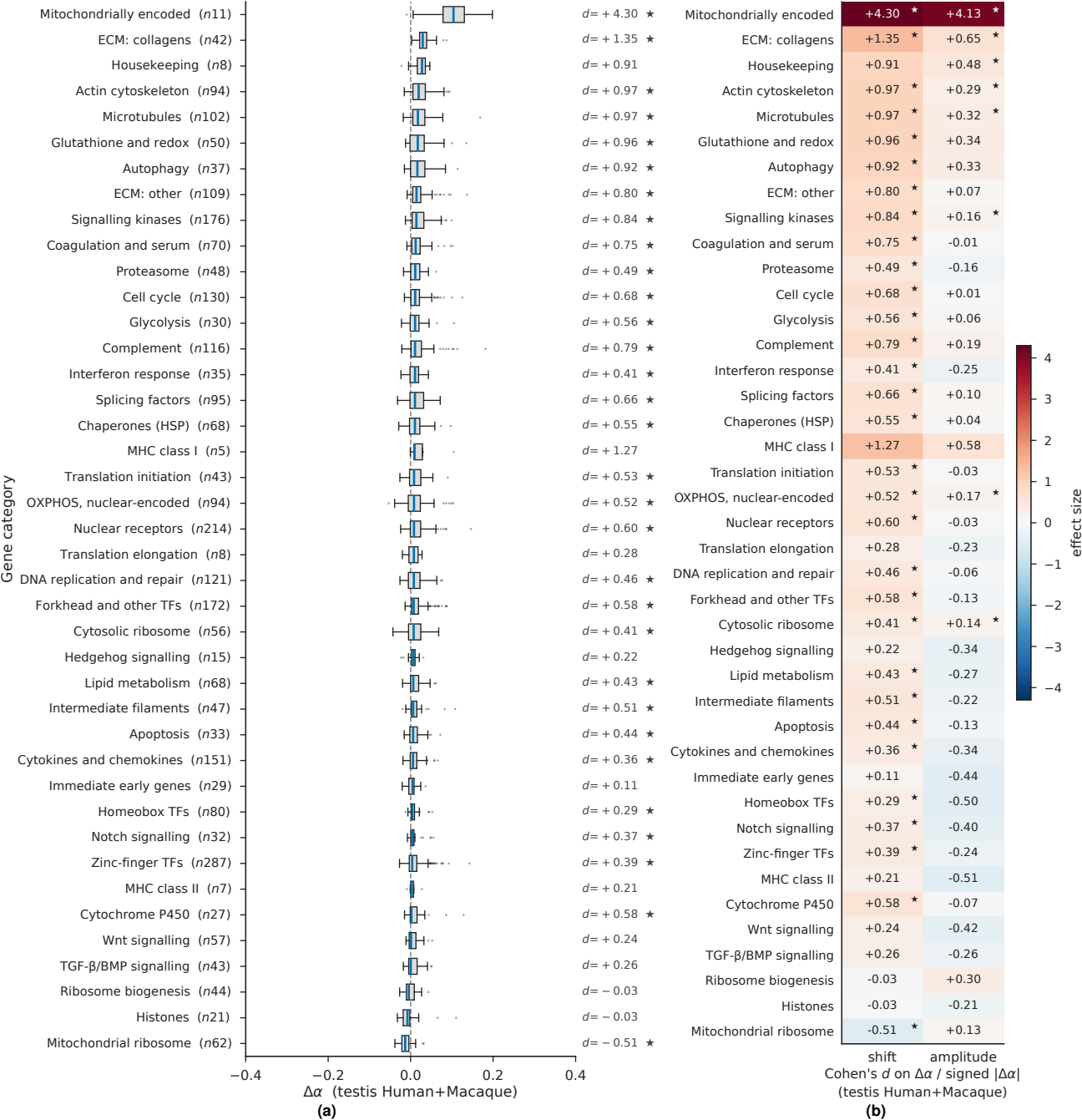
Adjusted: rank partial correlation given the other covariates (identity, expression, detection, CDS length). * one seed. n/a: B2b is not defined for these cohorts.

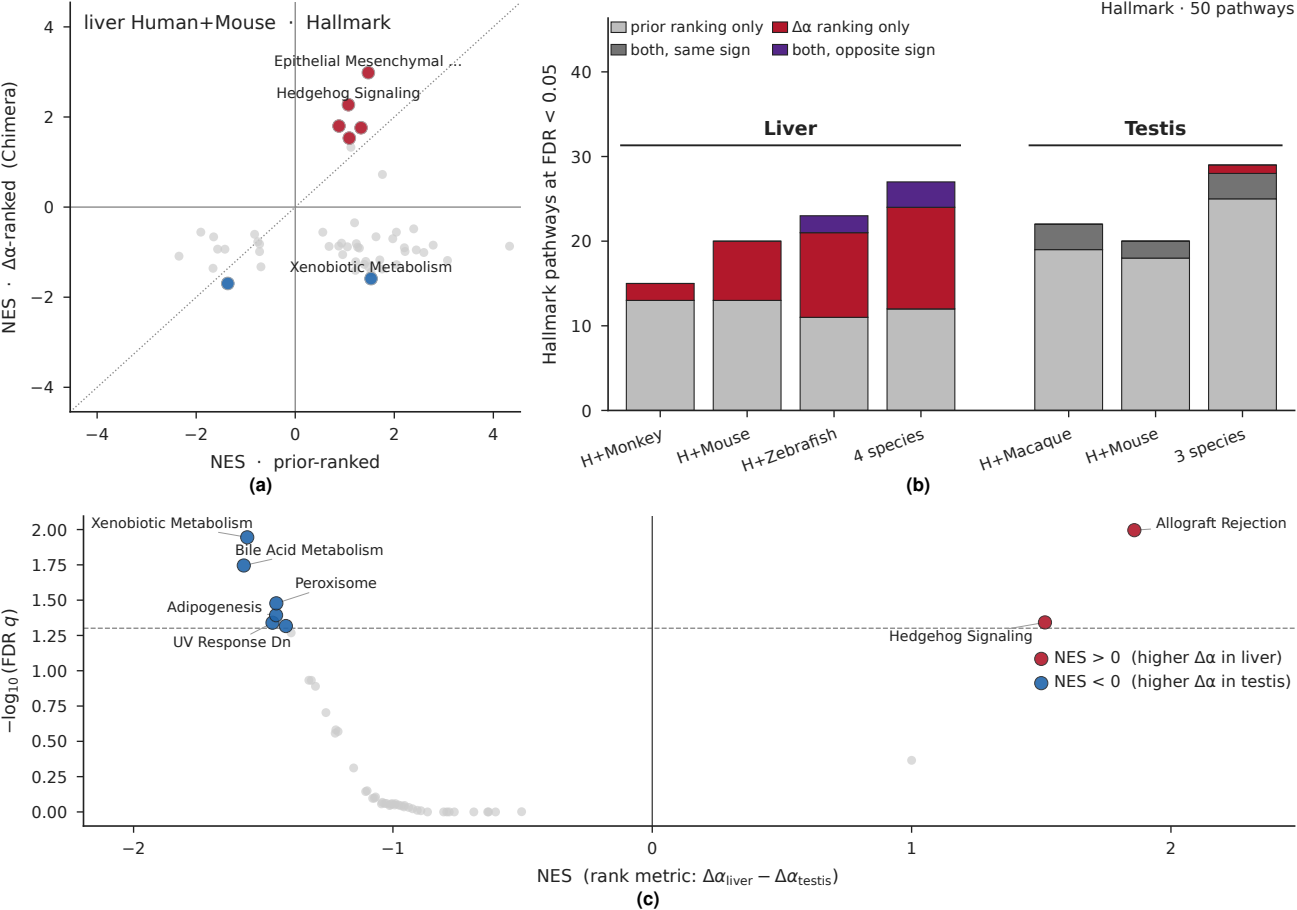

Liver H+Mouse
Liver 4-species
Testis H+Mouse
Δα
S_g
dN/dS (log10)
Protein identity π
LOEUF (LoF tolerance)
Expression (log CP10K)
Cell-type specificity τ
log CDS length
B1b (profile correlation)
B2b (specificity cosine)
ρ -0.13
n 7,692
ρ -0.02
n 7,692
ρ -0.12
n 8,173
ρ +0.03
n 8,173
ρ +0.12
n 13,578
ρ -0.05
n 13,578
ρ +0.10
n 8,257
ρ -0.03
n 8,257
ρ -0.01
n 8,879
ρ -0.13
n 8,879
ρ -0.10
n 14,731
ρ +0.09
n 14,731
ρ -0.06
n 7,974
ρ -0.01
n 7,974
ρ +0.08
n 8,474
ρ +0.18
n 8,474
ρ -0.16
n 14,069
ρ -0.19
n 14,069
ρ -0.27
n 8,257
ρ -0.21
n 8,257
ρ -0.71
n 8,879
ρ -0.70
n 8,879
ρ +0.74
n 14,731
ρ +0.56
n 14,731
ρ +0.28
n 8,257
ρ +0.23
n 8,257
ρ +0.59
n 8,879
ρ +0.60
n 8,879
ρ -0.38
n 14,731
ρ -0.30
n 14,731
ρ +0.03
n 8,255
ρ +0.03
n 8,255
ρ -0.06
n 8,799
ρ -0.12
n 8,799
ρ +0.21
n 14,702
ρ +0.17
n 14,702
ρ +0.16
n 6,018
ρ +0.25
n 6,018
ρ -0.01
n 7,100
ρ +0.13
n 7,100
ρ +0.14
n 11,692
ρ +0.33
n 11,692
ρ +0.00
n 6,018
ρ +0.04
n 6,018
not defined for 4 species
not defined for 4 species
ρ +0.16
n 11,692
ρ +0.21
n 11,692

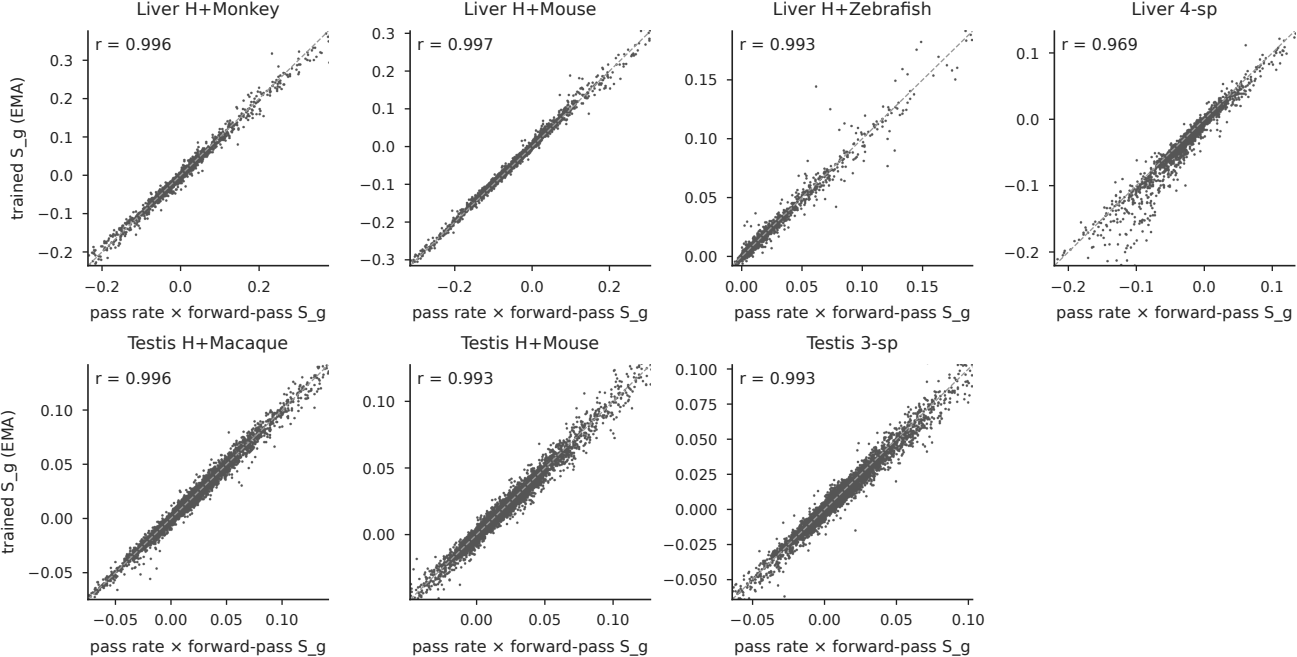

ZPBP2
log2(1 + CPM)
9
8
7
6
5
4
Human
Mouse
Macaque
SG
SC
rSD
eSD
Sertoli
other somatic
ALAS1
11
10
9
8
7
6
5
4
3
Mouse
Macaque
Human
Hepato.
Cholangio.
Endo.
Fibro.

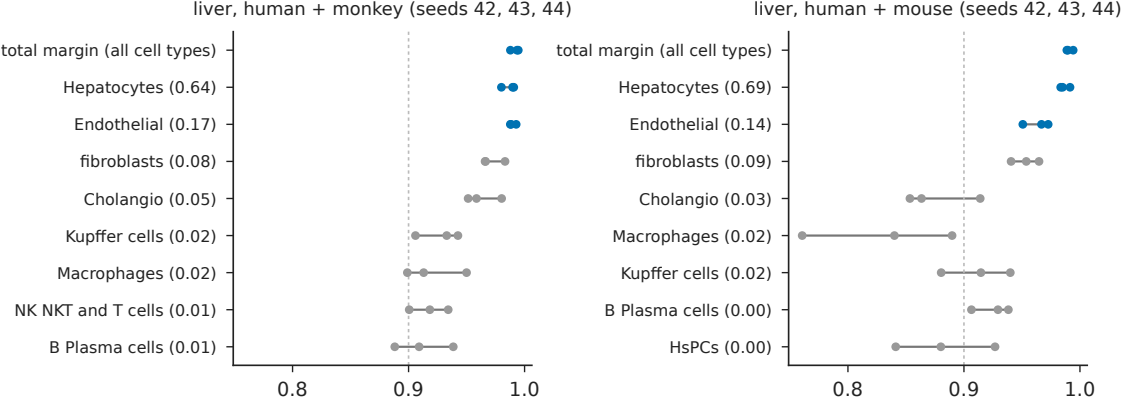

Gene category
Cytoskeleton & ECM
n184
d = −0.11
Signaling pathways
n190
d = −0.15
Immunity & inflammation
n130
d = −0.12
Transcription factors
n405
d = −0.31 ★
Cell cycle
n100
d = −0.40 ★
Protein homeostasis
n113
d = −0.54 ★
Chromatin & DNA processing
n186
d = −0.58 ★
Mitochondria & OXPHOS
n75
d = −1.26 ★
Metabolism
n102
d = −1.12 ★
Translation & ribosome
n179
d = −1.15 ★
-0.20
-0.10
0.00
0.10
0.20
Δα (liver 4-species)
(a)
Gene category
Cytoskeleton & ECM
-0.11
+0.12
Signaling pathways
-0.15
+0.23
Immunity & inflammation
-0.12
+0.04
Transcription factors
★
-0.31
+0.08
Cell cycle
★
-0.40
+0.17
Protein homeostasis
★
-0.54
+0.20
Chromatin & DNA processing
★
-0.58
+0.13
Mitochondria & OXPHOS
★
-1.26
★
-0.63
Metabolism
★
-1.12
★
-0.50
Translation & ribosome
★
-1.15
★
-0.49
shift
amplitude
1.0
0.5
0.0
−0.5
−1.0
effect size
Cohen's d on Δα / signed |Δα|
(liver 4-species)
(b)
Gene category
Cytoskeleton & ECM
n394
d = +0.91 ★
Mitochondria & OXPHOS
n105
d = +0.92 ★
Protein homeostasis
n153
d = +0.62 ★
Cell cycle
n130
d = +0.68 ★
Signaling pathways
n323
d = +0.58 ★
Immunity & inflammation
n343
d = +0.50 ★
Metabolism
n175
d = +0.63 ★
Chromatin & DNA processing
n237
d = +0.50 ★
Transcription factors
n753
d = +0.48 ★
Translation & ribosome
n213
d = +0.07
-0.05
0.00
0.05
0.10
Δα (testis Human+Macaque)
(c)
Gene category
Cytoskeleton & ECM
★
+0.91
★
+0.21
Mitochondria & OXPHOS
★
+0.92
★
+0.58
Protein homeostasis
★
+0.62
+0.05
Cell cycle
★
+0.68
+0.01
Signaling pathways
★
+0.58
-0.08
Immunity & inflammation
★
+0.50
-0.15
Metabolism
★
+0.63
-0.01
Chromatin & DNA processing
★
+0.50
+0.03
Transcription factors
★
+0.48
-0.18
Translation & ribosome
+0.07
-0.08
shift
amplitude
1.0
0.5
0.0
−0.5
−1.0
effect size
Cohen's d on Δα / signed |Δα|
(testis Human+Macaque)
(d)


Gene category
ECM: collagens (n = 16)
ECM: other (n = 55)
Hedgehog signaling (n = 10)
Homeobox TFs (n = 58)
TGF-β/BMP signaling (n = 18)
Cytokines and chemokines (n = 34)
Immediate early genes (n = 19)
Signaling kinases (n = 105)
Forkhead and other TFs (n = 112)
Cytochrome P450 (n = 15)
Intermediate filaments (n = 12)
Interferon response (n = 18)
Wnt signaling (n = 43)
Complement (n = 53)
Microtubules (n = 56)
Notch signaling (n = 14)
Chaperones (HSP) (n = 47)
DNA replication and repair (n = 112)
Cell cycle (n = 100)
Zinc-finger TFs (n = 114)
Nuclear receptors (n = 121)
Actin cytoskeleton (n = 45)
Apoptosis (n = 19)
Glycolysis (n = 17)
Translation initiation (n = 30)
Splicing factors (n = 70)
Proteasome (n = 39)
Housekeeping (n = 7)
Translation elongation (n = 7)
Autophagy (n = 27)
OXPHOS, nuclear-encoded (n = 66)
Glutathione and redox (n = 30)
Coagulation and serum (n = 29)
Ribosome biogenesis (n = 40)
Lipid metabolism (n = 40)
Mitochondrial ribosome (n = 58)
Cytosolic ribosome (n = 44)
Mitochondrially encoded (n = 9)
d = +0.40
d = +0.51 ★
d = +0.40
d = +0.19 ★
d = −0.08
d = +0.22
d = −0.07
d = −0.23
d = −0.15
d = −0.76
d = −0.59
d = −0.26
d = −0.04
d = −0.29
d = −0.49 ★
d = −0.44
d = −0.43 ★
d = −0.53 ★
d = −0.40 ★
d = −0.53 ★
d = −0.49 ★
d = −0.43 ★
d = −0.51
d = −0.92 ★
d = −0.55 ★
d = −0.60 ★
d = −0.45 ★
d = −0.82 ★
d = −0.75 ★
d = −0.84 ★
d = −0.78 ★
d = −0.99 ★
d = −1.55 ★
d = −0.92 ★
d = −1.44 ★
d = −1.38 ★
d = −1.53 ★
d = −4.79 ★
−0.4
−0.2
0.0
0.2
0.4
Δα (liver 4 species)
(a)
shift
amplitude
+0.40
+0.32 ★
+0.51 ★
−0.20
+0.40
−0.02
+0.19 ★
−0.52
−0.08
+0.47
+0.22
+0.05
−0.07
+0.26
−0.23
+0.18
−0.15
+0.14
−0.76
−0.22
−0.59
+0.04
−0.26
+0.29
−0.04
+0.32
−0.29
−0.10
−0.49 ★
+0.13
−0.44
+0.16
−0.43 ★
+0.33
−0.53 ★
+0.18
−0.40 ★
+0.17
−0.53 ★
+0.06
−0.49 ★
−0.16
−0.43 ★
+0.17
−0.51
−0.13
−0.92 ★
−0.22
−0.55 ★
+0.21
−0.60 ★
+0.14
−0.45 ★
+0.34
−0.82
−0.10
−0.75
−0.05
−0.84 ★
−0.20
−0.78 ★
−0.06
−0.99 ★
−0.30 ★
−1.55 ★
−1.45 ★
−0.92 ★
−0.20 ★
−1.44 ★
−0.88 ★
−1.38 ★
−0.76 ★
−1.53 ★
−0.95 ★
−4.79 ★
−4.84 ★
4
2
0
−2
−4
effect size
Cohen's d on Δα / signed |Δα|
(liver 4 species)
(b)

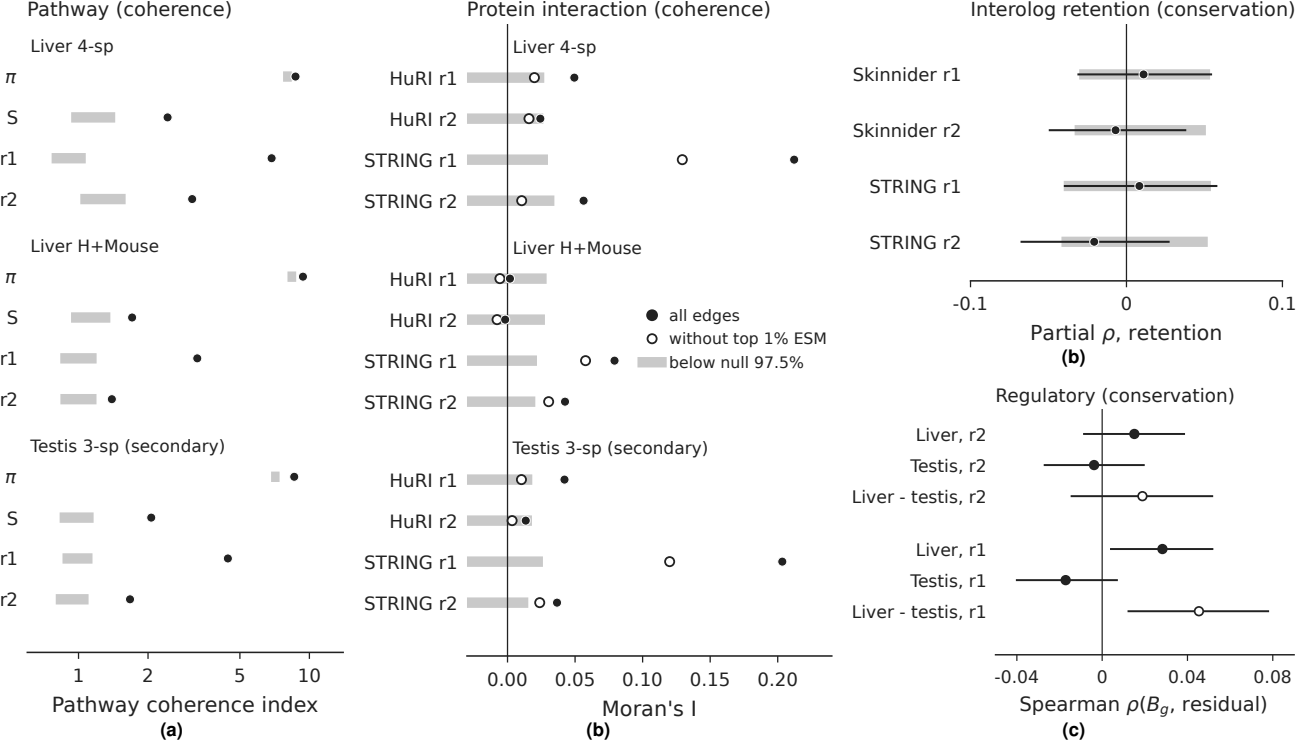

Gene category
Mitochondrially encoded (n11) d = +4.30 ★
ECM: collagens (n42) d = +1.35 ★
Housekeeping (n8) d = +0.91
Actin cytoskeleton (n94) d = +0.97 ★
Microtubules (n102) d = +0.97 ★
Glutathione and redox (n50) d = +0.96 ★
Autophagy (n37) d = +0.92 ★
ECM: other (n109) d = +0.80 ★
Signalling kinases (n176) d = +0.84 ★
Coagulation and serum (n70) d = +0.75 ★
Proteasome (n48) d = +0.49 ★
Cell cycle (n130) d = +0.68 ★
Glycolysis (n30) d = +0.56 ★
Complement (n116) d = +0.79 ★
Interferon response (n35) d = +0.41 ★
Splicing factors (n95) d = +0.66 ★
Chaperones (HSP) (n68) d = +0.55 ★
MHC class I (n5) d = +1.27
Translation initiation (n43) d = +0.53 ★
OXPHOS, nuclear-encoded (n94) d = +0.52 ★
Nuclear receptors (n214) d = +0.60 ★
Translation elongation (n8) d = +0.28
DNA replication and repair (n121) d = +0.46 ★
Forkhead and other TFs (n172) d = +0.58 ★
Cytosolic ribosome (n56) d = +0.41 ★
Hedgehog signalling (n15) d = +0.22
Lipid metabolism (n68) d = +0.43 ★
Intermediate filaments (n47) d = +0.51 ★
Apoptosis (n33) d = +0.44 ★
Cytokines and chemokines (n151) d = +0.36 ★
Immediate early genes (n29) d = +0.11
Homeobox TFs (n80) d = +0.29 ★
Notch signalling (n32) d = +0.37 ★
Zinc-finger TFs (n287) d = +0.39 ★
MHC class II (n7) d = +0.21
Cytochrome P450 (n27) d = +0.58 ★
Wnt signalling (n57) d = +0.24
TGF-β/BMP signalling (n43) d = +0.26
Ribosome biogenesis (n44) d = −0.03
Histones (n21) d = −0.03
Mitochondrial ribosome (n62) d = −0.51 ★
−0.4
−0.2
0.0
0.2
0.4
Δα (testis Human+Macaque)
(a)
Mitochondrially encoded +4.30 ★ +4.13 ★
ECM: collagens +1.35 ★ +0.65 ★
Housekeeping +0.91 +0.48 ★
Actin cytoskeleton +0.97 ★ +0.29 ★
Microtubules +0.97 ★ +0.32 ★
Glutathione and redox +0.96 ★ +0.34
Autophagy +0.92 ★ +0.33
ECM: other +0.80 ★ +0.07
Signalling kinases +0.84 ★ +0.16 ★
Coagulation and serum +0.75 ★ -0.01
Proteasome +0.49 ★ -0.16
Cell cycle +0.68 ★ +0.01
Glycolysis +0.56 ★ +0.06
Complement +0.79 ★ +0.19
Interferon response +0.41 ★ -0.25
Splicing factors +0.66 ★ +0.10
Chaperones (HSP) +0.55 ★ +0.04
MHC class I +1.27 +0.58
Translation initiation +0.53 ★ -0.03
OXPHOS, nuclear-encoded +0.52 ★ +0.17 ★
Nuclear receptors +0.60 ★ -0.03
Translation elongation +0.28 -0.23
DNA replication and repair +0.46 ★ -0.06
Forkhead and other TFs +0.58 ★ -0.13
Cytosolic ribosome +0.41 ★ +0.14 ★
Hedgehog signalling +0.22 -0.34
Lipid metabolism +0.43 ★ -0.27
Intermediate filaments +0.51 ★ -0.22
Apoptosis +0.44 ★ -0.13
Cytokines and chemokines +0.36 ★ -0.34
Immediate early genes +0.11 -0.44
Homeobox TFs +0.29 ★ -0.50
Notch signalling +0.37 ★ -0.40
Zinc-finger TFs +0.39 ★ -0.24
MHC class II +0.21 -0.51
Cytochrome P450 +0.58 ★ -0.07
Wnt signalling +0.24 -0.42
TGF-β/BMP signalling +0.26 -0.26
Ribosome biogenesis -0.03 +0.30
Histones -0.03 -0.21
Mitochondrial ribosome -0.51 ★ +0.13
shift
amplitude
Cohen's d on Δα / signed |Δα|
(testis Human+Macaque)
(b)
effect size
4
3
2
1
0
−1
−2
−3
−4

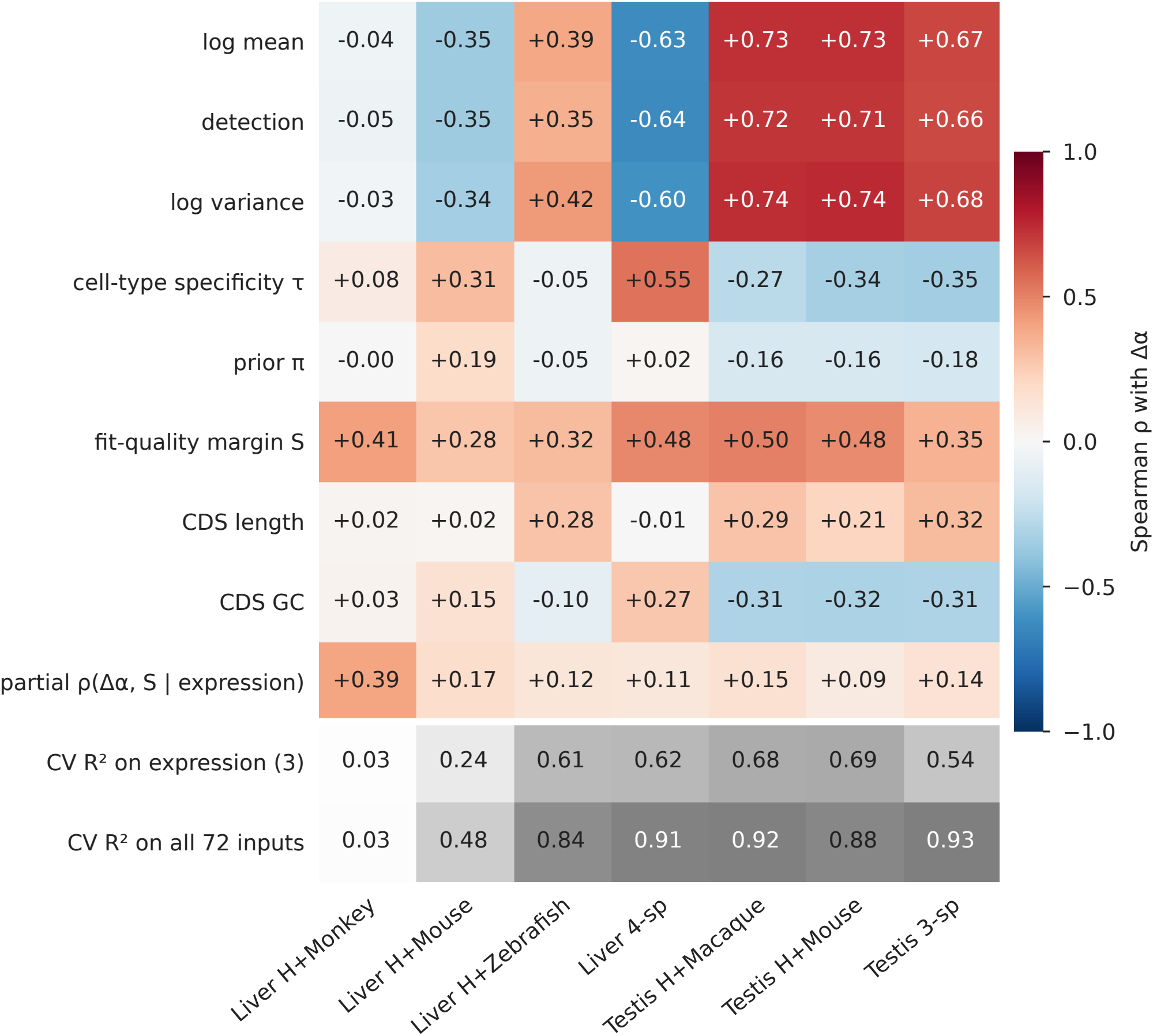

liver Human+Mouse · Hallmark
Epithelial Mesenchymal ...
Hedgehog Signaling
Xenobiotic Metabolism
NES · Δα-ranked (Chimera)
NES · prior-ranked
(a)
Hallmark · 50 pathways
prior ranking only
Δα ranking only
both, same sign
both, opposite sign
Liver
Testis
Hallmark pathways at FDR < 0.05
H+Monkey
H+Mouse
H+Zebrafish
4 species
H+Macaque
H+Mouse
3 species
(b)
Xenobiotic Metabolism
Bile Acid Metabolism
Peroxisome
Adipogenesis
UV Response Dn
Allograft Rejection
Hedgehog Signaling
NES > 0 (higher Δα in liver)
NES < 0 (higher Δα in testis)
−log10 (FDR q)
NES (rank metric: Δα_liver − Δα_testis)
(c)

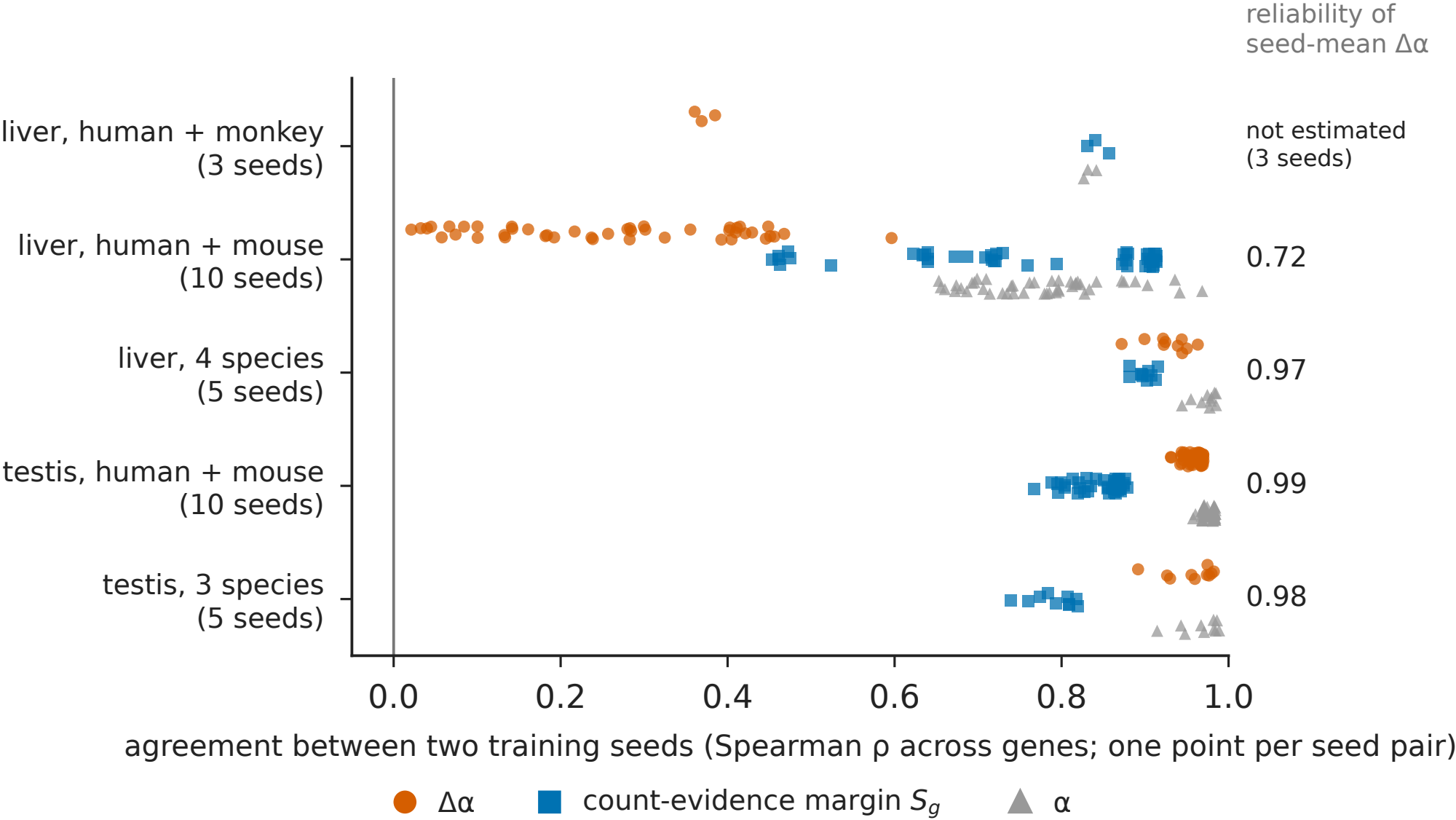

Liver H+Monkey
r = 0.996
Liver H+Mouse
r = 0.997
Liver H+Zebrafish
r = 0.993
Liver 4-sp
r = 0.969
Testis H+Macaque
r = 0.996
Testis H+Mouse
r = 0.993
Testis 3-sp
r = 0.993
trained S_g (EMA)
pass rate × forward-pass S_g

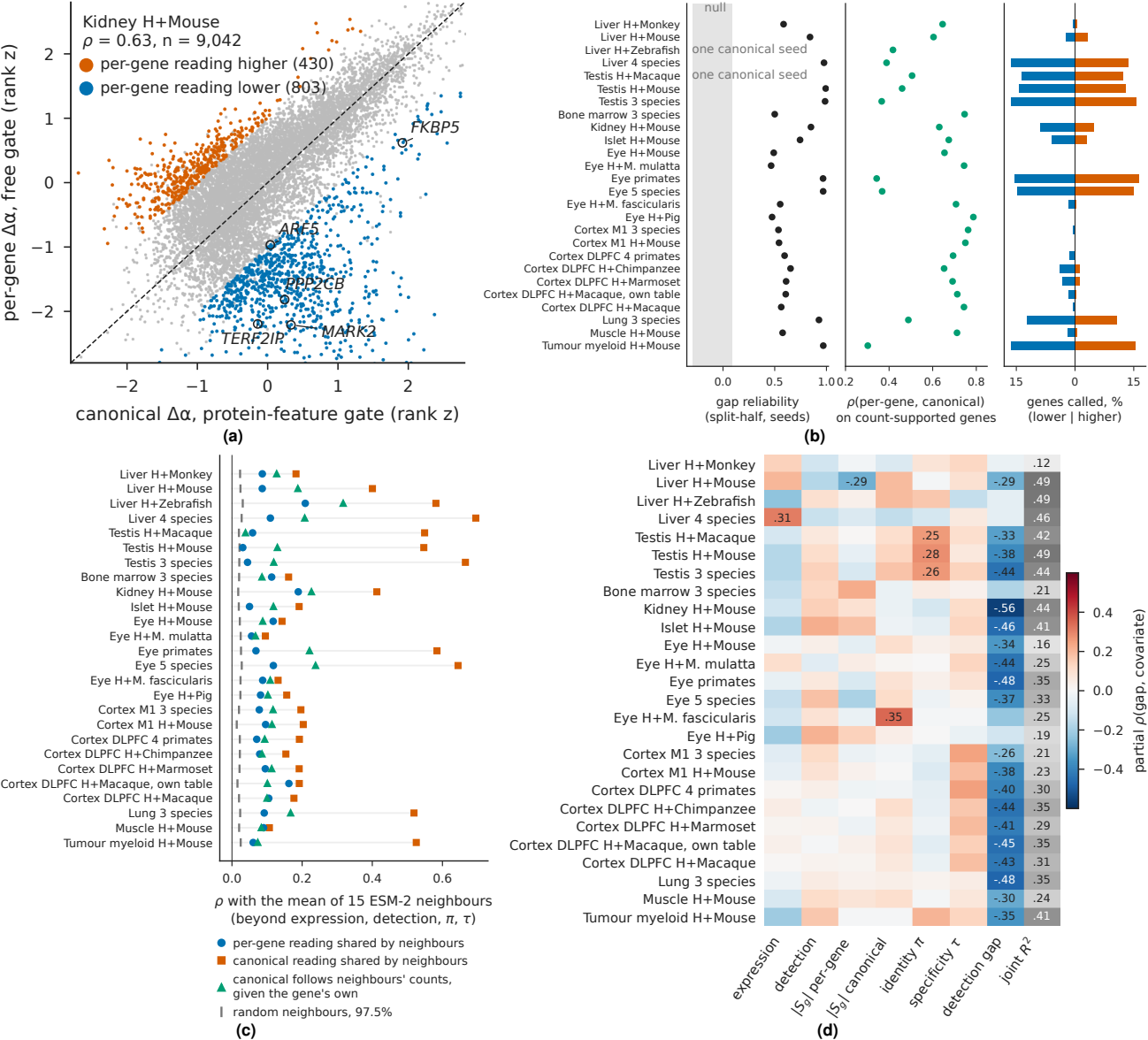


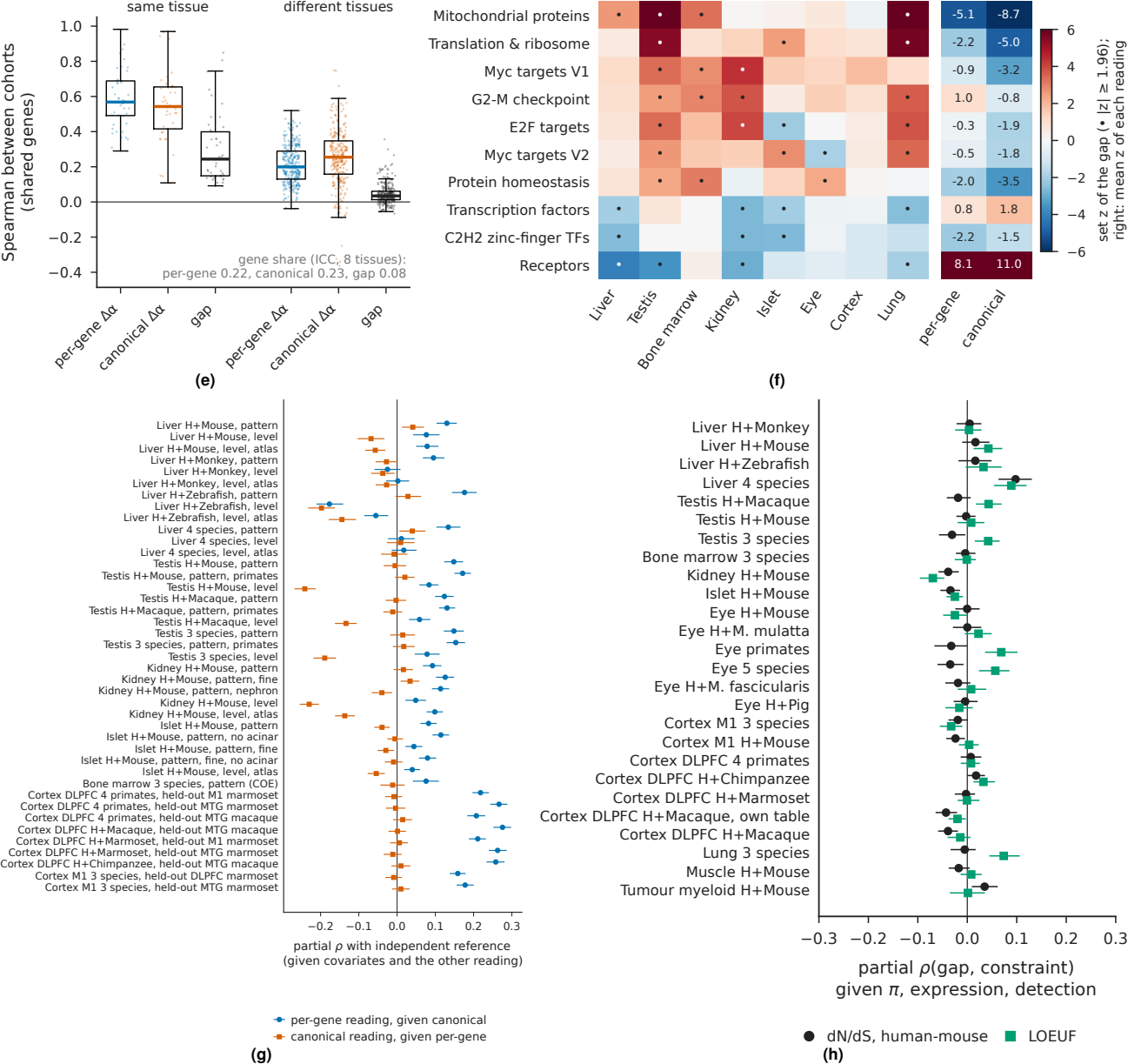
agreement between seeds of a cell type's contribution to S_g (Spearman ρ across genes;
points: seed pairs; in brackets: cell-type weight; blue: weight at least 0.10)

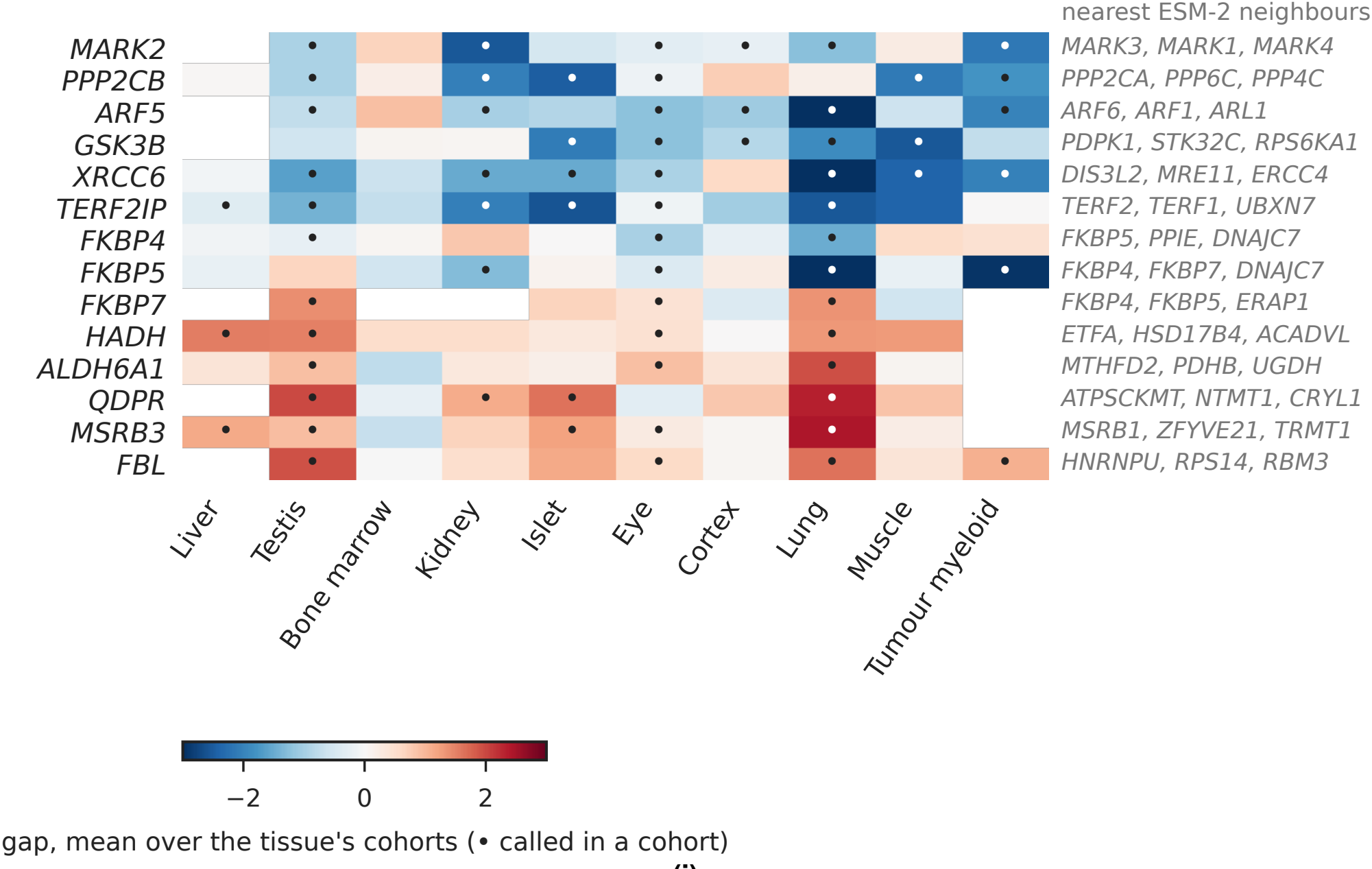

Pathway (coherence)
Liver 4-sp
π
S
r1
r2
Liver H+Mouse
π
S
r1
r2
Testis 3-sp (secondary)
π
S
r1
r2
1
2
5
10
Pathway coherence index
(a)
Protein interaction (coherence)
Liver 4-sp
HuRI r1
HuRI r2
STRING r1
STRING r2
Liver H+Mouse
HuRI r1
HuRI r2
STRING r1
STRING r2
Testis 3-sp (secondary)
HuRI r1
HuRI r2
STRING r1
STRING r2
all edges
without top 1% ESM
below null 97.5%
0.00
0.05
0.10
0.15
0.20
Moran's I
(b)
Interolog retention (conservation)
Skinnider r1
Skinnider r2
STRING r1
STRING r2
-0.1
0
0.1
Partial ρ, retention
(b)
Regulatory (conservation)
Liver, r2
Testis, r2
Liver - testis, r2
Liver, r1
Testis, r1
Liver - testis, r1
-0.04
0
0.04
0.08
Spearman ρ(B_g, residual)
(c)


log mean
detection
log variance
cell-type specificity τ
prior π
fit-quality margin S
CDS length
CDS GC
partial ρ(Δα, S | expression)
CV R² on expression (3)
CV R² on all 72 inputs
-0.04 -0.35 +0.39 -0.63 +0.73 +0.73 +0.67
-0.05 -0.35 +0.35 -0.64 +0.72 +0.71 +0.66
-0.03 -0.34 +0.42 -0.60 +0.74 +0.74 +0.68
+0.08 +0.31 -0.05 +0.55 -0.27 -0.34 -0.35
-0.00 +0.19 -0.05 +0.02 -0.16 -0.16 -0.18
+0.41 +0.28 +0.32 +0.48 +0.50 +0.48 +0.35
+0.02 +0.02 +0.28 -0.01 +0.29 +0.21 +0.32
+0.03 +0.15 -0.10 +0.27 -0.31 -0.32 -0.31
+0.39 +0.17 +0.12 +0.11 +0.15 +0.09 +0.14
0.03 0.24 0.61 0.62 0.68 0.69 0.54
0.03 0.48 0.84 0.91 0.92 0.88 0.93
Liver H+Monkey
Liver H+Mouse
Liver H+Zebrafish
Liver 4-sp
Testis H+Macaque
Testis H+Mouse
Testis 3-sp
1.0
0.5
0.0
−0.5
−1.0
Spearman ρ with Δα


reliability of
seed-mean Δα
liver, human + monkey
(3 seeds)
not estimated
(3 seeds)
liver, human + mouse
(10 seeds)
0.72
liver, 4 species
(5 seeds)
0.97
testis, human + mouse
(10 seeds)
0.99
testis, 3 species
(5 seeds)
0.98
0.0
0.2
0.4
0.6
0.8
1.0
agreement between two training seeds (Spearman ρ across genes; one point per seed pair)
Δα
count-evidence margin S_g
α

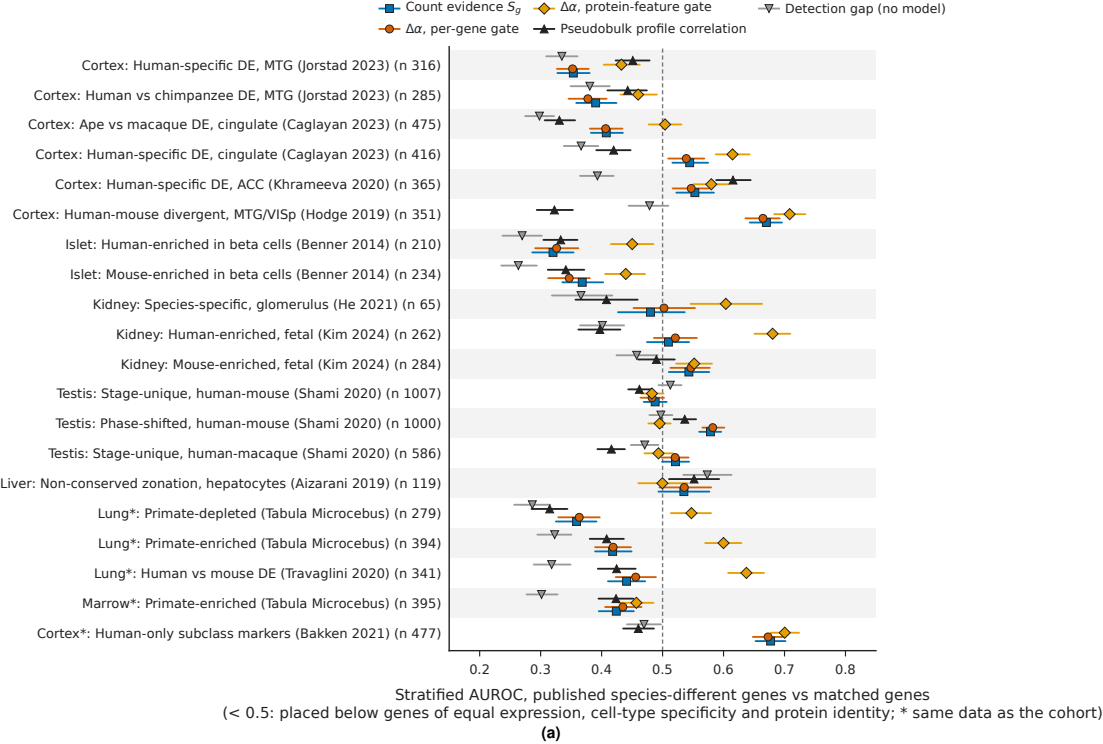

Kidney H+Mouse
ρ = 0.63, n = 9,042
per-gene reading higher (430)
per-gene reading lower (803)
FKBP5
ARL5
PPP2CB
TERF2IP
MARK2
per-gene Δα, free gate (rank z)
canonical Δα, protein-feature gate (rank z)
(a)
null
one canonical seed
one canonical seed
Liver H+Monkey
Liver H+Mouse
Liver H+Zebrafish
Liver 4 species
Testis H+Macaque
Testis H+Mouse
Testis 3 species
Bone marrow 3 species
Kidney H+Mouse
Islet H+Mouse
Eye H+Mouse
Eye H+M. mulatta
Eye primates
Eye 5 species
Eye H+M. fascicularis
Eye H+Pig
Cortex M1 3 species
Cortex M1 H+Mouse
Cortex DLPFC 4 primates
Cortex DLPFC H+Chimpanzee
Cortex DLPFC H+Marmoset
Cortex DLPFC H+Macaque, own table
Cortex DLPFC H+Macaque
Lung 3 species
Muscle H+Mouse
Tumour myeloid H+Mouse
gap reliability (split-half, seeds)
ρ(per-gene, canonical) on count-supported genes
genes called, % (lower | higher)
(b)
ρ with the mean of 15 ESM-2 neighbours (beyond expression, detection, π, τ)
per-gene reading shared by neighbours
canonical reading shared by neighbours
canonical follows neighbours' counts, given the gene's own
random neighbours, 97.5%
(c)
expression
detection
|S_g| per-gene
|S_g| canonical
identity π
specificity τ
detection gap
joint R²
partial ρ(gap, covariate)
(d)


same tissue
different tissues
Spearman between cohorts (shared genes)
gene share (ICC, 8 tissues):
per-gene 0.22, canonical 0.23, gap 0.08
per-gene Δα
canonical Δα
gap
(e)
Mitochondrial proteins
Translation & ribosome
Myc targets V1
G2-M checkpoint
E2F targets
Myc targets V2
Protein homeostasis
Transcription factors
C2H2 zinc-finger TFs
Receptors
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
per-gene
canonical
set z of the gap (• |z| ≥ 1.96);
right: mean z of each reading
(f)
partial ρ with independent reference
(given covariates and the other reading)
per-gene reading, given canonical
canonical reading, given per-gene
(g)
partial ρ(gap, constraint)
given π, expression, detection
dN/dS, human-mouse
LOEUF
(h)

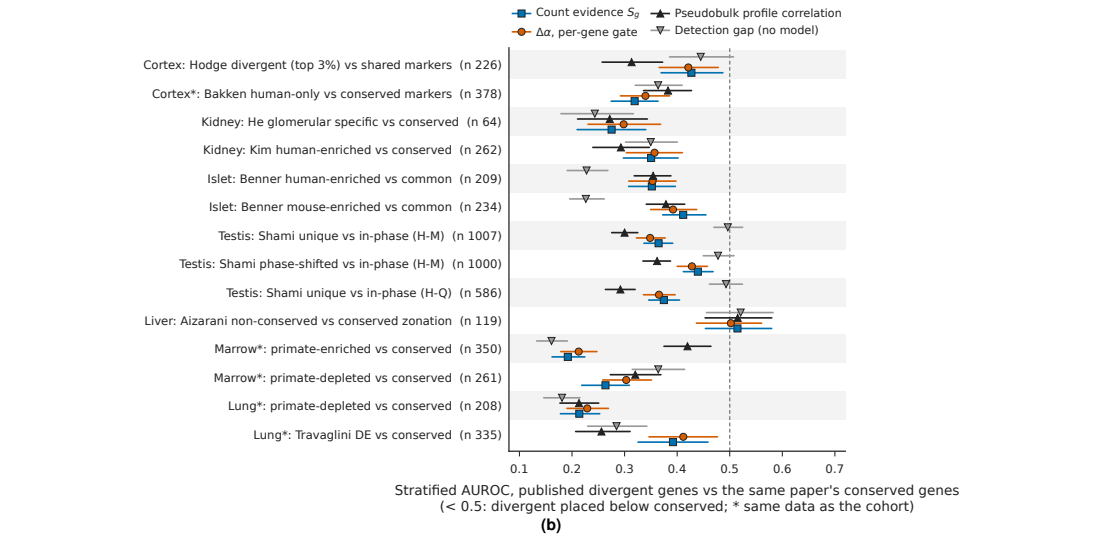


gap, mean over the tissue's cohorts (• called in a cohort)

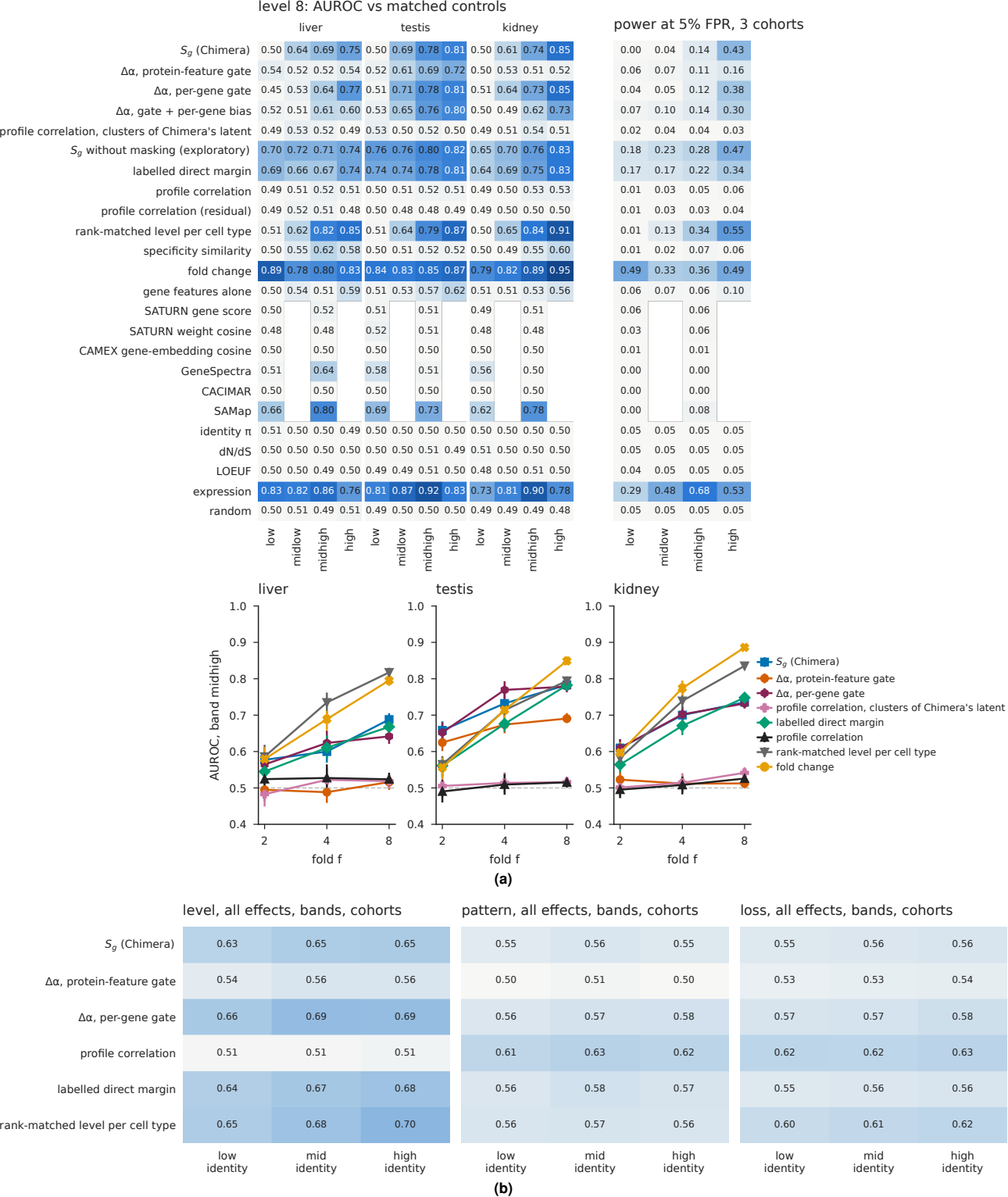
(i)


Count evidence S_g
$\Delta\alpha$, protein-feature gate
Detection gap (no model)
$\Delta\alpha$, per-gene gate
Pseudobulk profile correlation
Cortex: Human-specific DE, MTG (Jorstad 2023) (n 316)
Cortex: Human vs chimpanzee DE, MTG (Jorstad 2023) (n 285)
Cortex: Ape vs macaque DE, cingulate (Caglayan 2023) (n 475)
Cortex: Human-specific DE, cingulate (Caglayan 2023) (n 416)
Cortex: Human-specific DE, ACC (Khrameeva 2020) (n 365)
Cortex: Human-mouse divergent, MTG/VISp (Hodge 2019) (n 351)
Islet: Human-enriched in beta cells (Benner 2014) (n 210)
Islet: Mouse-enriched in beta cells (Benner 2014) (n 234)
Kidney: Species-specific, glomerulus (He 2021) (n 65)
Kidney: Human-enriched, fetal (Kim 2024) (n 262)
Kidney: Mouse-enriched, fetal (Kim 2024) (n 284)
Testis: Stage-unique, human-mouse (Shami 2020) (n 1007)
Testis: Phase-shifted, human-mouse (Shami 2020) (n 1000)
Testis: Stage-unique, human-macaque (Shami 2020) (n 586)
Liver: Non-conserved zonation, hepatocytes (Aizarani 2019) (n 119)
Lung*: Primate-depleted (Tabula Microcebus) (n 279)
Lung*: Primate-enriched (Tabula Microcebus) (n 394)
Lung*: Human vs mouse DE (Travaglini 2020) (n 341)
Marrow*: Primate-enriched (Tabula Microcebus) (n 395)
Cortex*: Human-only subclass markers (Bakken 2021) (n 477)
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Stratified AUROC, published species-different genes vs matched genes
(< 0.5: placed below genes of equal expression, cell-type specificity and protein identity; * same data as the cohort)

(a)


Count evidence S_g
Pseudobulk profile correlation
$\Delta\alpha$, per-gene gate
Detection gap (no model)
Cortex: Hodge divergent (top 3%) vs shared markers (n 226)
Cortex*: Bakken human-only vs conserved markers (n 378)
Kidney: He glomerular specific vs conserved (n 64)
Kidney: Kim human-enriched vs conserved (n 262)
Islet: Benner human-enriched vs common (n 209)
Islet: Benner mouse-enriched vs common (n 234)
Testis: Shami unique vs in-phase (H-M) (n 1007)
Testis: Shami phase-shifted vs in-phase (H-M) (n 1000)
Testis: Shami unique vs in-phase (H-Q) (n 586)
Liver: Aizarani non-conserved vs conserved zonation (n 119)
Marrow*: primate-enriched vs conserved (n 350)
Marrow*: primate-depleted vs conserved (n 261)
Lung*: primate-depleted vs conserved (n 208)
Lung*: Travaglini DE vs conserved (n 335)
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Stratified AUROC, published divergent genes vs the same paper's conserved genes
(< 0.5: divergent placed below conserved; * same data as the cohort)

(b)

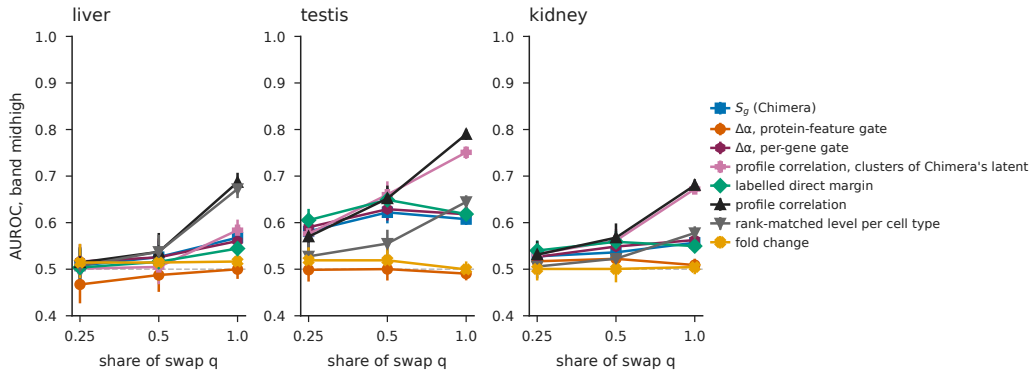

level 8: AUROC vs matched controls
liver
testis
kidney
power at 5% FPR, 3 cohorts
S_g (Chimera) 0.50 0.64 0.69 0.75 0.50 0.69 0.78 0.81 0.50 0.61 0.74 0.85 0.00 0.04 0.14 0.43
Δα, protein-feature gate 0.54 0.52 0.52 0.54 0.52 0.61 0.69 0.72 0.50 0.53 0.51 0.52 0.06 0.07 0.11 0.16
Δα, per-gene gate 0.45 0.53 0.64 0.77 0.51 0.71 0.78 0.81 0.51 0.64 0.73 0.85 0.04 0.05 0.12 0.38
Δα, gate + per-gene bias 0.52 0.51 0.61 0.60 0.53 0.65 0.76 0.80 0.50 0.49 0.62 0.73 0.07 0.10 0.14 0.30
profile correlation, clusters of Chimera's latent 0.49 0.53 0.52 0.49 0.53 0.50 0.52 0.50 0.49 0.51 0.54 0.51 0.02 0.04 0.04 0.03
S_g without masking (exploratory) 0.70 0.72 0.71 0.74 0.76 0.76 0.80 0.82 0.65 0.70 0.76 0.83 0.18 0.23 0.28 0.47
labelled direct margin 0.69 0.66 0.67 0.74 0.74 0.74 0.78 0.81 0.64 0.69 0.75 0.83 0.17 0.17 0.22 0.34
profile correlation 0.49 0.51 0.52 0.51 0.50 0.51 0.52 0.51 0.49 0.50 0.53 0.53 0.01 0.03 0.05 0.06
profile correlation (residual) 0.49 0.52 0.51 0.48 0.50 0.48 0.48 0.49 0.49 0.50 0.50 0.50 0.01 0.03 0.03 0.04
rank-matched level per cell type 0.51 0.62 0.82 0.85 0.51 0.64 0.79 0.87 0.50 0.65 0.84 0.91 0.01 0.13 0.34 0.55
specificity similarity 0.50 0.55 0.62 0.58 0.50 0.51 0.52 0.52 0.50 0.49 0.55 0.60 0.01 0.02 0.07 0.06
fold change 0.89 0.78 0.80 0.83 0.84 0.83 0.85 0.87 0.79 0.82 0.89 0.95 0.49 0.33 0.36 0.49
gene features alone 0.50 0.54 0.51 0.59 0.51 0.53 0.57 0.62 0.51 0.51 0.53 0.56 0.06 0.07 0.06 0.10
SATURN gene score 0.50 0.52 0.51 0.51 0.49 0.51 0.06 0.06
SATURN weight cosine 0.48 0.48 0.52 0.51 0.48 0.48 0.03 0.06
CAMEX gene-embedding cosine 0.50 0.50 0.50 0.50 0.50 0.50 0.01 0.01
GeneSpectra 0.51 0.64 0.58 0.51 0.56 0.50 0.00 0.00
CACIMAR 0.50 0.50 0.50 0.50 0.50 0.50 0.00 0.00
SAMap 0.66 0.80 0.69 0.73 0.62 0.78 0.00 0.08
identity π 0.51 0.50 0.50 0.49 0.50 0.50 0.50 0.50 0.50 0.50 0.50 0.50 0.05 0.05 0.05 0.05
dN/dS 0.50 0.50 0.50 0.50 0.50 0.50 0.51 0.49 0.51 0.50 0.50 0.50 0.05 0.05 0.05 0.05
LOEUF 0.50 0.50 0.49 0.50 0.49 0.49 0.51 0.50 0.48 0.50 0.51 0.50 0.04 0.05 0.05 0.05
expression 0.83 0.82 0.86 0.76 0.81 0.87 0.92 0.83 0.73 0.81 0.90 0.78 0.29 0.48 0.68 0.53
random 0.50 0.51 0.49 0.51 0.49 0.50 0.50 0.50 0.49 0.49 0.49 0.48 0.05 0.05 0.05 0.05
low midlow midhigh high low midlow midhigh high low midlow midhigh high low midlow midhigh high
liver
testis
kidney
AUROC, band midhigh
fold f
S_g (Chimera)
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labelled direct margin
profile correlation
rank-matched level per cell type
fold change
(a)
level, all effects, bands, cohorts
pattern, all effects, bands, cohorts
loss, all effects, bands, cohorts
S_g (Chimera) 0.63 0.65 0.65 0.55 0.56 0.55 0.55 0.56 0.56
Δα, protein-feature gate 0.54 0.56 0.56 0.50 0.51 0.50 0.53 0.53 0.54
Δα, per-gene gate 0.66 0.69 0.69 0.56 0.57 0.58 0.57 0.57 0.58
profile correlation 0.51 0.51 0.51 0.61 0.63 0.62 0.62 0.62 0.63
labelled direct margin 0.64 0.67 0.68 0.56 0.58 0.57 0.55 0.56 0.56
rank-matched level per cell type 0.65 0.68 0.70 0.56 0.57 0.56 0.60 0.61 0.62
low identity
mid identity
high identity
(b)

pattern 1: AUROC vs matched controls

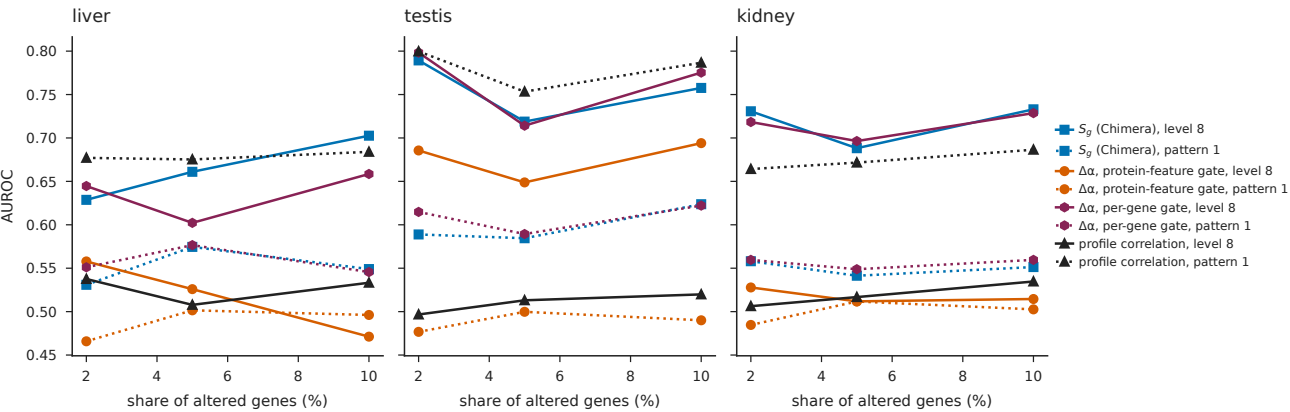
	liver				testis				kidney			
	low	midlow	midhigh	high	low	midlow	midhigh	high	low	midlow	midhigh	high
S_g (Chimera)	0.50	0.57	0.57	0.61	0.50	0.56	0.61	0.64	0.50	0.53	0.56	0.56
Δα, protein-feature gate	0.48	0.47	0.50	0.52	0.50	0.49	0.49	0.52	0.51	0.50	0.51	0.50
Δα, per-gene gate	0.50	0.57	0.56	0.66	0.51	0.57	0.62	0.63	0.51	0.54	0.56	0.56
Δα, gate + per-gene bias	0.50	0.51	0.53	0.65	0.50	0.50	0.53	0.59	0.50	0.50	0.53	0.52
profile correlation, clusters of Chimera's latent	0.56	0.53	0.58	0.82	0.65	0.72	0.75	0.77	0.62	0.67	0.67	0.76
S_g without masking (exploratory)	0.55	0.59	0.57	0.62	0.64	0.62	0.64	0.64	0.55	0.57	0.57	0.56
labelled direct margin	0.55	0.58	0.54	0.61	0.61	0.60	0.62	0.62	0.53	0.55	0.55	0.54
profile correlation	0.55	0.63	0.69	0.80	0.64	0.75	0.79	0.81	0.57	0.64	0.68	0.78
profile correlation (residual)	0.55	0.61	0.68	0.80	0.64	0.75	0.79	0.81	0.57	0.63	0.68	0.78
rank-matched level per cell type	0.55	0.63	0.67	0.69	0.59	0.64	0.64	0.66	0.54	0.58	0.58	0.55
specificity similarity	0.58	0.61	0.66	0.71	0.63	0.70	0.74	0.76	0.57	0.62	0.64	0.67
fold change	0.50	0.53	0.52	0.52	0.52	0.51	0.50	0.51	0.50	0.52	0.50	0.49
gene features alone	0.47	0.52	0.50	0.52	0.53	0.48	0.50	0.53	0.50	0.50	0.52	0.50
SATURN gene score	0.48		0.49		0.52		0.52		0.49		0.52	
SATURN weight cosine	0.52		0.51		0.52		0.50		0.49		0.52	
CAMEX gene-embedding cosine	0.50		0.50		0.50		0.50		0.50		0.50	
GeneSpectra	0.50		0.52		0.46		0.48		0.48		0.49	
CACIMAR	0.50		0.50		0.50		0.50		0.50		0.50	
SAMap	0.62		0.64		0.59		0.66		0.63		0.66	
identity π	0.49	0.50	0.49	0.51	0.49	0.50	0.50	0.50	0.50	0.50	0.50	0.50
dN/dS	0.49	0.48	0.49	0.49	0.49	0.52	0.49	0.49	0.49	0.51	0.50	0.51
LOEUF	0.52	0.45	0.47	0.50	0.52	0.50	0.50	0.50	0.52	0.51	0.51	0.50
expression	0.55	0.53	0.52	0.47	0.50	0.51	0.49	0.50	0.52	0.51	0.50	0.49
random	0.52	0.47	0.49	0.50	0.48	0.52	0.51	0.48	0.50	0.49	0.49	0.47

power at 5% FPR, 3 cohorts

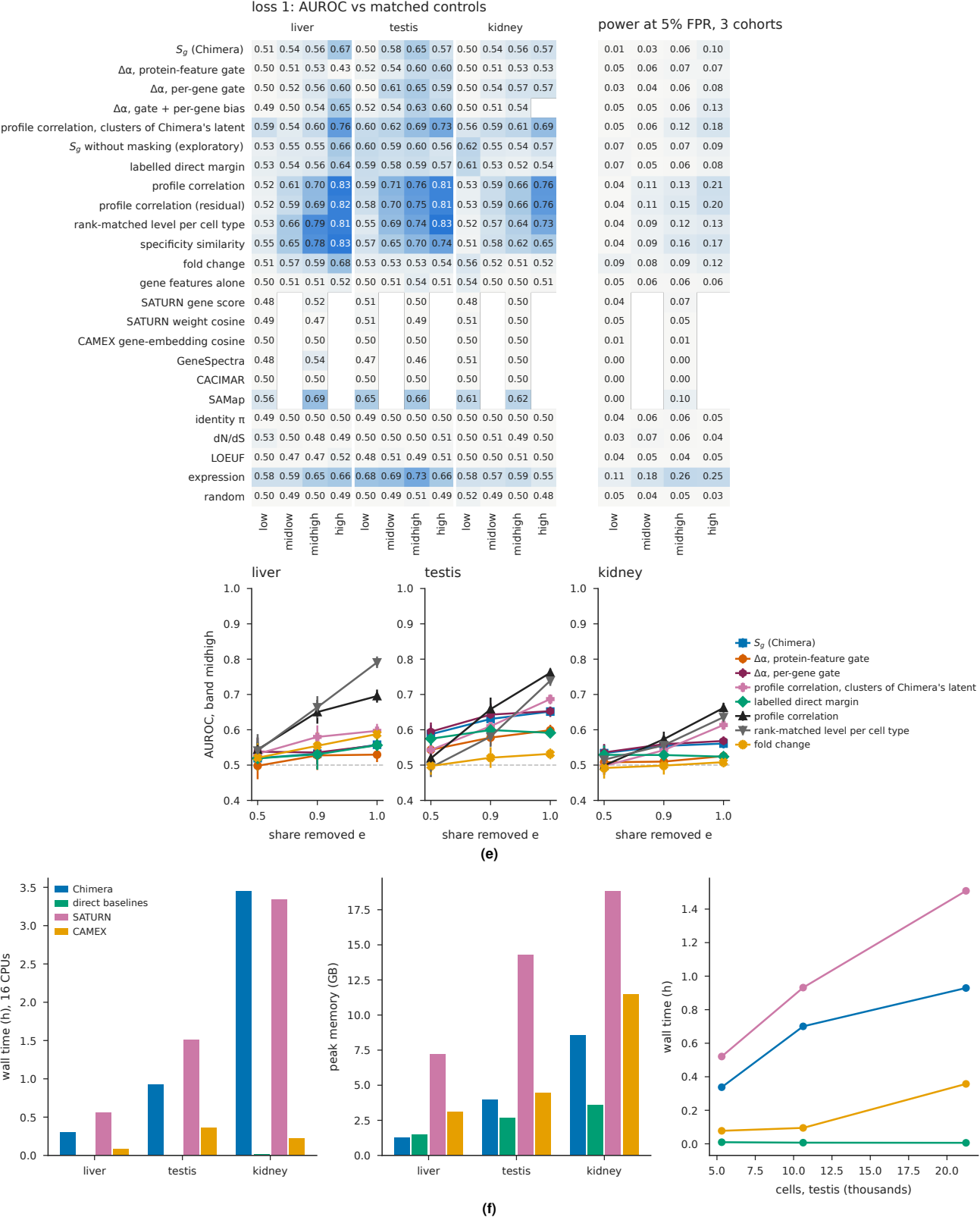
low	midlow	midhigh	high
0.01	0.07	0.08	0.08
0.04	0.05	0.05	0.06
0.03	0.06	0.08	0.08
0.05	0.05	0.06	0.08
0.07	0.10	0.11	0.21
0.07	0.10	0.08	0.08
0.06	0.07	0.07	0.07
0.06	0.10	0.12	0.16
0.06	0.11	0.13	0.17
0.08	0.12	0.10	0.10
0.11	0.13	0.16	0.17
0.06	0.05	0.06	0.05
0.07	0.05	0.05	0.07
0.05		0.05	
0.06		0.05	
0.01		0.02	
0.00		0.00	
0.00		0.00	
0.00		0.09	
0.04	0.06	0.05	0.05
0.03	0.06	0.05	0.07
0.05	0.04	0.05	0.05
0.06	0.07	0.06	0.05
0.06	0.05	0.05	0.05

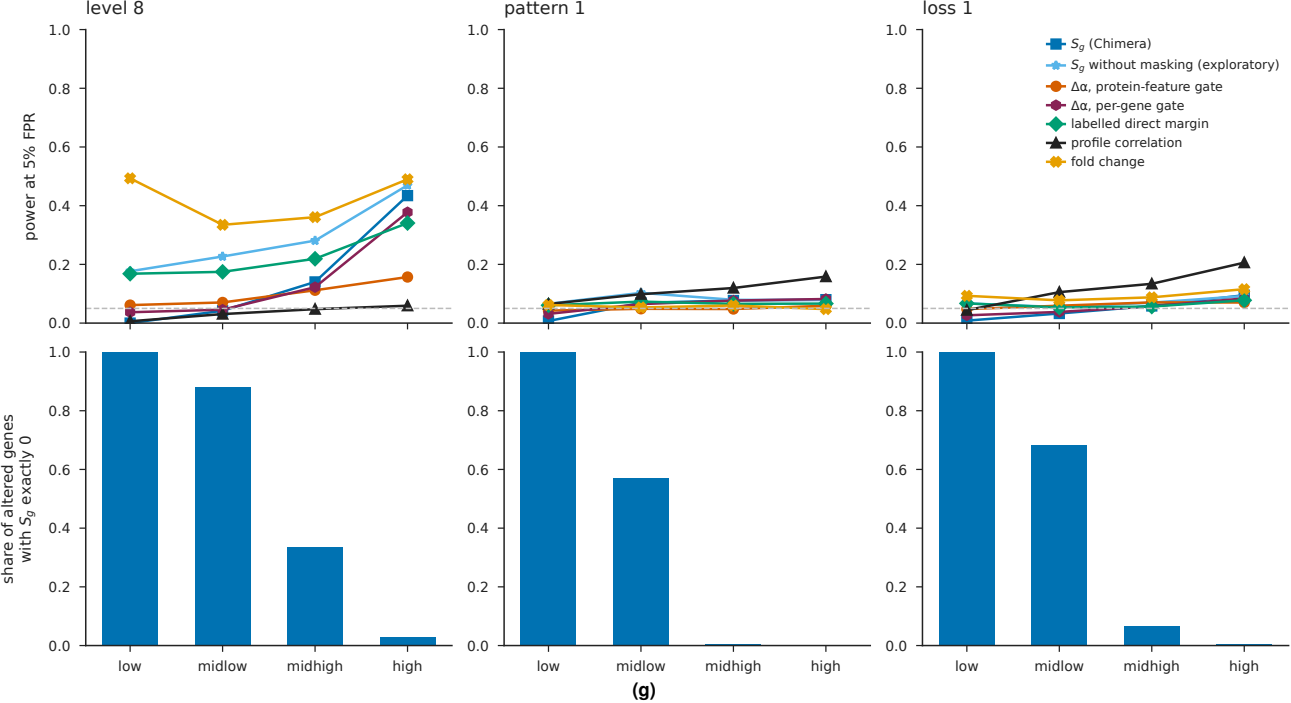


(c)

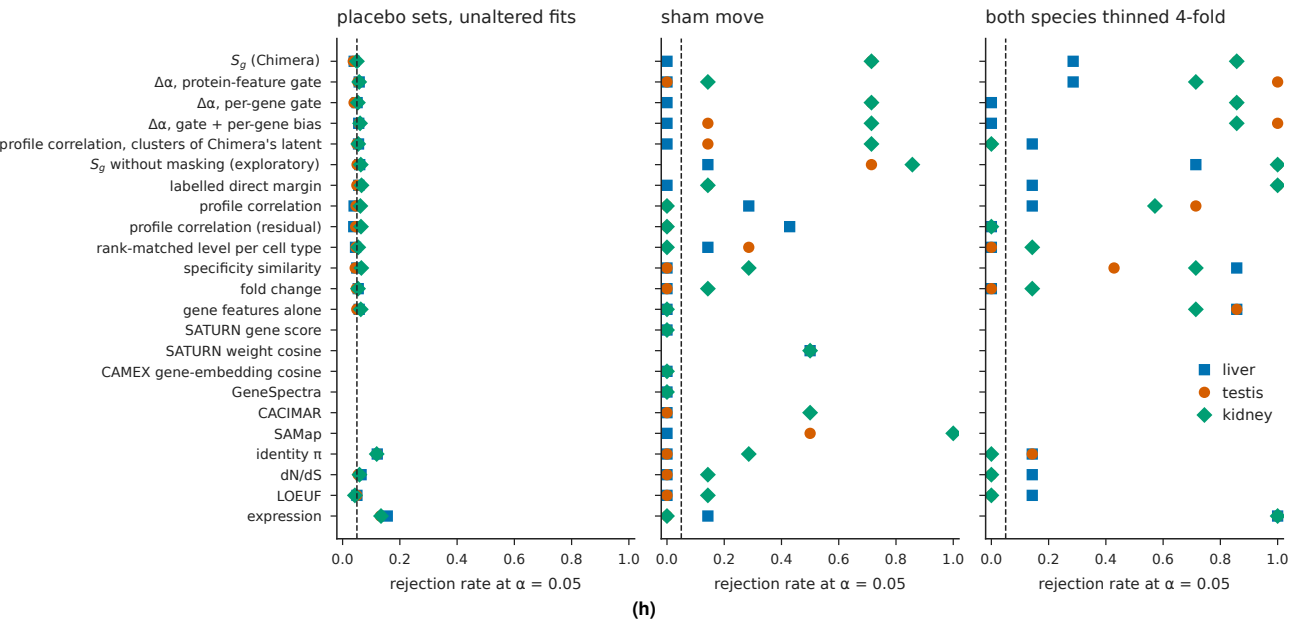


(d)

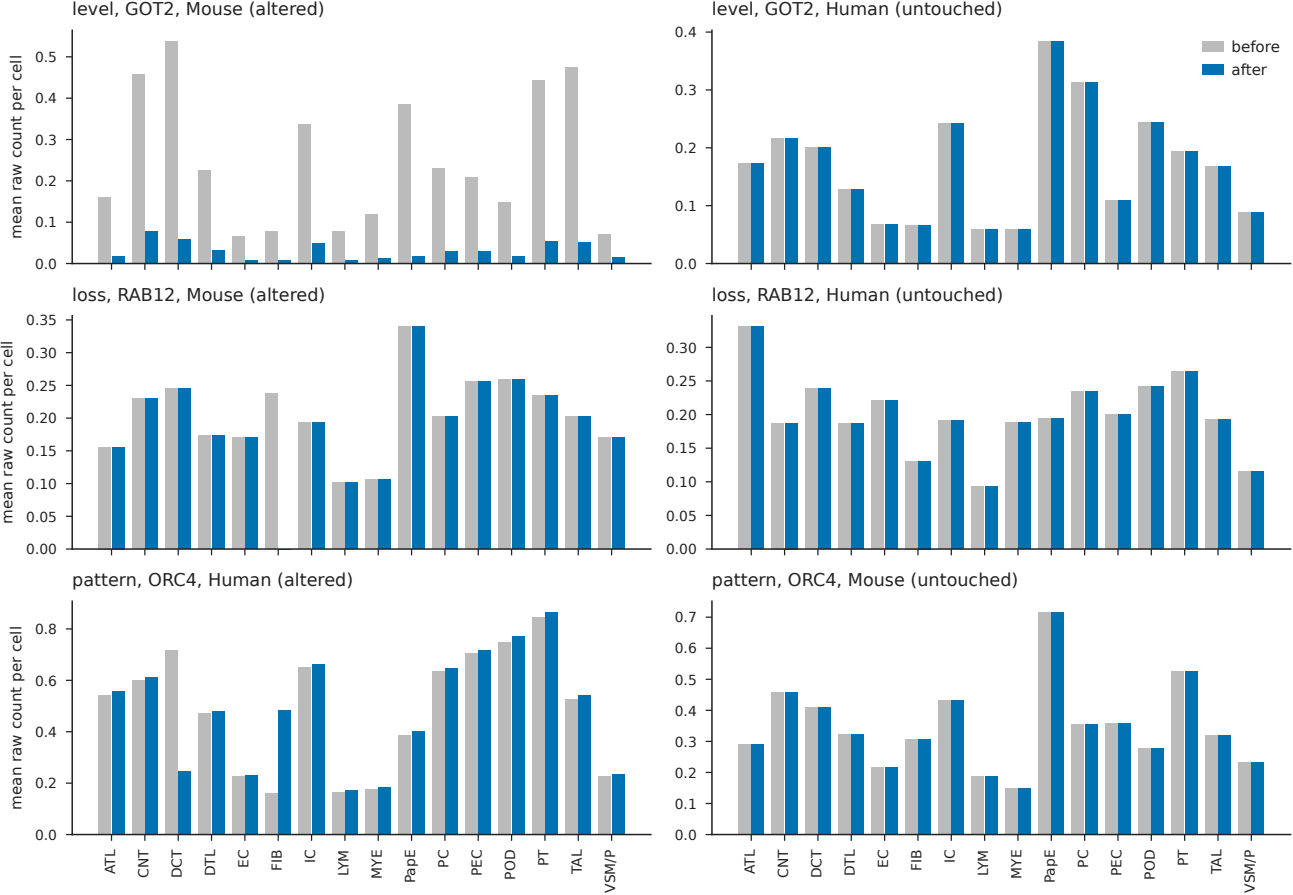

loss 1: AUROC vs matched controls
liver
testis
kidney
power at 5% FPR, 3 cohorts
S_g (Chimera) 0.51 0.54 0.56 0.67 0.50 0.58 0.65 0.57 0.50 0.54 0.56 0.57 0.01 0.03 0.06 0.10
Δα, protein-feature gate 0.50 0.51 0.53 0.43 0.52 0.54 0.60 0.60 0.50 0.51 0.53 0.53 0.05 0.06 0.07 0.07
Δα, per-gene gate 0.50 0.52 0.56 0.60 0.50 0.61 0.65 0.59 0.50 0.54 0.57 0.57 0.03 0.04 0.06 0.08
Δα, gate + per-gene bias 0.49 0.50 0.54 0.65 0.52 0.54 0.63 0.60 0.50 0.51 0.54 0.05 0.05 0.06 0.13
profile correlation, clusters of Chimera's latent 0.59 0.54 0.60 0.76 0.60 0.62 0.69 0.73 0.56 0.59 0.61 0.69 0.05 0.06 0.12 0.18
S_g without masking (exploratory) 0.53 0.55 0.55 0.66 0.60 0.59 0.60 0.56 0.62 0.55 0.54 0.57 0.07 0.05 0.07 0.09
labelled direct margin 0.53 0.54 0.56 0.64 0.59 0.58 0.59 0.57 0.61 0.53 0.52 0.54 0.07 0.05 0.06 0.08
profile correlation 0.52 0.61 0.70 0.83 0.59 0.71 0.76 0.81 0.53 0.59 0.66 0.76 0.04 0.11 0.13 0.21
profile correlation (residual) 0.52 0.59 0.69 0.82 0.58 0.70 0.75 0.81 0.53 0.59 0.66 0.76 0.04 0.11 0.15 0.20
rank-matched level per cell type 0.53 0.66 0.79 0.81 0.55 0.69 0.74 0.83 0.52 0.57 0.64 0.73 0.04 0.09 0.12 0.13
specificity similarity 0.55 0.65 0.78 0.83 0.57 0.65 0.70 0.74 0.51 0.58 0.62 0.65 0.04 0.09 0.16 0.17
fold change 0.51 0.57 0.59 0.68 0.53 0.53 0.53 0.54 0.56 0.52 0.51 0.52 0.09 0.08 0.09 0.12
gene features alone 0.50 0.51 0.51 0.52 0.50 0.51 0.54 0.51 0.54 0.50 0.50 0.51 0.05 0.06 0.06 0.06
SATURN gene score 0.48 0.52 0.51 0.50 0.48 0.50 0.04 0.07
SATURN weight cosine 0.49 0.47 0.51 0.49 0.51 0.50 0.05 0.05
CAMEX gene-embedding cosine 0.50 0.50 0.50 0.50 0.50 0.50 0.01 0.01
GeneSpectra 0.48 0.54 0.47 0.46 0.51 0.50 0.00 0.00
CACIMAR 0.50 0.50 0.50 0.50 0.50 0.50 0.00 0.00
SAMap 0.56 0.69 0.65 0.66 0.61 0.62 0.00 0.10
identity π 0.49 0.50 0.50 0.50 0.49 0.50 0.50 0.50 0.50 0.50 0.50 0.50 0.04 0.06 0.06 0.05
dN/dS 0.53 0.50 0.48 0.49 0.50 0.50 0.50 0.51 0.50 0.51 0.49 0.50 0.03 0.07 0.06 0.04
LOEUF 0.50 0.47 0.47 0.52 0.48 0.51 0.49 0.51 0.50 0.50 0.51 0.50 0.04 0.05 0.04 0.05
expression 0.58 0.59 0.65 0.66 0.68 0.69 0.73 0.66 0.58 0.57 0.59 0.55 0.11 0.18 0.26 0.25
random 0.50 0.49 0.50 0.49 0.50 0.49 0.51 0.49 0.52 0.49 0.50 0.48 0.05 0.04 0.05 0.03
low
midlow
midhigh
high
liver
testis
kidney
AUROC, band midhigh
share removed e
S_g (Chimera)
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labelled direct margin
profile correlation
rank-matched level per cell type
fold change
(e)
wall time (h), 16 CPUs
Chimera
direct baselines
SATURN
CAMEX
peak memory (GB)
wall time (h)
cells, testis (thousands)
(f)


level 8
pattern 1
loss 1
power at 5% FPR
share of altered genes with S_g exactly 0
S_g (Chimera)
S_g without masking (exploratory)
Δα, protein-feature gate
Δα, per-gene gate
labelled direct margin
profile correlation
fold change
low
midlow
midhigh
high

(g)

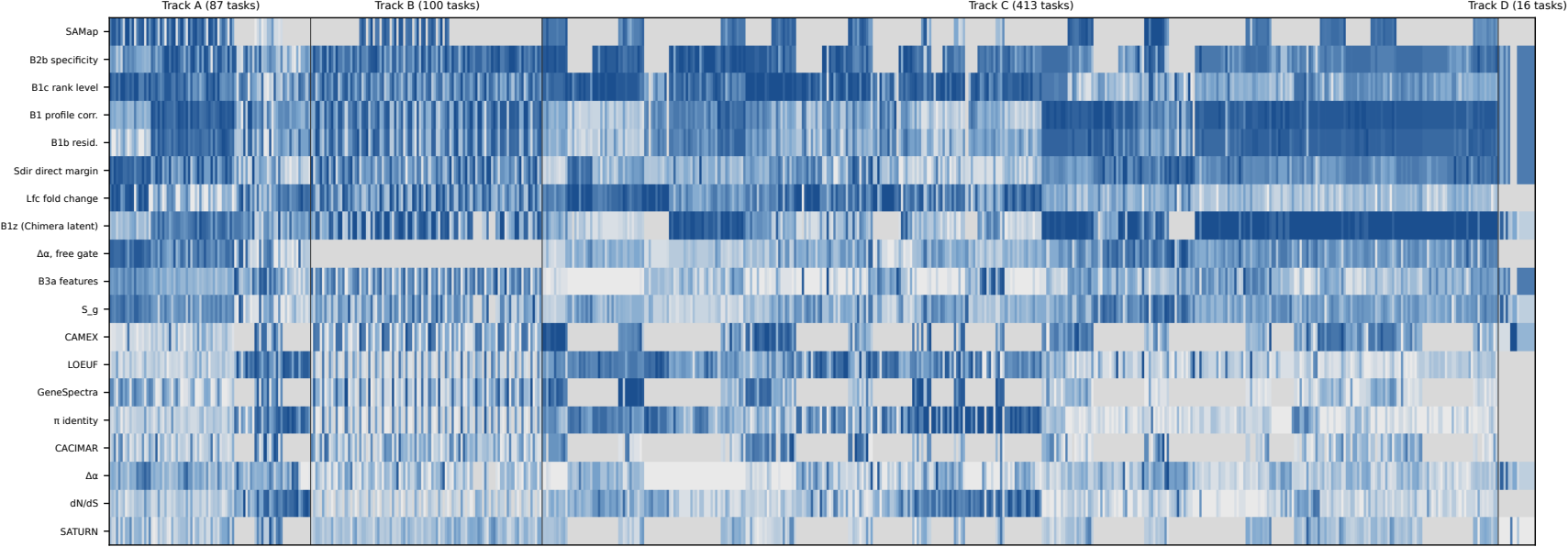

placebo sets, unaltered fits
sham move
both species thinned 4-fold
S_g (Chimera)
Δα, protein-feature gate
Δα, per-gene gate
Δα, gate + per-gene bias
profile correlation, clusters of Chimera's latent
S_g without masking (exploratory)
labelled direct margin
profile correlation
profile correlation (residual)
rank-matched level per cell type
specificity similarity
fold change
gene features alone
SATURN gene score
SATURN weight cosine
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SAMap
identity π
dN/dS
LOEUF
expression
liver
testis
kidney
rejection rate at α = 0.05

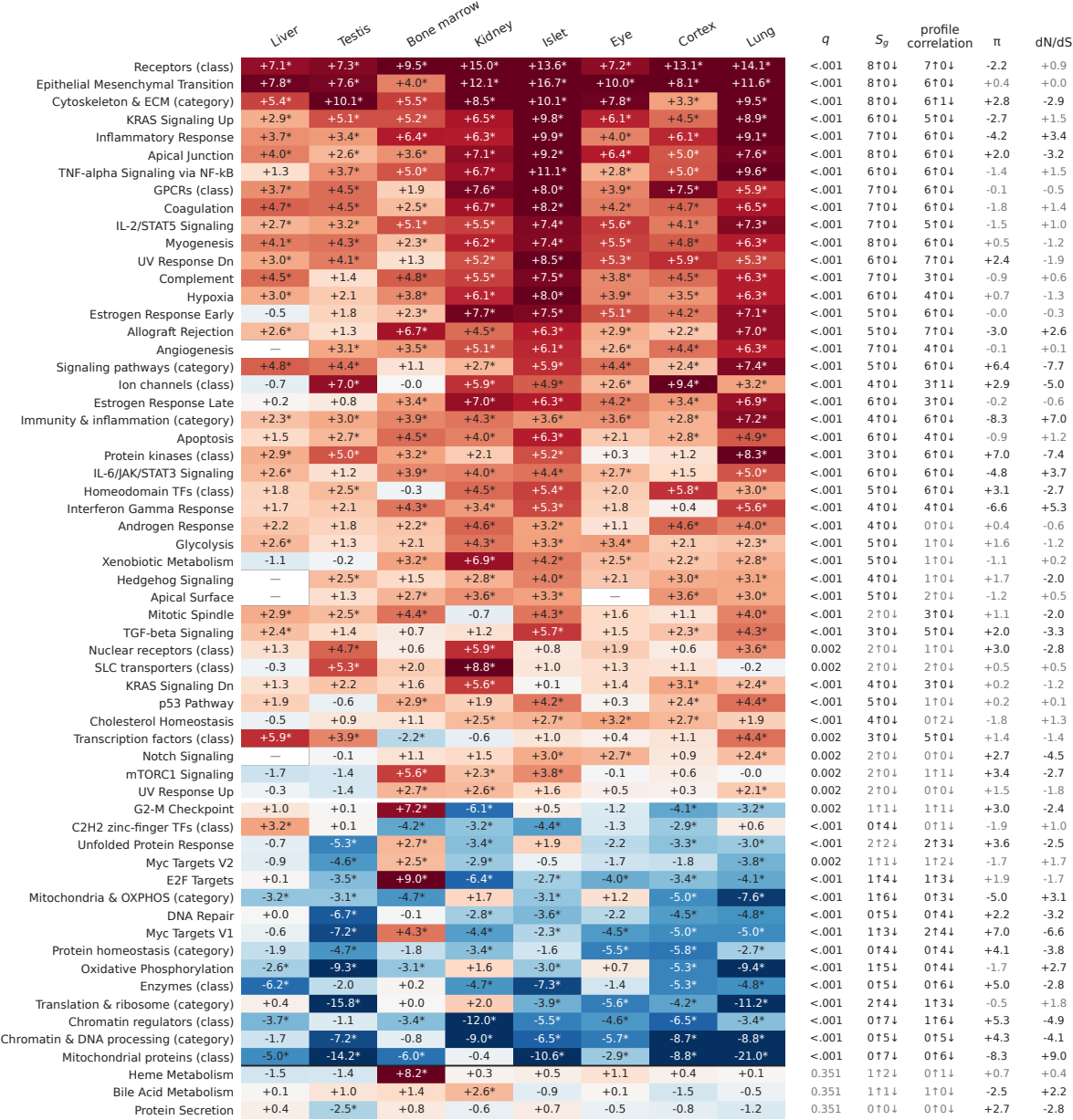
(h)

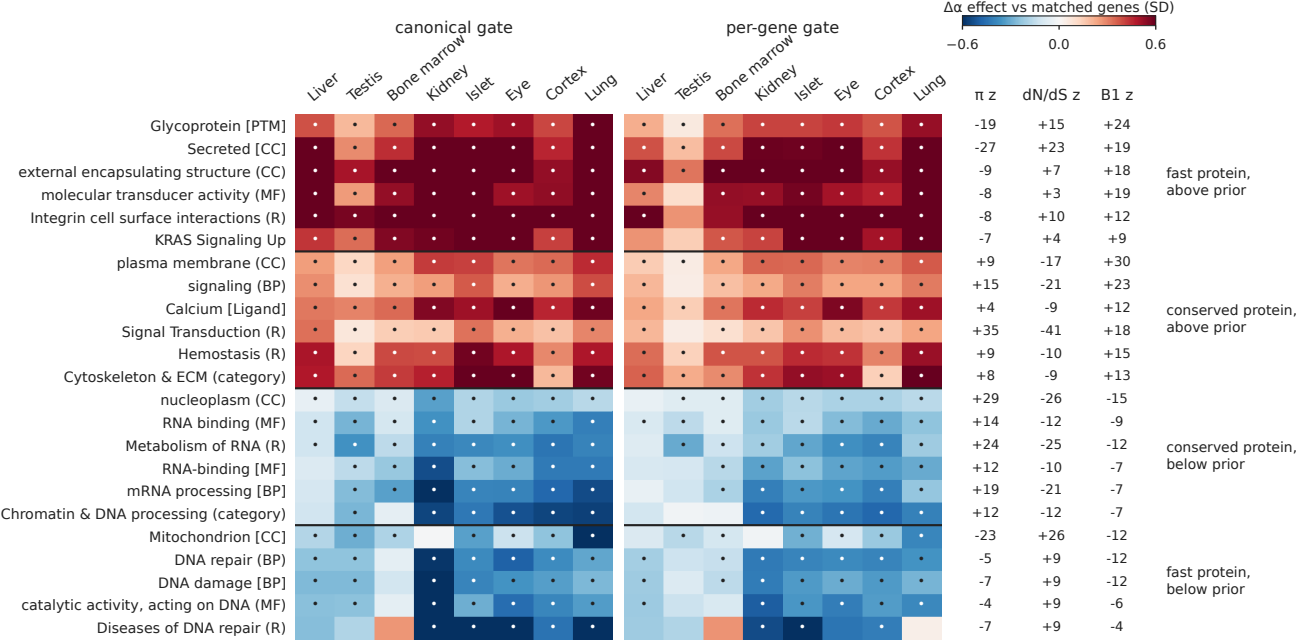

level, GOT2, Mouse (altered)
level, GOT2, Human (untouched)
loss, RAB12, Mouse (altered)
loss, RAB12, Human (untouched)
pattern, ORC4, Human (altered)
pattern, ORC4, Mouse (untouched)
mean raw count per cell
before
after
ATL
CNT
DCT
DTL
EC
FIB
IC
LYM
MYE
PapE
PC
PEC
POD
PT
TAL
VSM/P

(i)

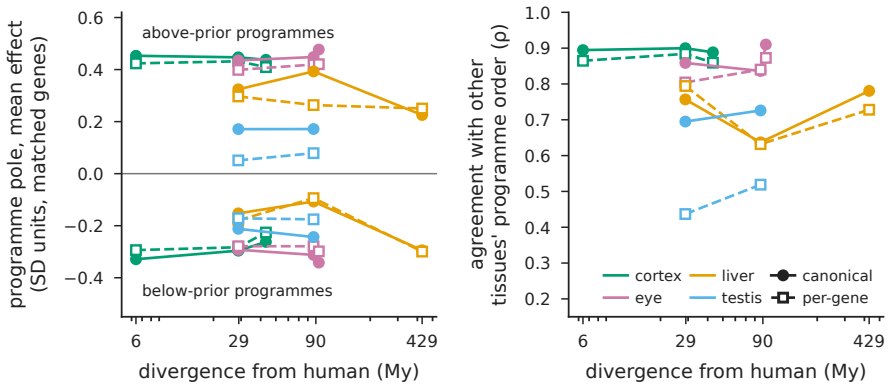
Row labels: blue Chimera readouts, black direct baselines, ochre sequence scores, purple other tools. Circle size and colour: mean within-task scaled rank (large, dark = better); hatched: not applicable or not run; red f/t: runs finished of tried.


Track A (87 tasks)
Track B (100 tasks)
Track C (413 tasks)
Track D (16 tasks)
SAMap
B2b specificity
B1c rank level
B1 profile corr.
B1b resid.
Sdir direct margin
Lfc fold change
B1z (Chimera latent)
Δα, free gate
B3a features
S_g
CAMEX
LOEUF
GeneSpectra
π identity
CACIMAR
Δα
dN/dS
SATURN

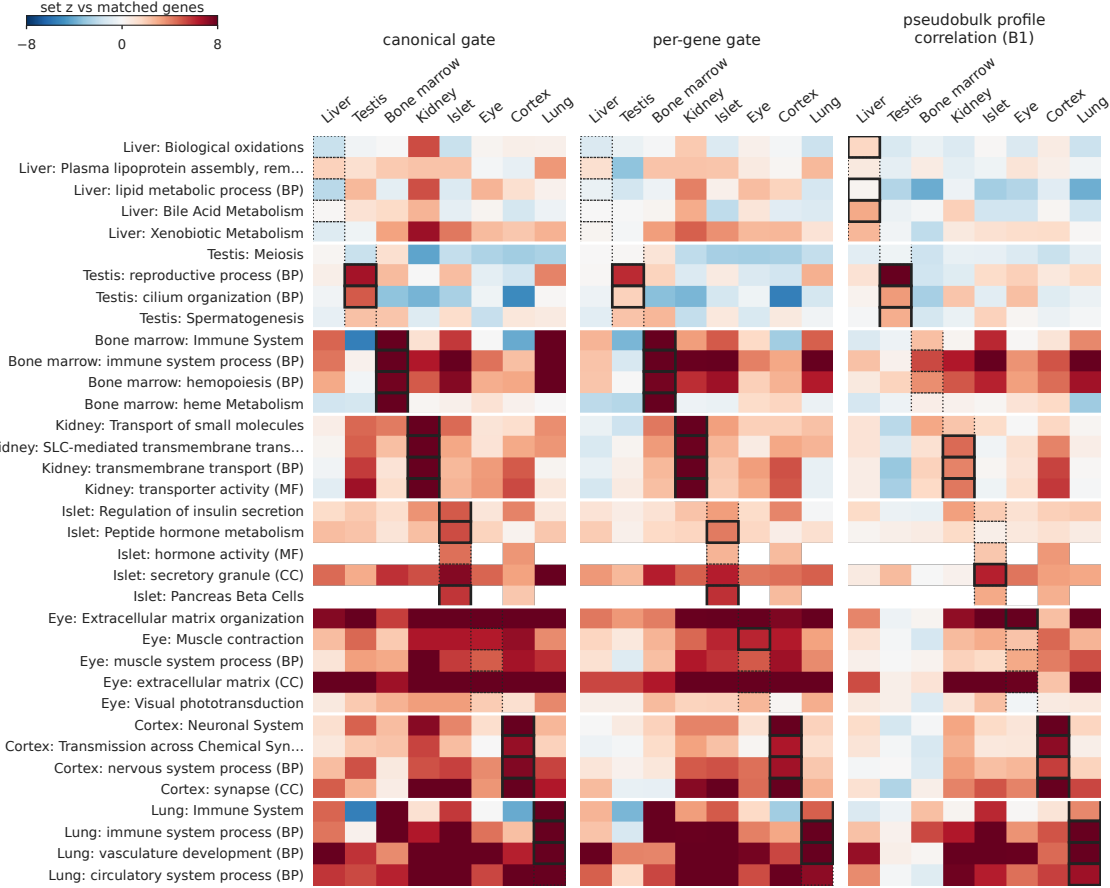

Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
q
S_g
profile correlation
π
dN/dS
Receptors (class) +7.1* +7.3* +9.5* +15.0* +13.6* +7.2* +13.1* +14.1* <.001 8↑0↓ 7↑0↓ -2.2 +0.9
Epithelial Mesenchymal Transition +7.8* +7.6* +4.0* +12.1* +16.7* +10.0* +8.1* +11.6* <.001 8↑0↓ 6↑0↓ +0.4 +0.0
Cytoskeleton & ECM (category) +5.4* +10.1* +5.5* +8.5* +10.1* +7.8* +3.3* +9.5* <.001 8↑0↓ 6↑1↓ +2.8 -2.9
KRAS Signaling Up +2.9* +5.1* +5.2* +6.5* +9.8* +6.1* +4.5* +8.9* <.001 6↑0↓ 5↑0↓ -2.7 +1.5
Inflammatory Response +3.7* +3.4* +6.4* +6.3* +9.9* +4.0* +6.1* +9.1* <.001 7↑0↓ 6↑0↓ -4.2 +3.4
Apical Junction +4.0* +2.6* +3.6* +7.1* +9.2* +6.4* +5.0* +7.6* <.001 8↑0↓ 6↑0↓ +2.0 -3.2
TNF-alpha Signaling via NF-kB +1.3 +3.7* +5.0* +6.7* +11.1* +2.8* +5.0* +9.6* <.001 6↑0↓ 6↑0↓ -1.4 +1.5
GPCRs (class) +3.7* +4.5* +1.9 +7.6* +8.0* +3.9* +7.5* +5.9* <.001 7↑0↓ 6↑0↓ -0.1 -0.5
Coagulation +4.7* +4.5* +2.5* +6.7* +8.2* +4.2* +4.7* +6.5* <.001 7↑0↓ 6↑0↓ -1.8 +1.4
IL-2/STAT5 Signaling +2.7* +3.2* +5.1* +5.5* +7.4* +5.6* +4.1* +7.3* <.001 7↑0↓ 5↑0↓ -1.5 +1.0
Myogenesis +4.1* +4.3* +2.3* +6.2* +7.4* +5.5* +4.8* +6.3* <.001 8↑0↓ 6↑0↓ +0.5 -1.2
UV Response Dn +3.0* +4.1* +1.3 +5.2* +8.5* +5.3* +5.9* +5.3* <.001 6↑0↓ 7↑0↓ +2.4 -1.9
Complement +4.5* +1.4 +4.8* +5.5* +7.5* +3.8* +4.5* +6.3* <.001 7↑0↓ 3↑0↓ -0.9 +0.6
Hypoxia +3.0* +2.1 +3.8* +6.1* +8.0* +3.9* +3.5* +6.3* <.001 6↑0↓ 4↑0↓ +0.7 -1.3
Estrogen Response Early -0.5 +1.8 +2.3* +7.7* +7.5* +5.1* +4.2* +7.1* <.001 5↑0↓ 6↑0↓ -0.0 -0.3
Allograft Rejection +2.6* +1.3 +6.7* +4.5* +6.3* +2.9* +2.2* +7.0* <.001 5↑0↓ 7↑0↓ -3.0 +2.6
Angiogenesis — +3.1* +3.5* +5.1* +6.1* +2.6* +4.4* +6.3* <.001 7↑0↓ 4↑0↓ -0.1 +0.1
Signaling pathways (category) +4.8* +4.4* +1.1 +2.7* +5.9* +4.4* +2.4* +7.4* <.001 5↑0↓ 6↑0↓ +6.4 -7.7
Ion channels (class) -0.7 +7.0* -0.0 +5.9* +4.9* +2.6* +9.4* +3.2* <.001 4↑0↓ 3↑1↓ +2.9 -5.0
Estrogen Response Late +0.2 +0.8 +3.4* +7.0* +6.3* +4.2* +3.4* +6.9* <.001 6↑0↓ 3↑0↓ -0.2 -0.6
Immunity & inflammation (category) +2.3* +3.0* +3.9* +4.3* +3.6* +3.6* +2.8* +7.2* <.001 4↑0↓ 6↑0↓ -8.3 +7.0
Apoptosis +1.5 +2.7* +4.5* +4.0* +6.3* +2.1 +2.8* +4.9* <.001 6↑0↓ 4↑0↓ -0.9 +1.2
Protein kinases (class) +2.9* +5.0* +3.2* +2.1 +5.2* +0.3 +1.2 +8.3* <.001 3↑0↓ 6↑0↓ +7.0 -7.4
IL-6/JAK/STAT3 Signaling +2.6* +1.2 +3.9* +4.0* +4.4* +2.7* +1.5 +5.0* <.001 6↑0↓ 6↑0↓ -4.8 +3.7
Homeodomain TFs (class) +1.8 +2.5* -0.3 +4.5* +5.4* +2.0 +5.8* +3.0* <.001 5↑0↓ 6↑0↓ +3.1 -2.7
Interferon Gamma Response +1.7 +2.1 +4.3* +3.4* +5.3* +1.8 +0.4 +5.6* <.001 4↑0↓ 4↑0↓ -6.6 +5.3
Androgen Response +2.2 +1.8 +2.2* +4.6* +3.2* +1.1 +4.6* +4.0* <.001 4↑0↓ 0↑0↓ +0.4 -0.6
Glycolysis +2.6* +1.3 +2.1 +4.3* +3.3* +3.4* +2.1 +2.3* <.001 5↑0↓ 1↑0↓ +1.6 -1.2
Xenobiotic Metabolism -1.1 -0.2 +3.2* +6.9* +4.2* +2.5* +2.2* +2.8* <.001 5↑0↓ 1↑0↓ -1.1 +0.2
Hedgehog Signaling — +2.5* +1.5 +2.8* +4.0* +2.1 +3.0* +3.1* <.001 4↑0↓ 1↑0↓ +1.7 -2.0
Apical Surface — +1.3 +2.7* +3.6* +3.3* — +3.6* +3.0* <.001 5↑0↓ 2↑0↓ -1.2 +0.5
Mitotic Spindle +2.9* +2.5* +4.4* -0.7 +4.3* +1.6 +1.1 +4.0* <.001 2↑0↓ 3↑0↓ +1.1 -2.0
TGF-beta Signaling +2.4* +1.4 +0.7 +1.2 +5.7* +1.5 +2.3* +4.3* <.001 3↑0↓ 5↑0↓ +2.0 -3.3
Nuclear receptors (class) +1.3 +4.7* +0.6 +5.9* +0.8 +1.9 +0.6 +3.6* 0.002 2↑0↓ 1↑0↓ +3.0 -2.8
SLC transporters (class) -0.3 +5.3* +2.0 +8.8* +1.0 +1.3 +1.1 -0.2 0.002 2↑0↓ 2↑0↓ +0.5 +0.5
KRAS Signaling Dn +1.3 +2.2 +1.6 +5.6* +0.1 +1.4 +3.1* +2.4* <.001 4↑0↓ 3↑0↓ +0.2 -1.2
p53 Pathway +1.9 -0.6 +2.9* +1.9 +4.2* +0.3 +2.4* +4.4* <.001 5↑0↓ 1↑0↓ +0.2 +0.1
Cholesterol Homeostasis -0.5 +0.9 +1.1 +2.5* +2.7* +3.2* +2.7* +1.9 <.001 4↑0↓ 0↑2↓ -1.8 +1.3
Transcription factors (class) +5.9* +3.9* -2.2* -0.6 +1.0 +0.4 +1.1 +4.4* 0.002 3↑0↓ 5↑0↓ +1.4 -1.4
Notch Signaling — -0.1 +1.1 +1.5 +3.0* +2.7* +0.9 +2.4* 0.002 2↑0↓ 0↑0↓ +2.7 -4.5
mTORC1 Signaling -1.7 -1.4 +5.6* +2.3* +3.8* -0.1 +0.6 -0.0 0.002 2↑0↓ 1↑1↓ +3.4 -2.7
UV Response Up -0.3 -1.4 +2.7* +2.6* +1.6 +0.5 +0.3 +2.1* 0.002 2↑0↓ 0↑0↓ +1.5 -1.8
G2-M Checkpoint +1.0 +0.1 +7.2* -6.1* +0.5 -1.2 -4.1* -3.2* 0.002 1↑1↓ 1↑1↓ +3.0 -2.4
C2H2 zinc-finger TFs (class) +3.2* +0.1 -4.2* -3.2* -4.4* -1.3 -2.9* +0.6 <.001 0↑4↓ 0↑1↓ -1.9 +1.0
Unfolded Protein Response -0.7 -5.3* +2.7* -3.4* +1.9 -2.2 -3.3* -3.0* <.001 2↑2↓ 2↑3↓ +3.6 -2.5
Myc Targets V2 -0.9 -4.6* +2.5* -2.9* -0.5 -1.7 -1.8 -3.8* 0.002 1↑1↓ 1↑2↓ -1.7 +1.7
E2F Targets +0.1 -3.5* +9.0* -6.4* -2.7* -4.0* -3.4* -4.1* <.001 1↑4↓ 1↑3↓ +1.9 -1.7
Mitochondria & OXPHOS (category) -3.2* -3.1* -4.7* +1.7 -3.1* +1.2 -5.0* -7.6* <.001 1↑6↓ 0↑3↓ -5.0 +3.1
DNA Repair +0.0 -6.7* -0.1 -2.8* -3.6* -2.2 -4.5* -4.8* <.001 0↑5↓ 0↑4↓ +2.2 -3.2
Myc Targets V1 -0.6 -7.2* +4.3* -4.4* -2.3* -4.5* -5.0* -5.0* <.001 1↑3↓ 2↑4↓ +7.0 -6.6
Protein homeostasis (category) -1.9 -4.7* -1.8 -3.4* -1.6 -5.5* -5.8* -2.7* <.001 0↑4↓ 0↑4↓ +4.1 -3.8
Oxidative Phosphorylation -2.6* -9.3* -3.1* +1.6 -3.0* +0.7 -5.3* -9.4* <.001 1↑5↓ 0↑4↓ -1.7 +2.7
Enzymes (class) -6.2* -2.0 +0.2 -4.7* -7.3* -1.4 -5.3* -4.8* <.001 0↑5↓ 0↑6↓ +5.0 -2.8
Translation & ribosome (category) +0.4 -15.8* +0.0 +2.0 -3.9* -5.6* -4.2* -11.2* <.001 2↑4↓ 1↑3↓ -0.5 +1.8
Chromatin regulators (class) -3.7* -1.1 -3.4* -12.0* -5.5* -4.6* -6.5* -3.4* <.001 0↑7↓ 1↑6↓ +5.3 -4.9
Chromatin & DNA processing (category) -1.7 -7.2* -0.8 -9.0* -6.5* -5.7* -8.7* -8.8* <.001 0↑5↓ 0↑5↓ +4.3 -4.1
Mitochondrial proteins (class) -5.0* -14.2* -6.0* -0.4 -10.6* -2.9* -8.8* -21.0* <.001 0↑7↓ 0↑6↓ -8.3 +9.0
Heme Metabolism -1.5 -1.4 +8.2* +0.3 +0.5 +1.1 +0.4 +0.1 0.351 1↑2↓ 0↑1↓ +0.7 +0.4
Bile Acid Metabolism +0.1 +1.0 +1.4 +2.6* -0.9 +0.1 -1.5 -0.5 0.351 1↑1↓ 1↑0↓ -2.5 +2.2
Protein Secretion +0.4 -2.5* +0.8 -0.6 +0.7 -0.5 -0.8 -1.2 0.351 0↑0↓ 0↑0↓ +2.7 -2.8


canonical gate
per-gene gate
Δα effect vs matched genes (SD)
−0.6
0.0
0.6
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
π z
dN/dS z
B1 z
Glycoprotein [PTM] -19 +15 +24
Secreted [CC] -27 +23 +19
external encapsulating structure (CC) -9 +7 +18
molecular transducer activity (MF) -8 +3 +19
Integrin cell surface interactions (R) -8 +10 +12
KRAS Signaling Up -7 +4 +9
plasma membrane (CC) +9 -17 +30
signaling (BP) +15 -21 +23
Calcium [Ligand] +4 -9 +12
Signal Transduction (R) +35 -41 +18
Hemostasis (R) +9 -10 +15
Cytoskeleton & ECM (category) +8 -9 +13
nucleoplasm (CC) +29 -26 -15
RNA binding (MF) +14 -12 -9
Metabolism of RNA (R) +24 -25 -12
RNA-binding [MF] +12 -10 -7
mRNA processing [BP] +19 -21 -7
Chromatin & DNA processing (category) +12 -12 -7
Mitochondrion [CC] -23 +26 -12
DNA repair (BP) -5 +9 -12
DNA damage [BP] -7 +9 -12
catalytic activity, acting on DNA (MF) -4 +9 -6
Diseases of DNA repair (R) -7 +9 -4
fast protein, above prior
conserved protein, above prior
conserved protein, below prior
fast protein, below prior

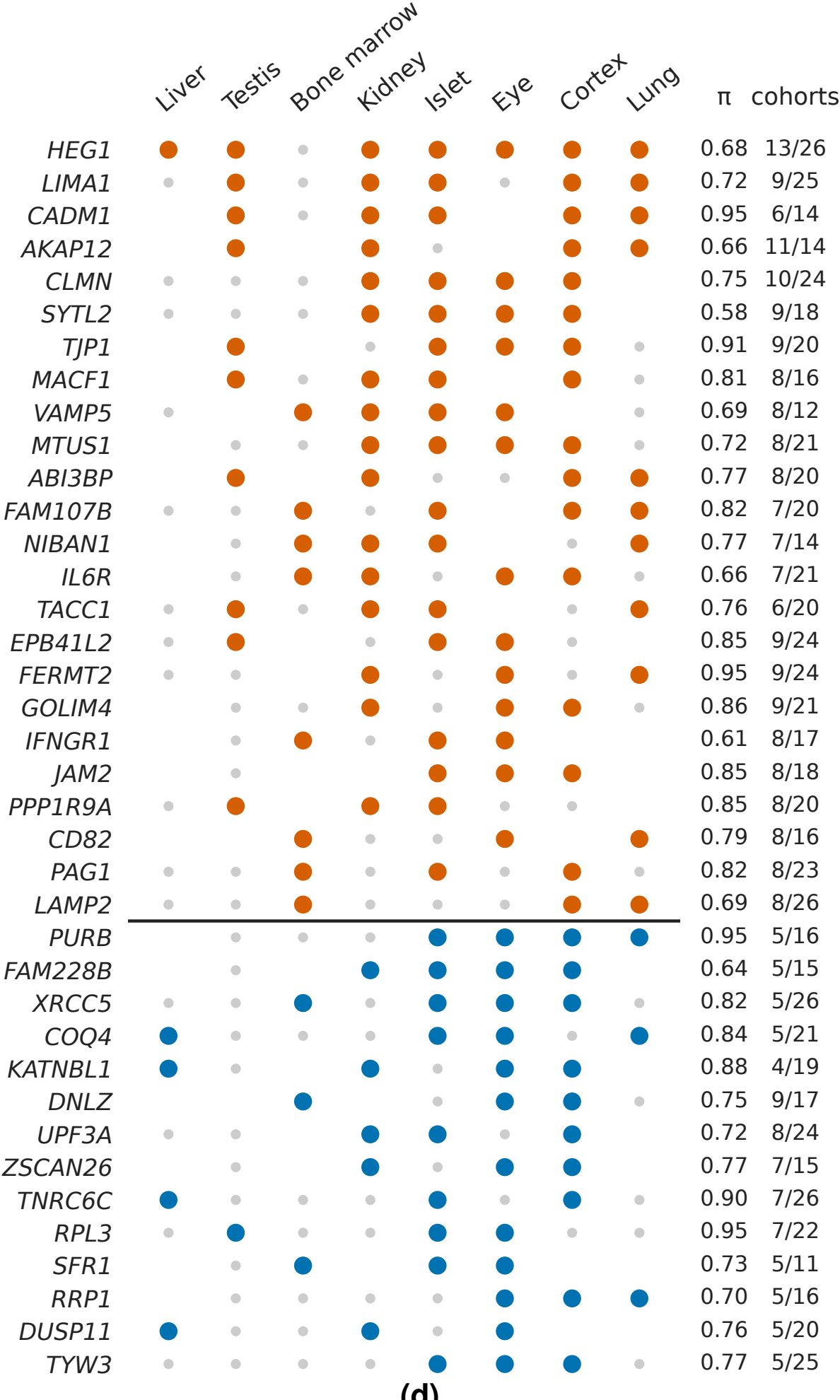
(a)


programme pole, mean effect (SD units, matched genes)
above-prior programmes
below-prior programmes
0.6
0.4
0.2
0.0
−0.2
−0.4
6
29
90
429
divergence from human (My)
agreement with other tissues' programme order (ρ)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
cortex
liver
eye
testis
canonical
per-gene

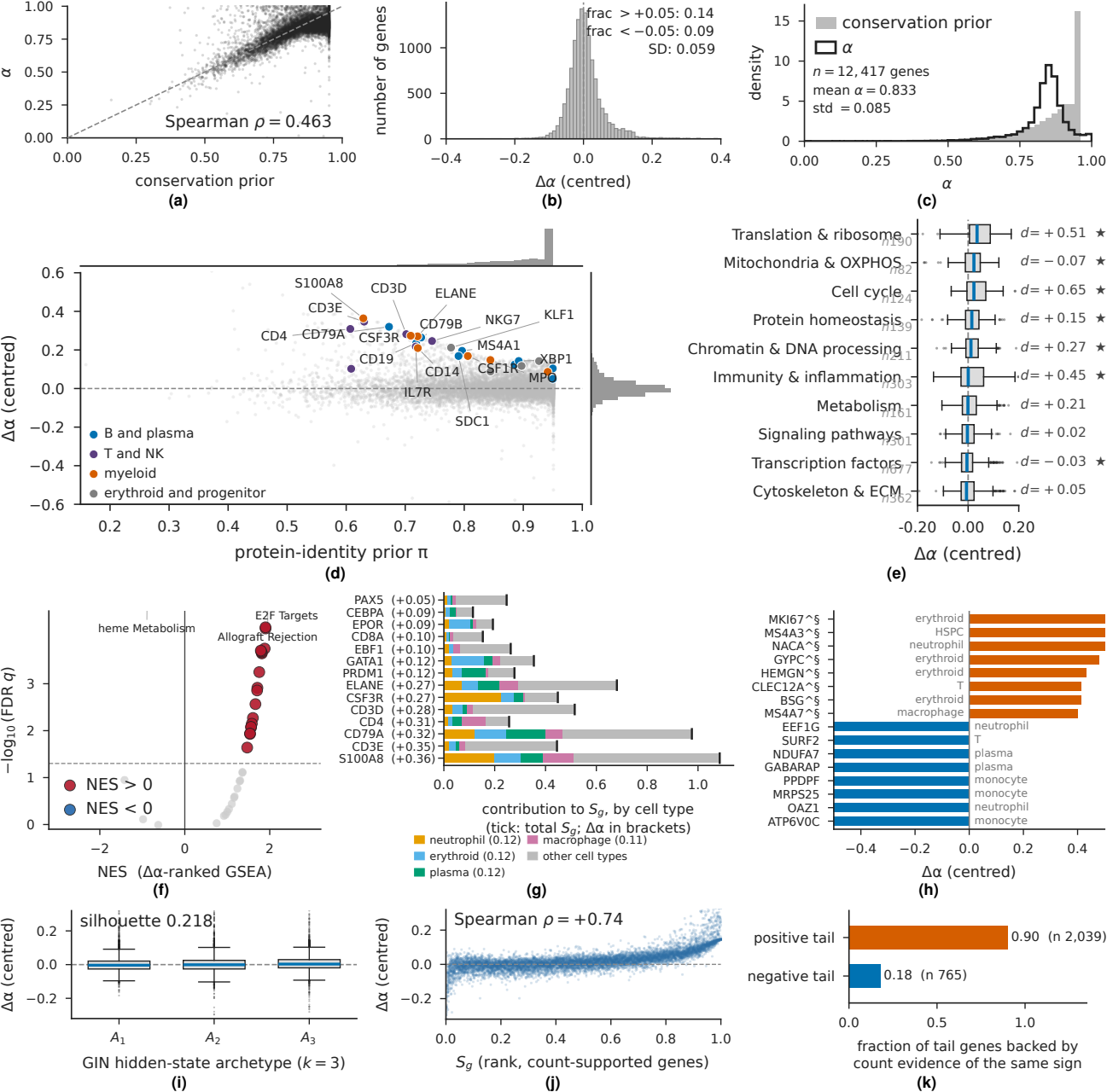
(b)

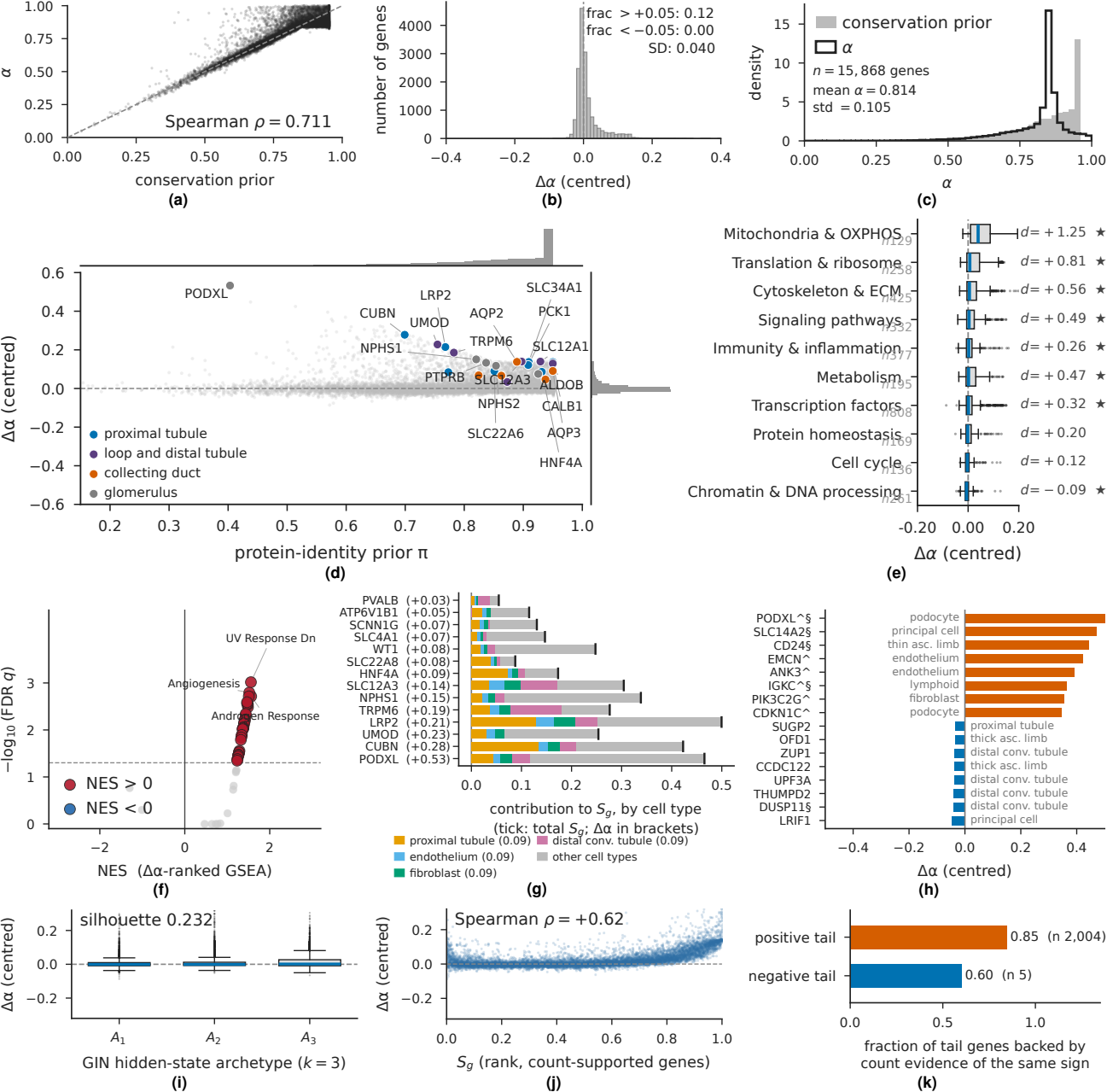

set z vs matched genes
−8
0
8
canonical gate
per-gene gate
pseudobulk profile correlation (B1)
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
Liver: Biological oxidations
Liver: Plasma lipoprotein assembly, rem...
Liver: lipid metabolic process (BP)
Liver: Bile Acid Metabolism
Liver: Xenobiotic Metabolism
Testis: Meiosis
Testis: reproductive process (BP)
Testis: cilium organization (BP)
Testis: Spermatogenesis
Bone marrow: Immune System
Bone marrow: immune system process (BP)
Bone marrow: hemopoiesis (BP)
Bone marrow: heme Metabolism
Kidney: Transport of small molecules
Kidney: SLC-mediated transmembrane trans...
Kidney: transmembrane transport (BP)
Kidney: transporter activity (MF)
Islet: Regulation of insulin secretion
Islet: Peptide hormone metabolism
Islet: hormone activity (MF)
Islet: secretory granule (CC)
Islet: Pancreas Beta Cells
Eye: Extracellular matrix organization
Eye: Muscle contraction
Eye: muscle system process (BP)
Eye: extracellular matrix (CC)
Eye: Visual phototransduction
Cortex: Neuronal System
Cortex: Transmission across Chemical Syn...
Cortex: nervous system process (BP)
Cortex: synapse (CC)
Lung: Immune System
Lung: immune system process (BP)
Lung: vasculature development (BP)
Lung: circulatory system process (BP)

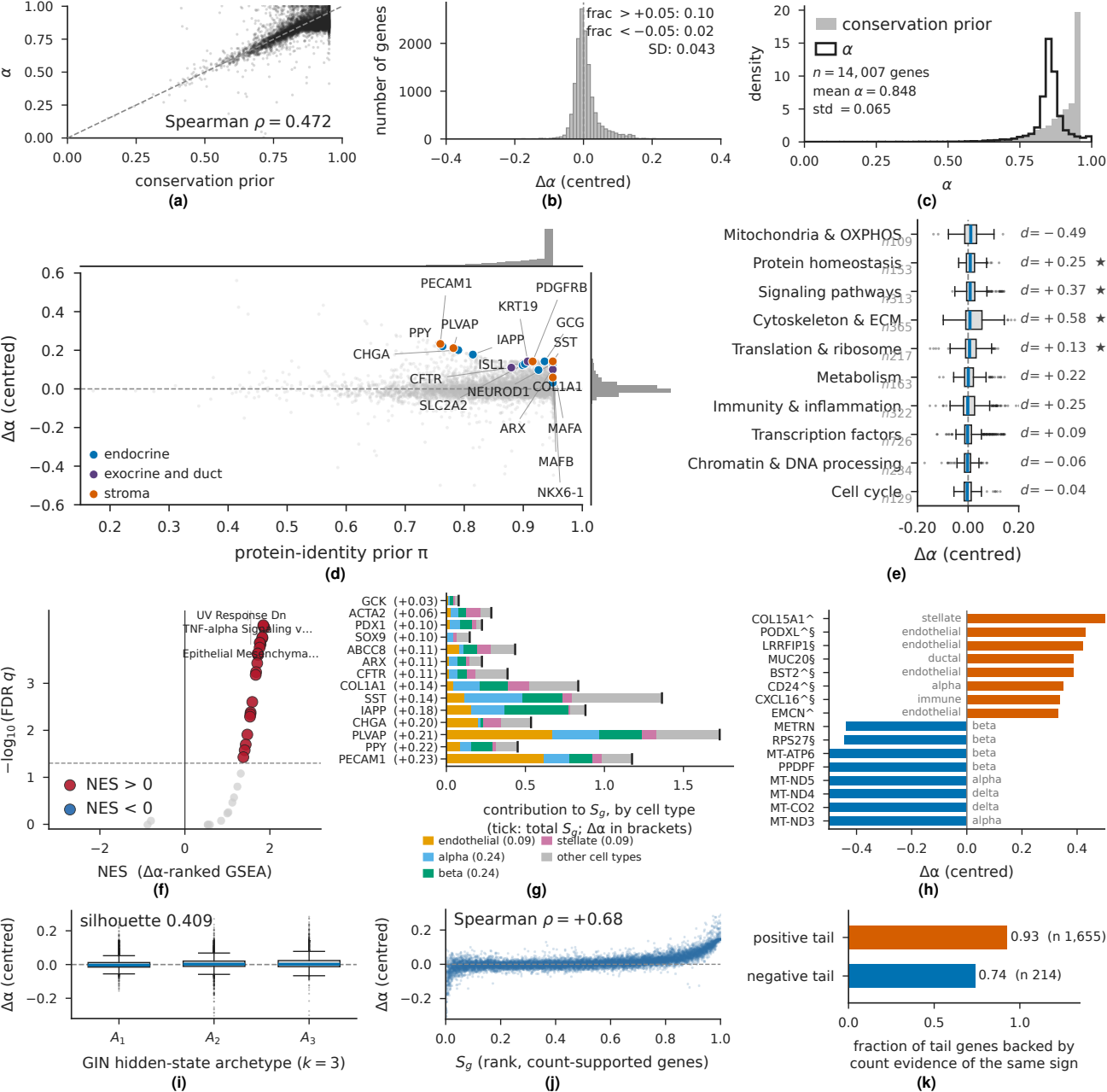
(c)

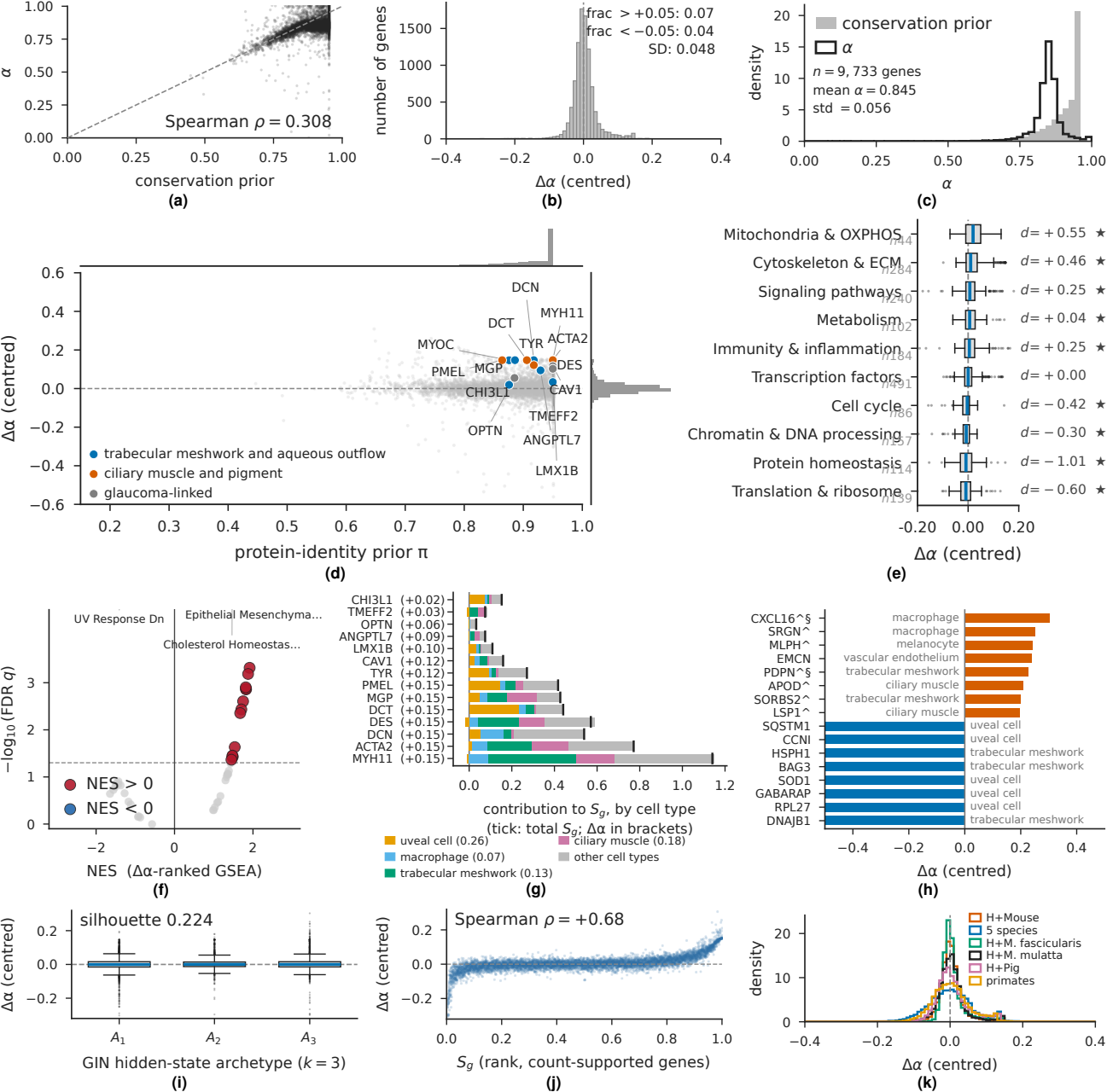

Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
π
cohorts
HEG1 0.68 13/26
LIMA1 0.72 9/25
CADM1 0.95 6/14
AKAP12 0.66 11/14
CLMN 0.75 10/24
SYTL2 0.58 9/18
TJP1 0.91 9/20
MACF1 0.81 8/16
VAMP5 0.69 8/12
MTUS1 0.72 8/21
ABI3BP 0.77 8/20
FAM107B 0.82 7/20
NIBAN1 0.77 7/14
IL6R 0.66 7/21
TACC1 0.76 6/20
EPB41L2 0.85 9/24
FERMT2 0.95 9/24
GOLIM4 0.86 9/21
IFNGR1 0.61 8/17
JAM2 0.85 8/18
PPP1R9A 0.85 8/20
CD82 0.79 8/16
PAG1 0.82 8/23
LAMP2 0.69 8/26
PURB 0.95 5/16
FAM228B 0.64 5/15
XRCC5 0.82 5/26
COQ4 0.84 5/21
KATNBL1 0.88 4/19
DNLZ 0.75 9/17
UPF3A 0.72 8/24
ZSCAN26 0.77 7/15
TNRC6C 0.90 7/26
RPL3 0.95 7/22
SFR1 0.73 5/11
RRP1 0.70 5/16
DUSP11 0.76 5/20
TYW3 0.77 5/25

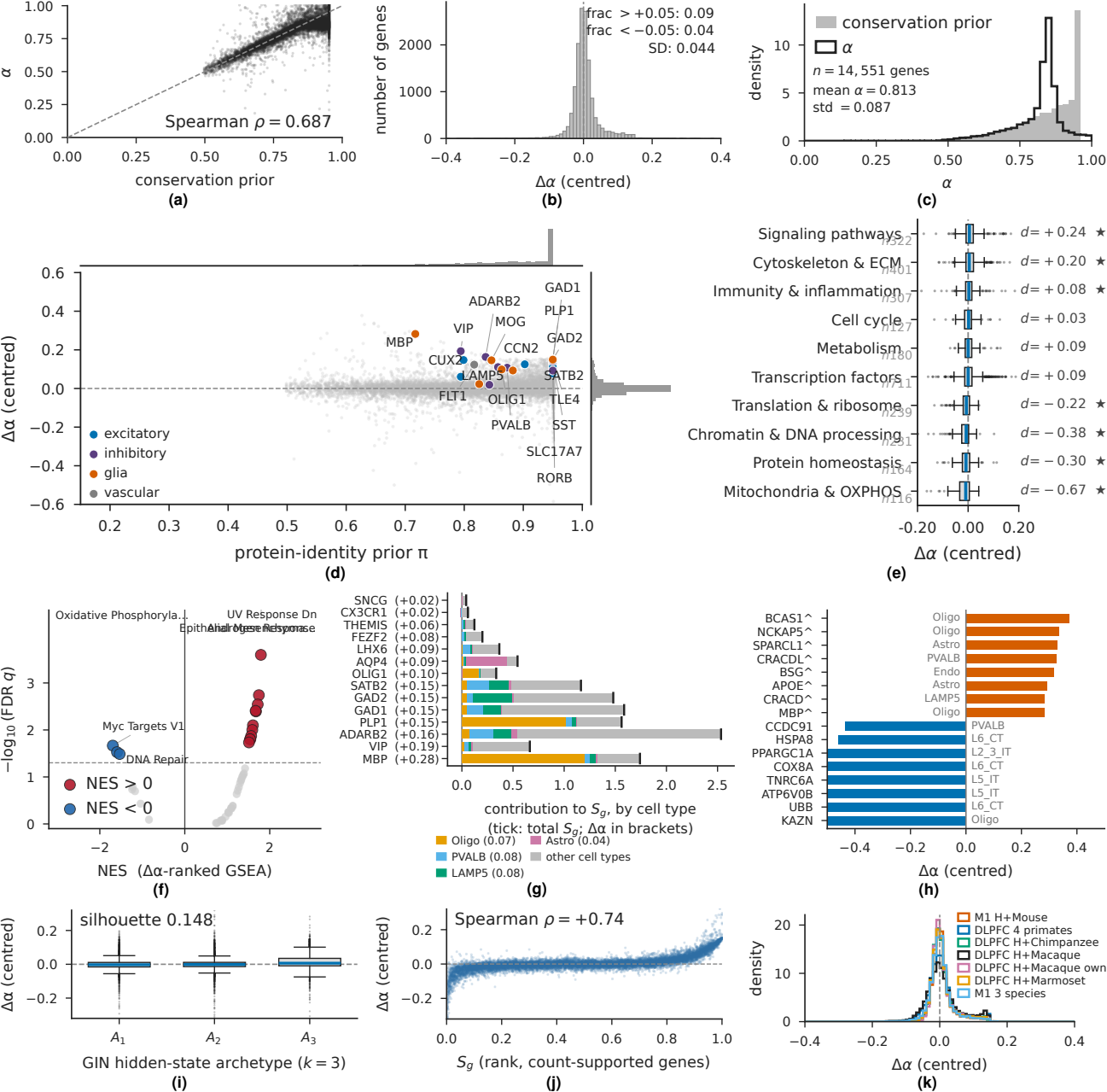
(d)

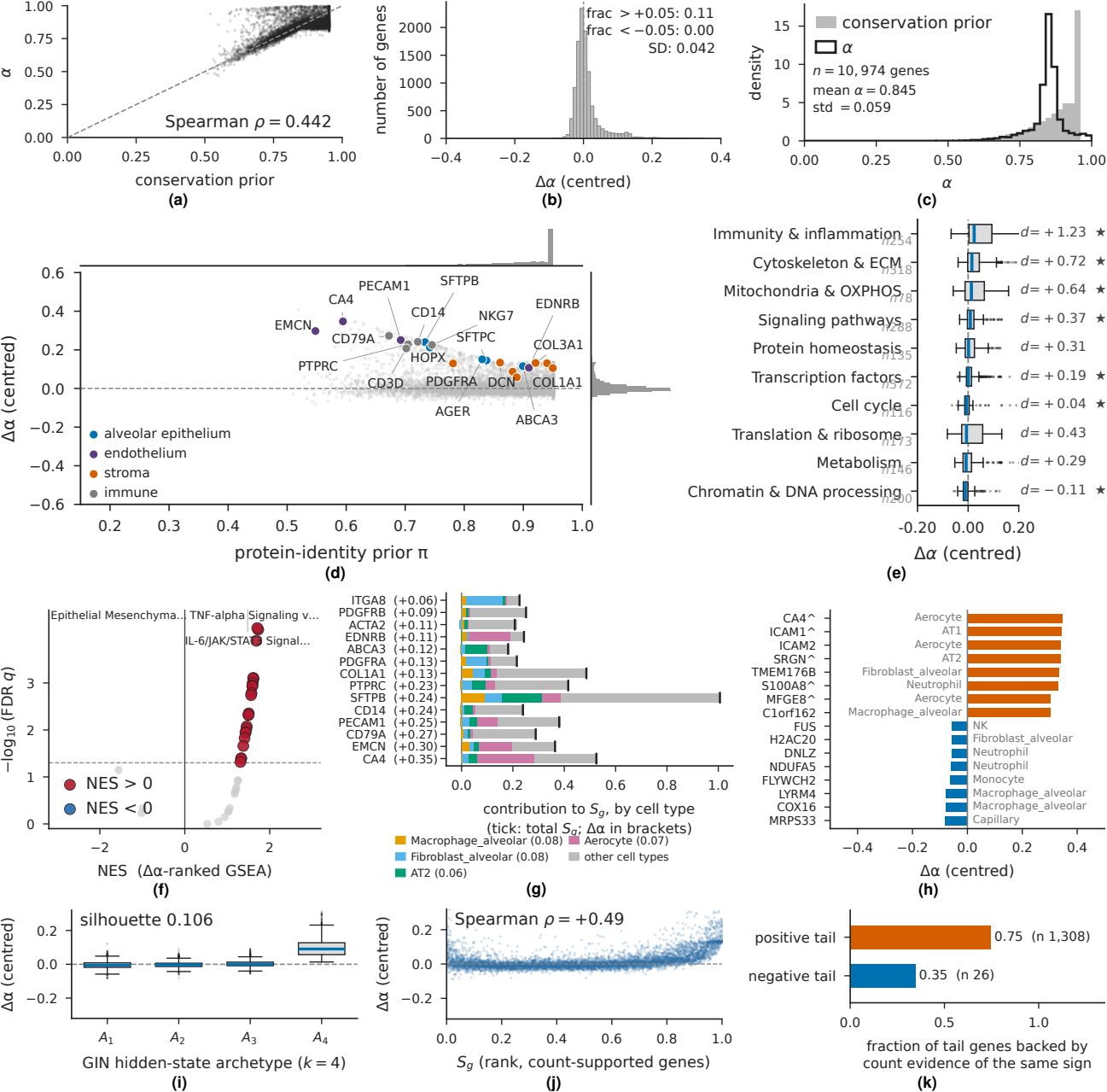

Spearman ρ = 0.463
conservation prior
α
(a)
frac > +0.05: 0.14
frac < −0.05: 0.09
SD: 0.059
number of genes
Δα (centred)
(b)
conservation prior
α
n = 12,417 genes
mean α = 0.833
std = 0.085
density
(c)
S100A8
CD3D
ELANE
CD3E
CD4
CD79A
CSF3R
CD79B
NKG7
KLF1
MS4A1
CD19
CD14
CSF1R
XBP1
MPO
IL7R
SDC1
B and plasma
T and NK
myeloid
erythroid and progenitor
Δα (centred)
protein-identity prior π
(d)
Translation & ribosome n190 d = +0.51 ★
Mitochondria & OXPHOS n82 d = −0.07 ★
Cell cycle n124 d = +0.65 ★
Protein homeostasis n139 d = +0.15 ★
Chromatin & DNA processing n211 d = +0.27 ★
Immunity & inflammation n303 d = +0.45 ★
Metabolism n161 d = +0.21
Signaling pathways n301 d = +0.02
Transcription factors n677 d = −0.03 ★
Cytoskeleton & ECM n362 d = +0.05
Δα (centred)
(e)
heme Metabolism
E2F Targets
Allograft Rejection
NES > 0
NES < 0
−log10(FDR q)
NES (Δα-ranked GSEA)
(f)
PAX5 (+0.05)
CEBPA (+0.09)
EPOR (+0.09)
CD8A (+0.10)
EBF1 (+0.10)
GATA1 (+0.12)
PRDM1 (+0.12)
ELANE (+0.27)
CSF3R (+0.27)
CD3D (+0.28)
CD4 (+0.31)
CD79A (+0.32)
CD3E (+0.35)
S100A8 (+0.36)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
neutrophil (0.12)
macrophage (0.11)
erythroid (0.12)
other cell types
plasma (0.12)
(g)
MKI67^§ erythroid
MS4A3^§ HSPC
NACA^§ neutrophil
GYPC^§ erythroid
HEMGN^§ erythroid
CLEC12A^§ T
BSG^§ erythroid
MS4A7^§ macrophage
EEF1G neutrophil
SURF2 T
NDUFA7 plasma
GABARAP plasma
PPDPF monocyte
MRPS25 monocyte
OAZ1 neutrophil
ATP6V0C monocyte
Δα (centred)
(h)
silhouette 0.218
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.74
Δα (centred)
S_g (rank, count-supported genes)
(j)
positive tail
0.90 (n 2,039)
negative tail
0.18 (n 765)
fraction of tail genes backed by
count evidence of the same sign
(k)

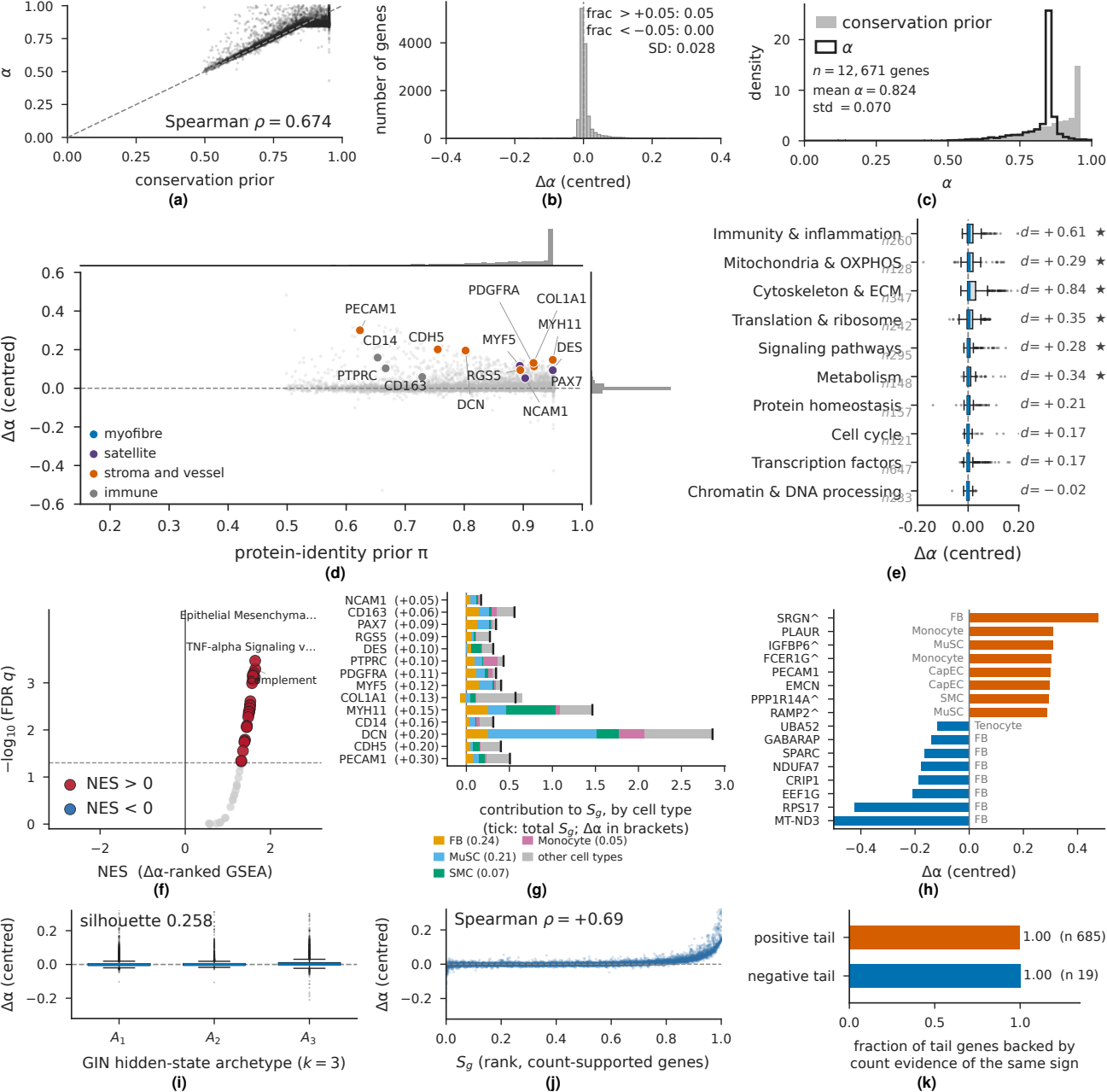

Spearman ρ = 0.711
conservation prior
α
(a)
frac > +0.05: 0.12
frac < −0.05: 0.00
SD: 0.040
number of genes
Δα (centred)
(b)
conservation prior
α
n = 15,868 genes
mean α = 0.814
std = 0.105
density
α
(c)
PODXL
LRP2
SLC34A1
CUBN
UMOD
AQP2
PCK1
NPHS1
TRPM6
SLC12A1
PTPRB
SLC12A3
ALDOB
NPHS2
CALB1
SLC22A6
AQP3
HNF4A
proximal tubule
loop and distal tubule
collecting duct
glomerulus
Δα (centred)
protein-identity prior π
(d)
Mitochondria & OXPHOS n129 d = +1.25 ★
Translation & ribosome n258 d = +0.81 ★
Cytoskeleton & ECM n425 d = +0.56 ★
Signaling pathways n932 d = +0.49 ★
Immunity & inflammation n377 d = +0.26 ★
Metabolism n195 d = +0.47 ★
Transcription factors n808 d = +0.32 ★
Protein homeostasis n169 d = +0.20
Cell cycle n136 d = +0.12
Chromatin & DNA processing n261 d = −0.09 ★
Δα (centred)
(e)
UV Response Dn
Angiogenesis
Androgen Response
NES > 0
NES < 0
−log10 (FDR q)
NES (Δα-ranked GSEA)
(f)
PVALB (+0.03)
ATP6V1B1 (+0.05)
SCNN1G (+0.07)
SLC4A1 (+0.07)
WT1 (+0.08)
SLC22A8 (+0.08)
HNF4A (+0.09)
SLC12A3 (+0.14)
NPHS1 (+0.15)
TRPM6 (+0.19)
LRP2 (+0.21)
UMOD (+0.23)
CUBN (+0.28)
PODXL (+0.53)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
proximal tubule (0.09)
distal conv. tubule (0.09)
endothelium (0.09)
other cell types
fibroblast (0.09)
(g)
PODXL^§ podocyte
SLC14A2§ principal cell
CD24§ thin asc. limb
EMCN^ endothelium
ANK3^ endothelium
IGKC^§ lymphoid
PIK3C2G^ fibroblast
CDKN1C^ podocyte
SUGP2 proximal tubule
OFD1 thick asc. limb
ZUP1 distal conv. tubule
CCDC122 thick asc. limb
UPF3A distal conv. tubule
THUMPD2 distal conv. tubule
DUSP11§ distal conv. tubule
LRIF1 principal cell
Δα (centred)
(h)
silhouette 0.232
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.62
Δα (centred)
S_g (rank, count-supported genes)
(j)
positive tail
0.85 (n 2,004)
negative tail
0.60 (n 5)
fraction of tail genes backed by count evidence of the same sign
(k)

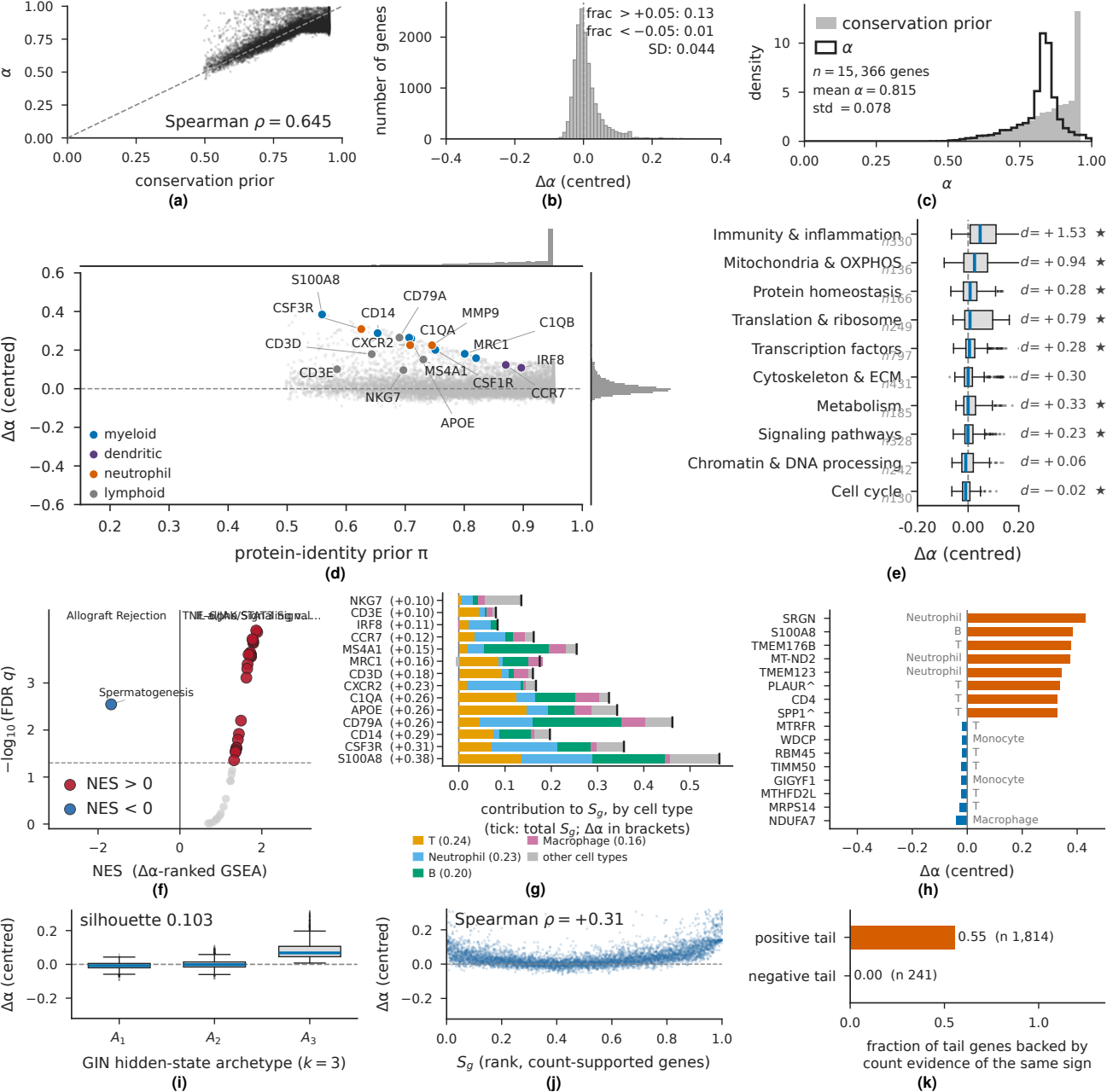

Spearman ρ = 0.472
conservation prior
(a)
frac > +0.05: 0.10
frac < −0.05: 0.02
SD: 0.043
number of genes
Δα (centred)
(b)
conservation prior
α
n = 14,007 genes
mean α = 0.848
std = 0.065
density
α
(c)
PECAM1
PDGFRB
KRT19
GCG
PPY
PLVAP
IAPP
SST
CHGA
ISL1
CFTR
NEUROD1
COL1A1
SLC2A2
ARX
MAFA
MAFB
NKX6-1
endocrine
exocrine and duct
stroma
Δα (centred)
protein-identity prior π
(d)
Mitochondria & OXPHOS n109 d = −0.49
Protein homeostasis n153 d = +0.25 ★
Signaling pathways n313 d = +0.37 ★
Cytoskeleton & ECM n365 d = +0.58 ★
Translation & ribosome n217 d = +0.13 ★
Metabolism n163 d = +0.22
Immunity & inflammation n322 d = +0.25
Transcription factors n726 d = +0.09
Chromatin & DNA processing n294 d = −0.06
Cell cycle n129 d = −0.04
Δα (centred)
(e)
UV Response Dn
TNF-alpha Signaling v...
Epithelial Mesenchyma...
NES > 0
NES < 0
−log10 (FDR q)
NES (Δα-ranked GSEA)
(f)
GCK (+0.03)
ACTA2 (+0.06)
PDX1 (+0.10)
SOX9 (+0.10)
ABCC8 (+0.11)
ARX (+0.11)
CFTR (+0.11)
COL1A1 (+0.14)
SST (+0.14)
IAPP (+0.18)
CHGA (+0.20)
PLVAP (+0.21)
PPY (+0.22)
PECAM1 (+0.23)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
endothelial (0.09)
stellate (0.09)
alpha (0.24)
other cell types
beta (0.24)
(g)
COL15A1^ stellate
PODXL^§ endothelial
LRRFIP1§ endothelial
MUC20§ ductal
BST2^§ endothelial
CD24^§ alpha
CXCL16^§ immune
EMCN^ endothelial
METRN beta
RPS27§ beta
MT-ATP6 beta
PPDPF beta
MT-ND5 alpha
MT-ND4 delta
MT-CO2 delta
MT-ND3 alpha
Δα (centred)
(h)
silhouette 0.409
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.68
Δα (centred)
S_g (rank, count-supported genes)
(j)
positive tail
0.93 (n 1,655)
negative tail
0.74 (n 214)
fraction of tail genes backed by
count evidence of the same sign
(k)

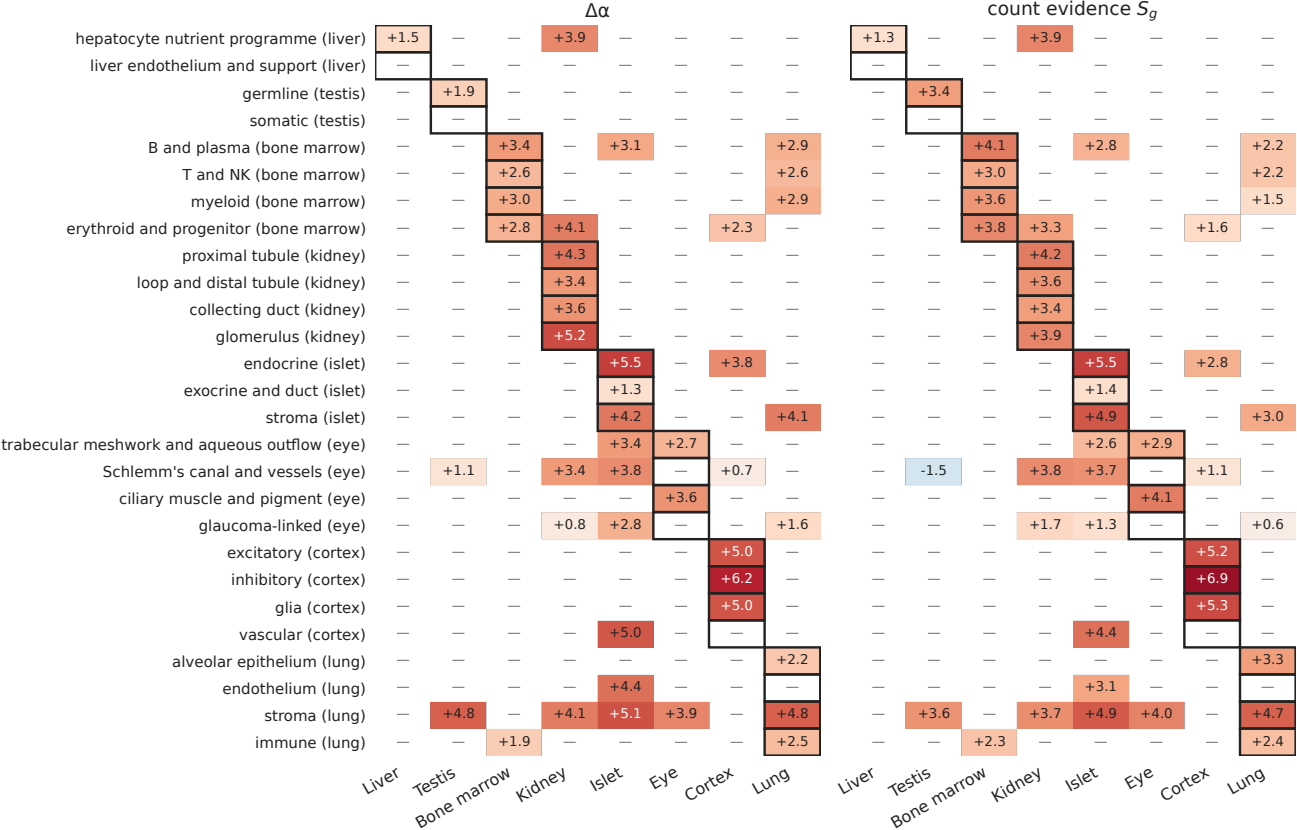

Spearman ρ = 0.308
conservation prior
α
(a)
frac > +0.05: 0.07
frac < −0.05: 0.04
SD: 0.048
number of genes
Δα (centred)
(b)
conservation prior
α
n = 9,733 genes
mean α = 0.845
std = 0.056
density
(c)
DCN
MYH11
DCT
ACTA2
MYOC
TYR
DES
PMEL
MGP
CAV1
CHI3L1
TMEFF2
OPTN
ANGPTL7
LMX1B
trabecular meshwork and aqueous outflow
ciliary muscle and pigment
glaucoma-linked
Δα (centred)
protein-identity prior π
(d)
Mitochondria & OXPHOS n44 d = +0.55 ★
Cytoskeleton & ECM n284 d = +0.46 ★
Signaling pathways n240 d = +0.25 ★
Metabolism n102 d = +0.04 ★
Immunity & inflammation n184 d = +0.25 ★
Transcription factors n491 d = +0.00
Cell cycle n86 d = −0.42 ★
Chromatin & DNA processing n257 d = −0.30 ★
Protein homeostasis n114 d = −1.01 ★
Translation & ribosome n159 d = −0.60 ★
Δα (centred)
(e)
UV Response Dn
Epithelial Mesenchyma...
Cholesterol Homeostas...
NES > 0
NES < 0
−log10(FDR q)
NES (Δα-ranked GSEA)
(f)
CHI3L1 (+0.02)
TMEFF2 (+0.03)
OPTN (+0.06)
ANGPTL7 (+0.09)
LMX1B (+0.10)
CAV1 (+0.12)
TYR (+0.12)
PMEL (+0.15)
MGP (+0.15)
DCT (+0.15)
DES (+0.15)
DCN (+0.15)
ACTA2 (+0.15)
MYH11 (+0.15)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
uveal cell (0.26)
ciliary muscle (0.18)
macrophage (0.07)
other cell types
trabecular meshwork (0.13)
(g)
CXCL16^§ macrophage
SRGN^ macrophage
MLPH^ melanocyte
EMCN vascular endothelium
PDPN^§ trabecular meshwork
APOD^ ciliary muscle
SORBS2^ trabecular meshwork
LSP1^ ciliary muscle
SQSTM1 uveal cell
CCNI uveal cell
HSPH1 trabecular meshwork
BAG3 trabecular meshwork
SOD1 uveal cell
GABARAP uveal cell
RPL27 uveal cell
DNAJB1 trabecular meshwork
Δα (centred)
(h)
silhouette 0.224
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.68
Δα (centred)
S_g (rank, count-supported genes)
(j)
H+Mouse
5 species
H+M. fascicularis
H+M. mulatta
H+Pig
primates
density
Δα (centred)
(k)

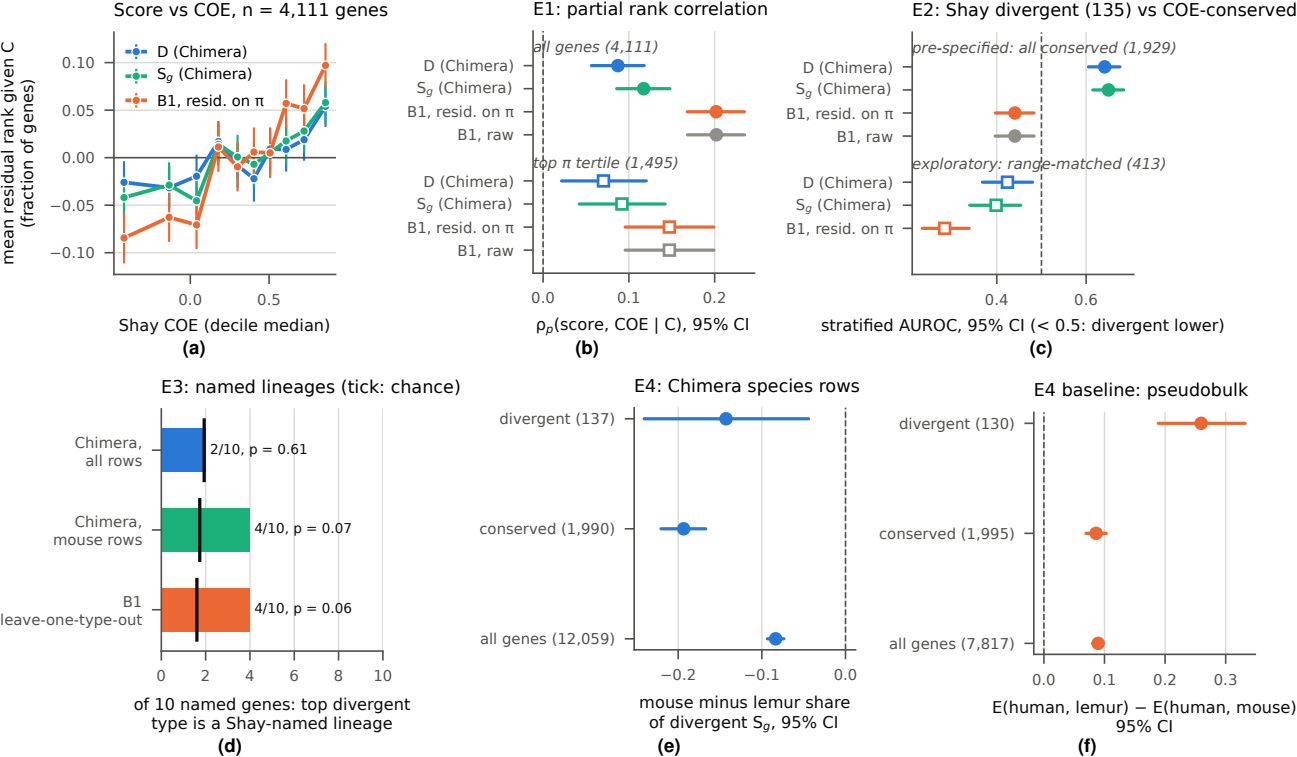

Spearman ρ = 0.687
conservation prior
α
(a)
frac > +0.05: 0.09
frac < −0.05: 0.04
SD: 0.044
number of genes
Δα (centred)
(b)
conservation prior
α
n = 14,551 genes
mean α = 0.813
std = 0.087
density
(c)
excitatory
inhibitory
glia
vascular
GAD1
ADARB2
PLP1
VIP
MOG
MBP
GAD2
CCN2
CUX2
LAMP5
SATB2
FLT1
OLIG1
TLE4
PVALB
SST
SLC17A7
RORB
Δα (centred)
protein-identity prior π
(d)
Signaling pathways n322 d = +0.24 ★
Cytoskeleton & ECM n401 d = +0.20 ★
Immunity & inflammation n307 d = +0.08 ★
Cell cycle n127 d = +0.03
Metabolism n180 d = +0.09
Transcription factors n711 d = +0.09
Translation & ribosome n239 d = −0.22 ★
Chromatin & DNA processing n251 d = −0.38 ★
Protein homeostasis n164 d = −0.30 ★
Mitochondria & OXPHOS n216 d = −0.67 ★
Δα (centred)
(e)
Oxidative Phosphorylа...
UV Response Dn
Myc Targets V1
DNA Repair
NES > 0
NES < 0
−log10 (FDR q)
NES (Δα-ranked GSEA)
(f)
SNCG (+0.02)
CX3CR1 (+0.02)
THEMIS (+0.06)
FEZF2 (+0.08)
LHX6 (+0.09)
AQP4 (+0.09)
OLIG1 (+0.10)
SATB2 (+0.15)
GAD2 (+0.15)
GAD1 (+0.15)
PLP1 (+0.15)
ADARB2 (+0.16)
VIP (+0.19)
MBP (+0.28)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
Oligo (0.07)
Astro (0.04)
PVALB (0.08)
other cell types
LAMP5 (0.08)
(g)
BCAS1^ Oligo
NCKAP5^ Oligo
SPARCL1^ Astro
CRACDL^ PVALB
BSG^ Endo
APOE^ Astro
CRACD^ LAMP5
MBP^ Oligo
CCDC91 PVALB
HSPA8 L6_CT
PPARGC1A L2_3_IT
COX8A L6_CT
TNRC6A L5_IT
ATP6V0B L5_IT
UBB L6_CT
KAZN Oligo
Δα (centred)
(h)
silhouette 0.148
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.74
Δα (centred)
S_g (rank, count-supported genes)
(j)
M1 H+Mouse
DLPFC 4 primates
DLPFC H+Chimpanzee
DLPFC H+Macaque
DLPFC H+Macaque own
DLPFC H+Marmoset
M1 3 species
density
Δα (centred)
(k)

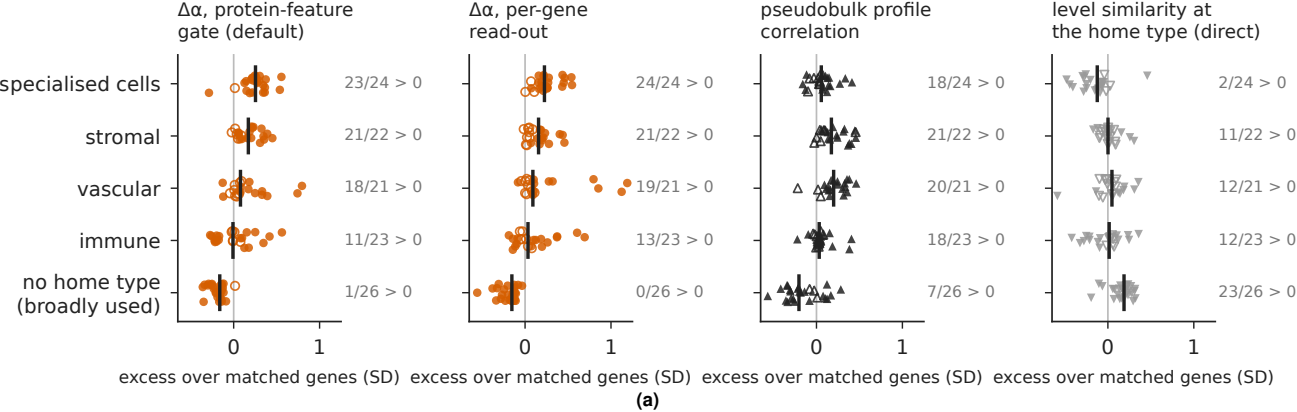

Spearman ρ = 0.442
conservation prior
α
(a)
frac > +0.05: 0.11
frac < −0.05: 0.00
SD: 0.042
number of genes
Δα (centred)
(b)
conservation prior
α
n = 10,974 genes
mean α = 0.845
std = 0.059
density
α
(c)
Δα (centred)
protein-identity prior π
alveolar epithelium
endothelium
stroma
immune
PECAM1
SFTPB
CA4
CD14
EDNRB
NKG7
EMCN
SFTPC
CD79A
COL3A1
HOPX
PTPRC
CD3D
PDGFRA
DCN
COL1A1
AGER
ABCA3
(d)
Immunity & inflammation n254 d = +1.23 ★
Cytoskeleton & ECM n318 d = +0.72 ★
Mitochondria & OXPHOS n78 d = +0.64 ★
Signaling pathways n288 d = +0.37 ★
Protein homeostasis n135 d = +0.31
Transcription factors n372 d = +0.19 ★
Cell cycle n116 d = +0.04 ★
Translation & ribosome n173 d = +0.43
Metabolism n146 d = +0.29
Chromatin & DNA processing n200 d = −0.11 ★
Δα (centred)
(e)
Epithelial Mesenchyma...
TNF-alpha Signaling v...
IL-6/JAK/STAT3 Signal...
NES > 0
NES < 0
−log10 (FDR q)
NES (Δα-ranked GSEA)
(f)
ITGA8 (+0.06)
PDGFRB (+0.09)
ACTA2 (+0.11)
EDNRB (+0.11)
ABCA3 (+0.12)
PDGFRA (+0.13)
COL1A1 (+0.13)
PTPRC (+0.23)
SFTPB (+0.24)
CD14 (+0.24)
PECAM1 (+0.25)
CD79A (+0.27)
EMCN (+0.30)
CA4 (+0.35)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
Macrophage_alveolar (0.08)
Aerocyte (0.07)
Fibroblast_alveolar (0.08)
other cell types
AT2 (0.06)
(g)
CA4^
ICAM1^
ICAM2
SRGN^
TMEM176B
S100A8^
MFGE8^
C1orf162
FUS
H2AC20
DNLZ
NDUFA5
FLYWCH2
LYRM4
COX16
MRPS33
Aerocyte
AT1
Aerocyte
AT2
Fibroblast_alveolar
Neutrophil
Aerocyte
Macrophage_alveolar
NK
Fibroblast_alveolar
Neutrophil
Neutrophil
Monocyte
Macrophage_alveolar
Macrophage_alveolar
Capillary
Δα (centred)
(h)
silhouette 0.106
Δα (centred)
A_1
A_2
A_3
A_4
GIN hidden-state archetype ($k = 4$)
(i)
Spearman ρ = +0.49
Δα (centred)
S_g (rank, count-supported genes)
(j)
positive tail
0.75 (n 1,308)
negative tail
0.35 (n 26)
fraction of tail genes backed by count evidence of the same sign
(k)


Spearman ρ = 0.674
conservation prior
α
(a)
frac > +0.05: 0.05
frac < −0.05: 0.00
SD: 0.028
number of genes
Δα (centred)
(b)
conservation prior
α
n = 12,671 genes
mean α = 0.824
std = 0.070
density
(c)
PECAM1
CD14
CDH5
PTPRC
CD163
MYF5
RGS5
PDGFRA
COL1A1
MYH11
DES
PAX7
DCN
NCAM1
myofibre
satellite
stroma and vessel
immune
Δα (centred)
protein-identity prior π
(d)
Immunity & inflammation n260 d = +0.61 ★
Mitochondria & OXPHOS n128 d = +0.29 ★
Cytoskeleton & ECM n347 d = +0.84 ★
Translation & ribosome n242 d = +0.35 ★
Signaling pathways n295 d = +0.28 ★
Metabolism n148 d = +0.34 ★
Protein homeostasis n157 d = +0.21
Cell cycle n121 d = +0.17
Transcription factors n647 d = +0.17
Chromatin & DNA processing n93 d = −0.02
Δα (centred)
(e)
Epithelial Mesenchyma...
TNF-alpha Signaling v...
Complement
NES > 0
NES < 0
−log10 (FDR q)
NES (Δα-ranked GSEA)
(f)
NCAM1 (+0.05)
CD163 (+0.06)
PAX7 (+0.09)
RGS5 (+0.09)
DES (+0.10)
PTPRC (+0.10)
PDGFRA (+0.11)
MYF5 (+0.12)
COL1A1 (+0.13)
MYH11 (+0.15)
CD14 (+0.16)
DCN (+0.20)
CDH5 (+0.20)
PECAM1 (+0.30)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
FB (0.24)
Monocyte (0.05)
MuSC (0.21)
other cell types
SMC (0.07)
(g)
SRGN^ FB
PLAUR Monocyte
IGFBP6^ MuSC
FCER1G^ Monocyte
PECAM1 CapEC
EMCN CapEC
PPP1R14A^ SMC
RAMP2^ MuSC
UBA52 Tenocyte
GABARAP FB
SPARC FB
NDUFA7 FB
CRIP1 FB
EEF1G FB
RPS17 FB
MT-ND3 FB
Δα (centred)
(h)
silhouette 0.258
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.69
Δα (centred)
S_g (rank, count-supported genes)
(j)
positive tail 1.00 (n 685)
negative tail 1.00 (n 19)
fraction of tail genes backed by count evidence of the same sign
(k)

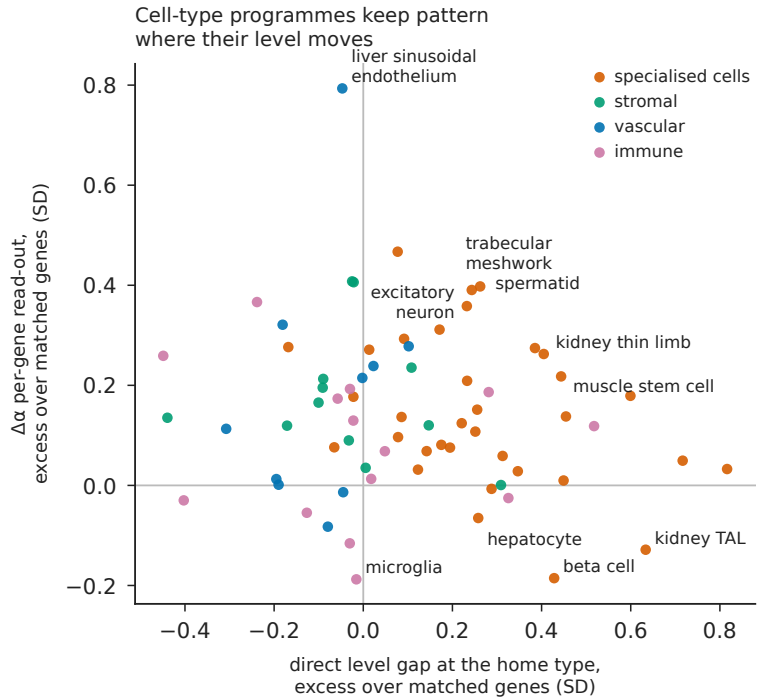

Spearman ρ = 0.645
conservation prior
α
(a)
frac > +0.05: 0.13
frac < −0.05: 0.01
SD: 0.044
number of genes
Δα (centred)
(b)
conservation prior
α
n = 15,366 genes
mean α = 0.815
std = 0.078
density
α
(c)
S100A8
CD79A
CSF3R
CD14
MMP9
C1QA
C1QB
CD3D
CXCR2
MRC1
IRF8
CD3E
MS4A1
CSF1R
CCR7
NKG7
APOE
myeloid
dendritic
neutrophil
lymphoid
Δα (centred)
protein-identity prior π
(d)
Immunity & inflammation n330 d = +1.53 ★
Mitochondria & OXPHOS n136 d = +0.94 ★
Protein homeostasis n166 d = +0.28 ★
Translation & ribosome n249 d = +0.79 ★
Transcription factors n797 d = +0.28 ★
Cytoskeleton & ECM n431 d = +0.30
Metabolism n185 d = +0.33 ★
Signaling pathways n328 d = +0.23 ★
Chromatin & DNA processing n242 d = +0.06
Cell cycle n130 d = −0.02 ★
Δα (centred)
(e)
Allograft Rejection
Spermatogenesis
NES > 0
NES < 0
−log10(FDR q)
NES (Δα-ranked GSEA)
(f)
NKG7 (+0.10)
CD3E (+0.10)
IRF8 (+0.11)
CCR7 (+0.12)
MS4A1 (+0.15)
MRC1 (+0.16)
CD3D (+0.18)
CXCR2 (+0.23)
C1QA (+0.26)
APOE (+0.26)
CD79A (+0.26)
CD14 (+0.29)
CSF3R (+0.31)
S100A8 (+0.38)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
T (0.24)
Macrophage (0.16)
Neutrophil (0.23)
other cell types
B (0.20)
(g)
SRGN Neutrophil
S100A8 B
TMEM176B T
MT-ND2 Neutrophil
TMEM123 Neutrophil
PLAUR^ T
CD4 T
SPP1^ T
MTRFR T
WDCP Monocyte
RBM45 T
TIMM50 T
GIGYF1 Monocyte
MTHFD2L T
MRPS14 T
NDUFA7 Macrophage
Δα (centred)
(h)
silhouette 0.103
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.31
Δα (centred)
S_g (rank, count-supported genes)
(j)
positive tail 0.55 (n 1,814)
negative tail 0.00 (n 241)
fraction of tail genes backed by count evidence of the same sign
(k)


Δα
count evidence S_g
hepatocyte nutrient programme (liver)
liver endothelium and support (liver)
germline (testis)
somatic (testis)
B and plasma (bone marrow)
T and NK (bone marrow)
myeloid (bone marrow)
erythroid and progenitor (bone marrow)
proximal tubule (kidney)
loop and distal tubule (kidney)
collecting duct (kidney)
glomerulus (kidney)
endocrine (islet)
exocrine and duct (islet)
stroma (islet)
trabecular meshwork and aqueous outflow (eye)
Schlemm's canal and vessels (eye)
ciliary muscle and pigment (eye)
glaucoma-linked (eye)
excitatory (cortex)
inhibitory (cortex)
glia (cortex)
vascular (cortex)
alveolar epithelium (lung)
endothelium (lung)
stroma (lung)
immune (lung)
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
+1.5
+3.9
+1.9
+3.4
+3.1
+2.9
+2.6
+3.0
+2.8
+4.1
+2.3
+4.3
+3.4
+3.6
+5.2
+5.5
+3.8
+1.3
+4.2
+4.1
+3.4
+2.7
+1.1
+0.8
+2.8
+1.6
+5.0
+6.2
+2.2
+4.4
+4.8
+5.1
+3.9
+1.3
+3.4
+4.1
+2.8
+2.2
+3.0
+3.6
+1.5
+3.8
+3.3
+1.6
+4.2
+3.9
+2.6
+2.9
-1.5
+3.7
+1.7
+1.3
+0.6
+4.9
+3.1
+3.6
+4.0
+2.3

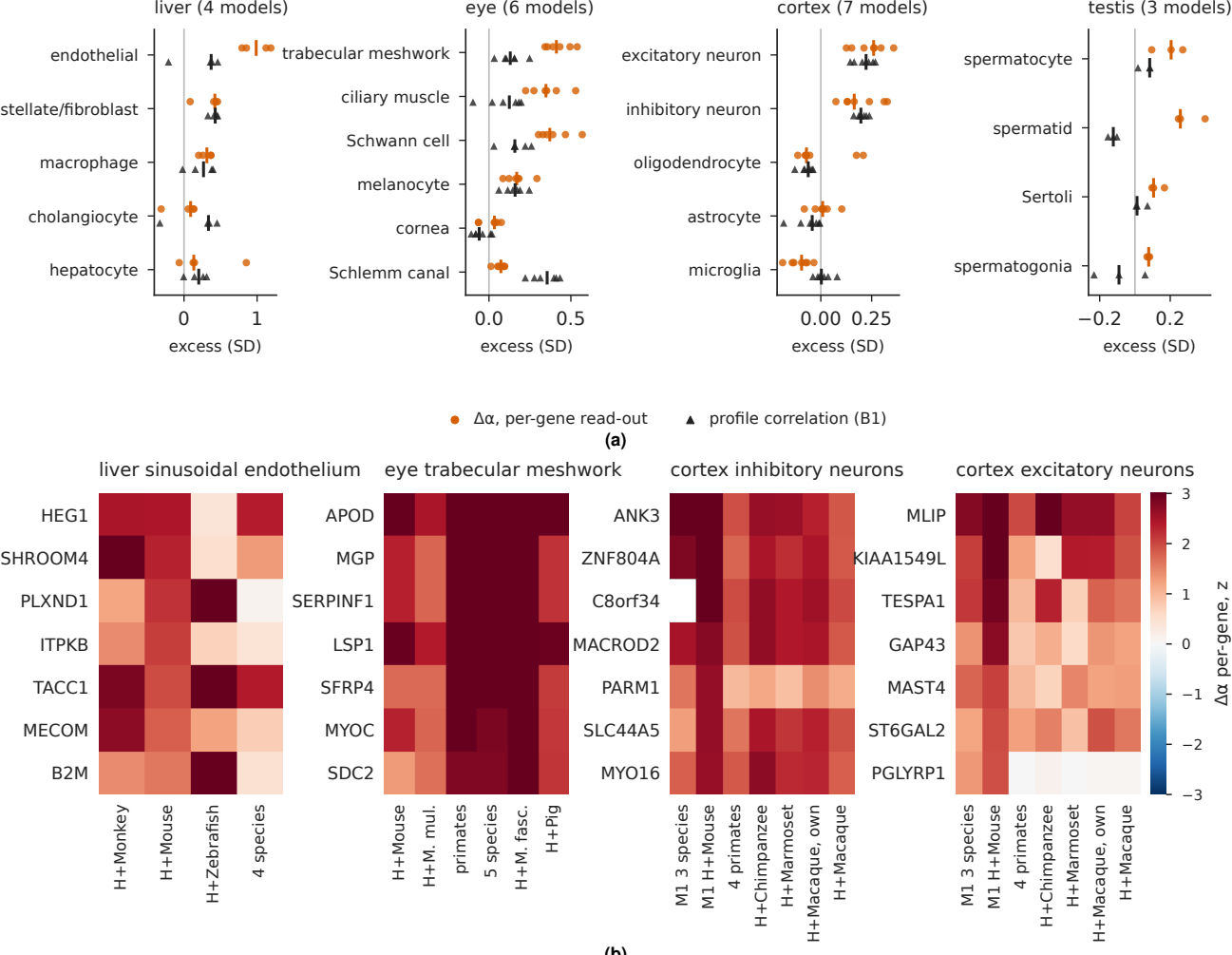

Score vs COE, n = 4,111 genes
D (Chimera)
S_g (Chimera)
B1, resid. on π
mean residual rank given C (fraction of genes)
Shay COE (decile median)
(a)
E1: partial rank correlation
all genes (4,111)
top π tertile (1,495)
D (Chimera)
S_g (Chimera)
B1, resid. on π
B1, raw
ρ_p(score, COE | C), 95% CI
(b)
E2: Shay divergent (135) vs COE-conserved
pre-specified: all conserved (1,929)
exploratory: range-matched (413)
stratified AUROC, 95% CI (< 0.5: divergent lower)
(c)
E3: named lineages (tick: chance)
Chimera, all rows
Chimera, mouse rows
B1 leave-one-type-out
2/10, p = 0.61
4/10, p = 0.07
4/10, p = 0.06
of 10 named genes: top divergent type is a Shay-named lineage
(d)
E4: Chimera species rows
divergent (137)
conserved (1,990)
all genes (12,059)
mouse minus lemur share of divergent S_g, 95% CI
(e)
E4 baseline: pseudobulk
divergent (130)
conserved (1,995)
all genes (7,817)
E(human, lemur) − E(human, mouse) 95% CI
(f)

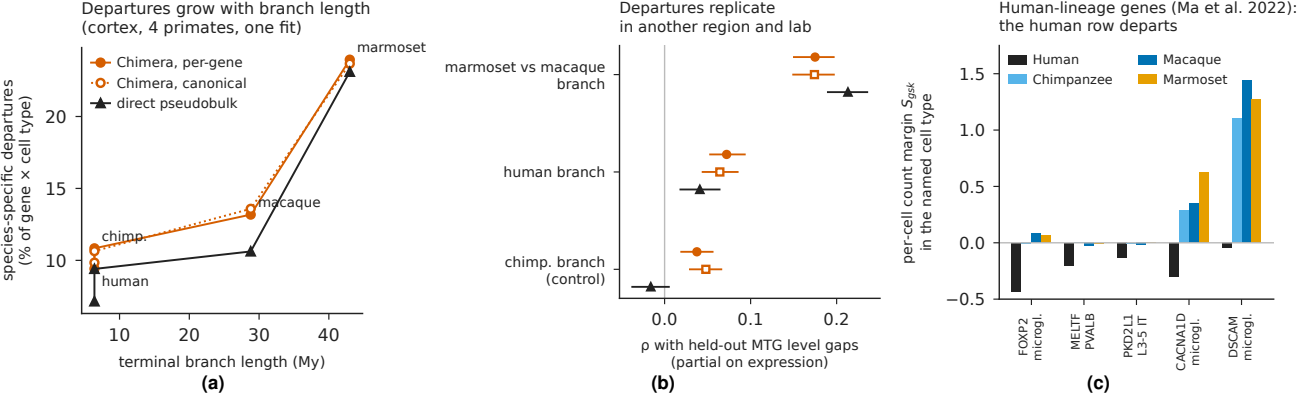

Δα, protein-feature gate (default)
Δα, per-gene read-out
pseudobulk profile correlation
level similarity at the home type (direct)
specialised cells
stromal
vascular
immune
no home type (broadly used)
23/24 > 0
21/22 > 0
18/21 > 0
11/23 > 0
1/26 > 0
24/24 > 0
21/22 > 0
19/21 > 0
13/23 > 0
0/26 > 0
18/24 > 0
21/22 > 0
20/21 > 0
18/23 > 0
7/26 > 0
2/24 > 0
11/22 > 0
12/21 > 0
12/23 > 0
23/26 > 0
0
1
excess over matched genes (SD)

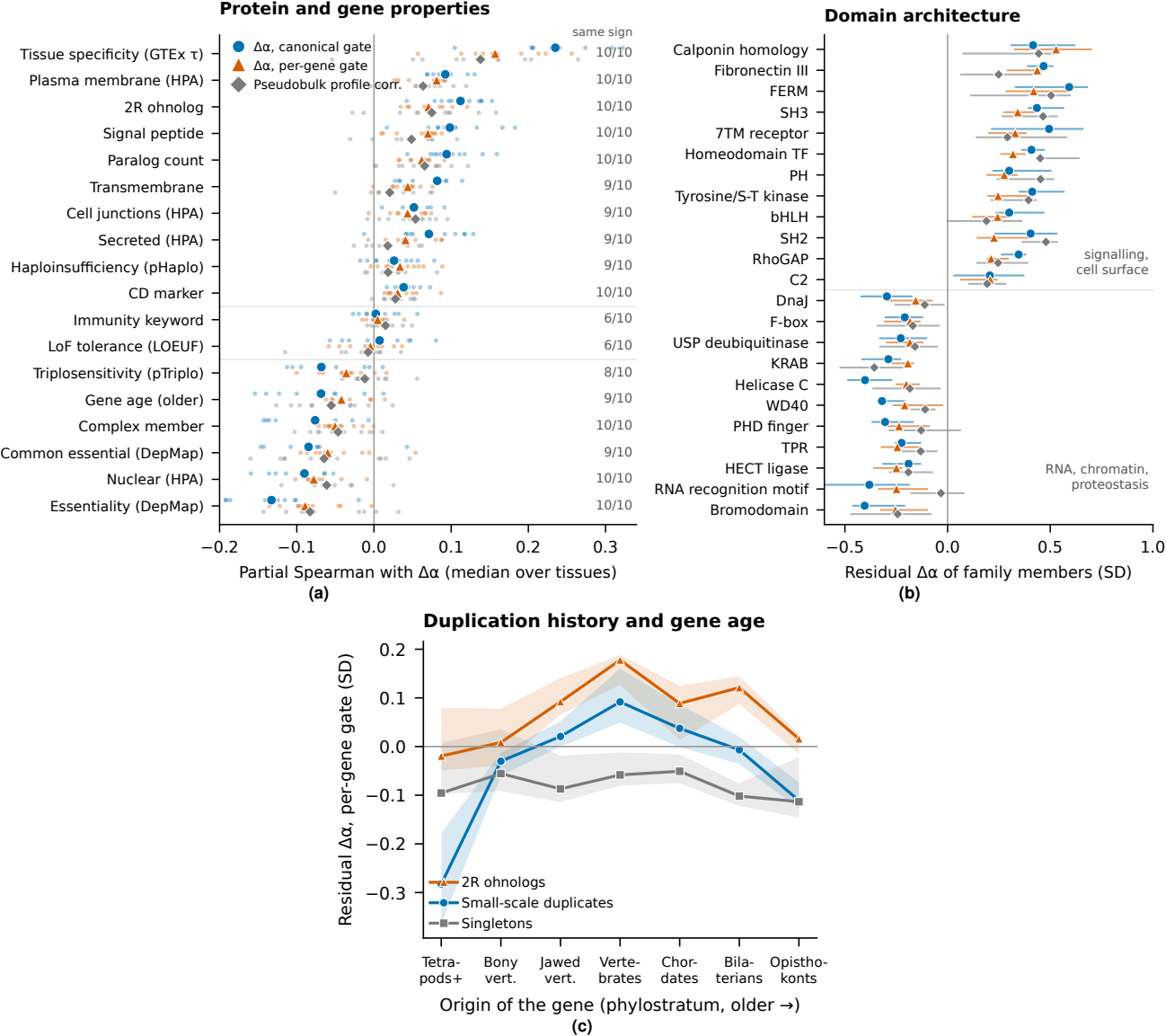
(a)

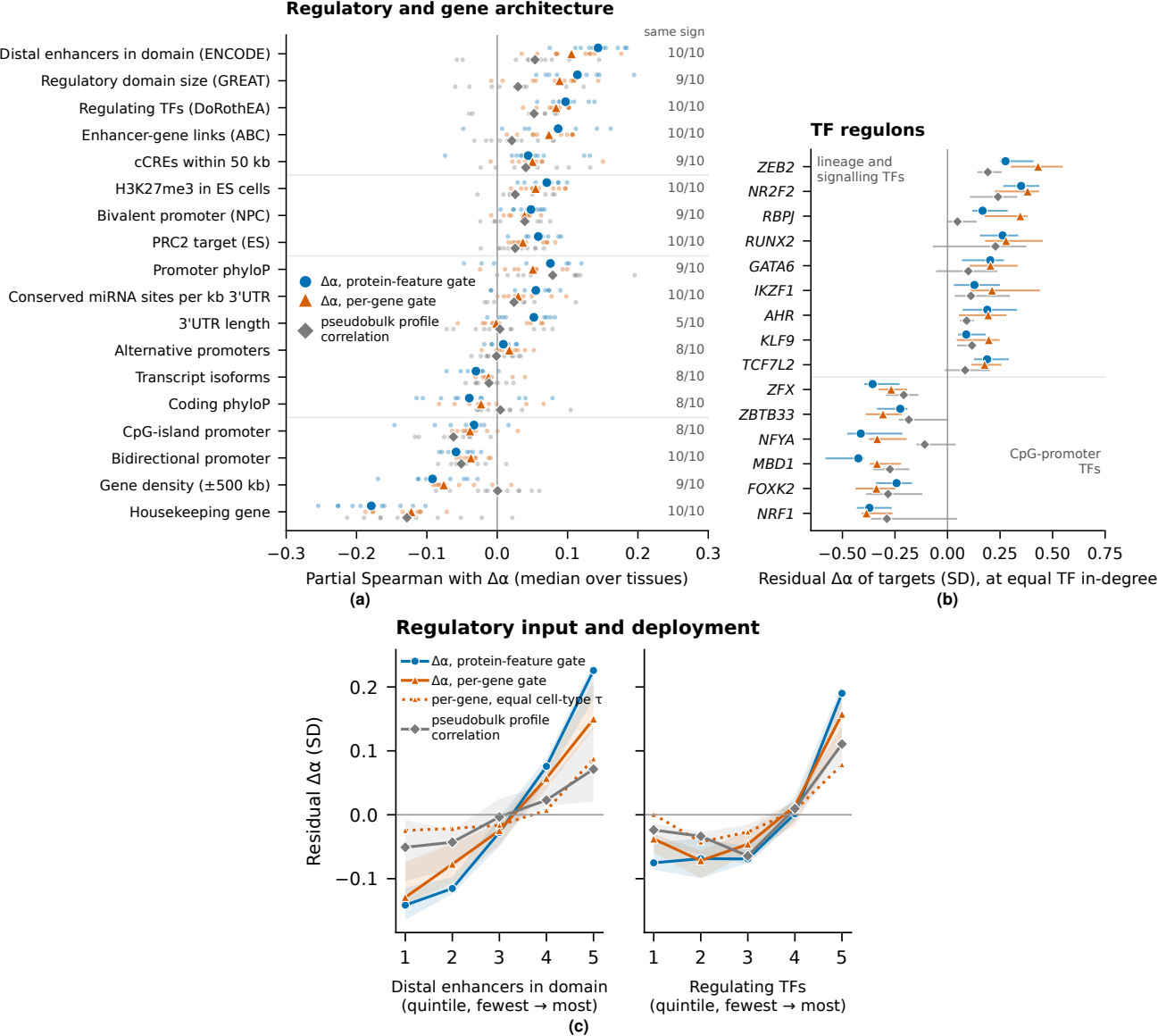

Cell-type programmes keep pattern where their level moves
liver sinusoidal endothelium
specialised cells
stromal
vascular
immune
trabecular meshwork
spermatid
excitatory neuron
kidney thin limb
muscle stem cell
hepatocyte
kidney TAL
beta cell
microglia
Δα per-gene read-out, excess over matched genes (SD)
direct level gap at the home type, excess over matched genes (SD)
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8

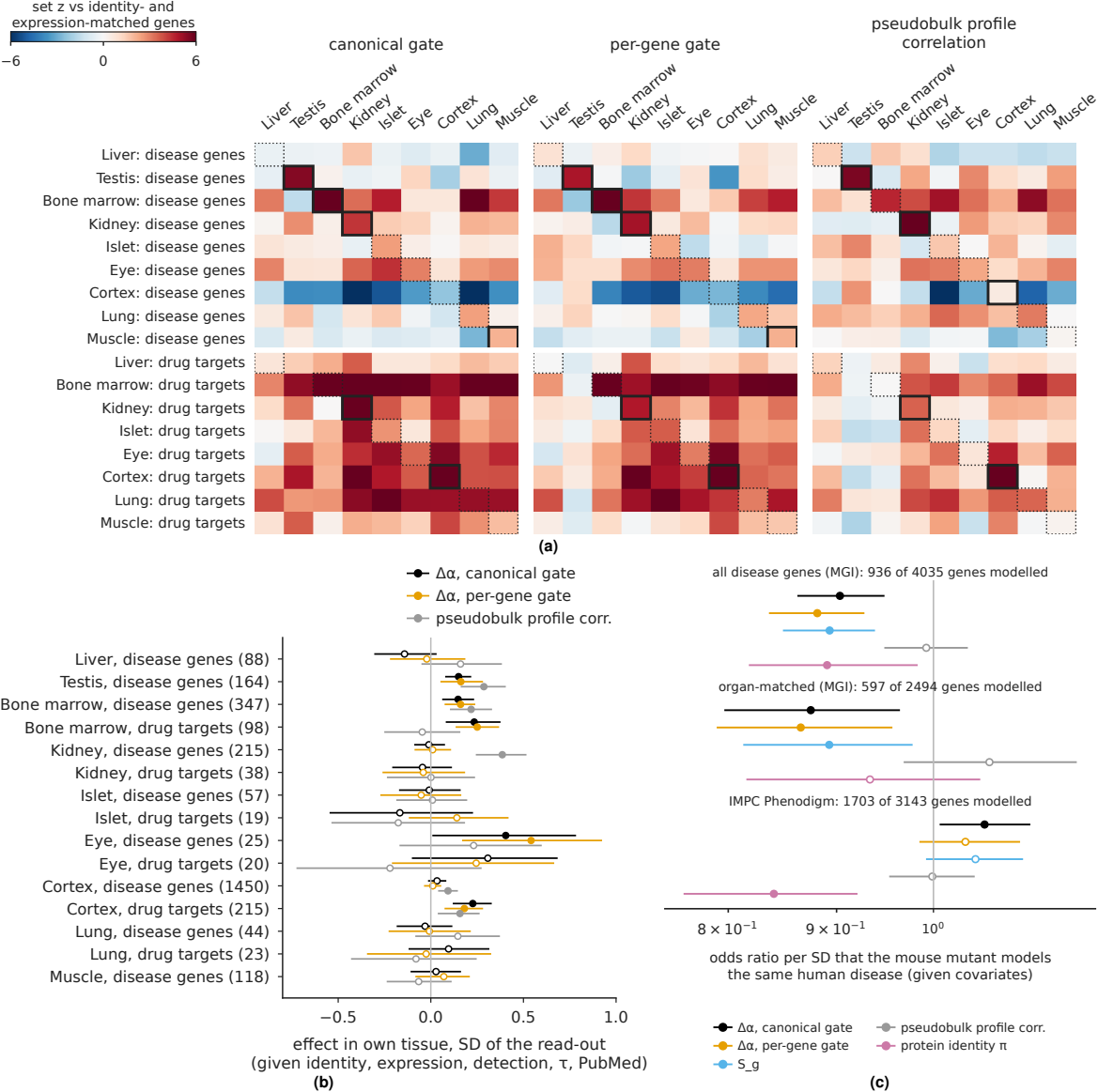
(b)

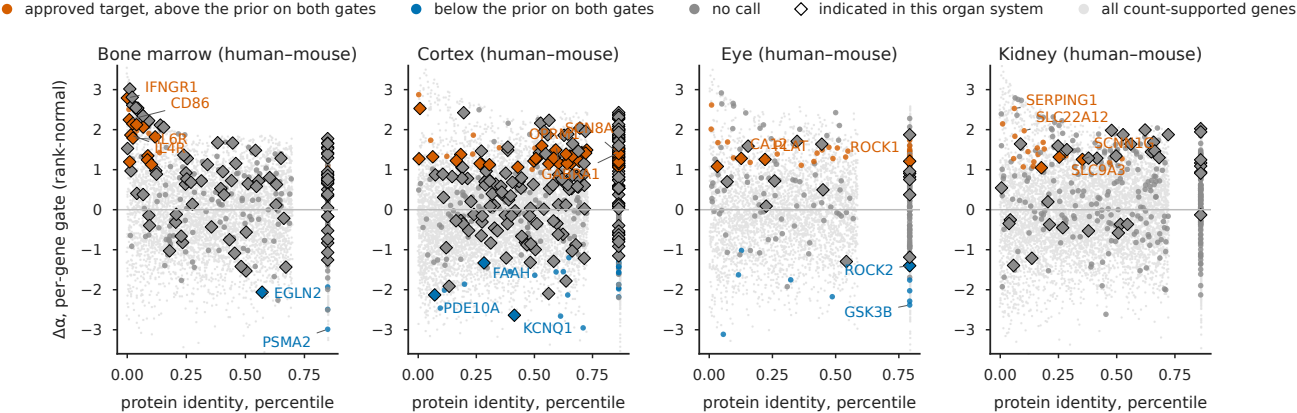

liver (4 models)
endothelial
stellate/fibroblast
macrophage
cholangiocyte
hepatocyte
0
1
excess (SD)
eye (6 models)
trabecular meshwork
ciliary muscle
Schwann cell
melanocyte
cornea
Schlemm canal
0.0
0.5
excess (SD)
cortex (7 models)
excitatory neuron
inhibitory neuron
oligodendrocyte
astrocyte
microglia
0.00
0.25
excess (SD)
testis (3 models)
spermatocyte
spermatid
Sertoli
spermatogonia
−0.2
0.2
excess (SD)
Δα, per-gene read-out
profile correlation (B1)
(a)
liver sinusoidal endothelium
HEG1
SHROOM4
PLXND1
ITPKB
TACC1
MECOM
B2M
H+Monkey
H+Mouse
H+Zebrafish
4 species
eye trabecular meshwork
APOD
MGP
SERPINF1
LSP1
SFRP4
MYOC
SDC2
H+Mouse
H+M. mul.
primates
5 species
H+M. fasc.
H+Pig
cortex inhibitory neurons
ANK3
ZNF804A
C8orf34
MACROD2
PARM1
SLC44A5
MYO16
M1 3 species
M1 H+Mouse
4 primates
H+Chimpanzee
H+Marmoset
H+Macaque, own
H+Macaque
cortex excitatory neurons
MLIP
KIAA1549L
TESPA1
GAP43
MAST4
ST6GAL2
PGLYRP1
M1 3 species
M1 H+Mouse
4 primates
H+Chimpanzee
H+Marmoset
H+Macaque, own
H+Macaque
3
2
1
0
−1
−2
−3
Δα per-gene, z
(b)


Departures grow with branch length
(cortex, 4 primates, one fit)
Chimera, per-gene
Chimera, canonical
direct pseudobulk
marmoset
macaque
chimp.
human
species-specific departures
(% of gene × cell type)
terminal branch length (My)
(a)
Departures replicate
in another region and lab
marmoset vs macaque
branch
human branch
chimp. branch
(control)
ρ with held-out MTG level gaps
(partial on expression)
(b)
Human-lineage genes (Ma et al. 2022):
the human row departs
Human
Chimpanzee
Macaque
Marmoset
per-cell count margin S_{gsk}
in the named cell type
FOXP2
microgl.
MELTF
PVALB
PKD2L1
L3-5 IT
CACNA1D
microgl.
DSCAM
microgl.
(c)

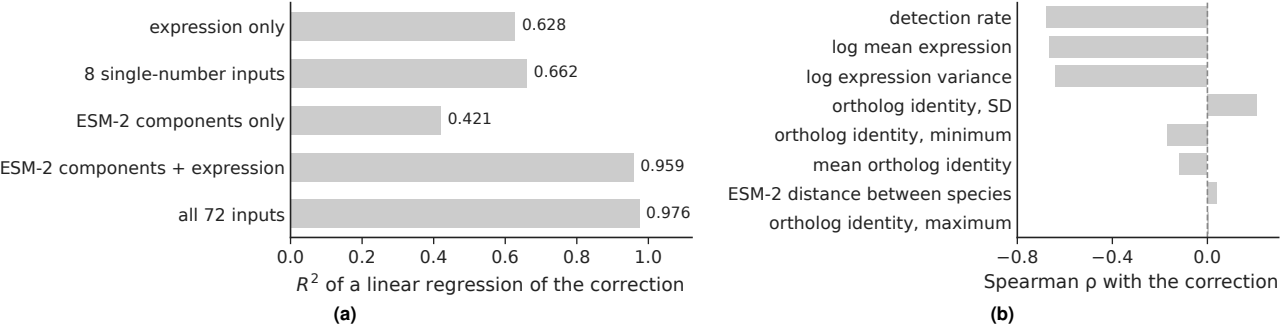

Protein and gene properties
same sign
Δα, canonical gate
Δα, per-gene gate
Pseudobulk profile corr.
Tissue specificity (GTEx τ)
Plasma membrane (HPA)
2R ohnolog
Signal peptide
Paralog count
Transmembrane
Cell junctions (HPA)
Secreted (HPA)
Haploinsufficiency (pHaplo)
CD marker
Immunity keyword
LoF tolerance (LOEUF)
Triplosensitivity (pTriplo)
Gene age (older)
Complex member
Common essential (DepMap)
Nuclear (HPA)
Essentiality (DepMap)
10/10
10/10
10/10
10/10
10/10
9/10
9/10
9/10
9/10
10/10
6/10
6/10
8/10
9/10
10/10
9/10
10/10
10/10
−0.2
−0.1
0.0
0.1
0.2
0.3
Partial Spearman with Δα (median over tissues)
(a)
Domain architecture
Calponin homology
Fibronectin III
FERM
SH3
7TM receptor
Homeodomain TF
PH
Tyrosine/S-T kinase
bHLH
SH2
RhoGAP
C2
DnaJ
F-box
USP deubiquitinase
KRAB
Helicase C
WD40
PHD finger
TPR
HECT ligase
RNA recognition motif
Bromodomain
signalling, cell surface
RNA, chromatin, proteostasis
−0.5
0.0
0.5
1.0
Residual Δα of family members (SD)
(b)
Duplication history and gene age
Residual Δα, per-gene gate (SD)
0.2
0.1
0.0
−0.1
−0.2
−0.3
2R ohnologs
Small-scale duplicates
Singletons
Tetra-pods+
Bony vert.
Jawed vert.
Verte-brates
Chor-dates
Bila-terians
Opistho-konts
Origin of the gene (phylostratum, older →)
(c)

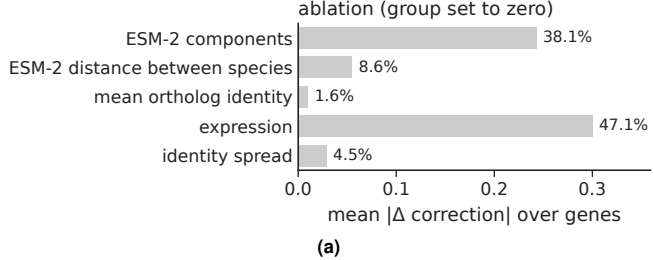

Regulatory and gene architecture
same sign
Distal enhancers in domain (ENCODE)
Regulatory domain size (GREAT)
Regulating TFs (DoRothEA)
Enhancer-gene links (ABC)
cCREs within 50 kb
H3K27me3 in ES cells
Bivalent promoter (NPC)
PRC2 target (ES)
Promoter phyloP
Conserved miRNA sites per kb 3'UTR
3'UTR length
Alternative promoters
Transcript isoforms
Coding phyloP
CpG-island promoter
Bidirectional promoter
Gene density (±500 kb)
Housekeeping gene
10/10
9/10
10/10
10/10
9/10
10/10
9/10
10/10
9/10
10/10
5/10
8/10
8/10
8/10
8/10
10/10
9/10
10/10
Δα, protein-feature gate
Δα, per-gene gate
pseudobulk profile correlation
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Partial Spearman with Δα (median over tissues)
(a)
TF regulons
lineage and signalling TFs
ZEB2
NR2F2
RBPJ
RUNX2
GATA6
IKZF1
AHR
KLF9
TCF7L2
ZFX
ZBTB33
NFYA
MBD1
FOXK2
NRF1
CpG-promoter TFs
−0.50
−0.25
0.00
0.25
0.50
0.75
Residual Δα of targets (SD), at equal TF in-degree
(b)
Regulatory input and deployment
Δα, protein-feature gate
Δα, per-gene gate
per-gene, equal cell-type τ
pseudobulk profile correlation
Residual Δα (SD)
0.2
0.1
0.0
−0.1
1
2
3
4
5
Distal enhancers in domain (quintile, fewest → most)
Regulating TFs (quintile, fewest → most)
(c)


set z vs identity- and expression-matched genes
−6
0
6
canonical gate
per-gene gate
pseudobulk profile correlation
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
Muscle
Liver: disease genes
Testis: disease genes
Bone marrow: disease genes
Kidney: disease genes
Islet: disease genes
Eye: disease genes
Cortex: disease genes
Lung: disease genes
Muscle: disease genes
Liver: drug targets
Bone marrow: drug targets
Kidney: drug targets
Islet: drug targets
Eye: drug targets
Cortex: drug targets
Lung: drug targets
Muscle: drug targets
(a)
Δα, canonical gate
Δα, per-gene gate
pseudobulk profile corr.
Liver, disease genes (88)
Testis, disease genes (164)
Bone marrow, disease genes (347)
Bone marrow, drug targets (98)
Kidney, disease genes (215)
Kidney, drug targets (38)
Islet, disease genes (57)
Islet, drug targets (19)
Eye, disease genes (25)
Eye, drug targets (20)
Cortex, disease genes (1450)
Cortex, drug targets (215)
Lung, disease genes (44)
Lung, drug targets (23)
Muscle, disease genes (118)
−0.5
0.0
0.5
1.0
effect in own tissue, SD of the read-out
(given identity, expression, detection, τ, PubMed)
(b)
all disease genes (MGI): 936 of 4035 genes modelled
organ-matched (MGI): 597 of 2494 genes modelled
IMPC Phenodigm: 1703 of 3143 genes modelled
8 × 10−1
9 × 10−1
100
odds ratio per SD that the mouse mutant models
the same human disease (given covariates)
Δα, canonical gate
Δα, per-gene gate
S_g
pseudobulk profile corr.
protein identity π
(c)

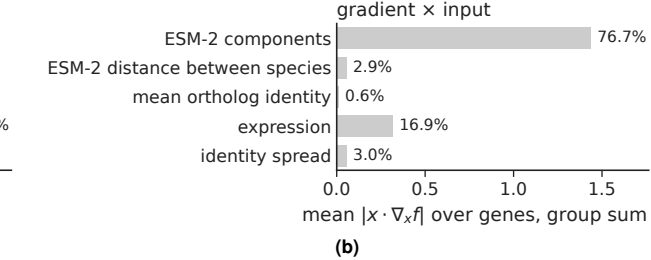

approved target, above the prior on both gates
below the prior on both gates
no call
indicated in this organ system
all count-supported genes
Bone marrow (human–mouse)
Cortex (human–mouse)
Eye (human–mouse)
Kidney (human–mouse)
Δα, per-gene gate (rank-normal)
protein identity, percentile
IFNGR1
CD86
IL6R
IL4R
EGLN2
PSMA2
CACNA1H
FAAH
PDE10A
KCNQ1
CA12
CLCN1
ROCK1
ROCK2
GSK3B
SERPING1
SLC22A12
SCNN1G
SLC9A3

(d)

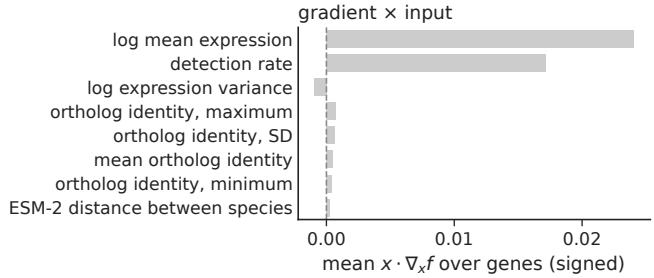

expression only
8 single-number inputs
ESM-2 components only
ESM-2 components + expression
all 72 inputs
0.628
0.662
0.421
0.959
0.976
0.0
0.2
0.4
0.6
0.8
1.0
R^2 of a linear regression of the correction
(a)
detection rate
log mean expression
log expression variance
ortholog identity, SD
ortholog identity, minimum
mean ortholog identity
ESM-2 distance between species
ortholog identity, maximum
−0.8
−0.4
0.0
Spearman ρ with the correction
(b)


ablation (group set to zero)
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
38.1%
8.6%
1.6%
47.1%
4.5%
0.0
0.1
0.2
0.3
mean |Δ correction| over genes

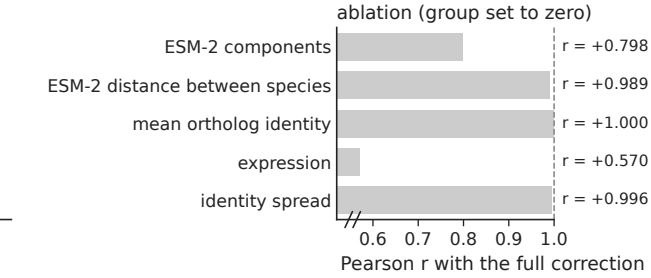
(a)


gradient × input
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
76.7%
2.9%
0.6%
16.9%
3.0%
0.0
0.5
1.0
1.5
mean |x · ∇ₓf| over genes, group sum

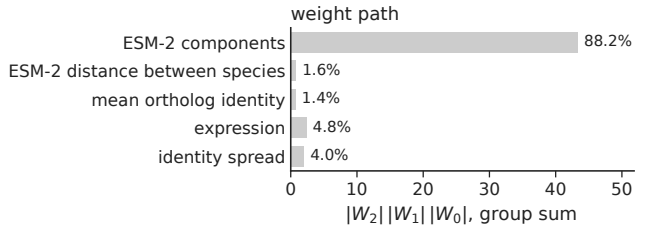
(b)


gradient × input
log mean expression
detection rate
log expression variance
ortholog identity, maximum
ortholog identity, SD
mean ortholog identity
ortholog identity, minimum
ESM-2 distance between species
0.00
0.01
0.02
mean x · ∇ₓf over genes (signed)

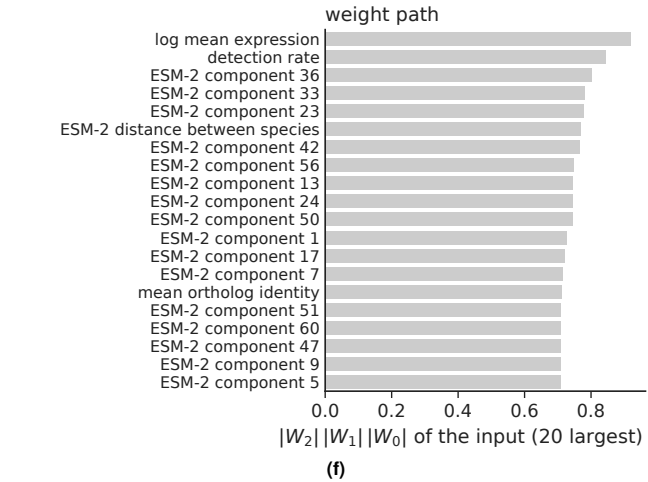
(c)


ablation (group set to zero)
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
r = +0.798
r = +0.989
r = +1.000
r = +0.570
r = +0.996
0.6
0.7
0.8
0.9
1.0
Pearson r with the full correction

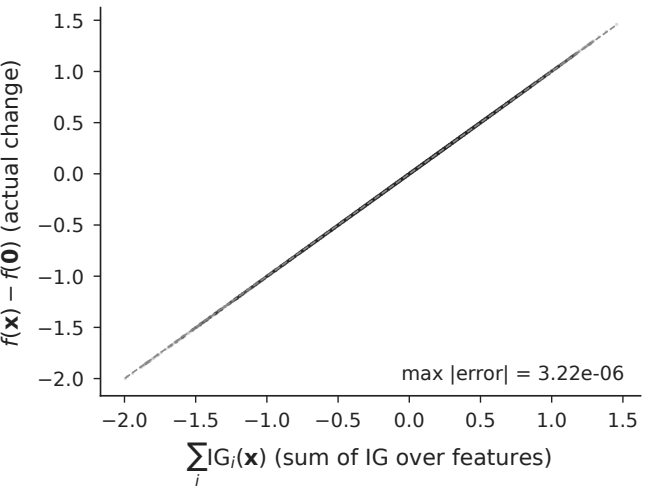
(d)


weight path
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
88.2%
1.6%
1.4%
4.8%
4.0%
0
10
20
30
40
50
|W₂| |W₁| |W₀|, group sum

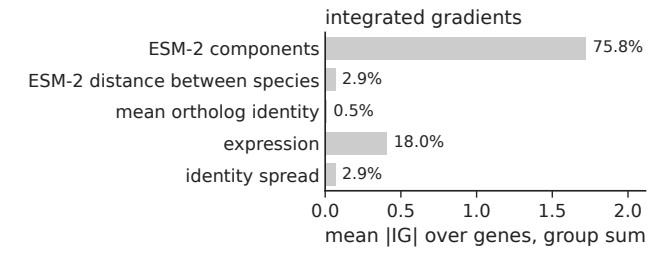
(e)


weight path
log mean expression
detection rate
ESM-2 component 36
ESM-2 component 33
ESM-2 component 23
ESM-2 distance between species
ESM-2 component 42
ESM-2 component 56
ESM-2 component 13
ESM-2 component 24
ESM-2 component 50
ESM-2 component 1
ESM-2 component 17
ESM-2 component 7
mean ortholog identity
ESM-2 component 51
ESM-2 component 60
ESM-2 component 47
ESM-2 component 9
ESM-2 component 5
0.0
0.2
0.4
0.6
0.8
|W₂| |W₁| |W₀| of the input (20 largest)

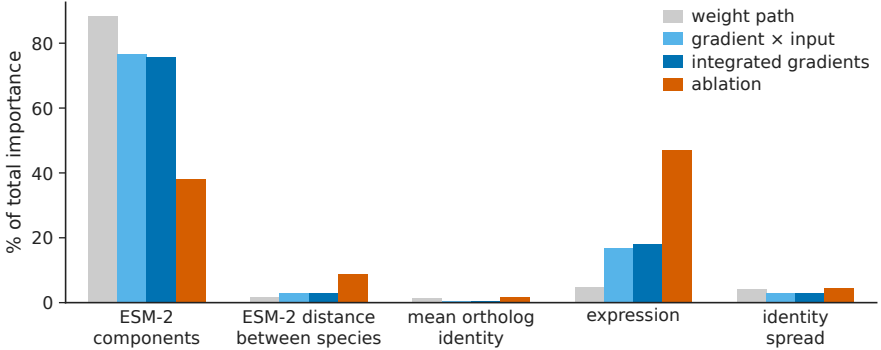
(f)


f(x) − f(0) (actual change)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
max |error| = 3.22e-06
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
∑ᵢ IGᵢ(x) (sum of IG over features)

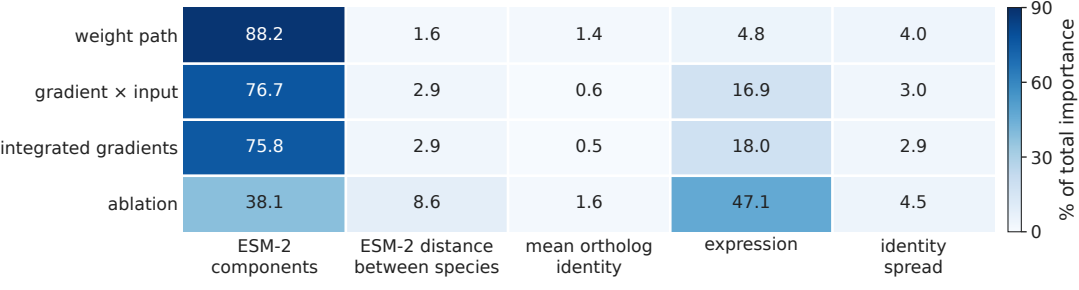
(g)


integrated gradients
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
75.8%
2.9%
0.5%
18.0%
2.9%
0.0
0.5
1.0
1.5
2.0
mean |IG| over genes, group sum

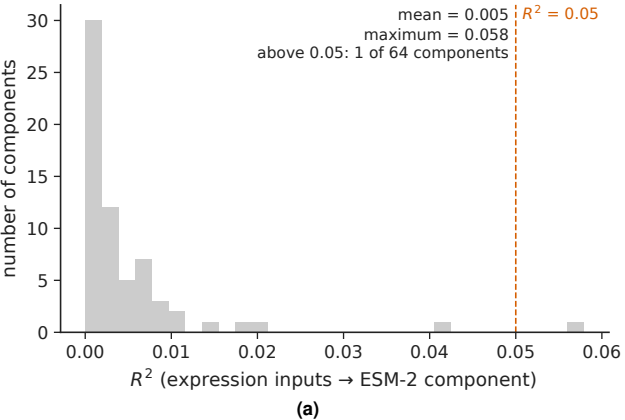
(h)


weight path
gradient × input
integrated gradients
ablation
% of total importance
0
20
40
60
80
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread

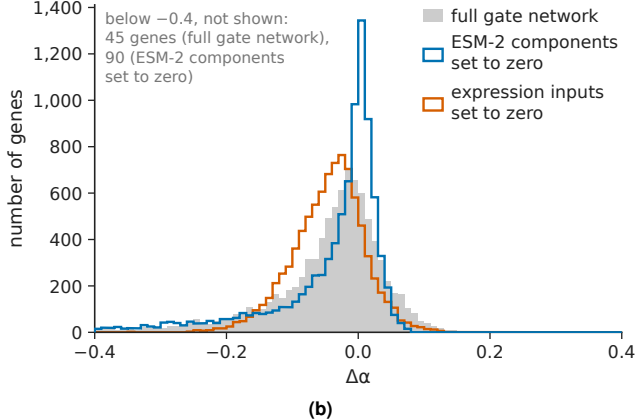
(a)


weight path
gradient × input
integrated gradients
ablation
88.2
1.6
1.4
4.8
4.0
76.7
2.9
0.6
16.9
3.0
75.8
2.9
0.5
18.0
2.9
38.1
8.6
1.6
47.1
4.5
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
90
60
30
0
% of total importance

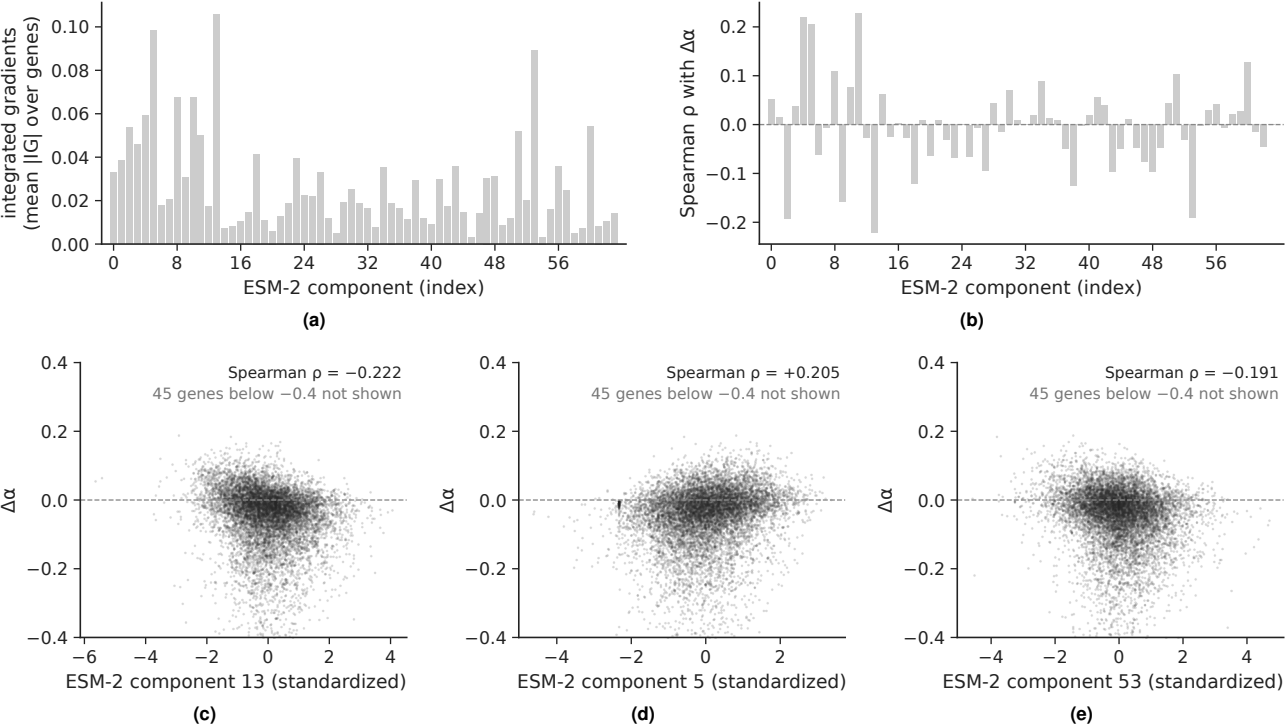
(b)

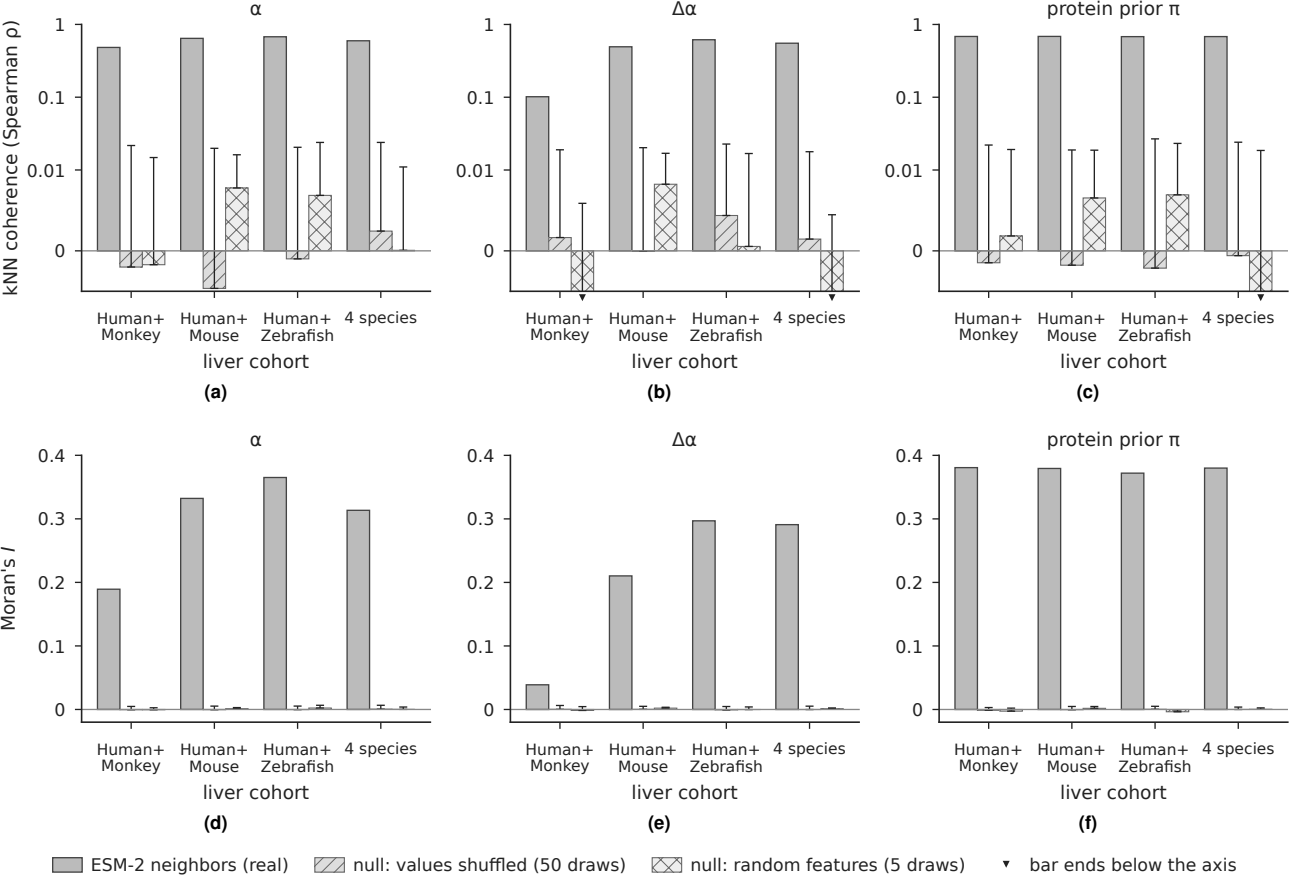

mean = 0.005
maximum = 0.058
above 0.05: 1 of 64 components
R^2 = 0.05
number of components
0
5
10
15
20
25
30
0.00
0.01
0.02
0.03
0.04
0.05
0.06
R^2 (expression inputs → ESM-2 component)

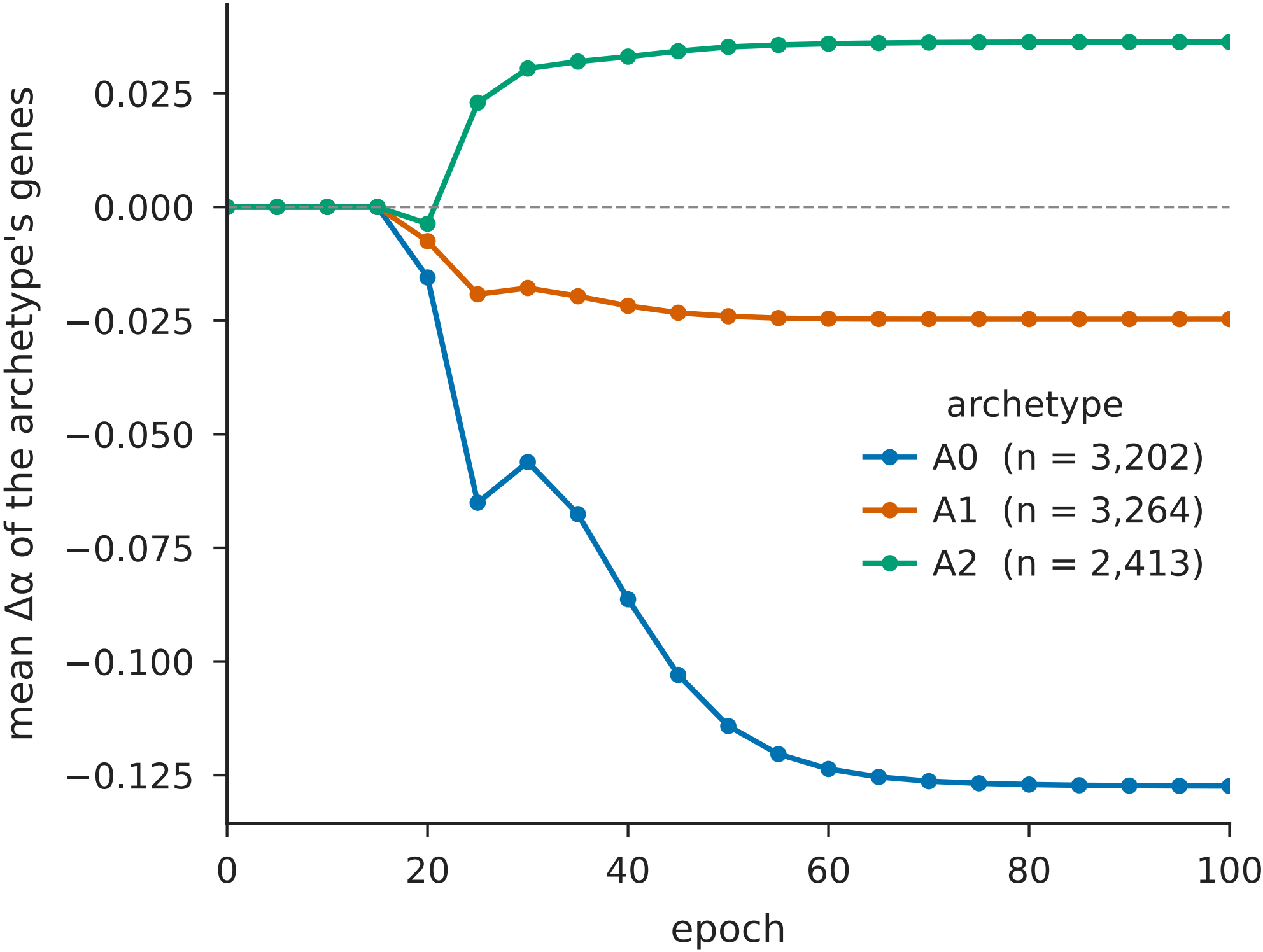
(a)

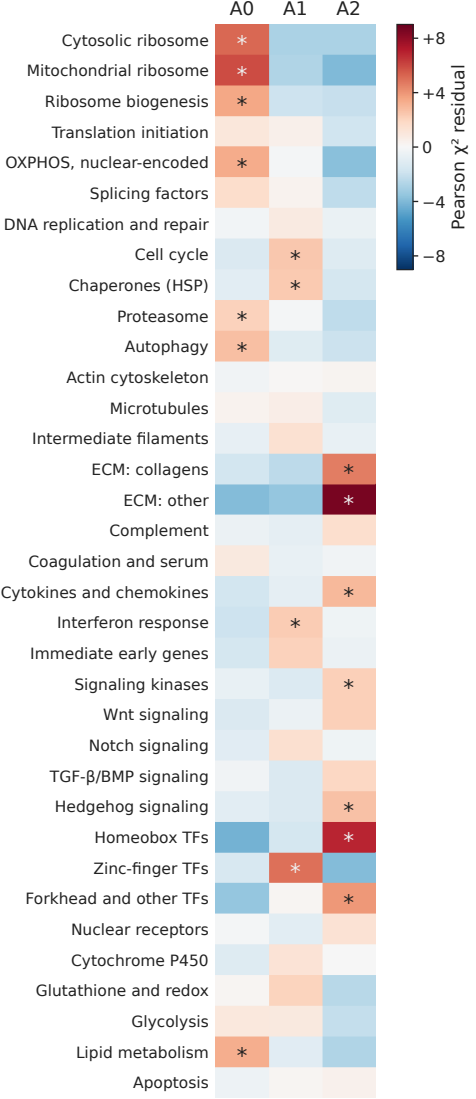

below −0.4, not shown:
45 genes (full gate network),
90 (ESM-2 components
set to zero)
full gate network
ESM-2 components
set to zero
expression inputs
set to zero
number of genes
0
200
400
600
800
1,000
1,200
1,400
−0.4
−0.2
0.0
0.2
0.4
Δα

(b)

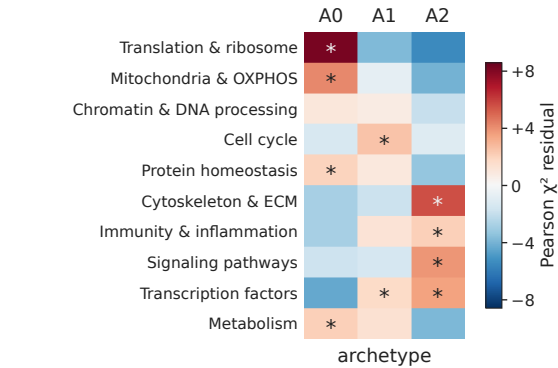

integrated gradients
(mean |IG| over genes)
0.10
0.08
0.06
0.04
0.02
0.00
0
8
16
24
32
40
48
56
ESM-2 component (index)
(a)
Spearman ρ with Δα
0.2
0.1
0.0
−0.1
−0.2
0
8
16
24
32
40
48
56
ESM-2 component (index)
(b)
Spearman ρ = −0.222
45 genes below −0.4 not shown
Δα
0.4
0.2
0.0
−0.2
−0.4
−6
−4
−2
0
2
4
ESM-2 component 13 (standardized)
(c)
Spearman ρ = +0.205
45 genes below −0.4 not shown
Δα
0.4
0.2
0.0
−0.2
−0.4
−4
−2
0
2
ESM-2 component 5 (standardized)
(d)
Spearman ρ = −0.191
45 genes below −0.4 not shown
Δα
0.4
0.2
0.0
−0.2
−0.4
−4
−2
0
2
4
ESM-2 component 53 (standardized)
(e)


α
Δα
protein prior π
kNN coherence (Spearman ρ)
1
0.1
0.01
0
Human+ Monkey
Human+ Mouse
Human+ Zebrafish
4 species
liver cohort
(a)
(b)
(c)
Moran's I
0.4
0.3
0.2
0.1
(d)
(e)
(f)
ESM-2 neighbors (real)
null: values shuffled (50 draws)
null: random features (5 draws)
bar ends below the axis

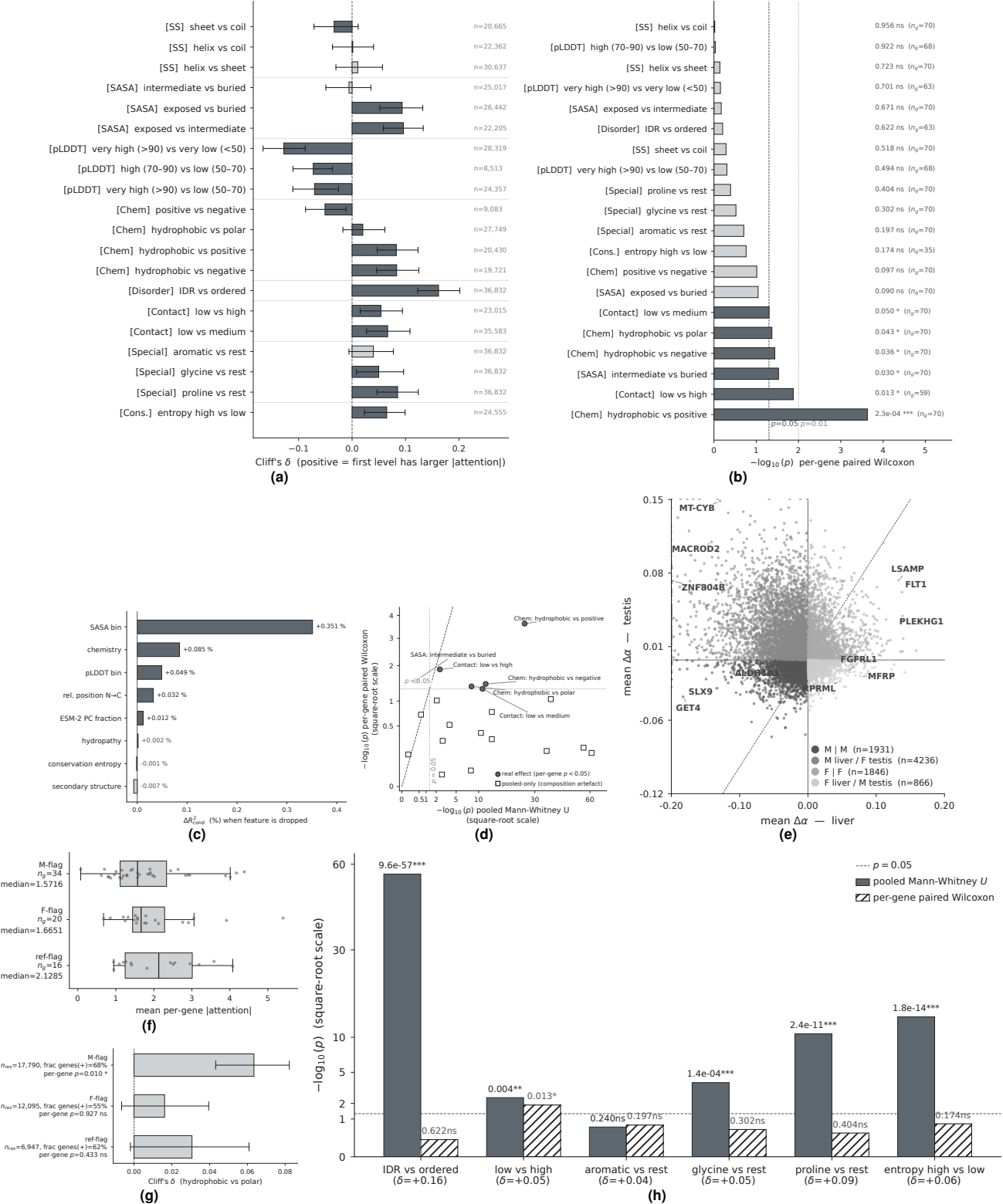

mean Δα of the archetype's genes
0.025
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
0
20
40
60
80
100
epoch
archetype
A0 (n = 3,202)
A1 (n = 3,264)
A2 (n = 2,413)

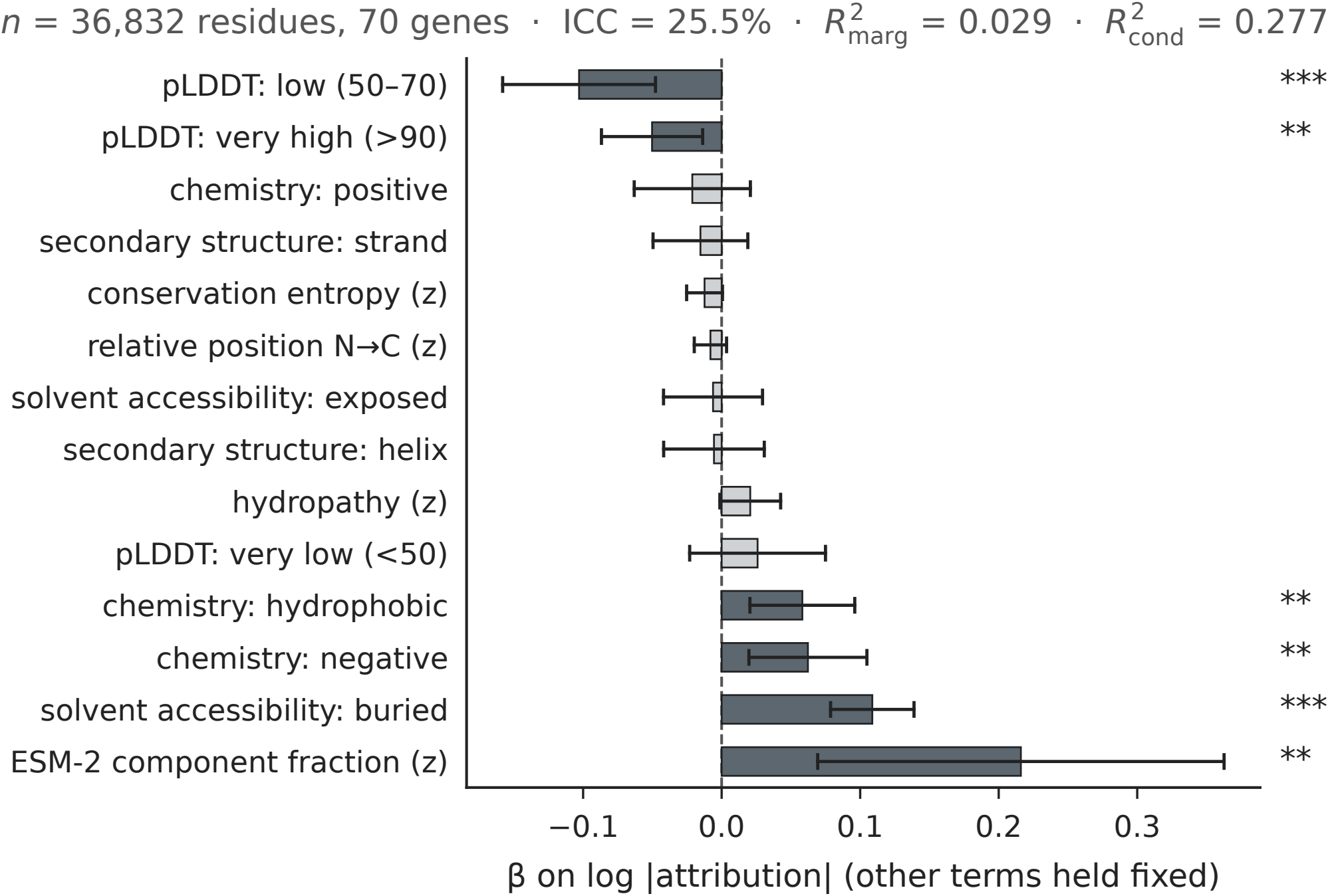

A0
A1
A2
Cytosolic ribosome
Mitochondrial ribosome
Ribosome biogenesis
Translation initiation
OXPHOS, nuclear-encoded
Splicing factors
DNA replication and repair
Cell cycle
Chaperones (HSP)
Proteasome
Autophagy
Actin cytoskeleton
Microtubules
Intermediate filaments
ECM: collagens
ECM: other
Complement
Coagulation and serum
Cytokines and chemokines
Interferon response
Immediate early genes
Signaling kinases
Wnt signaling
Notch signaling
TGF-β/BMP signaling
Hedgehog signaling
Homeobox TFs
Zinc-finger TFs
Forkhead and other TFs
Nuclear receptors
Cytochrome P450
Glutathione and redox
Glycolysis
Lipid metabolism
Apoptosis
archetype
+8
+4
0
−4
−8
Pearson χ² residual

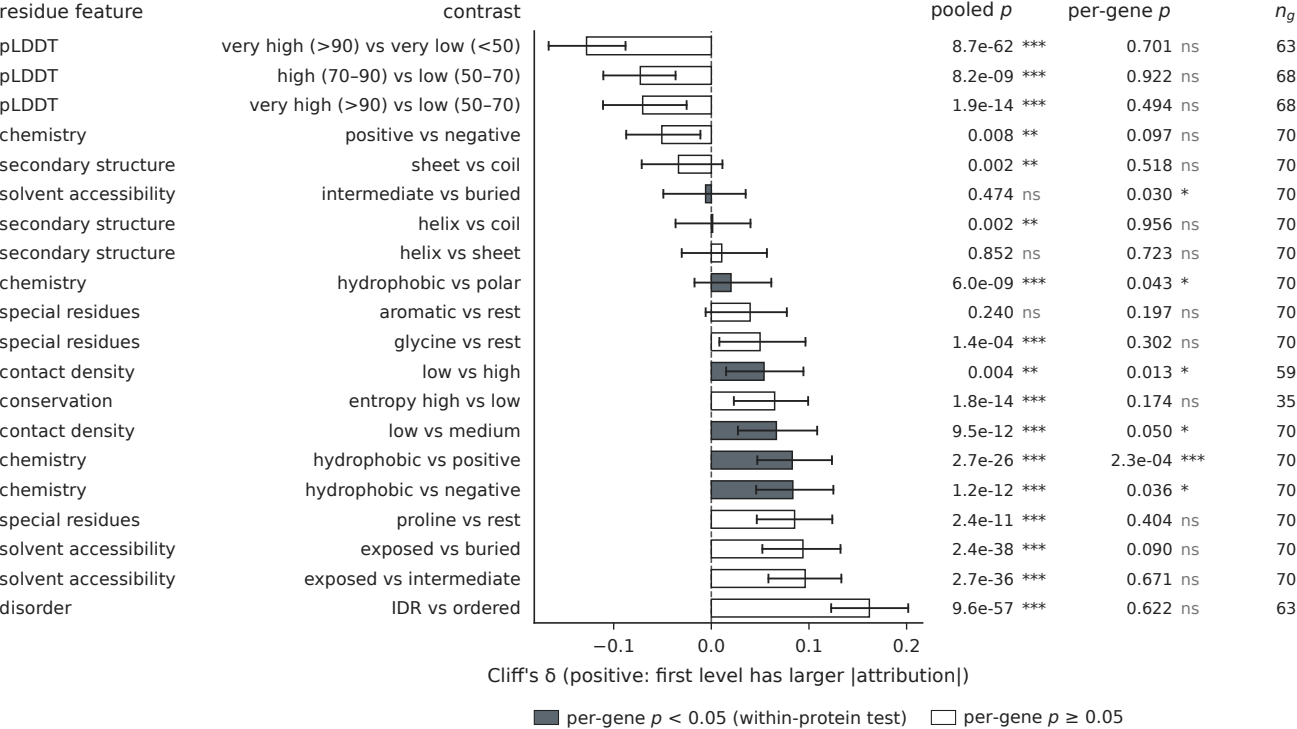
(a)

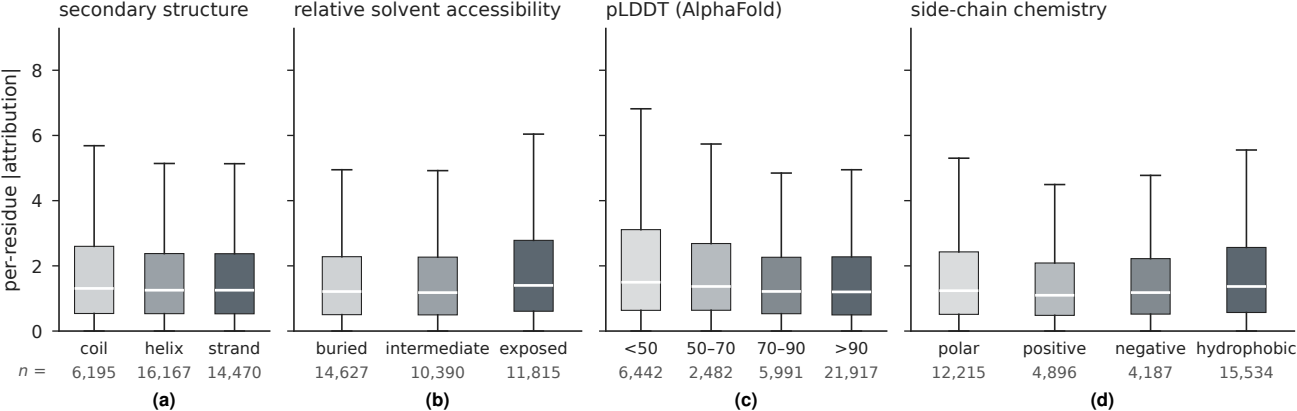

A0
A1
A2
Translation & ribosome
Mitochondria & OXPHOS
Chromatin & DNA processing
Cell cycle
Protein homeostasis
Cytoskeleton & ECM
Immunity & inflammation
Signaling pathways
Transcription factors
Metabolism
archetype
+8
+4
0
−4
−8
Pearson χ² residual

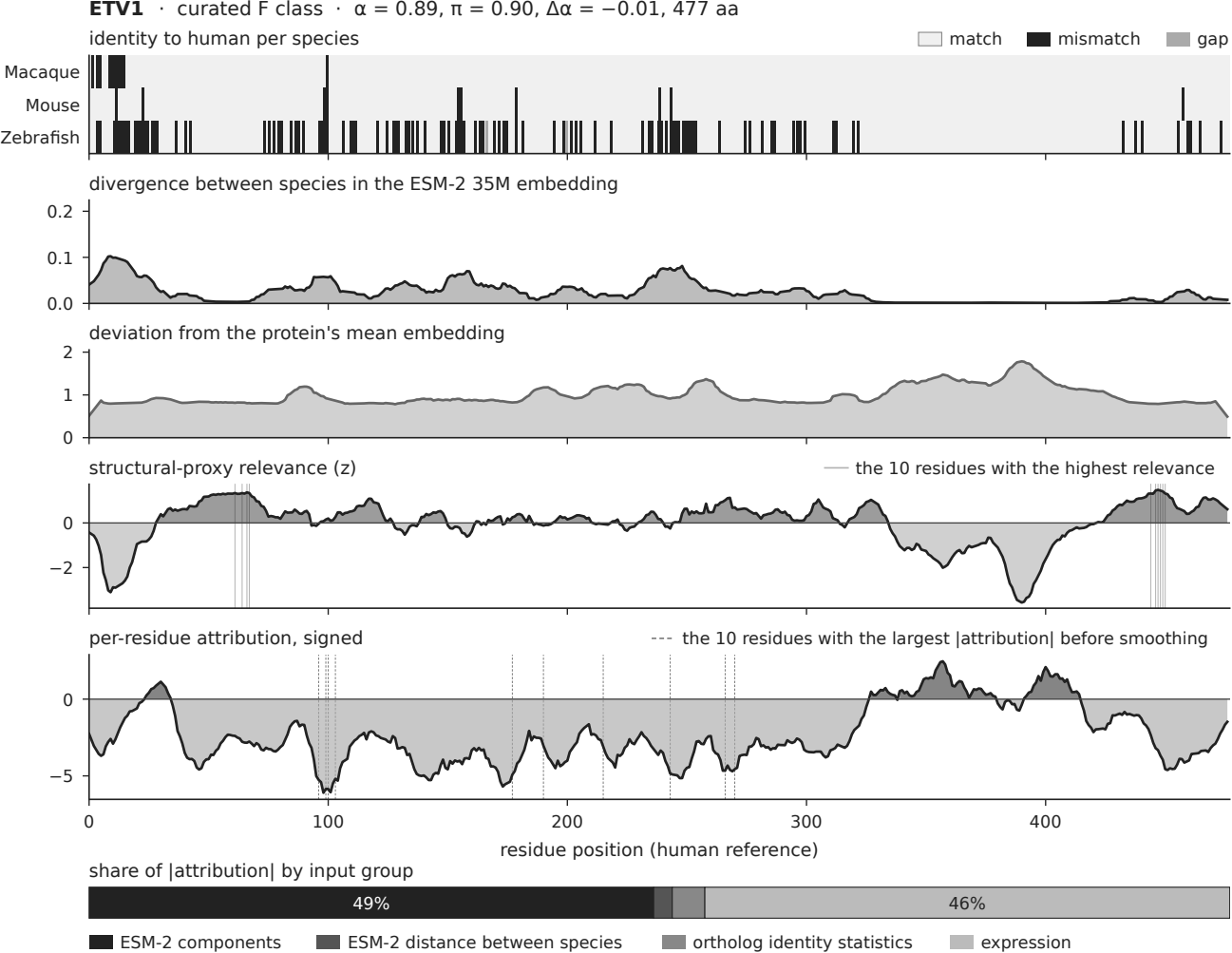
(b)

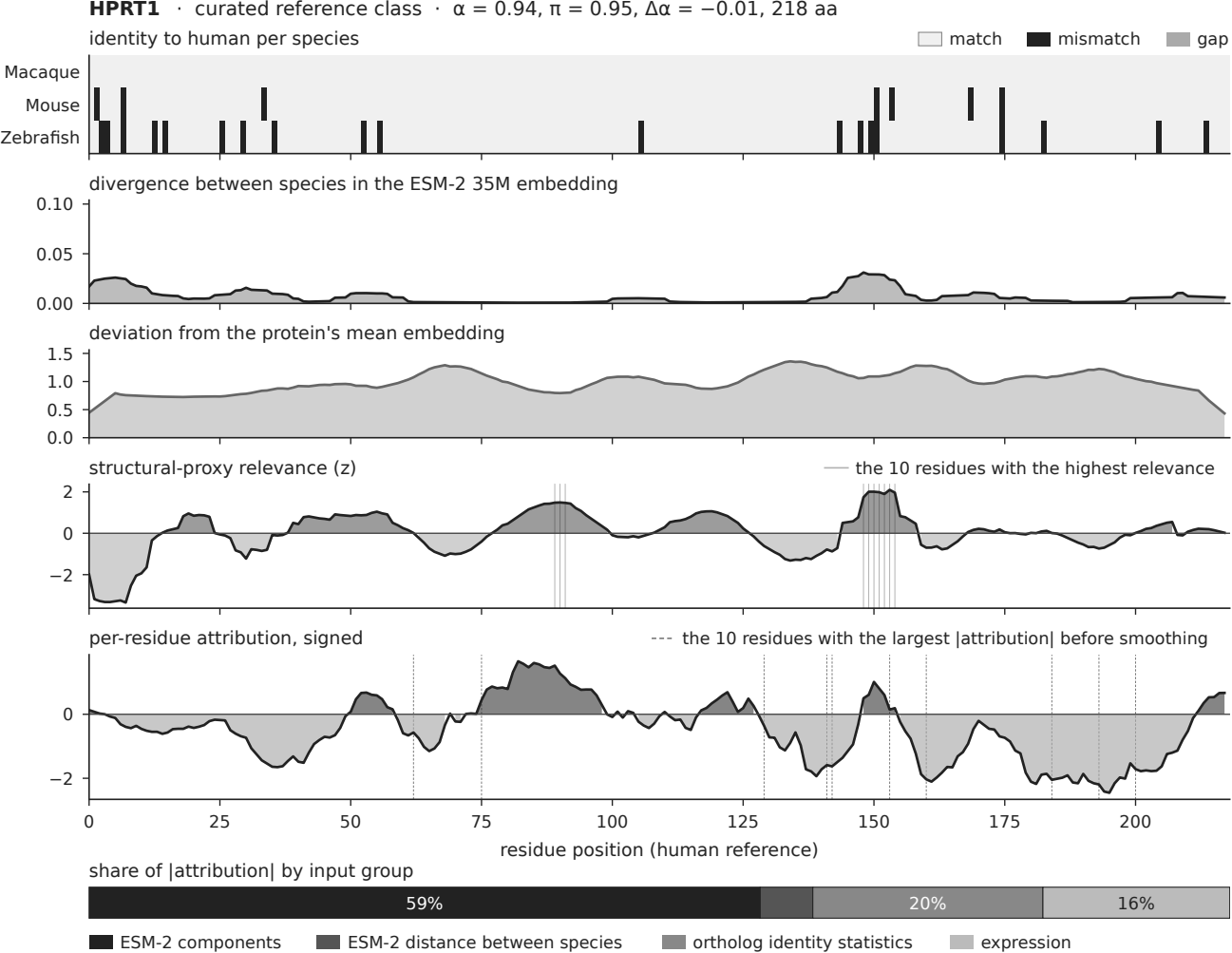

[SS] sheet vs coil
[SS] helix vs coil
[SS] helix vs sheet
[SASA] intermediate vs buried
[SASA] exposed vs buried
[SASA] exposed vs intermediate
[pLDDT] very high (>90) vs very low (<50)
[pLDDT] high (70–90) vs low (50–70)
[pLDDT] very high (>90) vs low (50–70)
[Chem] positive vs negative
[Chem] hydrophobic vs polar
[Chem] hydrophobic vs positive
[Chem] hydrophobic vs negative
[Disorder] IDR vs ordered
[Contact] low vs high
[Contact] low vs medium
[Special] aromatic vs rest
[Special] glycine vs rest
[Special] proline vs rest
[Cons.] entropy high vs low
n=20,665
n=22,362
n=30,637
n=25,017
n=26,442
n=22,205
n=28,319
n=8,513
n=24,357
n=9,083
n=27,749
n=20,430
n=19,721
n=36,832
n=23,015
n=35,583
n=36,832
n=36,832
n=36,832
n=24,555
−0.1
0.0
0.1
0.2
Cliff's δ (positive = first level has larger |attention|)
(a)
0.956 ns (n_g=70)
0.922 ns (n_g=68)
0.723 ns (n_g=70)
0.701 ns (n_g=63)
0.671 ns (n_g=70)
0.622 ns (n_g=63)
0.518 ns (n_g=70)
0.494 ns (n_g=68)
0.404 ns (n_g=70)
0.302 ns (n_g=70)
0.197 ns (n_g=70)
0.174 ns (n_g=35)
0.097 ns (n_g=70)
0.090 ns (n_g=70)
0.050 * (n_g=70)
0.043 * (n_g=70)
0.036 * (n_g=70)
0.030 * (n_g=70)
0.013 * (n_g=59)
2.3e-04 *** (n_g=70)
p=0.05 p=0.01
0
1
2
3
4
5
$-\log_{10}(p)$ per-gene paired Wilcoxon
(b)
SASA bin
chemistry
pLDDT bin
rel. position N→C
ESM-2 PC fraction
hydropathy
conservation entropy
secondary structure
+0.351 %
+0.085 %
+0.049 %
+0.032 %
+0.012 %
+0.002 %
-0.001 %
-0.007 %
0.0
0.1
0.2
0.3
0.4
ΔR^2_{cond} (%) when feature is dropped
(c)
Chem: hydrophobic vs positive
SASA: intermediate vs buried
Contact: low vs high
Chem: hydrophobic vs negative
Chem: hydrophobic vs polar
Contact: low vs medium
p = 0.05
p = 0.05
real effect (per-gene $p < 0.05$)
pooled-only (composition artefact)
0
0.5
1
2
3
4
0
0.5
1
2
5
10
30
60
$-\log_{10}(p)$ per-gene paired Wilcoxon (square-root scale)
$-\log_{10}(p)$ pooled Mann-Whitney U (square-root scale)
(d)
MT-CYB
MACROD2
ZNF804B
LSAMP
FLT1
PLEKHG1
FGFRL1
MFRP
ALDH1L1
RPRML
SLX9
GET4
M | M (n=1931)
M liver / F testis (n=4236)
F | F (n=1846)
F liver / M testis (n=866)
0.15
0.08
0.01
-0.06
-0.12
-0.20
-0.10
0.00
0.10
0.20
mean Δα — testis
mean Δα — liver
(e)
M-flag
n_g=34
median=1.5716
F-flag
n_g=20
median=1.6651
ref-flag
n_g=16
median=2.1285
0
1
2
3
4
5
mean per-gene |attention|
(f)
M-flag
n_{res}=17,790, frac genes(+)=68%
per-gene p=0.010 *
F-flag
n_{res}=12,095, frac genes(+)=55%
per-gene p=0.927 ns
ref-flag
n_{res}=6,947, frac genes(+)=62%
per-gene p=0.433 ns
0.00
0.02
0.04
0.06
0.08
Cliff's δ (hydrophobic vs polar)
(g)
$p = 0.05$
pooled Mann-Whitney U
per-gene paired Wilcoxon
9.6e-57***
0.622ns
0.004**
0.013*
0.240ns
0.197ns
1.4e-04***
0.302ns
2.4e-11***
0.404ns
1.8e-14***
0.174ns
60
30
10
5
2
1
0
$-\log_{10}(p)$ (square-root scale)
IDR vs ordered
(δ=+0.16)
low vs high
(δ=+0.05)
aromatic vs rest
(δ=+0.04)
glycine vs rest
(δ=+0.05)
proline vs rest
(δ=+0.09)
entropy high vs low
(δ=+0.06)
(h)

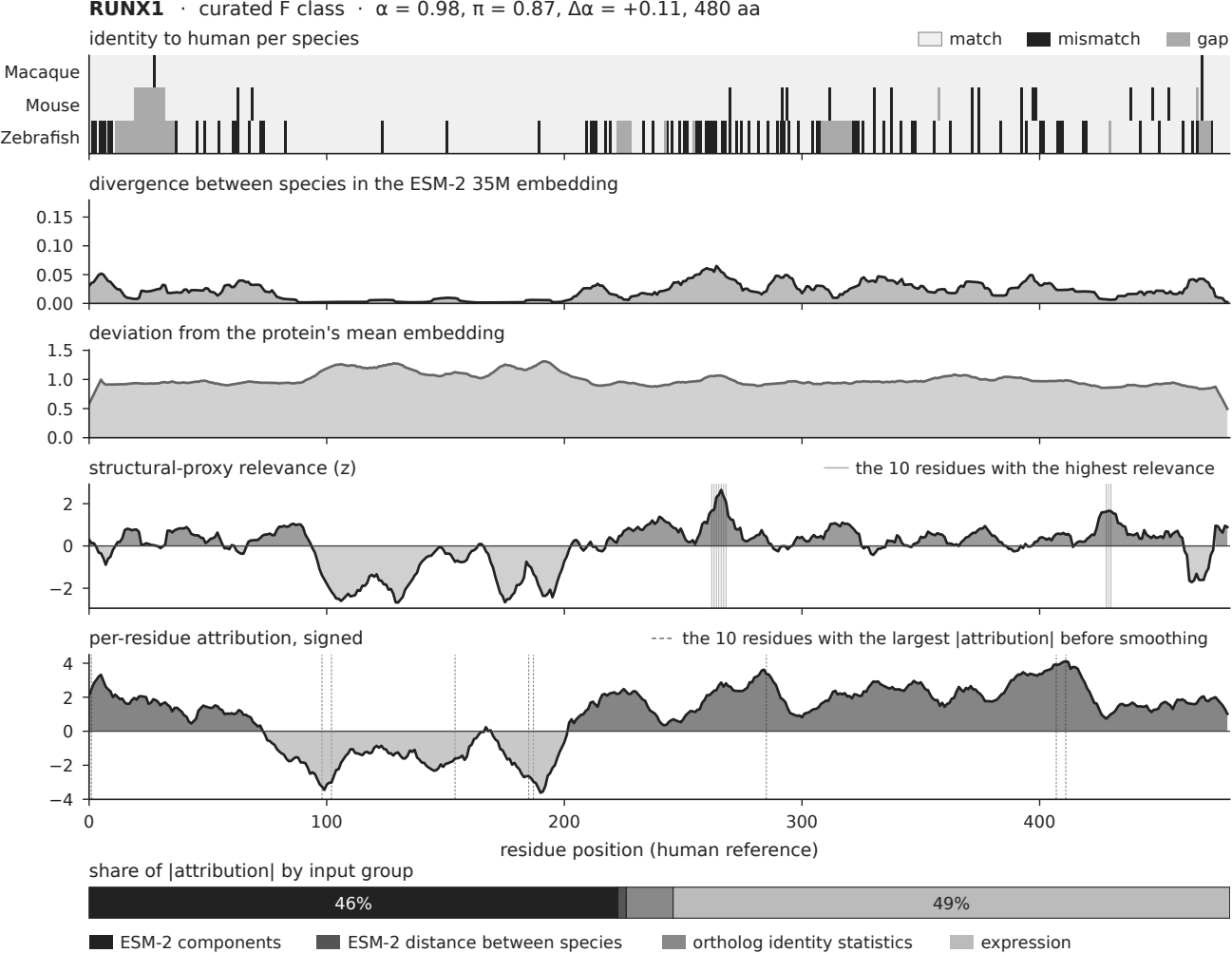

n = 36,832 residues, 70 genes · ICC = 25.5% · R^2_{marg} = 0.029 · R^2_{cond} = 0.277
pLDDT: low (50–70)
pLDDT: very high (>90)
chemistry: positive
secondary structure: strand
conservation entropy (z)
relative position N→C (z)
solvent accessibility: exposed
secondary structure: helix
hydropathy (z)
pLDDT: very low (<50)
chemistry: hydrophobic
chemistry: negative
solvent accessibility: buried
ESM-2 component fraction (z)
***
**
**
**
***
**
−0.1
0.0
0.1
0.2
0.3
β on log |attribution| (other terms held fixed)


residue feature
contrast
pooled p
per-gene p
n_g
pLDDT
very high (>90) vs very low (<50)
8.7e-62 ***
0.701 ns
63
pLDDT
high (70–90) vs low (50–70)
8.2e-09 ***
0.922 ns
68
pLDDT
very high (>90) vs low (50–70)
1.9e-14 ***
0.494 ns
68
chemistry
positive vs negative
0.008 **
0.097 ns
70
secondary structure
sheet vs coil
0.002 **
0.518 ns
70
solvent accessibility
intermediate vs buried
0.474 ns
0.030 *
70
secondary structure
helix vs coil
0.002 **
0.956 ns
70
secondary structure
helix vs sheet
0.852 ns
0.723 ns
70
chemistry
hydrophobic vs polar
6.0e-09 ***
0.043 *
70
special residues
aromatic vs rest
0.240 ns
0.197 ns
70
special residues
glycine vs rest
1.4e-04 ***
0.302 ns
70
contact density
low vs high
0.004 **
0.013 *
59
conservation
entropy high vs low
1.8e-14 ***
0.174 ns
35
contact density
low vs medium
9.5e-12 ***
0.050 *
70
chemistry
hydrophobic vs positive
2.7e-26 ***
2.3e-04 ***
70
chemistry
hydrophobic vs negative
1.2e-12 ***
0.036 *
70
special residues
proline vs rest
2.4e-11 ***
0.404 ns
70
solvent accessibility
exposed vs buried
2.4e-38 ***
0.090 ns
70
solvent accessibility
exposed vs intermediate
2.7e-36 ***
0.671 ns
70
disorder
IDR vs ordered
9.6e-57 ***
0.622 ns
63
−0.1
0.0
0.1
0.2
Cliff's δ (positive: first level has larger |attribution|)
per-gene $p < 0.05$ (within-protein test)
per-gene $p \geq 0.05$


secondary structure
relative solvent accessibility
pLDDT (AlphaFold)
side-chain chemistry
per-residue |attribution|
0
2
4
6
8
coil
helix
strand
buried
intermediate
exposed
<50
50–70
70–90
>90
polar
positive
negative
hydrophobic
n =
6,195
16,167
14,470
14,627
10,390
11,815
6,442
2,482
5,991
21,917
12,215
4,896
4,187
15,534
(a)
(b)
(c)
(d)


ETV1 · curated F class · α = 0.89, π = 0.90, Δα = −0.01, 477 aa
identity to human per species
match
mismatch
gap
Macaque
Mouse
Zebrafish
divergence between species in the ESM-2 35M embedding
0.2
0.1
0.0
deviation from the protein's mean embedding
2
1
0
structural-proxy relevance (z)
the 10 residues with the highest relevance
0
−2
per-residue attribution, signed
the 10 residues with the largest |attribution| before smoothing
0
−5
0
100
200
300
400
residue position (human reference)
share of |attribution| by input group
49%
46%
ESM-2 components
ESM-2 distance between species
ortholog identity statistics
expression


HPRT1 · curated reference class · α = 0.94, π = 0.95, Δα = −0.01, 218 aa
identity to human per species
match
mismatch
gap
Macaque
Mouse
Zebrafish
divergence between species in the ESM-2 35M embedding
0.10
0.05
0.00
deviation from the protein's mean embedding
1.5
1.0
0.5
0.0
structural-proxy relevance (z)
the 10 residues with the highest relevance
2
0
−2
per-residue attribution, signed
the 10 residues with the largest |attribution| before smoothing
0
−2
0
25
50
75
100
125
150
175
200
residue position (human reference)
share of |attribution| by input group
59%
20%
16%
ESM-2 components
ESM-2 distance between species
ortholog identity statistics
expression


RUNX1 · curated F class · α = 0.98, π = 0.87, Δα = +0.11, 480 aa
identity to human per species
match
mismatch
gap
Macaque
Mouse
Zebrafish
divergence between species in the ESM-2 35M embedding
0.15
0.10
0.05
0.00
deviation from the protein's mean embedding
1.5
1.0
0.5
0.0
structural-proxy relevance (z)
the 10 residues with the highest relevance
2
0
−2
per-residue attribution, signed
the 10 residues with the largest |attribution| before smoothing
4
2
0
−2
−4
0
100
200
300
400
residue position (human reference)
share of |attribution| by input group
46%
49%
ESM-2 components
ESM-2 distance between species
ortholog identity statistics
expression